

Breeze LES master report

Report updated: 2026-09-23T06:12:09.348 UTC. Original GABLS1 collection checked: 2026-09-22T03:08:26.728Z

DYCOMS: 15 completed cases. GABLS1: 15 of 15 planned cases have completed, audited local results.

This document combines the completed DYCOMS RF01 experiment (including the Pressel comparison) with the GABLS1 experiment. The GABLS run order is smallest grids first; incomplete runs contribute status information only, never scientific curves.

Read finite-output completion separately from physical fidelity: all three unbounded DYCOMS Centered2 cases developed negative total water. The DYCOMS chapter retains that audit.

GABLS1 uses a fixed 400 m cube and 9 h of surface cooling and geostrophic forcing. Its no-interior-closure configurations retain MOST wall stress and heat exchange. The question is whether numerical choices preserve resolved shear-driven turbulence on coarser grids.

Contents: Part I - completed DYCOMS and Pressel comparison; Part II - GABLS setup, progress, numerical references, then audited results as they arrive.

Part III adds GABLS3: all four corrected coarse cases have completed and passed export and physical-flux audits. All prior DYCOMS and GABLS1 information is retained. The requested persistent evaluation repository is [glwagner/BreezeEvaluation.jl](https://glwagner.github.io/BreezeEvaluation.jl).

Part II includes four matched SurfaceLayerDiffusivity GABLS1 runs at 12.5 m, plus the earlier 12.5 m Smagorinsky run for context. SLD strongly suppresses near-wall turbulence without improving the profile measures tested; Smagorinsky has almost no resolved vertical turbulence despite a smaller mean-wind error. Corrected native flux diagnostics show that SGS mixing carries 86–95% of first-level u-momentum transport. Historical flux exclusions remain attached to the original attempts. Part IV explains the mechanism and numerical validation.

New factor-3 comparison: the added mixing also largely switches off. Final-hour first-face variance is $0.06073 \text{ m}^2/\text{s}^2$ versus 0.00817 for factor 1; sampled zero-viscosity fraction is 98.47%. New plots show only factors 1 and 3, with fixed 1 m references. Earlier factor-2/10 studies remain in their historical sections.

A matched GABLS1 scheme-native WENO flux run directly measured the numerical transport correction. In the final hour its mean first-face u-momentum correction was 2.95% of covariance, while the heat correction was effectively zero. The new Julia figures compare both final-hour windows and retain the fixed 1 m LES median.

Breeze DYCOMS-II RF01: resolution and numerics

Complete: 15 cases analyzed

Data checked: 2026-09-19T11:58:17.096Z

All 15 requested four-hour cases are analyzed. This matrix does not establish faithful low-resolution DYCOMS turbulence: the four coarse WENO cases retain the wrong third-moment sign in the 600–760 m layer, and refining WENO9/no-SGS from 35 m to 10 m horizontal spacing substantially improves the third moment despite little improvement in variance-marker error. Fine WENO9/no-SGS has the smallest third-moment marker RMSE among the tested WENO cases. All three unbounded Centered2 cases develop negative total water and cannot establish physical fidelity. Cloud persistence alone is therefore insufficient. These are single-seed comparisons; the fine grid is not a proven converged solution, and the larger coarse domain confounds the resolution comparison.

Completed Centered2 runs contain physically inadmissible negative total water (sampled minima: coarse: -4.338 g/kg; canonical: -7.697 g/kg; fine: -7.401 g/kg). Their cloud and turbulence statistics are retained as diagnostics of this unbounded numerical configuration, not evidence of a physically faithful solution. The production extrema figure shows the full sampled time history; verification here means complete finite output, not physical validity.

Centered2 smoke audit: all three grids stayed finite over the first 600 simulated seconds, but canonical and fine cases produced negative total water. Across the 11 one-minute samples, minimum q_t was coarse: 1.0872 g/kg; canonical: -0.0605 g/kg; fine: -1.3679 g/kg. Passing the finite-value gate establishes neither moisture positivity nor physical validity. Production retains unbounded centered moisture; full-run extrema are reported separately in the production audit. These are smoke-test results, not completed four-hour Centered2 statistics.

At 10 m horizontal / 5 m vertical spacing with WENO9, adding Smagorinsky changes fourth-hour minimum w'^3 over 600–760 m from -0.0622 to -0.0304 m^3/s^3 , third-moment marker RMSE from 0.0159 to 0.0313 m^3/s^3 , and variance-marker RMSE from 0.0631 to 0.0769 m^2/s^2 . LWP changes from 55.48 to 53.23 g/m². Solver wall increases by 10.1%. This closure weakens the negative third moment and increases both marker errors in this paired single-realization comparison; it is not evidence against every possible closure or coefficient choice.

Breeze DYCOMS-II RF01: resolution and numerics

At 10 m horizontal / 5 m vertical resolution without SGS, WENO5 uses 70.5 solver wall minutes versus 92.8 for WENO9 (24.0% less). However, its fourth-hour third-moment RMSE against digitized in-situ markers is 0.0310 rather than 0.0159 m^3/s^3 . Its minimum w'^3 in 600–760 m is -0.0372 versus -0.0622 m^3/s^3 . Lower advection order saves wall time but weakens the negative third moment in this realization. Timings include output within the solve and exclude startup/compilation before the solve; they are not isolated kernel benchmarks.

Fine Centered2 with Smagorinsky completes in 70.2 solver wall minutes, 31.3% less than fine WENO9 with Smagorinsky. Its fourth-hour LWP is 44.17 versus 53.23 g/m^2 , and third-moment marker RMSE is 0.0382 versus 0.0313 m^3/s^3 . Negative total water prevents interpreting this cheaper run as an acceptable physical solution. Timing includes diagnostics and differing timestep counts; no GPU profiling was performed to identify the cost mechanism.

At fixed 5 m vertical spacing and fixed horizontal domain, changing WENO9/no-SGS from 35 m to 10 m horizontal spacing changes the fourth-hour minimum w'^3 over 600–760 m from -0.0126 to -0.0622 m^3/s^3 . The fine case is negative throughout 600–760 m, agreeing with the sign of the retained in-situ observations there.

For this same 35 m to 10 m comparison, unweighted RMSE against the digitized in-situ third-moment markers changes from 0.0454 to 0.0159 m^3/s^3 ; variance-marker RMSE changes from 0.0606 to 0.0631 m^2/s^2 . The third moment improves substantially, while variance does not improve by this metric. These are single realizations without observational error-bar weighting, not a convergence demonstration.

All 4 completed coarse WENO cases have positive fourth-hour w'^3 throughout 600–760 m, while the retained in-situ observations there are negative. Cloud persistence therefore does not establish faithful coarse-grid turbulence.

Coarse, WENO9 without closure: fourth-hour LWP 49.87 g/m^2 , cloud fraction 99.34%, peak w^2 0.395 m^2/s^2 , and Δq : 0.019 g/kg .

At coarse resolution without closure, WENO9 minus WENO5 changes fourth-hour LWP by 3.42 g/m^2 and peak w^2 by 0.099 m^2/s^2 . These are paired single-realization differences, not statistically established scheme effects.

At coarse resolution with WENO9, adding Smagorinsky changes fourth-hour LWP by -6.54 g/m^2 and peak w^2 by -0.038 m^2/s^2 . Third-moment marker RMSE changes from 0.0704 to 0.0595 m^3/s^3 .

Breeze DYCOMS-II RF01: resolution and numerics

Canonical, WENO9 without closure: fourth-hour LWP 61.69 g/m², cloud fraction 99.91%, peak w^2 0.521 m²/s², and Δq_t 0.003 g/kg.

At canonical resolution without closure, WENO9 minus WENO5 changes fourth-hour LWP by 2.09 g/m² and peak w^2 by 0.054 m²/s². These are paired single-realization differences, not statistically established scheme effects.

At canonical resolution with WENO9, adding Smagorinsky changes fourth-hour LWP by -9.50 g/m² and peak w^2 by -0.067 m²/s². Third-moment marker RMSE changes from 0.0454 to 0.0444 m³/s³.

Fine, WENO9 without closure: fourth-hour LWP 55.48 g/m², cloud fraction 99.95%, peak w^2 0.472 m²/s², and Δq_t 0.031 g/kg.

At fine resolution without closure, WENO9 minus WENO5 changes fourth-hour LWP by -2.06 g/m² and peak w^2 by -0.011 m²/s². These are paired single-realization differences, not statistically established scheme effects.

At fine resolution with WENO9, adding Smagorinsky changes fourth-hour LWP by -2.25 g/m² and peak w^2 by -0.027 m²/s². Third-moment marker RMSE changes from 0.0159 to 0.0313 m³/s³.

Coarse, Centered2 with Smagorinsky: sampled global q_t extrema over 0–4 h were -4.3385 to 15.0839 g/kg; 227 of 241 samples contained negative moisture. These are 60-second samples, not bounds on every timestep.

At coarse resolution with Smagorinsky, Centered2 minus WENO9 changes fourth-hour LWP by 5.52 g/m² and peak w^2 by 0.053 m²/s². Centered moisture is unbounded; this changes scheme family and bounding as well as formal order.

At coarse resolution with Smagorinsky, Centered2 minus WENO5 changes fourth-hour LWP by 6.59 g/m² and peak w^2 by 0.095 m²/s². Centered moisture is unbounded; this changes scheme family and bounding as well as formal order.

Canonical, Centered2 with Smagorinsky: sampled global q_t extrema over 0–4 h were -7.6972 to 17.4041 g/kg; 231 of 241 samples contained negative moisture. These are 60-second samples, not bounds on every timestep.

At canonical resolution with Smagorinsky, Centered2 minus WENO9 changes fourth-hour LWP by -16.49 g/m² and peak w^2 by -0.027 m²/s². Centered moisture is unbounded; this changes scheme family and bounding as well as formal order.

Breeze DYCOMS-II RF01: resolution and numerics

At canonical resolution with Smagorinsky, Centered2 minus WENO5 changes fourth-hour LWP by -15.33 g/m^2 and peak w^2 by $-0.006 \text{ m}^2/\text{s}^2$. Centered moisture is unbounded; this changes scheme family and bounding as well as formal order.

Fine, Centered2 with Smagorinsky: sampled global q_t extrema over 0–4 h were -7.4007 to 18.0746 g/kg ; 234 of 241 samples contained negative moisture. These are 60-second samples, not bounds on every timestep.

At fine resolution with Smagorinsky, Centered2 minus WENO9 changes fourth-hour LWP by -9.06 g/m^2 and peak w^2 by $-0.072 \text{ m}^2/\text{s}^2$. Centered moisture is unbounded; this changes scheme family and bounding as well as formal order.

At fine resolution with Smagorinsky, Centered2 minus WENO5 changes fourth-hour LWP by -7.54 g/m^2 and peak w^2 by $-0.082 \text{ m}^2/\text{s}^2$. Centered moisture is unbounded; this changes scheme family and bounding as well as formal order.

Fourth-hour metrics

Coarse / WENO9 / no SGS: LWP 49.87 g/m²; cloud 99.34%; peak w^2 0.395 m²/s²; Δq_t 0.019 g/kg; E 4.75 mm/s.

Coarse / WENO5 / no SGS: LWP 46.45 g/m²; cloud 98.91%; peak w^2 0.296 m²/s²; Δq_t 0.015 g/kg; E 4.70 mm/s.

Coarse / WENO9 / Smagorinsky: LWP 43.33 g/m²; cloud 99.07%; peak w^2 0.356 m²/s²; Δq_t 0.080 g/kg; E 5.21 mm/s.

Coarse / WENO5 / Smagorinsky: LWP 42.27 g/m²; cloud 98.52%; peak w^2 0.314 m²/s²; Δq_t 0.047 g/kg; E 4.93 mm/s.

Coarse / Centered2 / Smagorinsky: LWP 48.85 g/m²; cloud 99.44%; peak w^2 0.409 m²/s²; Δq_t 0.214 g/kg; E 7.27 mm/s.

Canonical / WENO9 / no SGS: LWP 61.69 g/m²; cloud 99.91%; peak w^2 0.521 m²/s²; Δq_t 0.003 g/kg; E 4.32 mm/s.

Canonical / WENO5 / no SGS: LWP 59.61 g/m²; cloud 99.92%; peak w^2 0.467 m²/s²; Δq_t 0.020 g/kg; E 5.19 mm/s.

Canonical / WENO9 / Smagorinsky: LWP 52.19 g/m²; cloud 99.37%; peak w^2 0.454 m²/s²; Δq_t 0.023 g/kg; E 5.24 mm/s.

Canonical / WENO5 / Smagorinsky: LWP 51.04 g/m²; cloud 98.95%; peak w^2 0.433 m²/s²; Δq_t 0.060 g/kg; E 5.24 mm/s.

Canonical / Centered2 / Smagorinsky: LWP 35.70 g/m²; cloud 99.59%; peak w^2 0.427 m²/s²; Δq_t 0.250 g/kg; E 6.57 mm/s.

Fine / WENO9 / no SGS: LWP 55.48 g/m²; cloud 99.95%; peak w^2 0.472 m²/s²; Δq_t 0.031 g/kg; E 5.34 mm/s.

Fine / WENO5 / no SGS: LWP 57.54 g/m²; cloud 99.90%; peak w^2 0.483 m²/s²; Δq_t 0.007 g/kg; E 5.24 mm/s.

Fine / WENO9 / Smagorinsky: LWP 53.23 g/m²; cloud 99.89%; peak w^2 0.445 m²/s²; Δq_t 0.036 g/kg; E 5.09 mm/s.

Fine / WENO5 / Smagorinsky: LWP 51.71 g/m²; cloud 99.77%; peak w^2 0.455 m²/s²; Δq_t 0.034 g/kg; E 5.48 mm/s.

Fine / Centered2 / Smagorinsky: LWP 44.17 g/m²; cloud 99.98%; peak w^2 0.373 m²/s²; Δq_t 0.154 g/kg; E 6.44 mm/s.

Fourth-hour metrics

Peak w^2 is the maximum of the fourth-hour mean native-face profile; entrainment includes subsidence. These are single-seed results.

Fine-grid averaging-window comparison

Compare the same statistic across averaging windows. Dimensional third moment and skewness differ in normalization as well as the windows used in the two papers. All minima below are over 600–760 m and may occur at different heights. Skewness is the ratio of time-averaged moments.

fine_none_weno9, 2-4 h: minimum w'^3 -0.0497 m³/s³; minimum skewness -0.1503.

fine_none_weno9, 3-4 h: minimum w'^3 -0.0622 m³/s³; minimum skewness -0.1928.

fine_none_weno5, 2-4 h: minimum w'^3 -0.0433 m³/s³; minimum skewness -0.1334.

fine_none_weno5, 3-4 h: minimum w'^3 -0.0372 m³/s³; minimum skewness -0.1109.

fine_smagorinsky_weno9, 2-4 h: minimum w'^3 -0.0419 m³/s³; minimum skewness -0.1394.

fine_smagorinsky_weno9, 3-4 h: minimum w'^3 -0.0304 m³/s³; minimum skewness -0.1026.

fine_smagorinsky_weno5, 2-4 h: minimum w'^3 -0.0163 m³/s³; minimum skewness -0.0538.

fine_smagorinsky_weno5, 3-4 h: minimum w'^3 -0.0160 m³/s³; minimum skewness -0.0521.

fine_smagorinsky_centered2, 2-4 h: minimum w'^3 -0.0236 m³/s³; minimum skewness -0.1206.

fine_smagorinsky_centered2, 3-4 h: minimum w'^3 -0.0272 m³/s³; minimum skewness -0.1309.

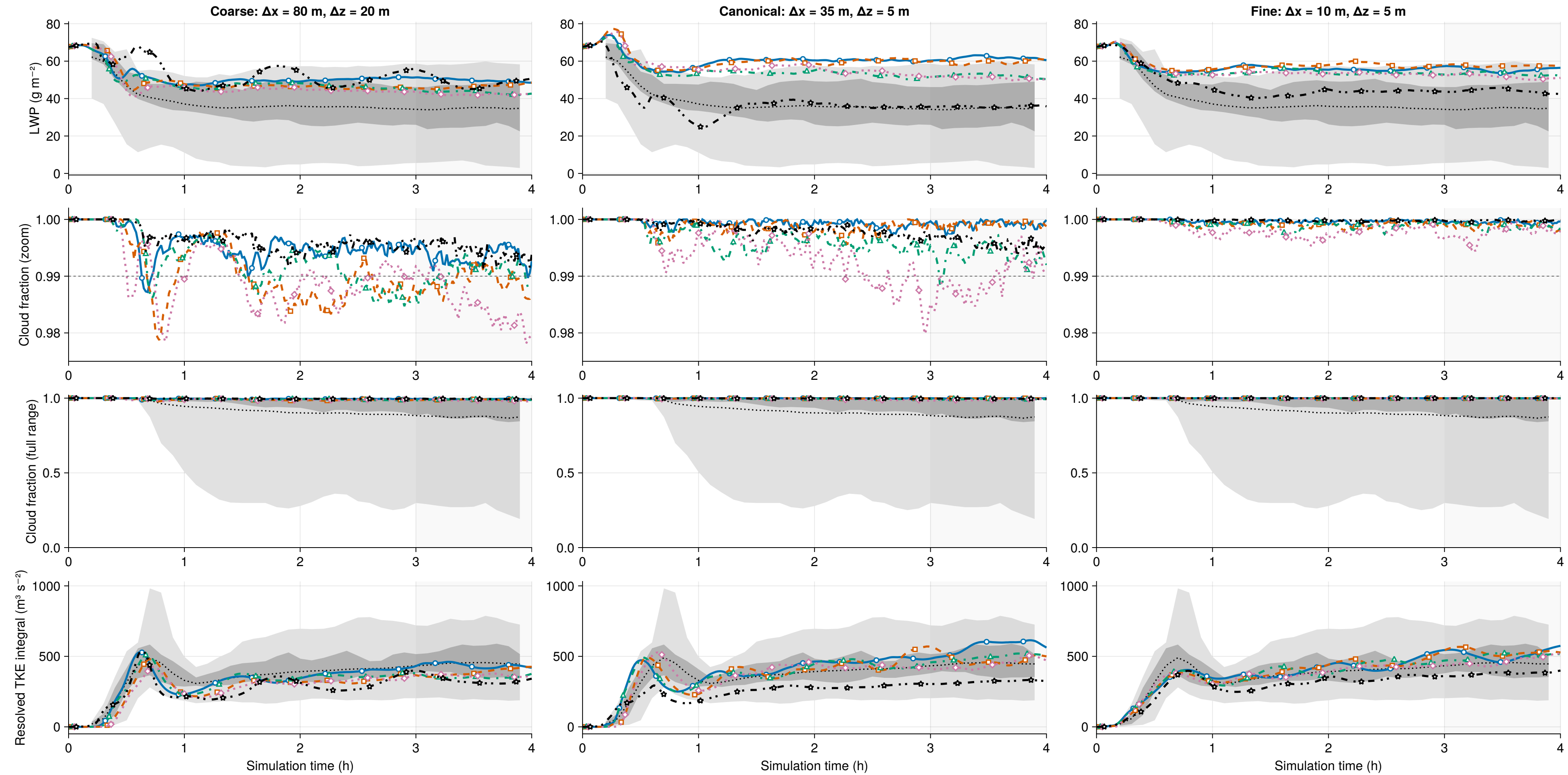
How to read these comparisons

All scientific figures are generated in Julia with CairoMakie. Numerical configurations use fixed Okabe-Ito colors, distinct dashes and marker shapes; different resolutions occupy panels with shared scales. The cloud-fraction zoom follows the simulated range, while a separate full-range panel preserves the paper ensemble spread.

Light gray is the paper's model-ensemble minimum–maximum; darker gray is its interquartile range. These are not observational uncertainty bands or validation tolerances. The mean and envelope were digitized from Figure 2 at approximately 0.2–3.9 h. Original pixel coordinates, calibration, extraction settings and an overlay are retained. Subpixel early cloud-fraction bands are left missing. Only rendered cloud-fraction band edges are clipped to the physical 0–1 range; CSV values preserve raw digitization. The zoom panel omits the paper bands to avoid magnifying raster-edge uncertainty.

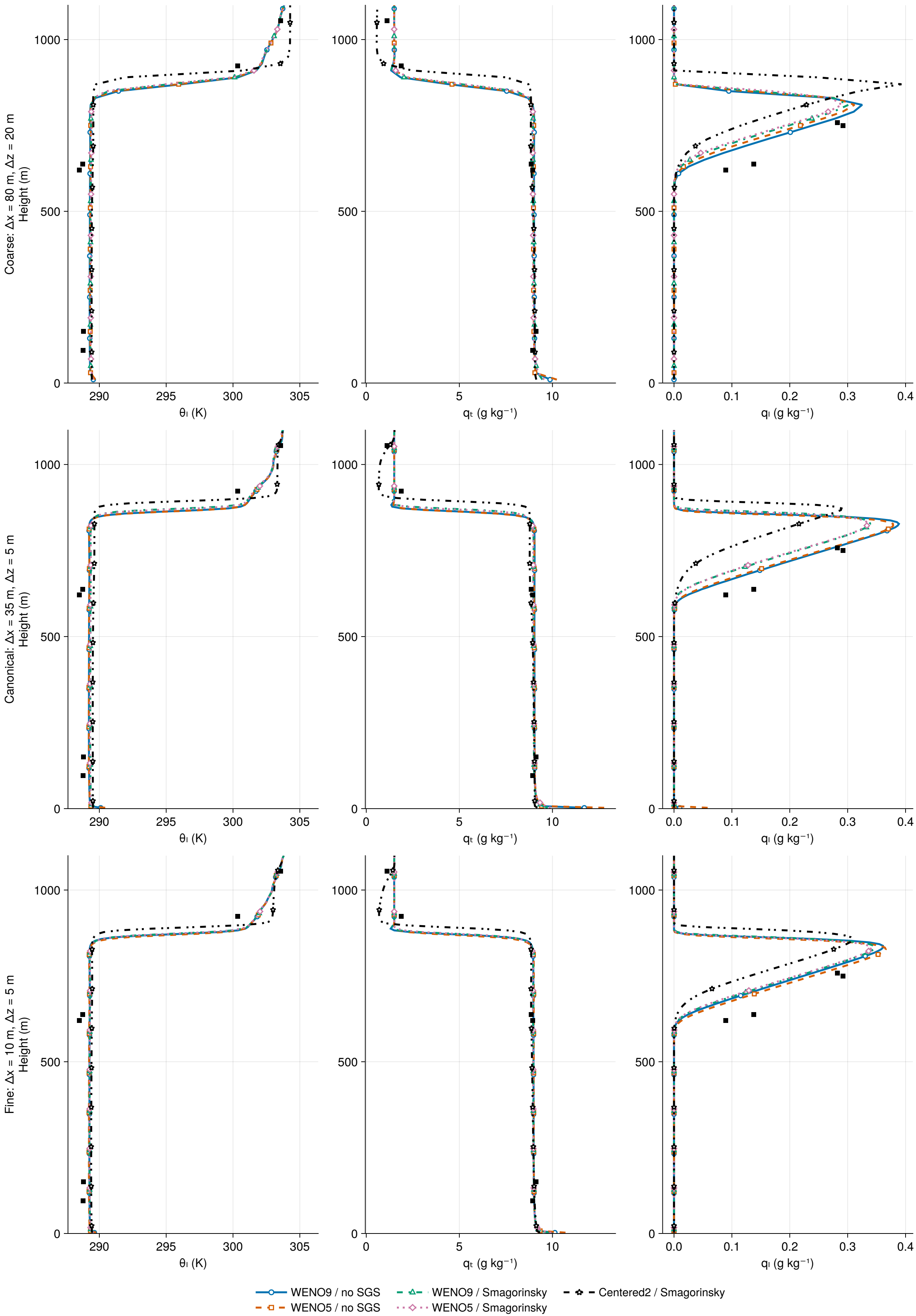
Paper observations are black markers in profile plots. Measurement error bars were not digitized. Numerical configurations use redundant color, line and symbol cues; resolution panels share scales.

Cloud persistence and turbulence evolution



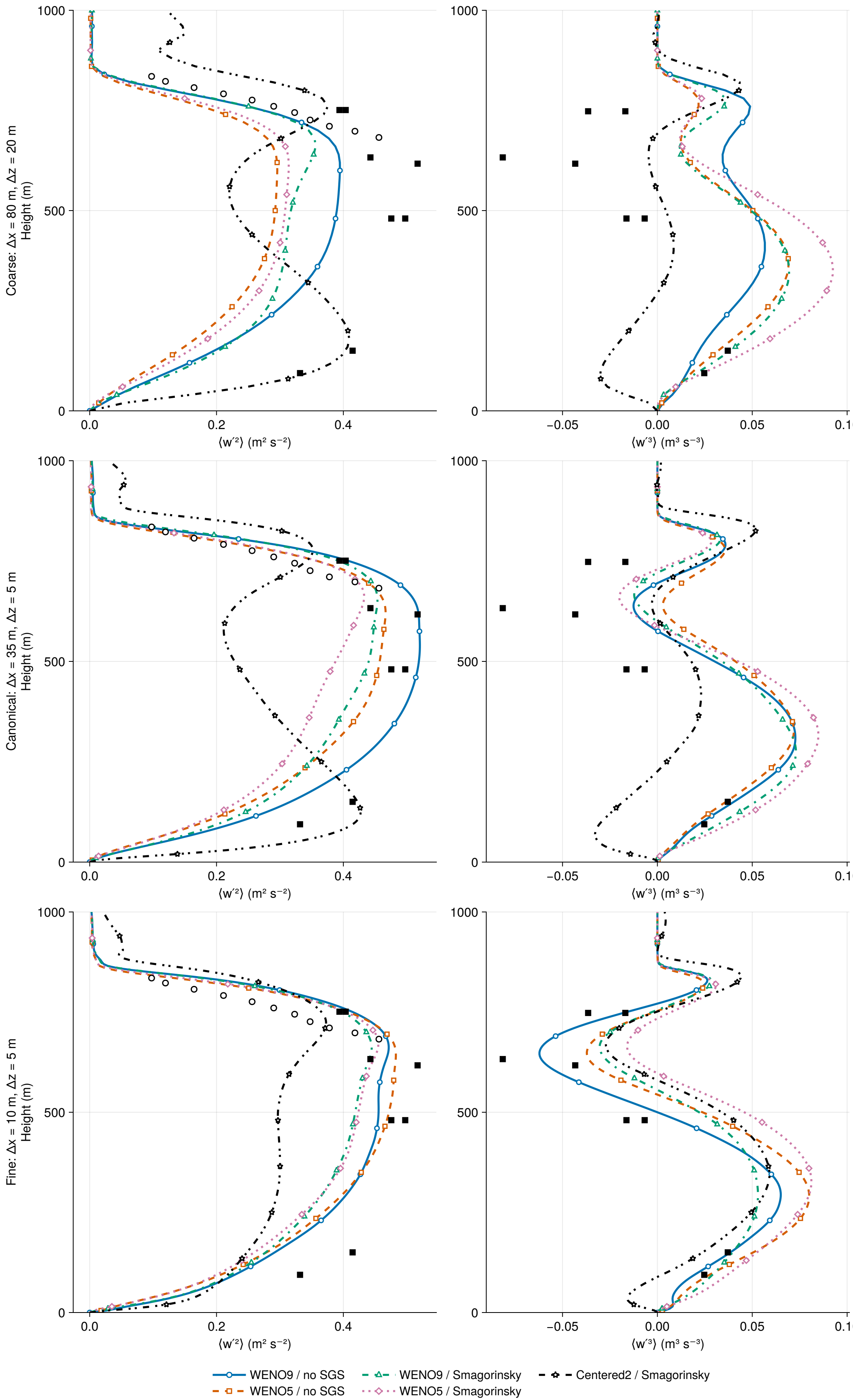
Gray bands: digitized MODEL spread (Stevens et al. 2005, Fig. 2), not observational uncertainty. Cloud-fraction zoom shows Breeze curves; the full-range row shows the paper envelope. Unresolved early shading is omitted. TKE is resolved only. All plots: Julia/CairoMakie.

Fourth-hour thermodynamic structure



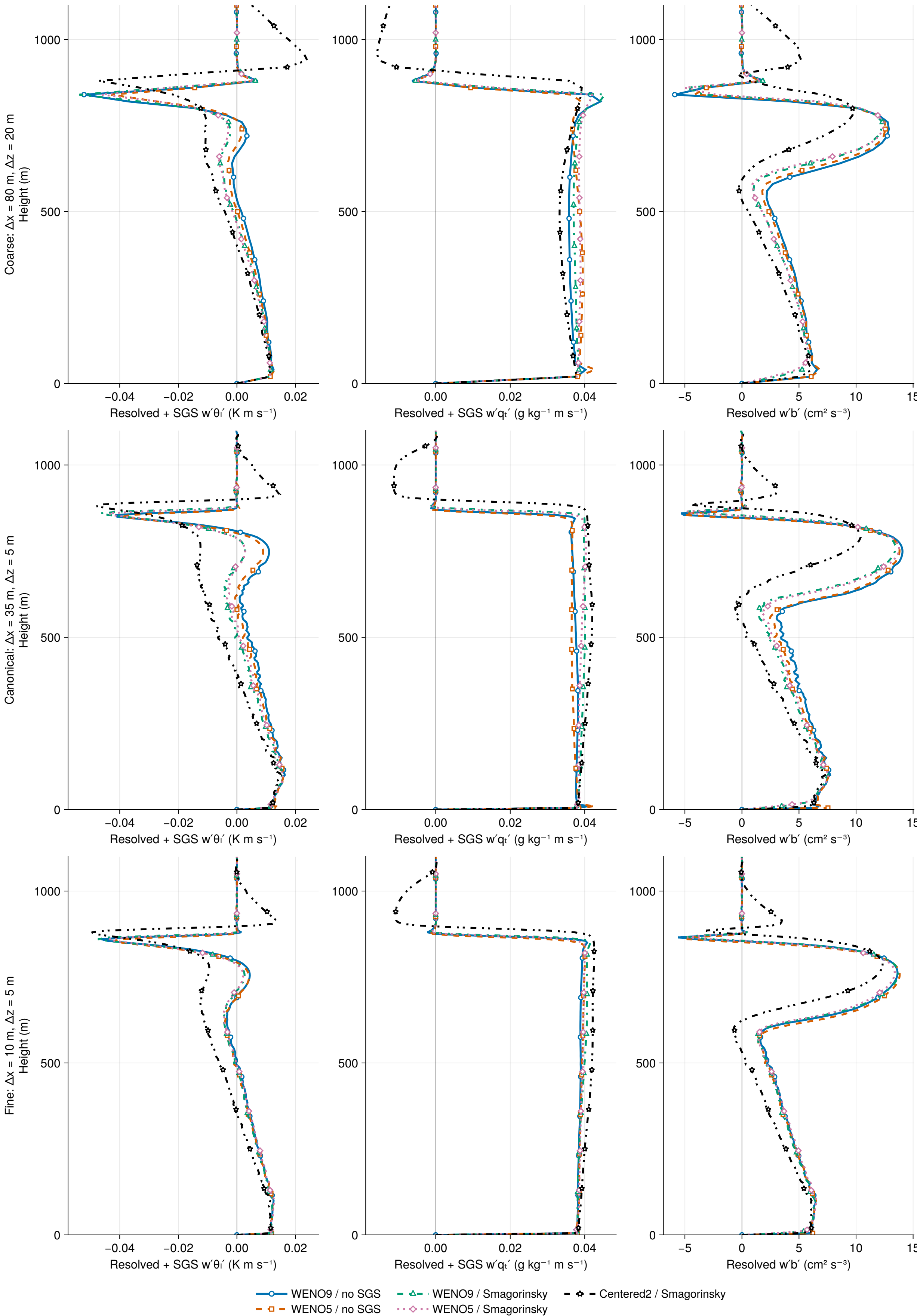
Fourth-hour means combine the two half-hour profile averages ending at 12600 and 14400 s. Black squares: retained paper Fig. 4 observations; measurement error bars were not digitized. Shared axis scales across resolutions.

Fourth-hour vertical-velocity moments



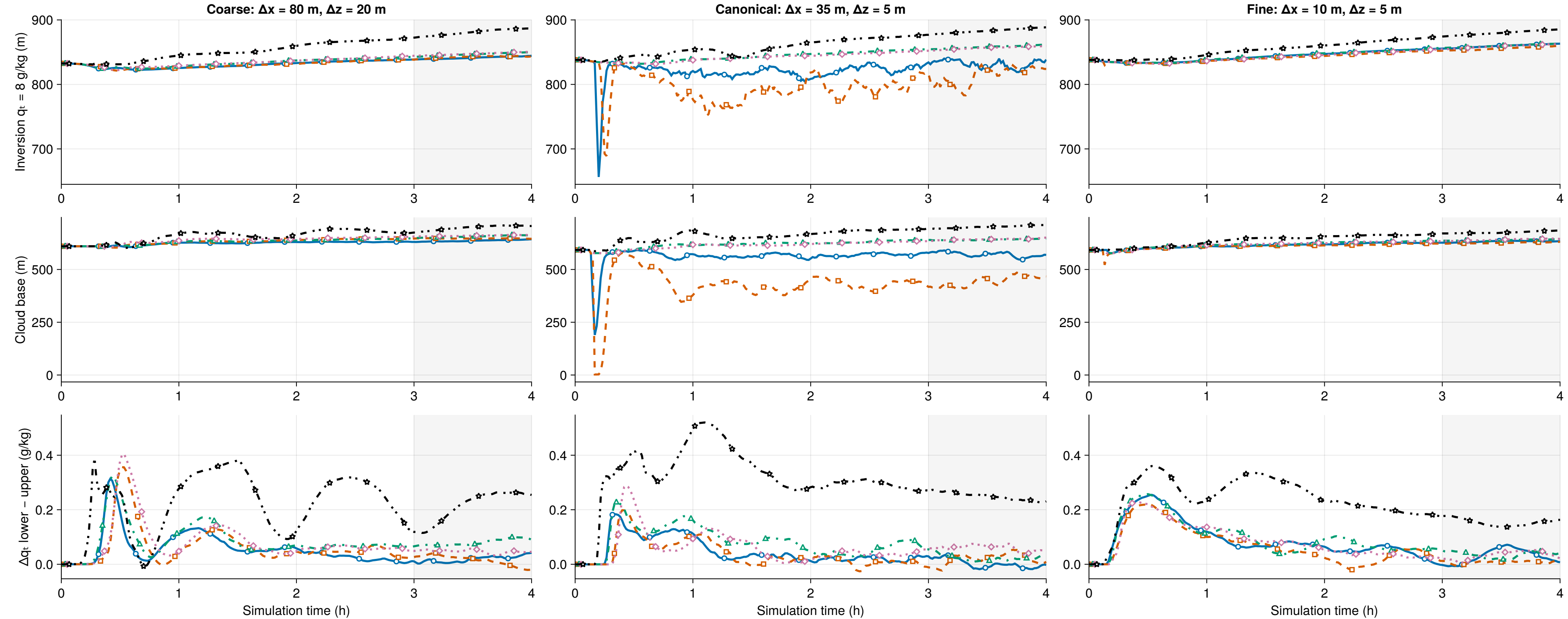
Fourth-hour means on native w faces. Black squares: retained in-situ observations; black open circles: radar (paper Fig. 5). Error bars were not digitized. Colors, line patterns and marker shapes identify numerical configurations; rows identify resolutions.

Fourth-hour scalar transport and buoyancy production



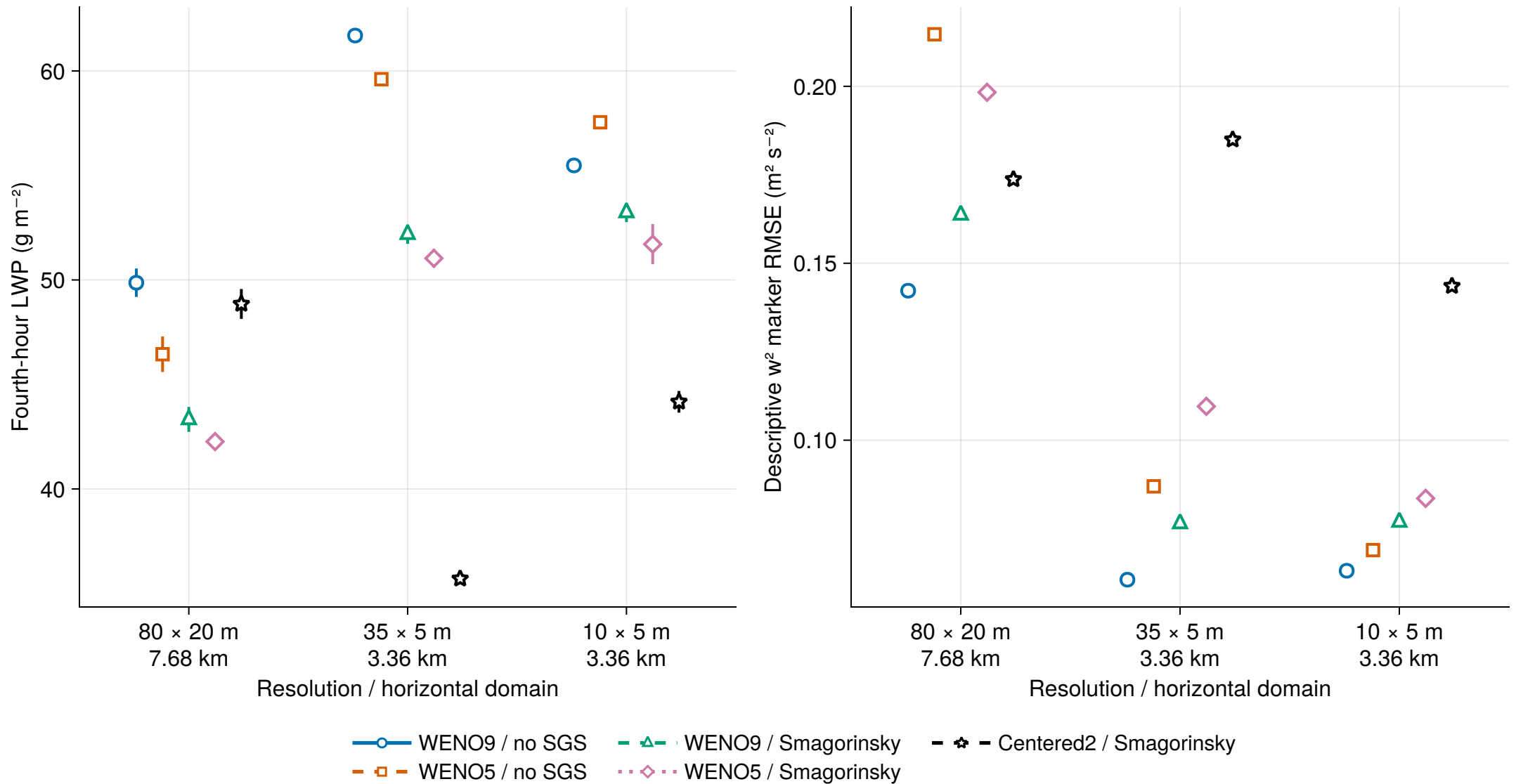
Scalar panels add resolved covariance and explicit SGS flux; buoyancy production is resolved only. Numerical transport is not separately measured, so resolved + SGS is not a complete discrete flux budget.

Cloud boundaries and decoupling



Cloud-base height is conditional on cloudy columns. Inversion height uses the interpolated q_t contour. $\Delta q_t = \text{mean}(100\text{--}200 \text{ m}) - \text{mean}(700\text{--}800 \text{ m})$, using exact geometric layer overlap.

Resolution and numerical choices



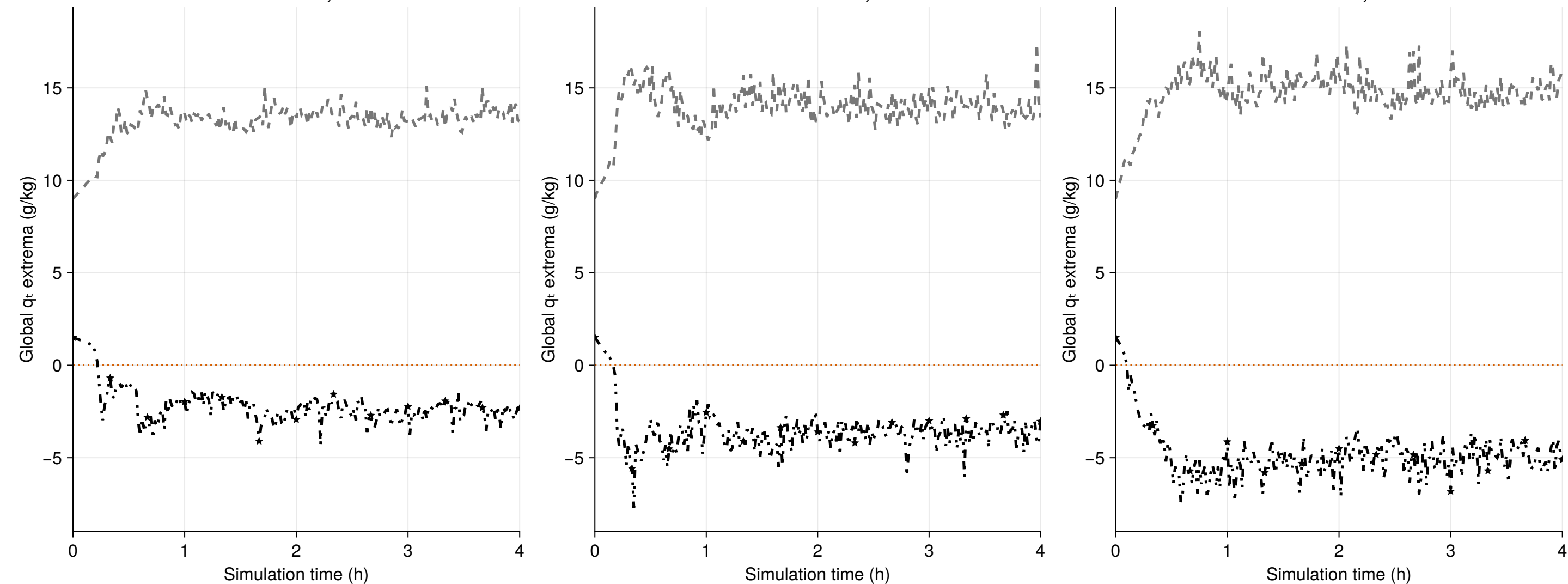
Only completed cases appear. LWP bars span two half-hour means, not confidence intervals. Marker RMSE is an unweighted descriptive distance to retained in-situ observations, not an uncertainty-normalized validation score.

Unbounded Centered2 + Smagorinsky: moisture extrema

Coarse: $\Delta x = 80$ m, $\Delta z = 20$ m

Canonical: $\Delta x = 35$ m, $\Delta z = 5$ m

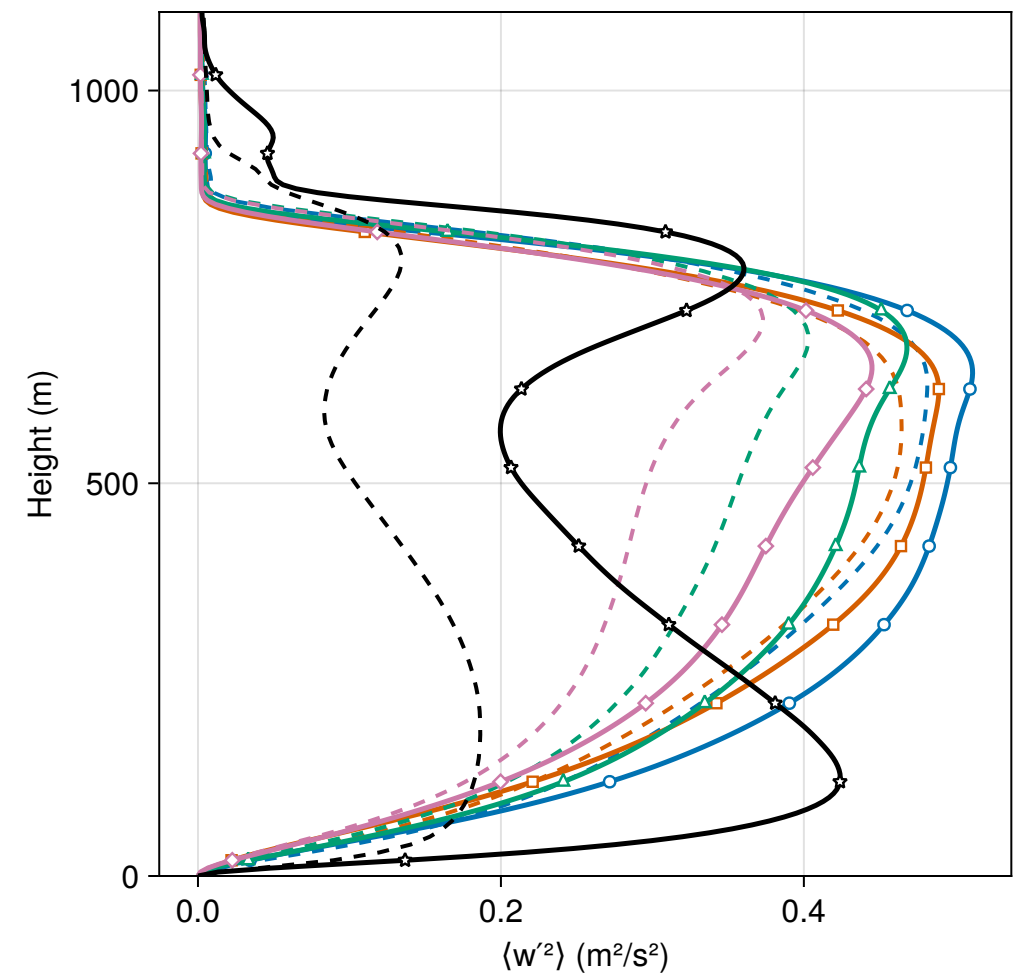
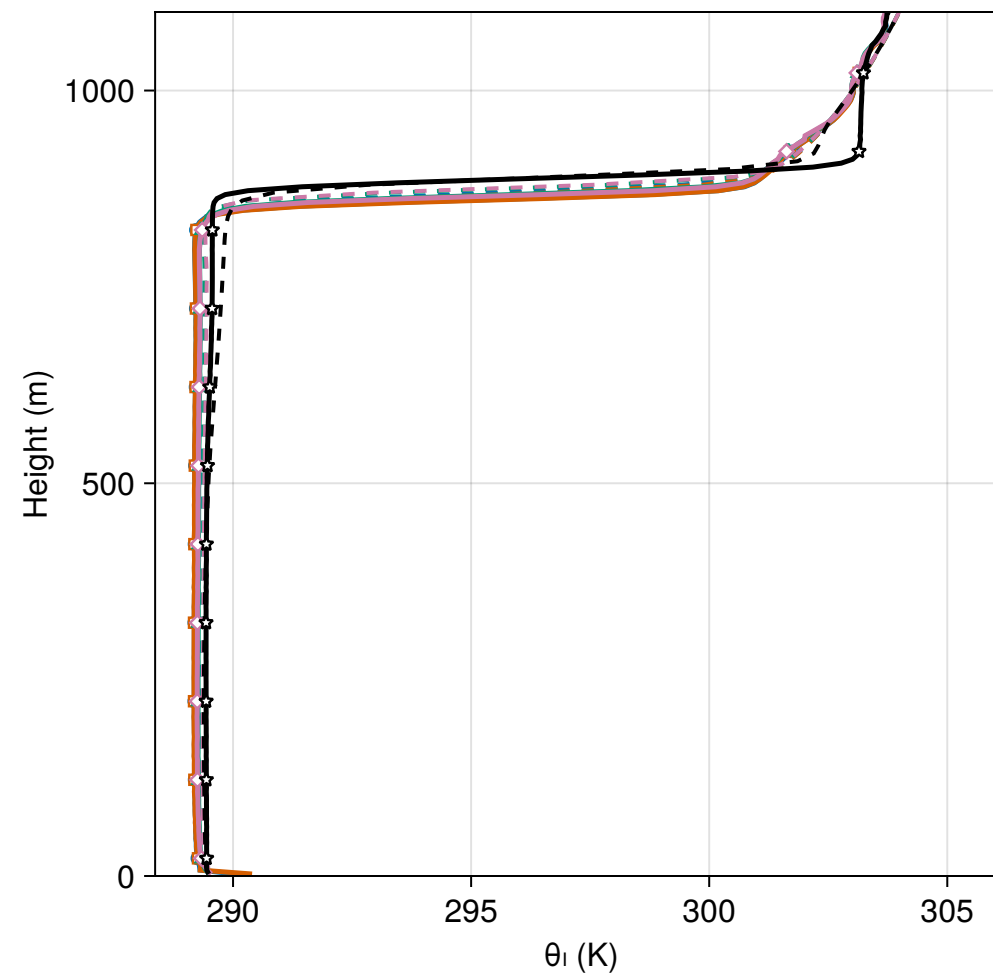
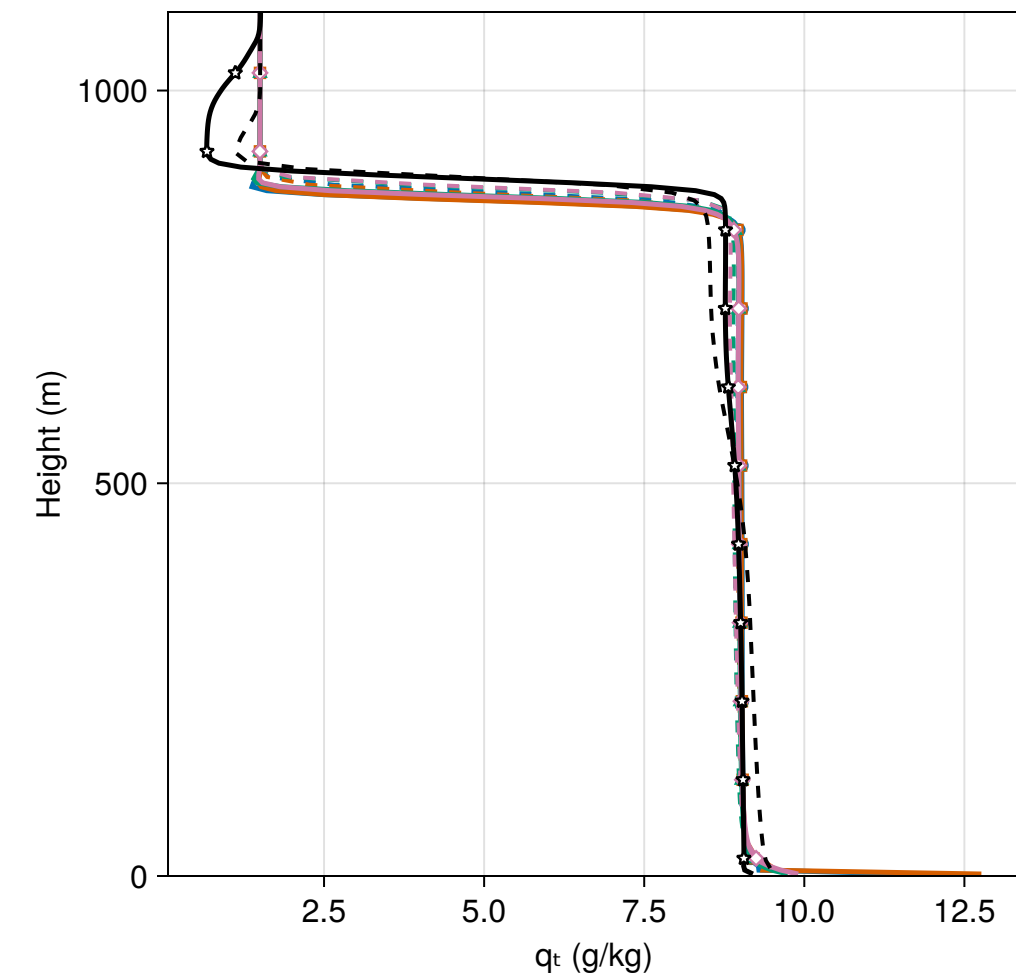
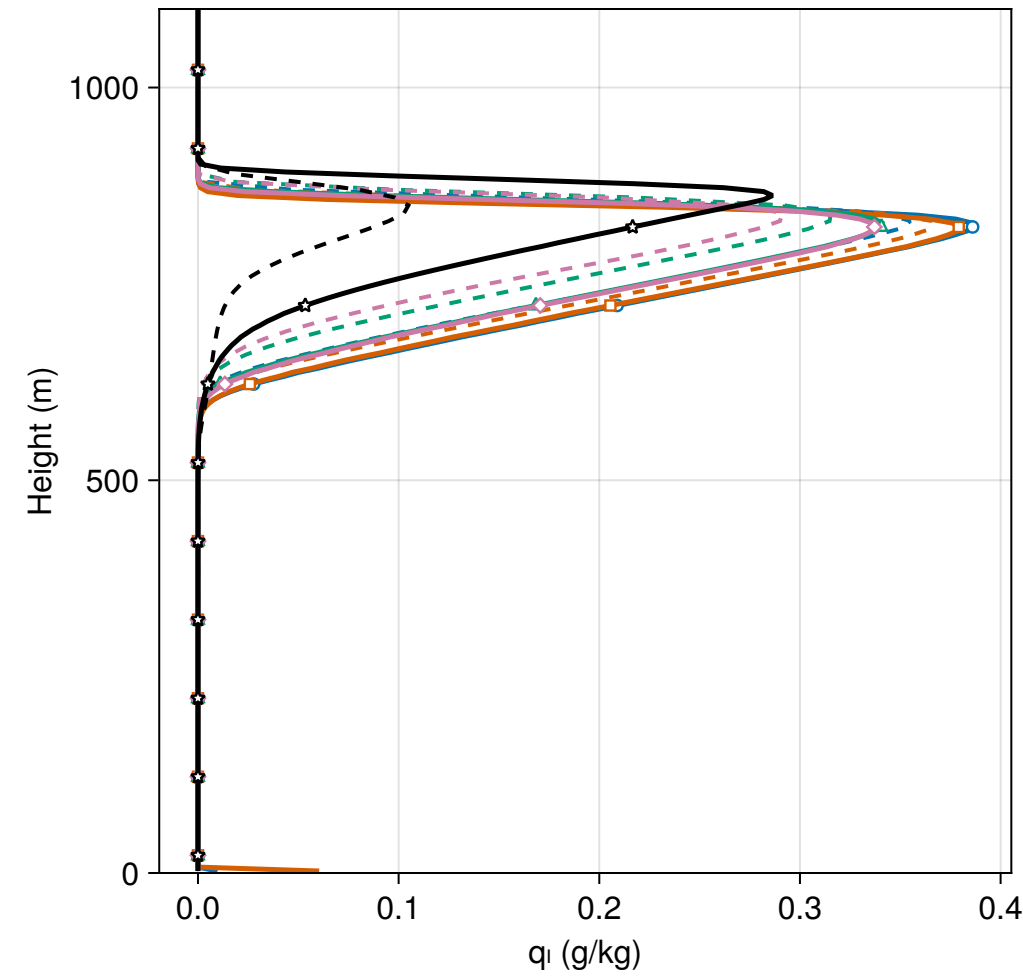
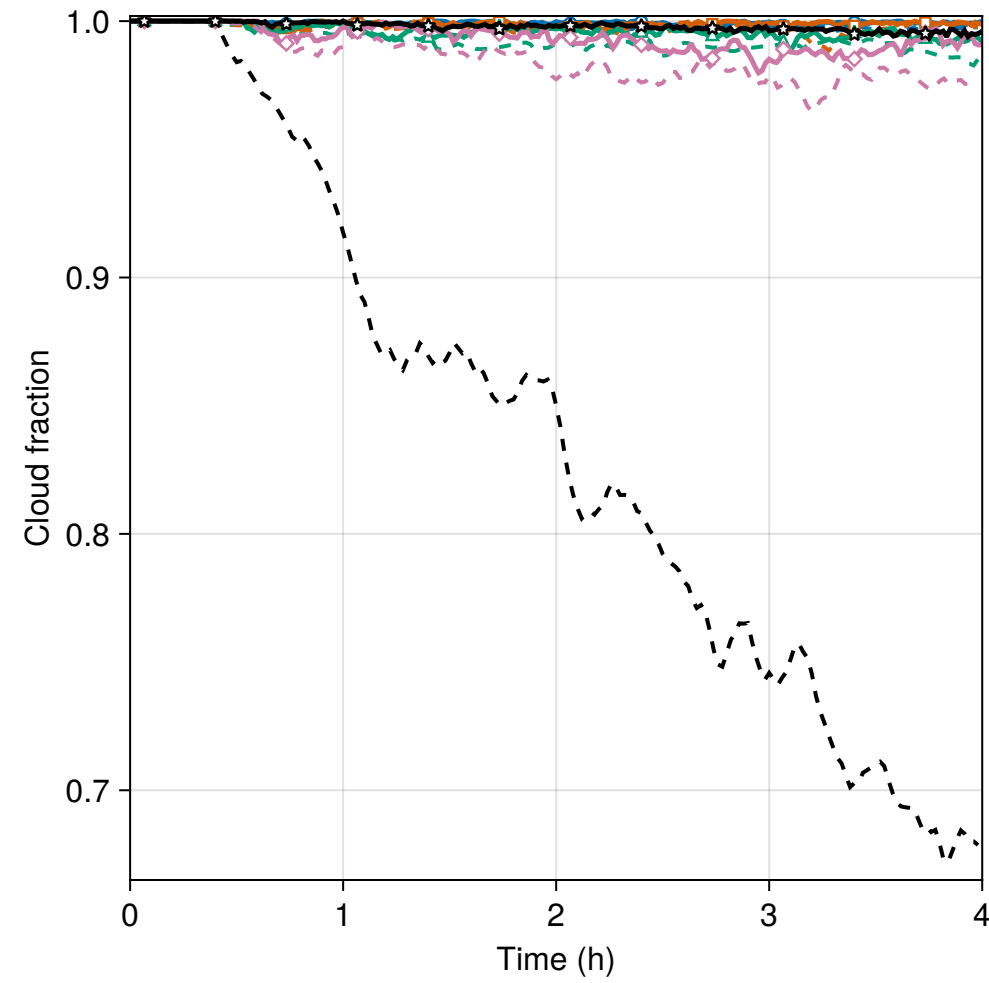
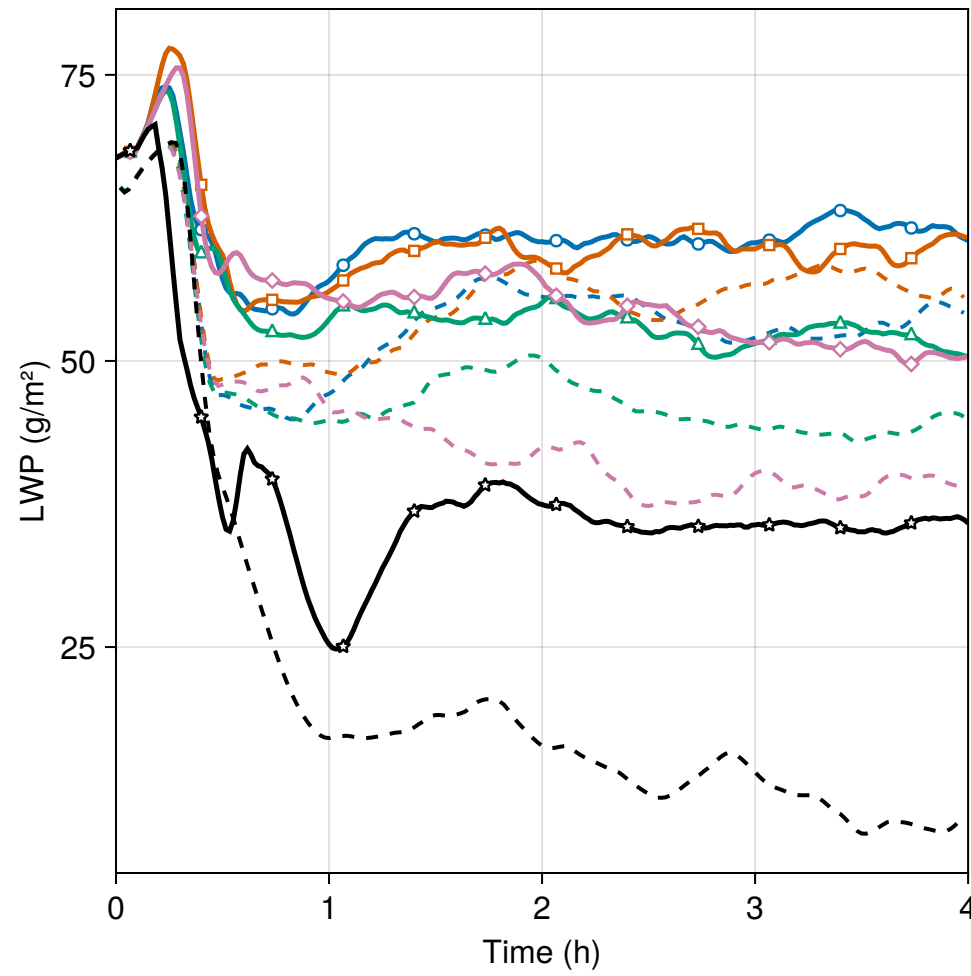
Fine: $\Delta x = 10$ m, $\Delta z = 5$ m



- · · · Minimum q_t (star markers) - - - Maximum q_t · · · · Zero total water

Global spatial extrema sampled every 60 s; excursions between samples are not bounded by these curves.
Negative total water is physically inadmissible. Finite output does not establish physical validity; no clipping or WENO fallback was applied.

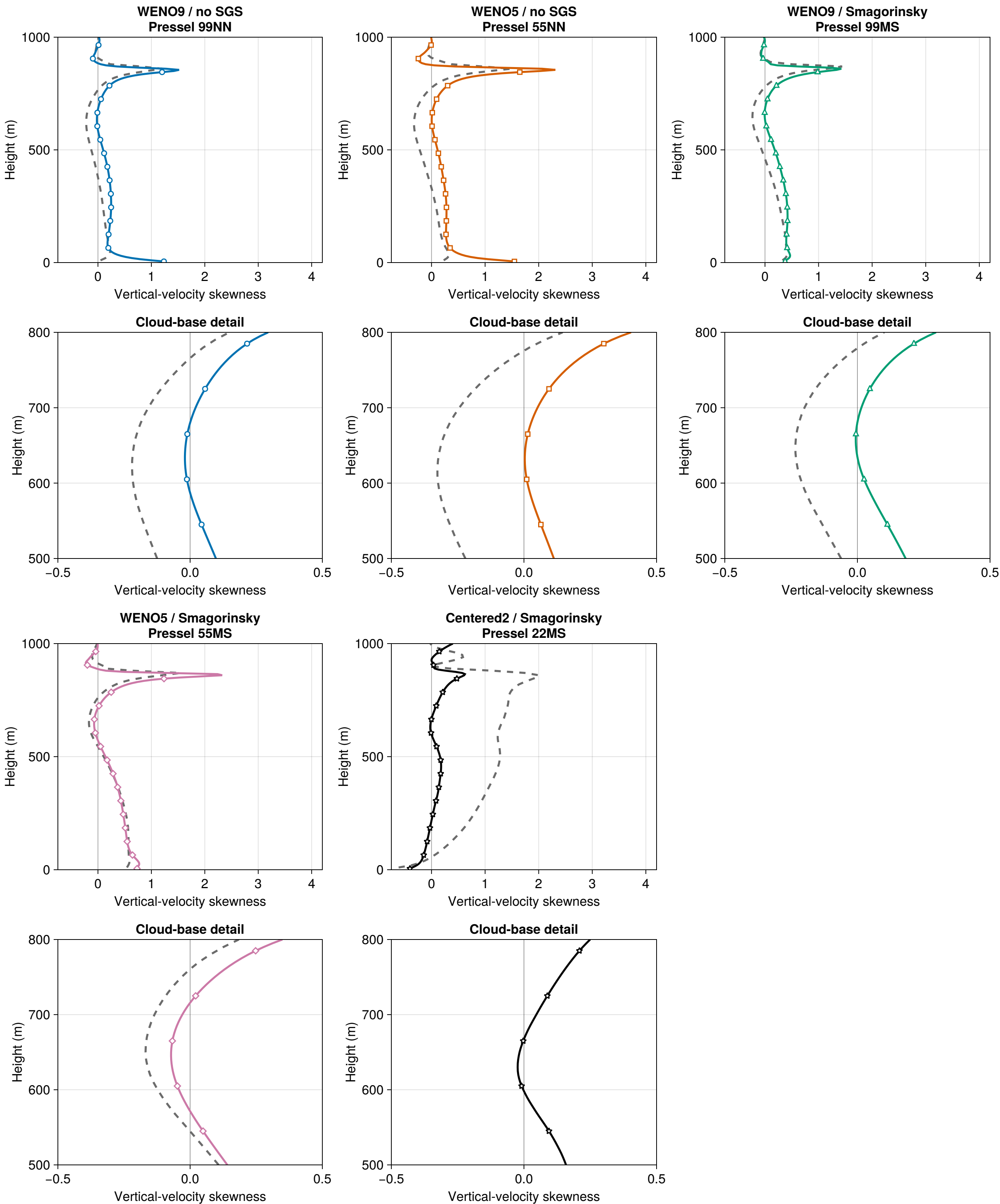
Canonical grid: Breeze and Pressel et al. (2017)



-- WENO9 / no SGS (99NN) -- WENO9 / Smagorinsky (99MS) -- Centered2 / Smagorinsky (22MS) --- Pressel model
-- WENO5 / no SGS (55NN) -- WENO5 / Smagorinsky (55MS) — Breeze

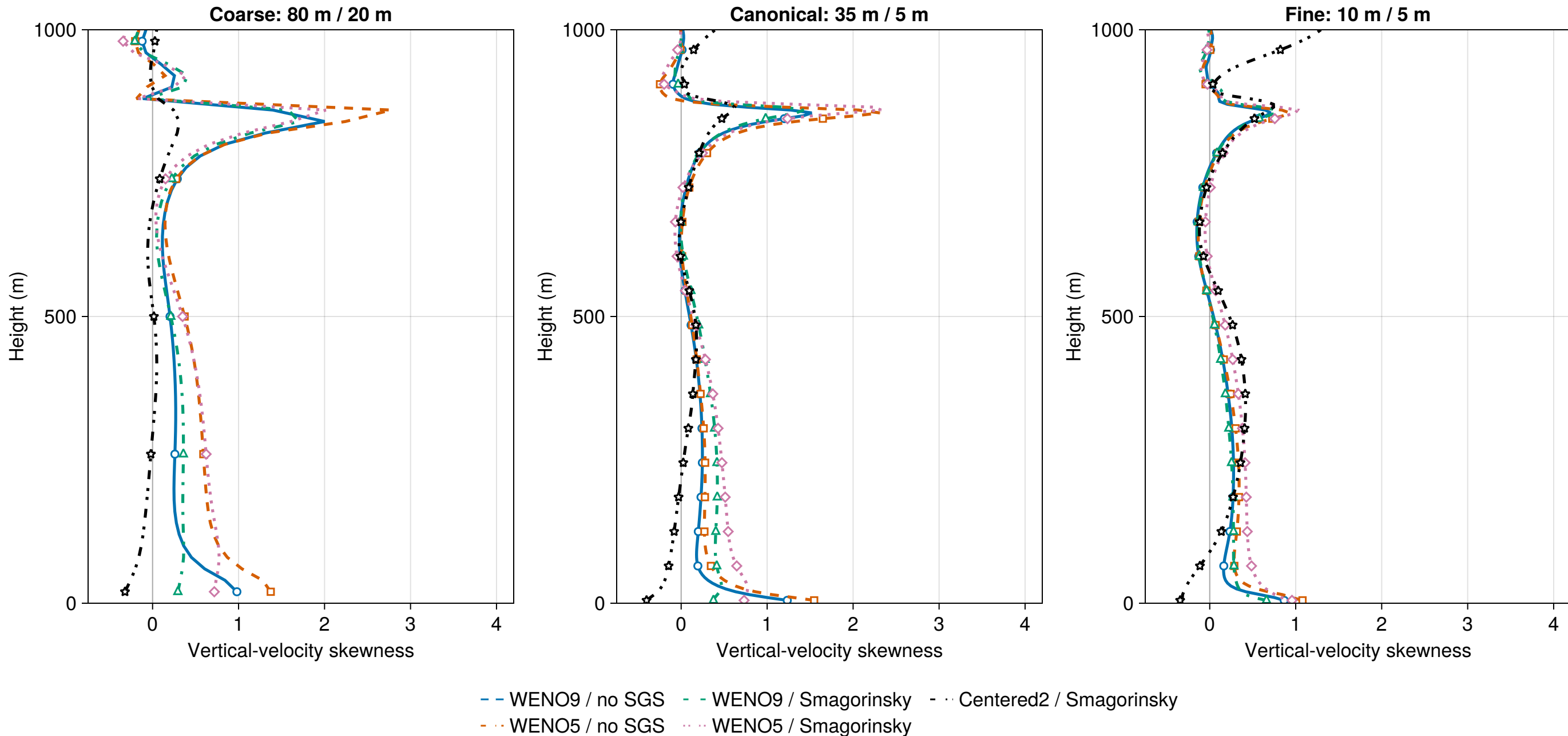
35 m horizontal / 5 m vertical; same domain. Profiles average 2–4 h. Cloud-fraction axis zoomed. Only completed Breeze cases shown.
 Scheme-family comparison: Cs, Pr, CFL, thermodynamics, surface treatment and moisture handling differ; this is not a replication.

Breeze and Pressel Figure 7: skewness, hours 2–4



Canonical grid: $\Delta x = 35$ m, $\Delta z = 5$ m. Breeze uses the ratio of time-averaged central moments. Pressel normalization/averaging order is unverified; these are model curves, not observations. Top: 0–1000 m. Bottom: magnified 500–800 m. Low-variance levels masked. Model settings differ.

Breeze skewness across resolutions: hours 2–4



Ratio of 2–4 h averaged central moments; not the time average of instantaneous skewness.
 Only completed exports shown. Coarse horizontal domain is larger. Variance $\leq 10^{-6} \text{ m}^2/\text{s}^2$ is masked; full-height values are in CSV.

Methods and limitations

Experimental design

Three grids: coarse $96 \times 96 \times 75$ at $80 \times 80 \times 20$ m in a $7.68 \times 7.68 \times 1.5$ km domain; canonical $96 \times 96 \times 300$ at $35 \times 35 \times 5$ m in $3.36 \times 3.36 \times 1.5$ km; fine $336 \times 336 \times 300$ at $10 \times 10 \times 5$ m in the same canonical domain. Fine $dz=5$ m is the stated implementation assumption. Each grid crosses WENO9/WENO5 with no closure/Smagorinsky-Lilly ($C_s=0.16$, $Pr=1$). WENO moisture uses bounds=(0,1) at both orders; order changes apply to all advected variables. Three additional runs apply Centered(order=2) to momentum and all scalars, including moisture, with the same Smagorinsky-Lilly settings: 15 cases total. Centered moisture is unbounded, so comparisons with WENO change scheme family and bounding as well as order. Finite diagnostics and moisture extrema must be checked before interpreting these cases.

Physics and initial conditions

Breeze uses anelastic dynamics, warm-phase saturation adjustment, fixed surface sensible/latent heat fluxes $15/115$ W/m², drag coefficient 0.0011, divergence $D=3.75 \times 10^{-6}$ s⁻¹, geostrophic winds (7,-5.5) m/s and the RF01 idealized longwave radiation. Initial inversion is 840 m, $\theta_l=289$ K and $q_t=9$ g/kg below, $\theta_l=297.5+(z-840)^{(1/3)}$ K and $q_t=1.5$ g/kg above. No drizzle or droplet sedimentation. All runs use Float32, seed 123 and four simulated hours. The fixed-flux moisture conversion uses $L_v=2.47 \times 10^6$ J/kg; Breeze retains its default thermodynamic latent-heat law with dry-air $c_p=1015$ J/(kg K), which differs from a strictly constant-mixture- c_p implementation.

Averaging and grid locations

Profiles are averages of instantaneous horizontally reduced statistics. The fourth hour combines the equal-duration bins ending at 12,600 and 14,400 s; it excludes the bin ending at 10,800 s. Scalar means are on centers; w variance and third central moment remain on native vertical faces. The third moment has units m³/s³, not normalized skewness. Series means use trapezoidal integration of 60 s samples between 10,800 and 14,400 s. Two half-hour LWP means give a descriptive temporal spread, not a confidence interval.

Methods and limitations

Cloud and entrainment definitions

Column cloud fraction uses $q_l > 10^{-6}$ kg/kg in any cell. Cloud-base means exclude clear columns. Inversion height uses the first downward $q_t = 8$ g/kg crossing with linear interpolation; its valid fraction is checked. The $\theta_{\text{ref}} = 295$ K contour and the nearest-cell height used internally by radiation are retained separately. Fourth-hour entrainment is $[z_i(4h) - z_i(3h)]/3600 + D \cdot \text{mean}(z_i)$, including subsidence. Decoupling is mean q_t over 100–200 m minus mean q_t over 700–800 m, with exact layer-overlap weights.

Reference data and comparison limits

Stevens et al. (2005), DOI 10.1175/MWR2930.1, supplies the RF01 comparison. Figure 2 is a model ensemble mean, not observations or an acceptance band; the paper has no observed LWP time series. Observed cloud cover exceeds 99%. Figure 4 and 5 markers are partial digitizations, with source pixels retained and error bars not extracted. Marker RMSE is an unweighted descriptive distance to retained in-situ points; it is not a likelihood, uncertainty-normalized score, or formal validation criterion. Radar markers are plotted separately. Resolved TKE is plotted against the historical ensemble for context; a prognostic SGS-TKE contribution is unavailable for these closures.

What this design can establish

The coarse domain has 5.22 times the canonical horizontal area, so coarse-versus-canonical differences combine changes in horizontal spacing, vertical spacing and domain size. Fine-versus-canonical holds domain and vertical spacing fixed. Closure/order comparisons share a seed within a grid, but one seed gives no ensemble uncertainty and different grids do not share identical perturbation fields. UCLA-0 in the paper disables scalar SGS mixing while retaining momentum treatment; it is not identical to Breeze closure=nothing. Neither matching LWP nor agreement with one fine simulation establishes convergence.

Methods and limitations

Numerical and output audit

The original WENO runs use an immutable source snapshot with per-file hashes and a pinned Julia 1.12.6 workspace. The Centered2 extension uses a separate versioned runner and provenance while preserving the original source. GPU smoke runs checked all grids and the diagnostic output path before production. Exports require completed jobs, end-of-run sentinels, finite numeric records, exactly nine profile times and 241 series times, 46 profile variables and 32 series variables. Native face/center coordinates and fixed fluxes are checked again locally. Replicated budget arrays, where present, are horizontally averaged by the exporter and their original shape/spread recorded. The TKE residual is a budget-closure residual, not a measured numerical dissipation.

Reproducibility and current run state

All active processing and figures use Julia. CairoMakie version 0.15.14; Julia 1.12.7. The companion Markdown/HTML report links to all CSVs, source hashes, digitization provenance and exact rebuild commands. Original restart checkpoints and JLD2 files remain on pcluster.

coarse_none_weno9: COMPLETED

coarse_none_weno5: COMPLETED

coarse_smagorinsky_weno9: COMPLETED

coarse_smagorinsky_weno5: COMPLETED

canonical_none_weno9: COMPLETED

canonical_none_weno5: COMPLETED

canonical_smagorinsky_weno9: COMPLETED

canonical_smagorinsky_weno5: COMPLETED

fine_none_weno9: COMPLETED

fine_none_weno5: COMPLETED

fine_smagorinsky_weno9: COMPLETED

fine_smagorinsky_weno5: COMPLETED

coarse_smagorinsky_centered2: COMPLETED

canonical_smagorinsky_centered2: COMPLETED

fine_smagorinsky_centered2: COMPLETED

Reference: Stevens et al. (2005), Monthly Weather Review 133, 1443–1462. DOI 10.1175/MWR2930.1.

GABLS1: setup and results

Part II: GABLS1

15/15 completed exports admitted to the scientific comparison. Checked
2026-09-22T03:08:26.728Z.

Grids: 12.5, 6.25, 3.125, 2, and 1 m isotropic; 32^3 , 64^3 , 128^3 , 200^3 , and 400^3 cells, all in the same 400 m cube. Small grids run first.

Each grid has exactly three configurations: WENO9 with no interior closure, WENO5 with no interior closure, and WENO9 with Smagorinsky. Smagorinsky $C_s=0.16$, $Pr=1$; paired seed 123 within each grid. These are controlled Breeze settings, not a reproduction of one named historical LES member.

Forcing: $U_g = 8$ m/s, $V_g = 0$, $f = 1.39e-4/s$. Initial potential temperature 265 K below 100 m and $265+0.01(z-100)$ K above. Initial temperature noise is uniform with 0.1 K peak-to-peak range below 50 m. Surface $\theta_{s,ref}=265-0.25t_{hours}$ K; $\theta_{ref}=263.5$ K is a separate buoyancy reference.

Wall: MOST exchange with $\kappa = 0.4$, $\beta_m = 4.8$, $\beta_h = 7.8$, roughness = 0.1 m for heat and momentum. Impermeable boundaries, free-slip upper lid. Vertical-velocity damping is zero below 300 m and ramps linearly to a 60 s relaxation time at 400 m. MOST uses a minimum wind of 0.01 m/s and caps z/L at 10, with a neutral fallback for nonpositive bulk Richardson number; their activity is diagnosed. Exact implementation, damping and timestep choices are recorded in production provenance.

Diagnostics: 541 instantaneous series records at 60 s intervals; 19 profile times with preceding 30 min means after instantaneous t_0 . Compare 7-8 h and 8-9 h. Final-hour profiles combine bins ending at 30600 and 32400 s; penultimate-hour bins end at 27000 and 28800 s.

Compare mean $u/v/\theta$, jet speed/height/turning, total-stress boundary-layer height, surface heat flux/ u_{star}/L , native-face $w_2/w_3/skewness$, resolved/SGS flux partitions and available TKE-budget terms. Missing SGS energy or budget terms remain unavailable; a residual is not numerical dissipation.

Beare et al. (2006), Boundary-Layer Meteorology 118, 247-272,
doi:10.1007/s10546-004-2820-6. Official reduced data archive contains 423 files. Native coordinates and missing values are retained; some NERSC headers give 474-534 min for nominal 8-9 h, so exact window matching is not universal.

GABLS1: setup and results

Cluster agents own implementation, scheduling and audited exports. Desktop monitoring is scheduled every 10 min to collect newly completed cases and rebuild this document. Actual submitted jobs and completed-export evidence are recorded in the status files, separate from the authorized matrix.

Production submission records: 15/15 cases have assigned Slurm jobs. Submitted is distinct from running or scientifically complete. The live queue snapshot below also includes any smoke tests.

Implementation choices: Breeze uses anelastic dynamics and Float32 arithmetic; the reference archive is predominantly Boussinesq. Base pressure is 100000 Pa so the prescribed surface temperature equals surface potential temperature; the corresponding reference density differs from the specification's rounded 1.3223 kg/m³ by about 8 parts per million. Smagorinsky-Lilly uses $C_s=0.16$, $C_b=1$, $Pr=1$.

At $\Delta=12.5$ m, WENO9/Smagorinsky has strongly suppressed resolved turbulence: peak final-hour w_2 is $1.12e-06$ m²/s², $1.21e-05$ times the no-closure value. Maximum magnitudes of the final-hour mean resolved and SGS momentum-flux vectors are $1.34e-07$ and 0.0415 m²/s². Its h is 140.6 m with only 0.242 m standard deviation across final-hour 60 s samples. Similar h or mean wind can therefore coexist with very different resolved turbulence and transport partition; h alone cannot establish fidelity. Skewness in this weak-variance regime must not be interpreted as evidence of vigorous turbulent motions.

At $\Delta=12.5$ m without interior closure, WENO9 versus WENO5 gives final-hour h 195.6 versus 136.1 m and peak resolved w_2 0.0927 versus 0.0574 m²/s². This is a scheme sensitivity at fixed grid and seed, not evidence of resolution convergence.

At $\Delta=6.25$ m without interior closure, WENO9 versus WENO5 gives final-hour h 187.8 versus 150.9 m and peak resolved w_2 0.0976 versus 0.0656 m²/s². This is a scheme sensitivity at fixed grid and seed, not evidence of resolution convergence.

At $\Delta=3.12$ m without interior closure, WENO9 versus WENO5 gives final-hour h 184.4 versus 162.9 m and peak resolved w_2 0.1008 versus 0.0696 m²/s². This is a scheme sensitivity at fixed grid and seed, not evidence of resolution convergence.

At $\Delta=2$ m without interior closure, WENO9 versus WENO5 gives final-hour h 178.5 versus 159.2 m and peak resolved w_2 0.0993 versus 0.0690 m²/s². This is a scheme sensitivity at fixed grid and seed, not evidence of resolution convergence.

At $\Delta=1$ m without interior closure, WENO9 versus WENO5 gives final-hour h 165.3 versus 159.4 m and peak resolved w_2 0.0893 versus 0.0681 m²/s². This is a scheme sensitivity at fixed grid and seed, not evidence of resolution convergence.

GABLS1: setup and results

n032_weno9_none: peak final-hour resolved w_2 ($0.0927 \text{ m}^2/\text{s}^2$) exceeds the largest archived model peak at the same spacing ($0.0602 \text{ m}^2/\text{s}^2$). Model spread is a comparison baseline, not observational truth; stronger resolved turbulence alone does not establish better fidelity.

n064_weno9_none: peak final-hour resolved w_2 ($0.0976 \text{ m}^2/\text{s}^2$) exceeds the largest archived model peak at the same spacing ($0.0779 \text{ m}^2/\text{s}^2$). Model spread is a comparison baseline, not observational truth; stronger resolved turbulence alone does not establish better fidelity.

n128_weno9_none: peak final-hour resolved w_2 ($0.1008 \text{ m}^2/\text{s}^2$) exceeds the largest archived model peak at the same spacing ($0.0904 \text{ m}^2/\text{s}^2$). Model spread is a comparison baseline, not observational truth; stronger resolved turbulence alone does not establish better fidelity.

n400_weno9_none: peak final-hour resolved w_2 ($0.0893 \text{ m}^2/\text{s}^2$) exceeds the largest archived model peak at the same spacing ($0.0851 \text{ m}^2/\text{s}^2$). Model spread is a comparison baseline, not observational truth; stronger resolved turbulence alone does not establish better fidelity.

n400_weno9_smagorinsky: peak final-hour resolved w_2 ($0.0985 \text{ m}^2/\text{s}^2$) exceeds the largest archived model peak at the same spacing ($0.0851 \text{ m}^2/\text{s}^2$). Model spread is a comparison baseline, not observational truth; stronger resolved turbulence alone does not establish better fidelity.

Resolution comparison for WENO9/none: reducing spacing from 12.5 to 1 m changes final-hour h from 195.6 to 165.3 m and peak resolved w_2 from 0.0927 to $0.08926 \text{ m}^2/\text{s}^2$. Surface heat flux changes from -15.52 to -8.05 W/m^2 . These compare the coarsest and finest completed grids for this configuration; similarity of selected quantities alone is not a convergence or fidelity test.

Resolution comparison for WENO5/none: reducing spacing from 12.5 to 1 m changes final-hour h from 136.1 to 159.4 m and peak resolved w_2 from 0.05744 to $0.06807 \text{ m}^2/\text{s}^2$. Surface heat flux changes from -11.79 to -7.47 W/m^2 . These compare the coarsest and finest completed grids for this configuration; similarity of selected quantities alone is not a convergence or fidelity test.

Resolution comparison for WENO9/smagorinsky: reducing spacing from 12.5 to 1 m changes final-hour h from 140.6 to 182.0 m and peak resolved w_2 from $1.12\text{e-}06$ to $0.09851 \text{ m}^2/\text{s}^2$. Surface heat flux changes from -9.17 to -9.12 W/m^2 . These compare the coarsest and finest completed grids for this configuration; similarity of selected quantities alone is not a convergence or fidelity test.

GABLS1: setup and results

For WENO9/Smagorinsky, the finest completed grid has $8.8\text{e}+04$ times the coarsest-grid peak final-hour resolved w^2 . At $\Delta=1$ m, maximum magnitudes of final-hour mean resolved and SGS momentum-flux vectors are 0.06483 and 0.04097 m^2/s^2 ; these maxima can occur at different heights and should not be treated as fractions of a common total. Final-hour h sample standard deviation is 2.05 m, and mean h changed by 2.57 m between 7–8 and 8–9 h. Read the variance evolution and flux profiles together: this resolution sensitivity does not establish statistical stationarity or identify a unique cause.

Final-hour values are descriptive results, not a fidelity ranking. Resolution comparisons must keep the scheme and closure fixed; archive ensemble membership changes with resolution.

The archive reports skewness directly; its averaging convention is not assumed identical to our ratio of time-averaged central moments. No archive third moment is reconstructed from independently averaged variance and skewness. Both Breeze skewness plots mask levels with final-hour variance at or below $1\text{e}-8$ m^2/s^2 ; the raw diagnostics remain available.

A fixed 1.0 m archive ensemble median (black dotted line) is repeated across resolution panels wherever a matching quantity is available, alongside same-grid archive spread. The 1.0 m archive has two models; pointwise member counts and source files are saved in `reference_data/fixed_1m_medians.json`. Linear totals and resolved TKE are computed per model before taking medians. No matching reference is fabricated for w^3 , time-mean stress magnitude, unmatched budget terms, wall-regime fractions, peak-variance time series or jet time series. The fixed reference aids resolution comparisons but is not an exact solution.

The 1.0 m IMUK archive contains conspicuously large upper-level SGS values, confirmed in the original files: SGS uw spans -0.872 to 0.869 m^2/s^2 and SGS shear production spans -0.283 to 0.793 m^2/s^3 . Tiny values approaching $1\text{e}-100$ abruptly become order-one values above roughly 330 m. This pattern suggests an archive exponent-format problem, but the intended values have not been established. Raw values and resulting median curves are retained, with affected flux/budget panels flagged. Do not interpret those upper-level excursions as a convergence target.

GABLS1: setup and results

Corrected production (v2) replaces the original campaign after a diagnostic index restriction truncated total-flux/stress profiles and exposed a boundary-layer-height contour-index error. The corrected source preserves the original physics and all three numerical configurations. Expanded CPU checks passed 278 kernel and 426 writer checks; GPU checks passed 268 kernel and 422 writer checks, including every raw profile extent. All 15 cases were resubmitted as arrays 7099-7103, beginning with 32³. Original outputs remain excluded from scientific comparisons and preserved as failure evidence. The collector and single cluster exporter now target v2.

The original 400³ WENO9/no-closure and WENO9/Smagorinsky attempts failed during startup and remain excluded. Their fixed initial timestep of 0.5 s implied an advective CFL of 4 for 8 m/s wind and 1 m cells. The separate startup_dt_v1 snapshot changes only initial dt to $\min(0.5, 0.5 \cdot dx/8)$, or 0.0625 s at 1 m. Both variants passed 72 s GPU tests with finite saved bounds: 852/832 iterations and final actual CFL 0.700088/0.700007. The unchanged wizard gave an effective first dt of 0.06875 s. Replacement jobs 7120_1 (WENO9/no closure) and 7120_2 (WENO9/Smagorinsky) were submitted with a one-job array limit; 7120_1 is running and 7120_2 queued at this snapshot. Original WENO5 job 7103_2 continues without duplication. Original failed source/logs/outputs are preserved in attempt history, and replacements require their own job identity, revision and source-hash audits. Passing the short startup tests supports the repair but does not prove full-run stability, uniquely establish the failure cause, or establish that coarser results are independent of startup settings.

n032_weno9_none: final-hour $h = 195.6$ m; $u_{star} = 0.281$ m/s; surface heat flux = -15.52 W/m²; peak mean $w2 = 0.0927$ m²/s². Between 7–8 and 8–9 h, h changed by -3.7 m and the $w2$ profile changed by 0.00438 m²/s² RMS. Simulation wall time 529.7 s (40114 iterations), excluding preceding compilation/startup.

n032_weno5_none: final-hour $h = 136.1$ m; $u_{star} = 0.242$ m/s; surface heat flux = -11.79 W/m²; peak mean $w2 = 0.05744$ m²/s². Between 7–8 and 8–9 h, h changed by -2.3 m and the $w2$ profile changed by 0.00379 m²/s² RMS. Simulation wall time 505.5 s (39025 iterations), excluding preceding compilation/startup.

n032_weno9_smagorinsky: final-hour $h = 140.6$ m; $u_{star} = 0.223$ m/s; surface heat flux = -9.17 W/m²; peak mean $w2 = 1.12e-06$ m²/s². Between 7–8 and 8–9 h, h changed by 0.1 m and the $w2$ profile changed by $4.74e-08$ m²/s² RMS. Simulation wall time 473.4 s (35312 iterations), excluding preceding compilation/startup.

n064_weno9_none: final-hour $h = 187.8$ m; $u_{star} = 0.279$ m/s; surface heat flux = -15.24 W/m²; peak mean $w2 = 0.09765$ m²/s². Between 7–8 and 8–9 h, h changed by -3.3 m and the $w2$ profile changed by 0.00154 m²/s² RMS. Simulation wall time 1050.0 s (80959 iterations), excluding preceding compilation/startup.

GABLS1: setup and results

n064_weno5_none: final-hour $h = 150.9$ m; $u_{star} = 0.248$ m/s; surface heat flux = -12.01 W/m²; peak mean $w2 = 0.06558$ m²/s². Between 7–8 and 8–9 h, h changed by -10.9 m and the $w2$ profile changed by 0.00248 m²/s² RMS. Simulation wall time 1029.6 s (79613 iterations), excluding preceding compilation/startup.

n064_weno9_smagorinsky: final-hour $h = 149.7$ m; $u_{star} = 0.234$ m/s; surface heat flux = -10.16 W/m²; peak mean $w2 = 0.0315$ m²/s². Between 7–8 and 8–9 h, h changed by -21.0 m and the $w2$ profile changed by 0.00986 m²/s² RMS. Simulation wall time 1020.2 s (79350 iterations), excluding preceding compilation/startup.

n128_weno9_none: final-hour $h = 184.4$ m; $u_{star} = 0.270$ m/s; surface heat flux = -13.56 W/m²; peak mean $w2 = 0.1008$ m²/s². Between 7–8 and 8–9 h, h changed by -1.3 m and the $w2$ profile changed by 0.000826 m²/s² RMS. Simulation wall time 4156.8 s (162036 iterations), excluding preceding compilation/startup.

n128_weno5_none: final-hour $h = 162.9$ m; $u_{star} = 0.245$ m/s; surface heat flux = -11.12 W/m²; peak mean $w2 = 0.0696$ m²/s². Between 7–8 and 8–9 h, h changed by 7.9 m and the $w2$ profile changed by 0.00189 m²/s² RMS. Simulation wall time 4058.5 s (160877 iterations), excluding preceding compilation/startup.

n128_weno9_smagorinsky: final-hour $h = 188.1$ m; $u_{star} = 0.277$ m/s; surface heat flux = -13.37 W/m²; peak mean $w2 = 0.08562$ m²/s². Between 7–8 and 8–9 h, h changed by 3.9 m and the $w2$ profile changed by 0.00112 m²/s² RMS. Simulation wall time 4172.6 s (161379 iterations), excluding preceding compilation/startup.

n200_weno9_none: final-hour $h = 178.5$ m; $u_{star} = 0.261$ m/s; surface heat flux = -11.87 W/m²; peak mean $w2 = 0.0993$ m²/s². Between 7–8 and 8–9 h, h changed by 3.3 m and the $w2$ profile changed by 0.000919 m²/s² RMS. Simulation wall time 9396.4 s (253504 iterations), excluding preceding compilation/startup.

n200_weno5_none: final-hour $h = 159.2$ m; $u_{star} = 0.240$ m/s; surface heat flux = -10.04 W/m²; peak mean $w2 = 0.06902$ m²/s². Between 7–8 and 8–9 h, h changed by 1.8 m and the $w2$ profile changed by 0.000677 m²/s² RMS. Simulation wall time 8387.3 s (252093 iterations), excluding preceding compilation/startup.

n200_weno9_smagorinsky: final-hour $h = 188.9$ m; $u_{star} = 0.274$ m/s; surface heat flux = -12.22 W/m²; peak mean $w2 = 0.09298$ m²/s². Between 7–8 and 8–9 h, h changed by 3.3 m and the $w2$ profile changed by 0.00138 m²/s² RMS. Simulation wall time 10660.3 s (252948 iterations), excluding preceding compilation/startup.

GABLS1: setup and results

n400_weno9_none: final-hour $h = 165.3$ m; $u_{star} = 0.241$ m/s; surface heat flux = -8.05 W/m²; peak mean $w_2 = 0.08926$ m²/s². Between 7–8 and 8–9 h, h changed by 2.4 m and the w_2 profile changed by 0.000512 m²/s² RMS. Simulation wall time 95437.1 s (510626 iterations), excluding preceding compilation/startup.

n400_weno5_none: final-hour $h = 159.4$ m; $u_{star} = 0.229$ m/s; surface heat flux = -7.47 W/m²; peak mean $w_2 = 0.06807$ m²/s². Between 7–8 and 8–9 h, h changed by 4.7 m and the w_2 profile changed by 0.000682 m²/s² RMS. Simulation wall time 71341.1 s (506162 iterations), excluding preceding compilation/startup.

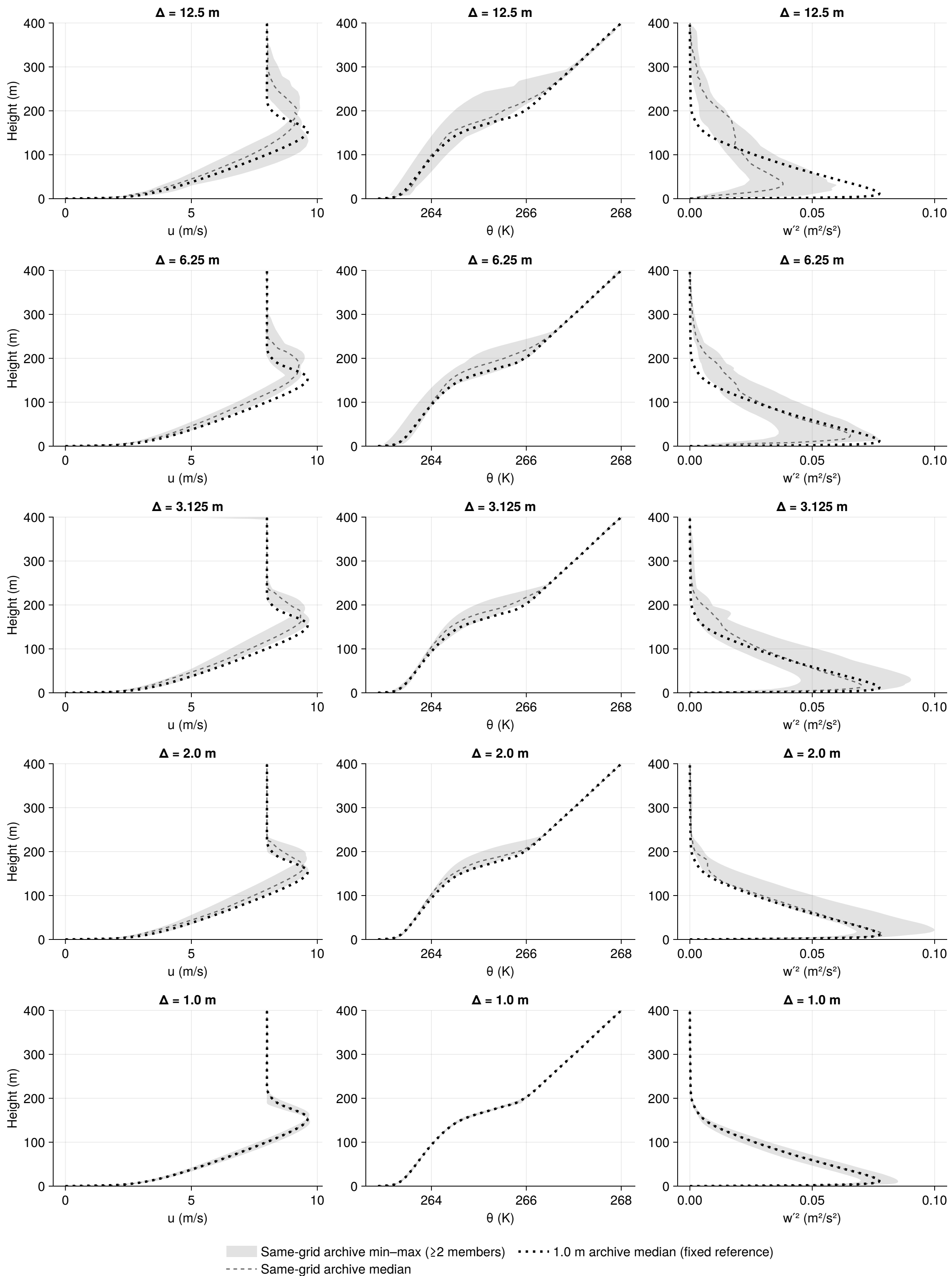
n400_weno9_smagorinsky: final-hour $h = 182.0$ m; $u_{star} = 0.266$ m/s; surface heat flux = -9.12 W/m²; peak mean $w_2 = 0.09851$ m²/s². Between 7–8 and 8–9 h, h changed by 2.6 m and the w_2 profile changed by 0.000469 m²/s² RMS. Simulation wall time 110783.2 s (514053 iterations), excluding preceding compilation/startup.

Current queue snapshot (job|state|elapsed|start|node or pending reason):

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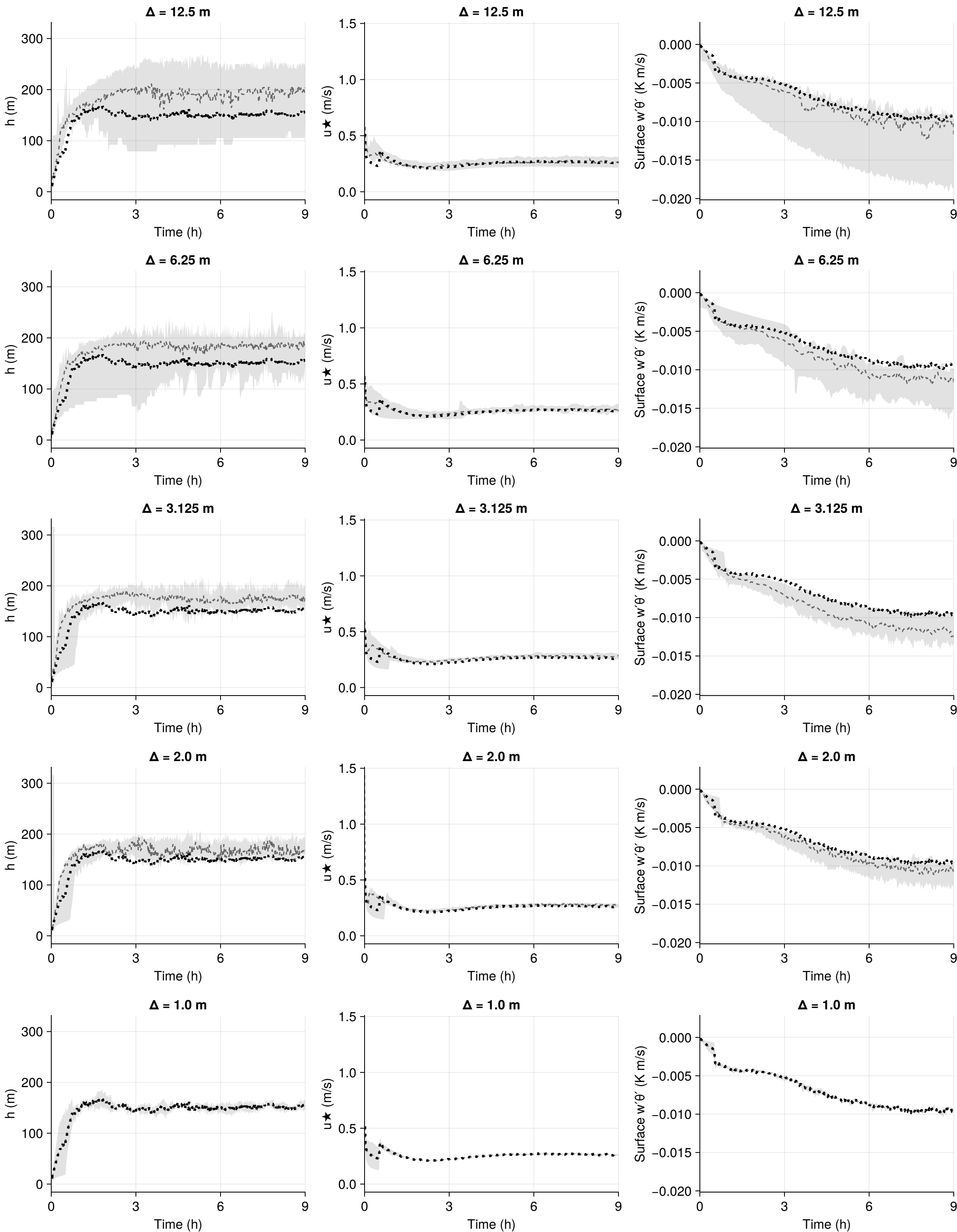
7343_2|PENDING|0:00|2026-09-22T05:51:20|(Resources) 7343_4|PENDING|0:00|N/A|(Resources)

GABLS1 reference archive: final-hour model spread



Archive: nominal final-hour profiles, including some NERSC windows of 474–534 min. Breeze curves, when present: 8–9 h means.
Native heights interpolated without extrapolation; model membership varies with height and grid. The spread is not observational uncertainty.
Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.

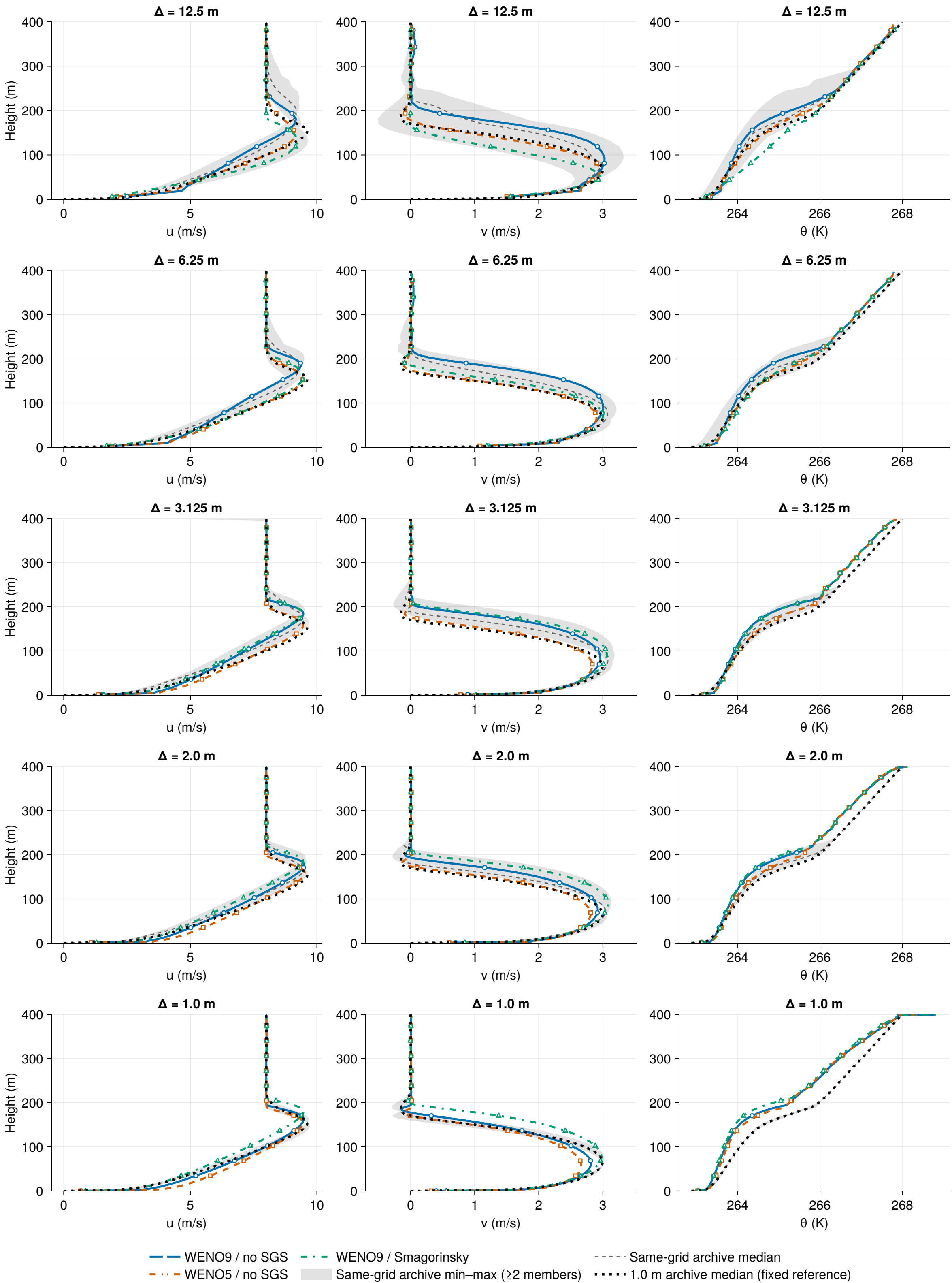
GABLS1 reference archive: evolution



Same-grid archive min-max (≥ 2 members)
 1.0 m archive median (fixed reference)
 Same-grid archive median

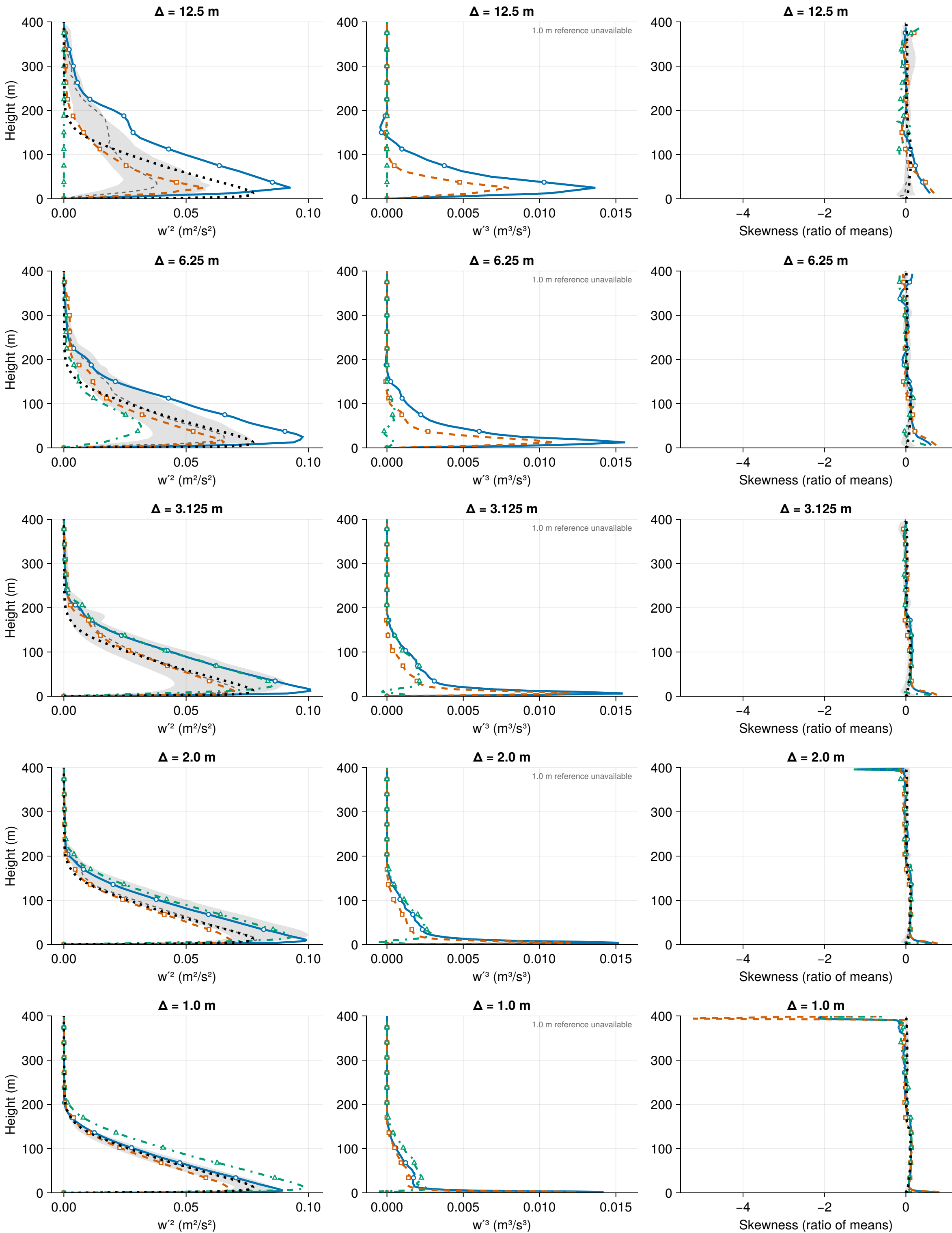
Archive series interpolated to common times without extrapolation; missing values omitted.
 Membership varies with grid and time. Black dots: the same 1.0 m ensemble median in every row. Heat flux is upward positive.

GABLS1: mean wind and temperature



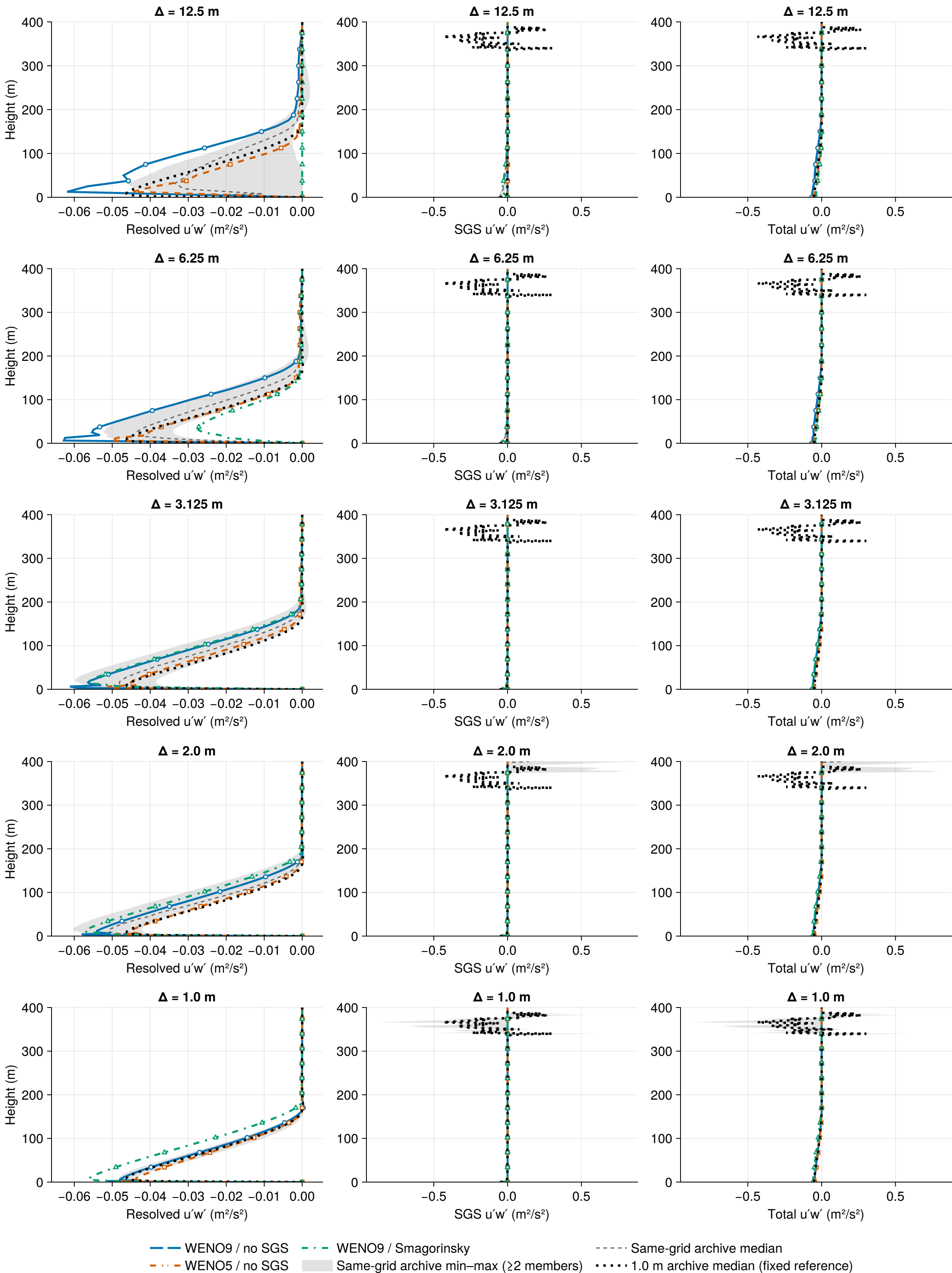
Archive: nominal final-hour profiles, including some NERSC windows of 474–534 min. Breeze curves, when present: 8–9 h means.
Native heights interpolated without extrapolation; model membership varies with height and grid. The spread is not observational uncertainty.
Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.

GABLS1: vertical velocity moments



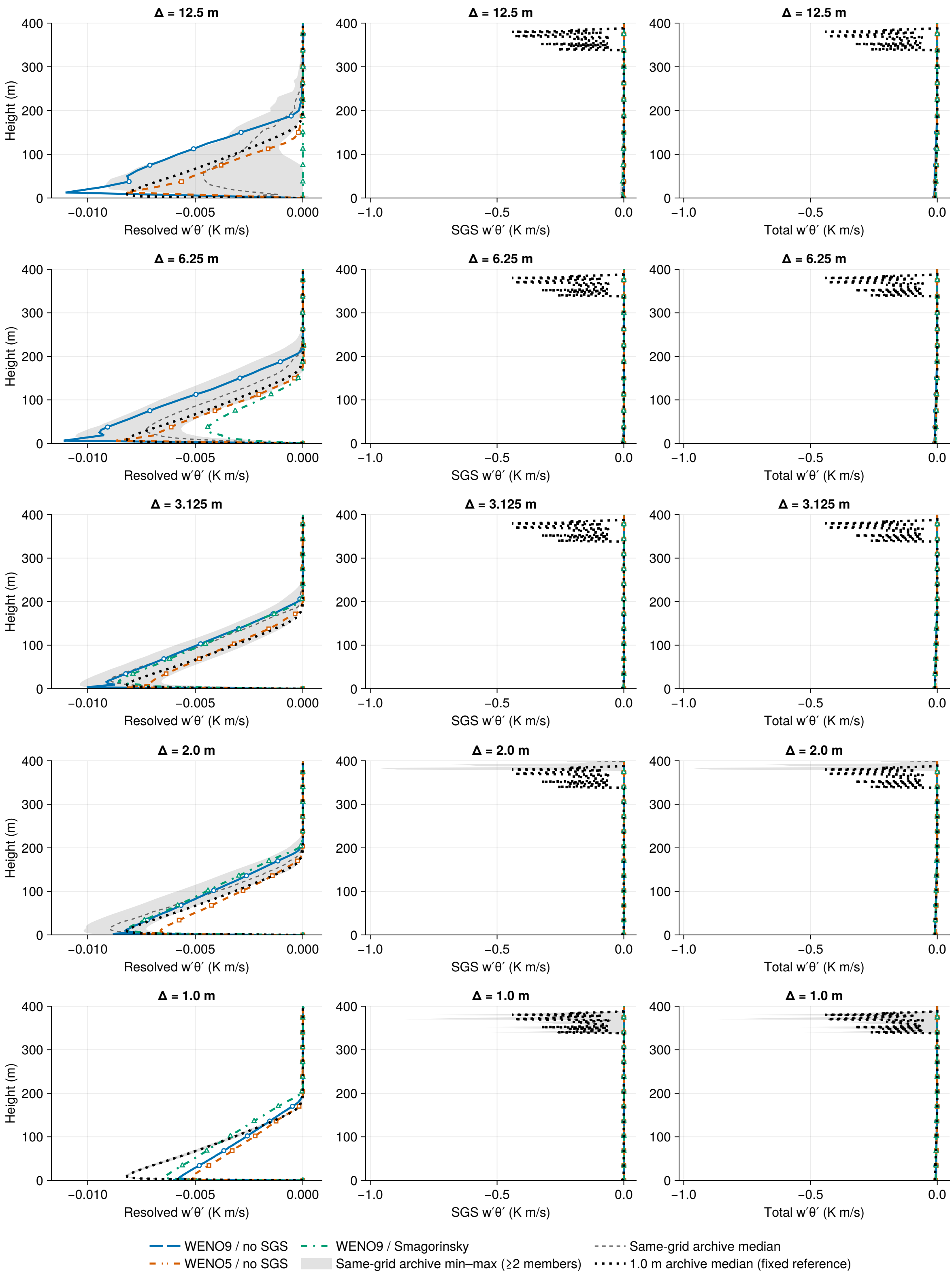
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Native heights interpolated without extrapolation; model membership varies with height and grid. The spread is not observational uncertainty.
Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.

GABLS1: streamwise momentum transport



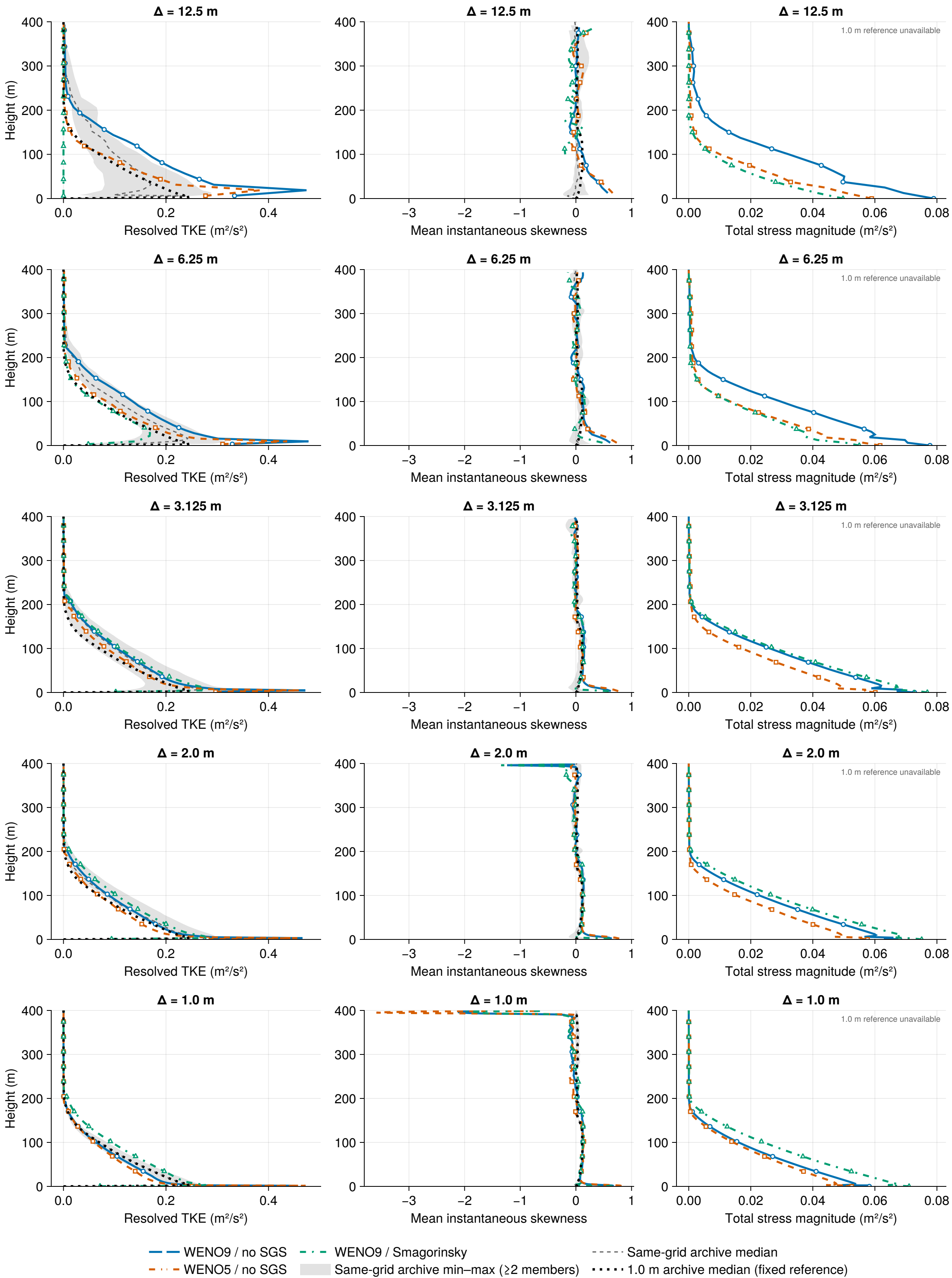
Archive: nominal final-hour profiles, including some NERSC windows of 474–534 min. Breeze curves, when present: 8–9 h means.
 Native heights interpolated without extrapolation; model membership varies with height and grid. The spread is not observational uncertainty.
 Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.
 Archive caution: 1.0 m IMUK SGS data contain upper-level anomalies; retained as supplied and flagged in the report.

GABLS1: vertical heat transport



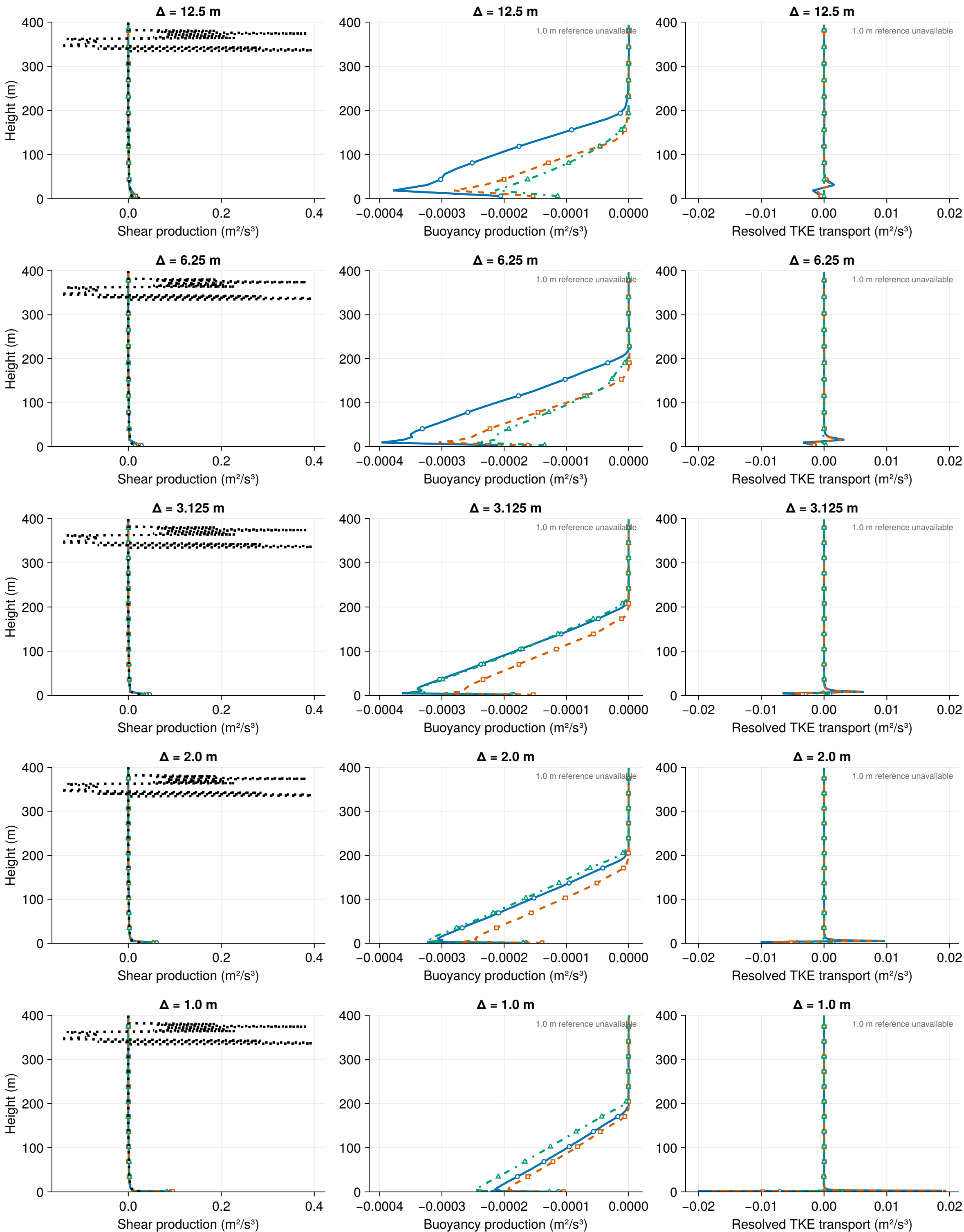
Archive: nominal final-hour profiles, including some NERSC windows of 474–534 min. Breeze curves, when present: 8–9 h means.
Native heights interpolated without extrapolation; model membership varies with height and grid. The spread is not observational uncertainty.
Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.
Archive caution: 1.0 m IMUK SGS data contain upper-level anomalies; retained as supplied and flagged in the report.

GABLS1: turbulence intensity, asymmetry and stress



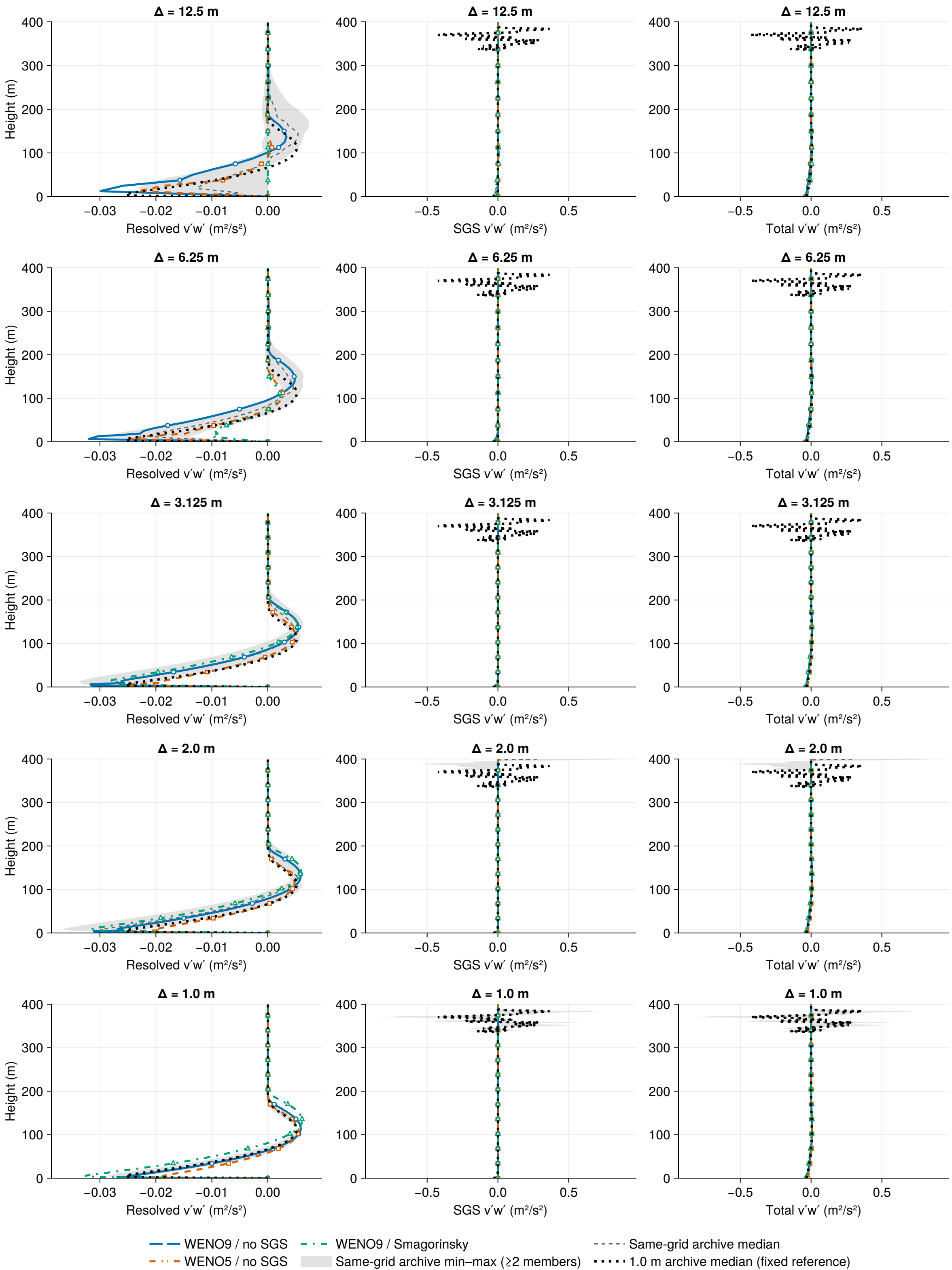
Archive: nominal final-hour profiles, including some NERSC windows of 474–534 min. Breeze curves, when present: 8–9 h means.
 Native heights interpolated without extrapolation; model membership varies with height and grid. The spread is not observational uncertainty.
 Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.

GABLS1: selected TKE budget terms



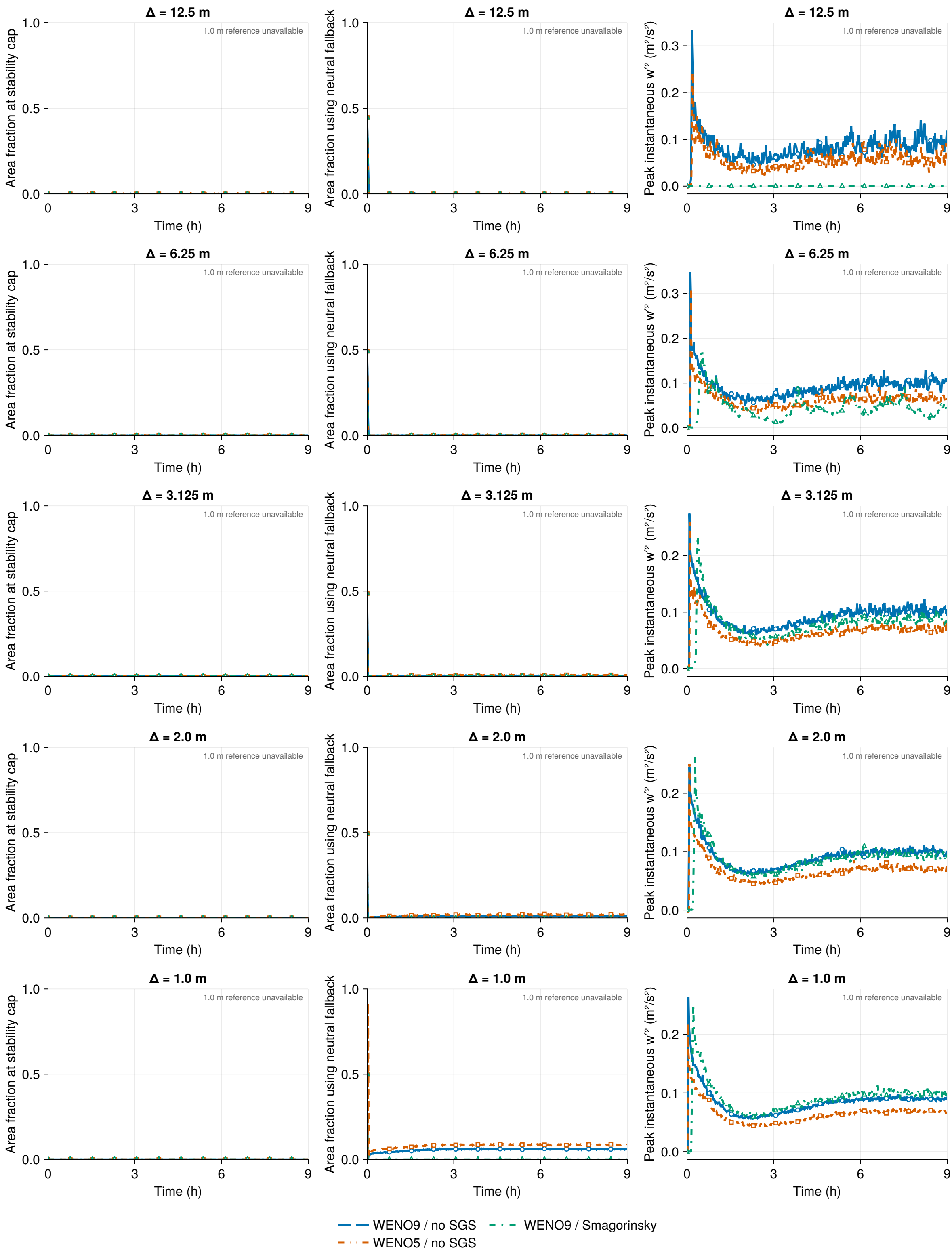
Breeze: 8–9 h means; only fully completed and audited exports. Shown budget terms do not form a closed budget.
A budget residual must not be interpreted as numerical dissipation; unavailable terms are not replaced by zero.
Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.
Archive caution: 1.0 m IMUK SGS data contain upper-level anomalies; retained as supplied and flagged in the report.

GABLS1: crosswind momentum transport



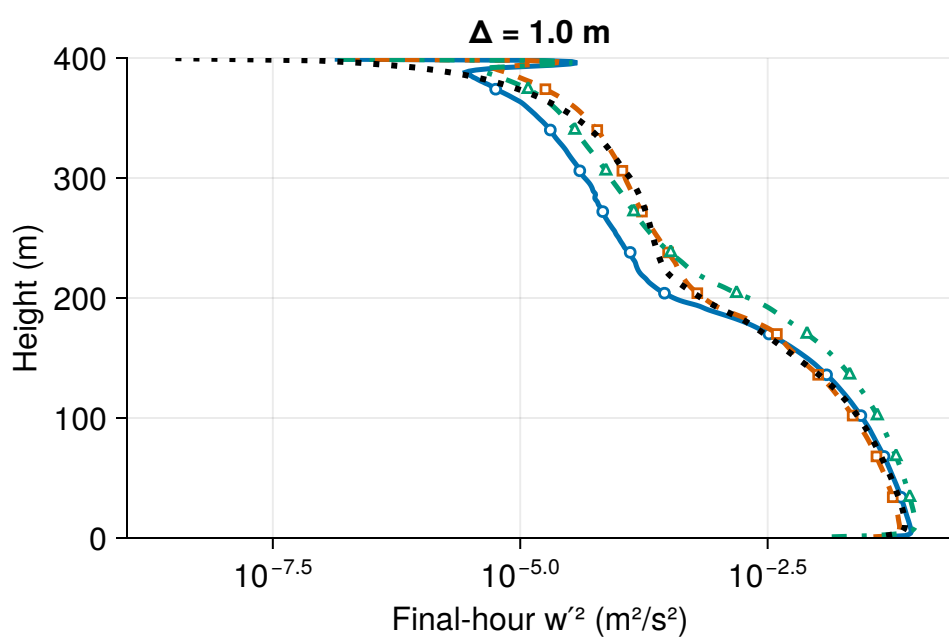
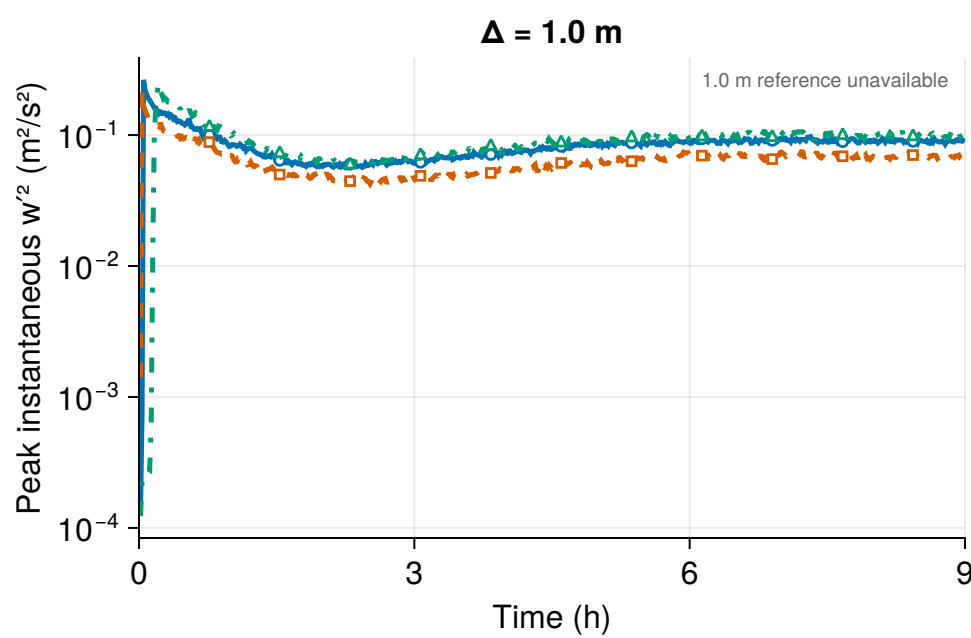
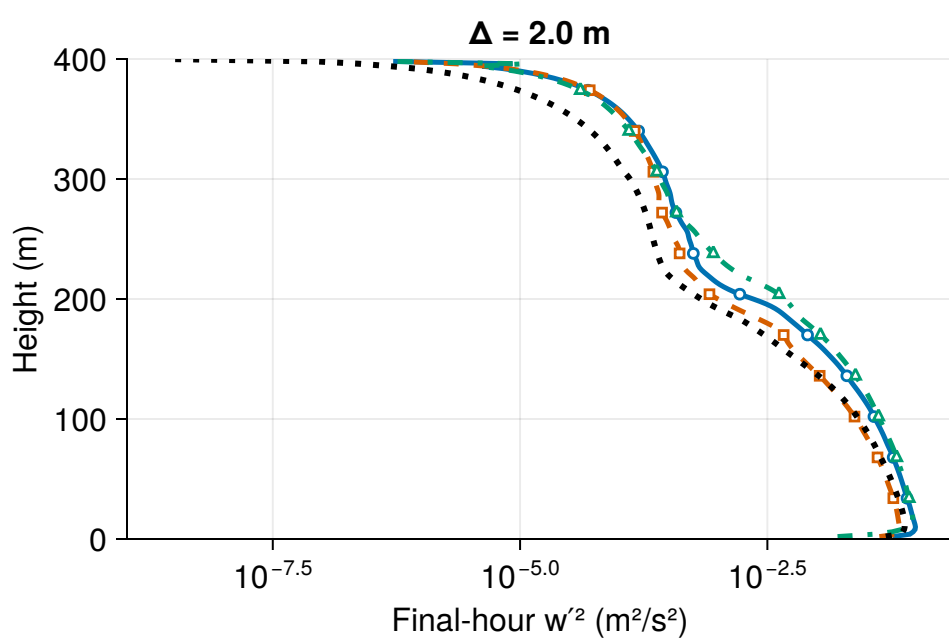
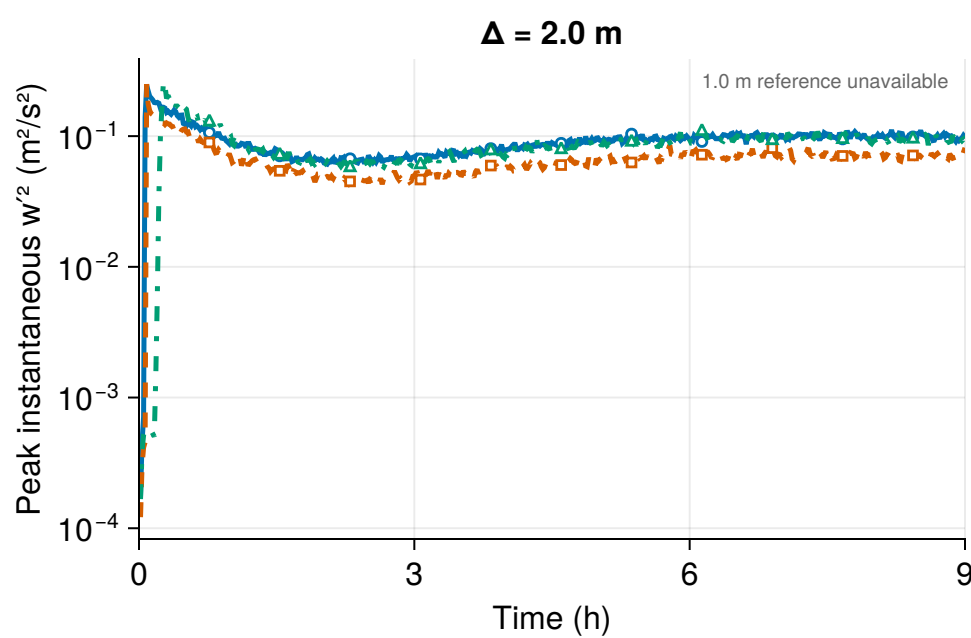
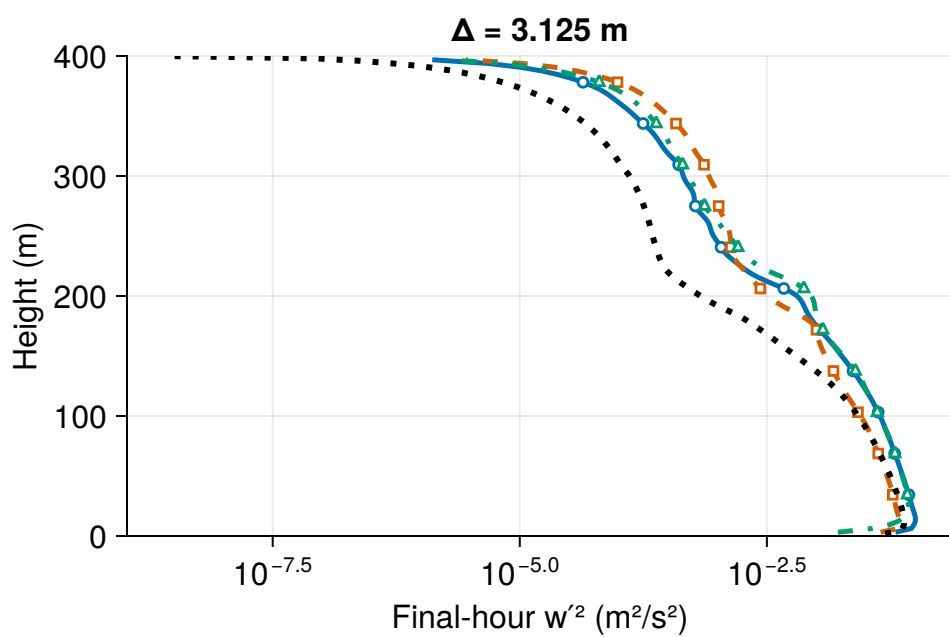
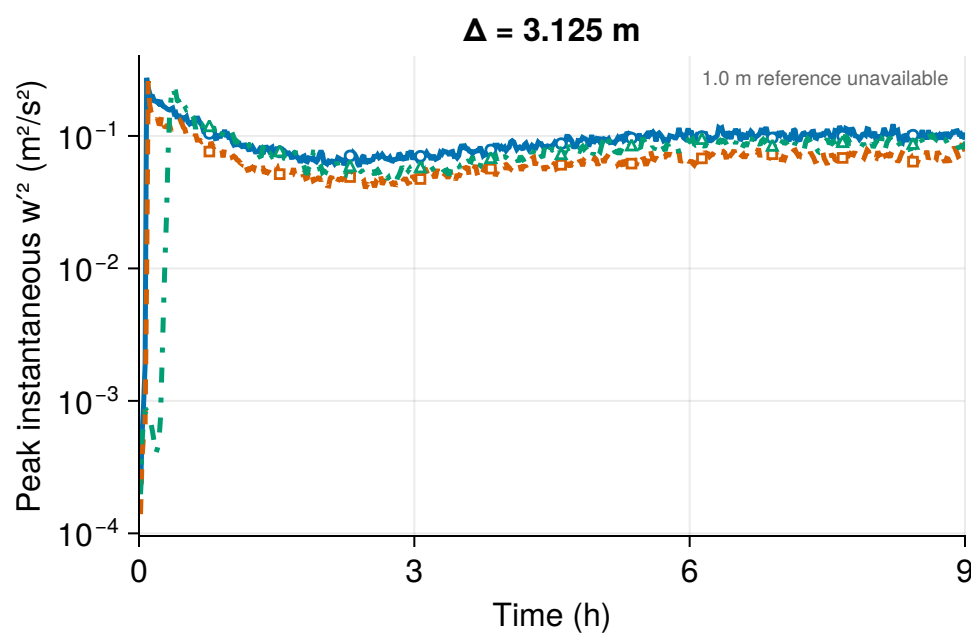
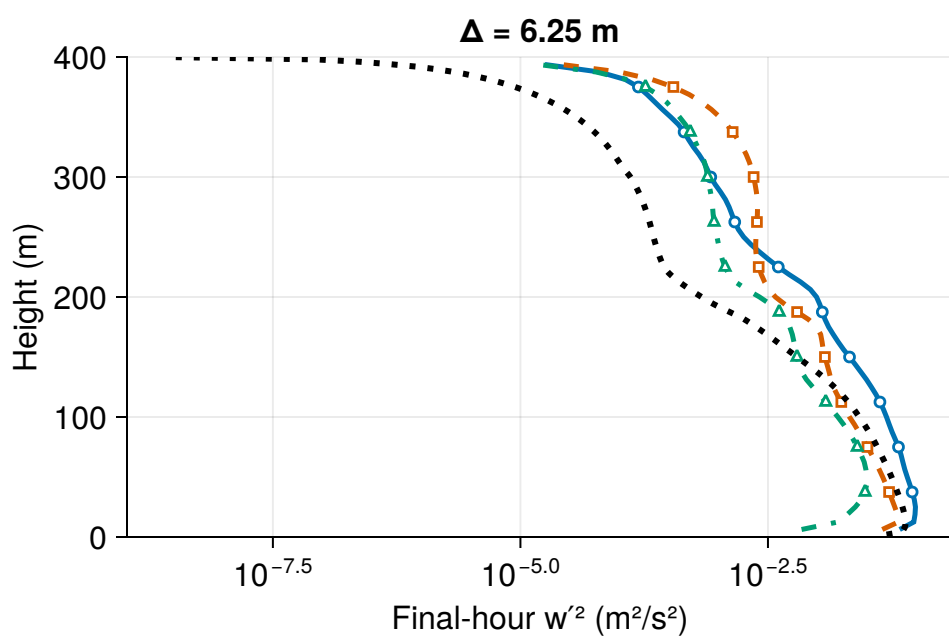
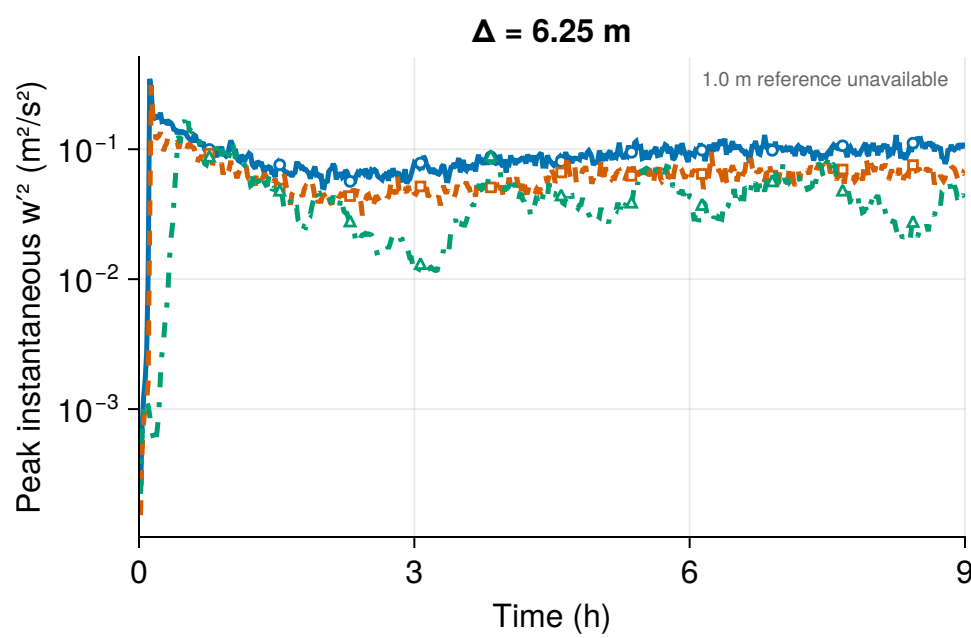
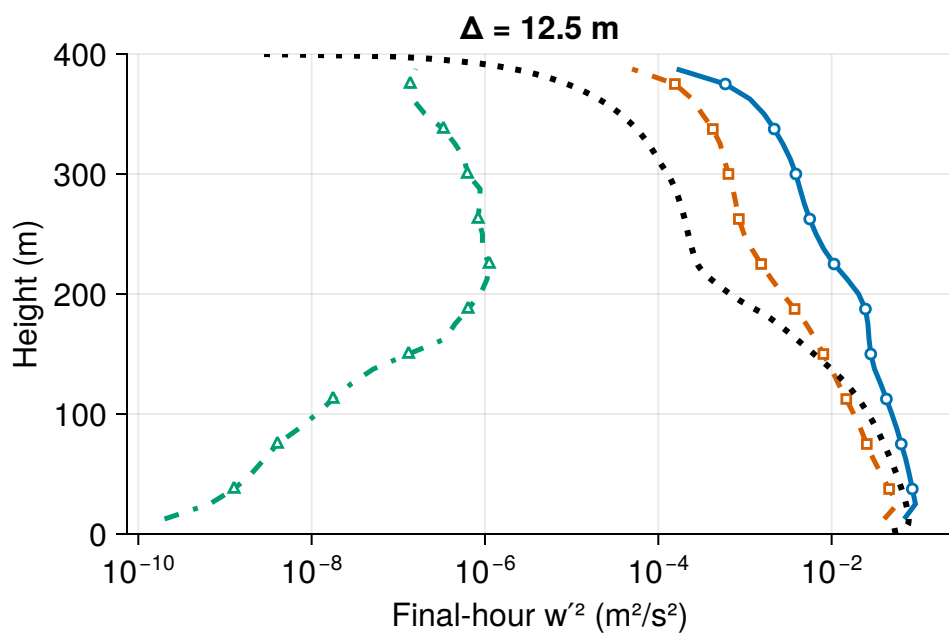
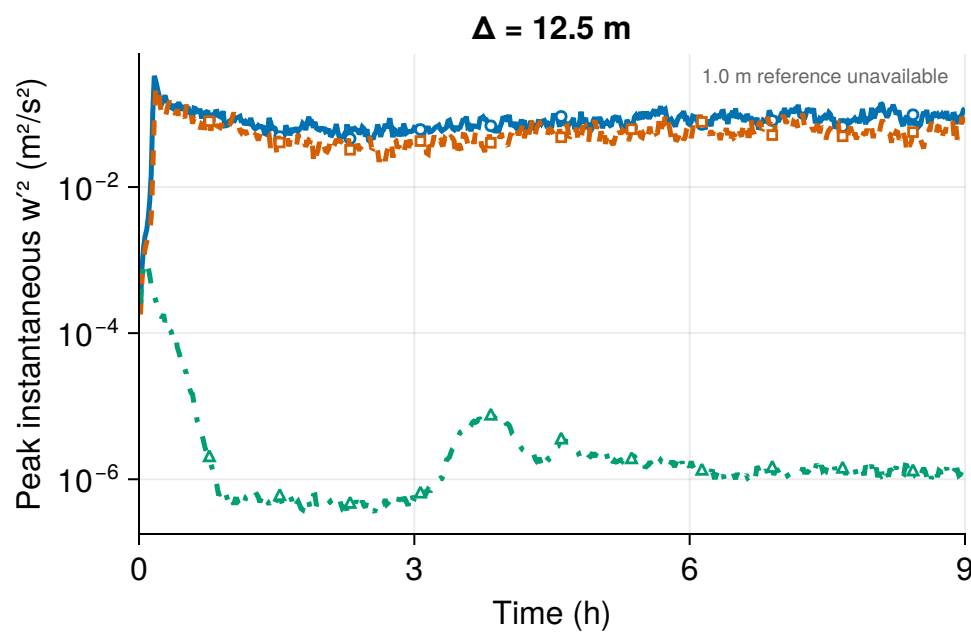
Archive: nominal final-hour profiles, including some NERSC windows of 474–534 min. Breeze curves, when present: 8–9 h means.
Native heights interpolated without extrapolation; model membership varies with height and grid. The spread is not observational uncertainty.
Black dots: fixed 1.0 m ensemble median; interpolation precedes the median. Unavailable references are labeled.
Archive caution: 1.0 m IMUK SGS data contain upper-level anomalies; retained as supplied and flagged in the report.

GABLS1: wall-model regimes and resolved turbulence



Instantaneous 60 s samples. The wall law caps ζ at 10 and uses neutral exchange for nonpositive bulk Richardson number. These fractions identify when wall-model limits participate in the result. No matching archive regime or peak-variance series is available.

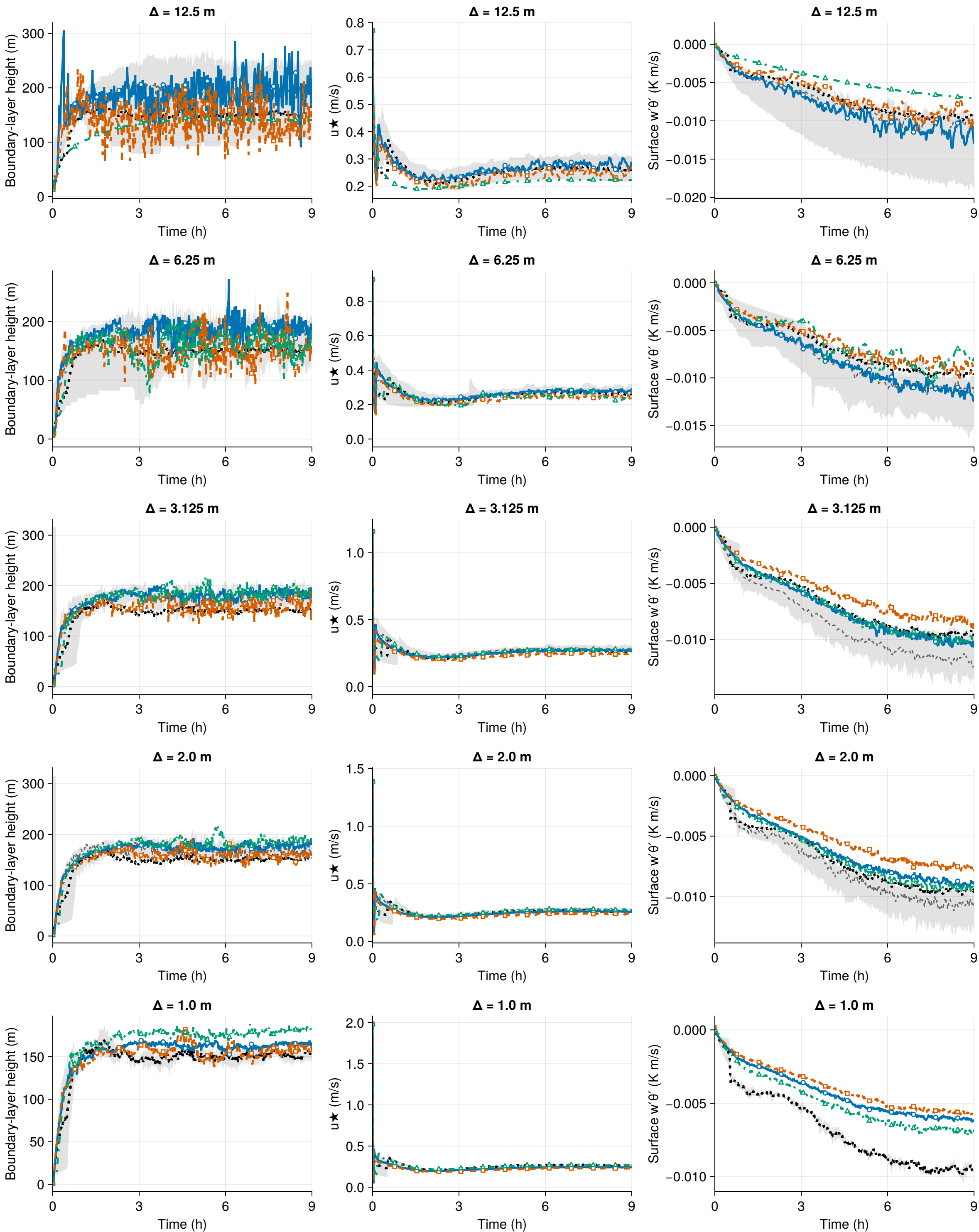
GABLS1: resolved vertical-velocity variance



— WENO9 / no SGS - - WENO9 / Smagorinsky
- - WENO5 / no SGS 1.0 m archive median (fixed reference)

Logarithmic variance axes distinguish weak resolved motions from sustained turbulence. Left: instantaneous profile maxima; right: 8–9 h means. Black dots: fixed 1.0 m median (right only). No archived peak-variance time series. Nonpositive values omitted without a substituted floor.

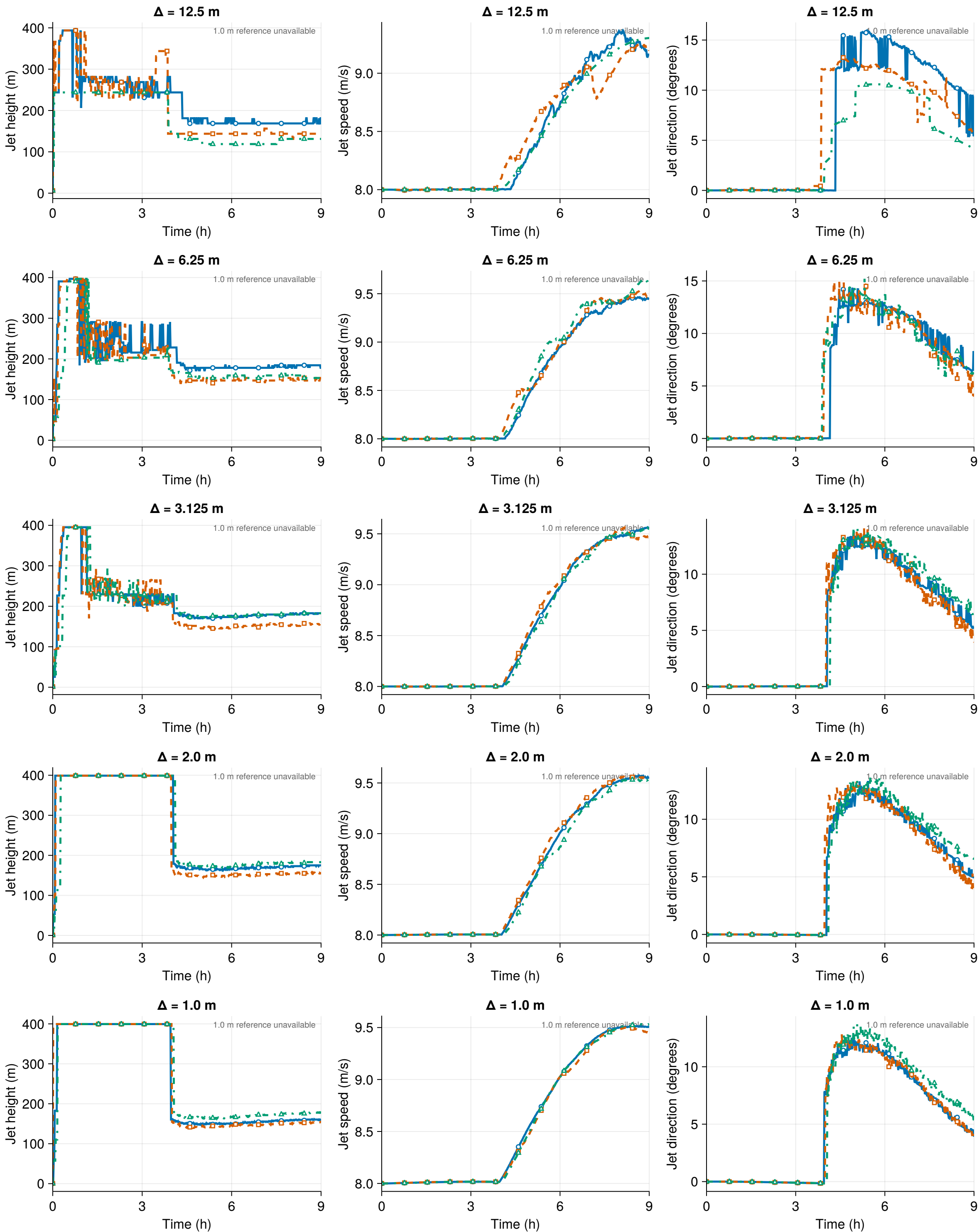
GABLS1: evolution



— WENO9 / no SGS - - - WENO9 / Smagorinsky - - - Same-grid archive median
 ··· WENO5 / no SGS Same-grid archive min-max (≥ 2 members) ···· 1.0 m archive median (fixed reference)

Breeze: instantaneous 60 s samples. Archive series interpolated to common times without extrapolation; model membership varies.
 h = height of 5% total surface stress / 0.95; invalid crossings omitted. Black dots: fixed 1.0 m median. Heat flux is upward positive.

GABLS1: low-level jet evolution



Instantaneous 60 s samples of the maximum horizontal-mean wind speed. Direction is $\text{atan}(v, u)$ relative to eastward geostrophic wind. Invalid jet diagnostics are omitted; a changing peak height can jump between adjacent grid levels. No archived jet time series is available.

SurfaceLayerDiffusivity: GABLS1 results

Corrected transport comparison | 21 September 2026 | four matched runs, one seed.

Question: can transport confined to one or two near-wall faces improve coarse LES? Four paired GABLS1 runs use WENO9 at 12.5 m resolution (32^3 cells), a 400 m cube, the same frozen source, and 9 h of integration.

Finding: all three SurfaceLayerDiffusivity variants move the mean u , mean θ and w^2 profiles farther from the fixed 1 m LES median by RMS error over native levels below 200 m. This result applies to this grid and these settings; it is not a general verdict on the closure.

Near-wall response: at 12.5 m, final-hour w^2 falls by 85–93%. Skewness changes from +0.56 in the control to -0.21, -0.25 and -0.42. Domain-integrated resolved TKE falls by 15–20%. The first-face variance is suppressed much more strongly than the integrated energy.

Surface exchange: final-hour u^* rises from 0.279 m/s to 0.292, 0.289 and 0.301 m/s, versus a reference mean of 0.262 m/s. Surface cooling also strengthens, moving farther from the reference. One-face 100 and 300 s responses are similar; two-face support increases local suppression.

Smagorinsky comparison: the earlier 12.5 m WENO9 run ($C_s=0.16$, $Pr=1$) has a final-hour peak w^2 of only $1.12e-6 \text{ m}^2/\text{s}^2$, versus 0.0864 in the control. Its mean- u error is smaller despite almost absent resolved turbulence. Teal curves show this historical context; its source differs from the four matched runs. Log axes reveal the contrast.

Next hypothesis: restrict SLD viscosity to u and v , retaining tracer diffusivity and all other settings. This isolates direct damping of w , but indirect effects through shear and stratification remain. No u/v -only result is included. Single-seed comparisons do not establish uncertainty; the LES median is not observational truth.

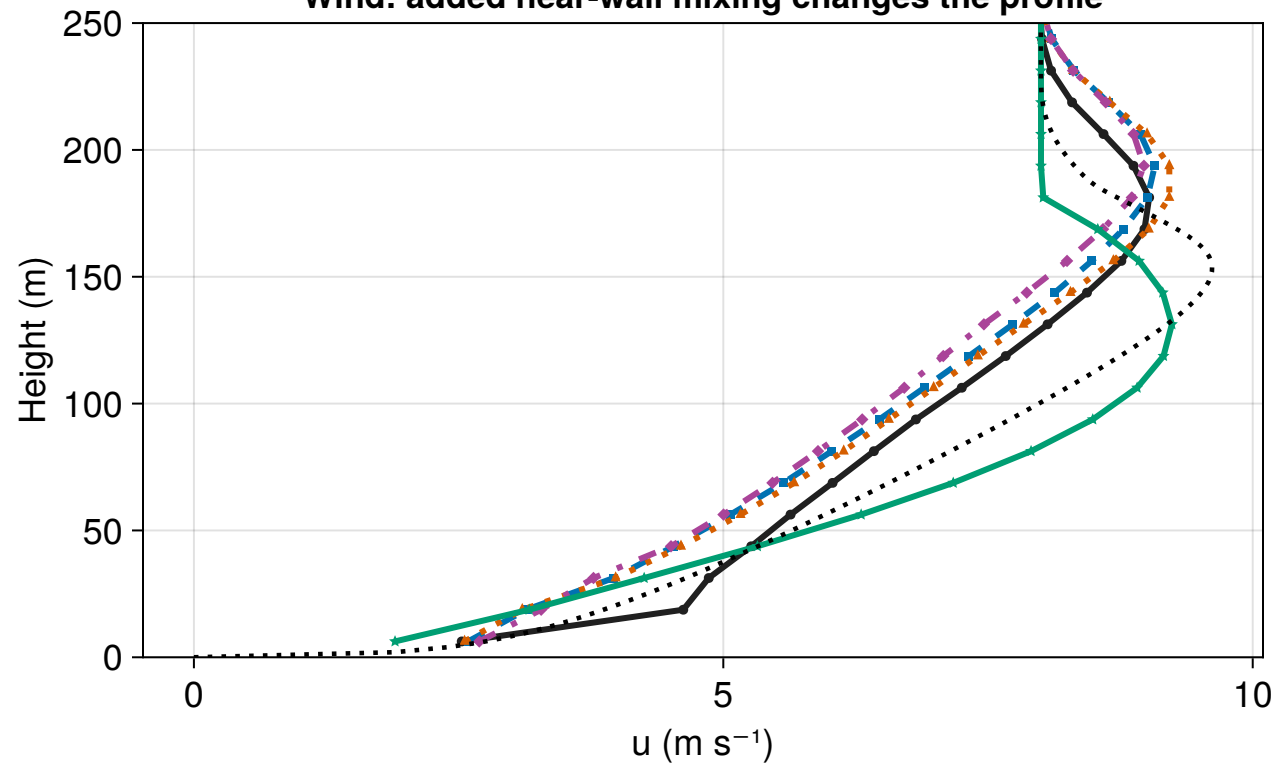
Transport split: corrected diagnostics show SGS mixing supplies 86–95% of first-level u -momentum transport in the SLD cases. Total flux magnitude rises by only 6–17%, while its resolved contribution falls sharply. Mean fields and resolved moments exactly reproduce the earlier runs. The new native coefficient–gradient products pass GPU and export consistency checks; this split does not measure WENO numerical mixing. Historical array 7156 fluxes remain excluded; current figures use corrected array 7293.

Reading the figures: native-height profiles are true 8–9 h means; timelines use instantaneous output. Colors, line styles and markers distinguish cases. Exact data, Julia source, reference hashes and diagnostic exclusions accompany this brief; all original DYCOMS/GABLS1 report material is retained.

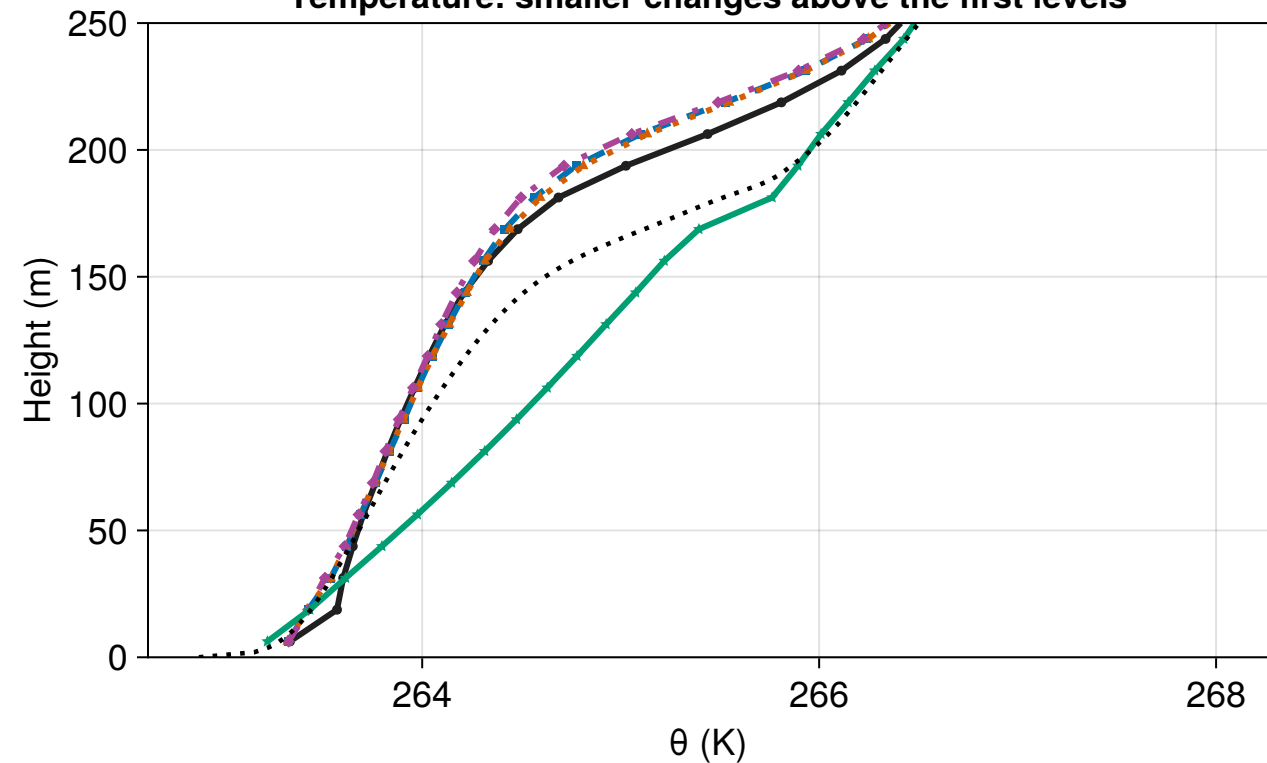
Surface-layer mixing versus coarse-grid Smagorinsky

GABLS1 · WENO9 · 12.5 m grid (32^3) · final hour, 8–9 h · four matched runs + earlier Smagorinsky

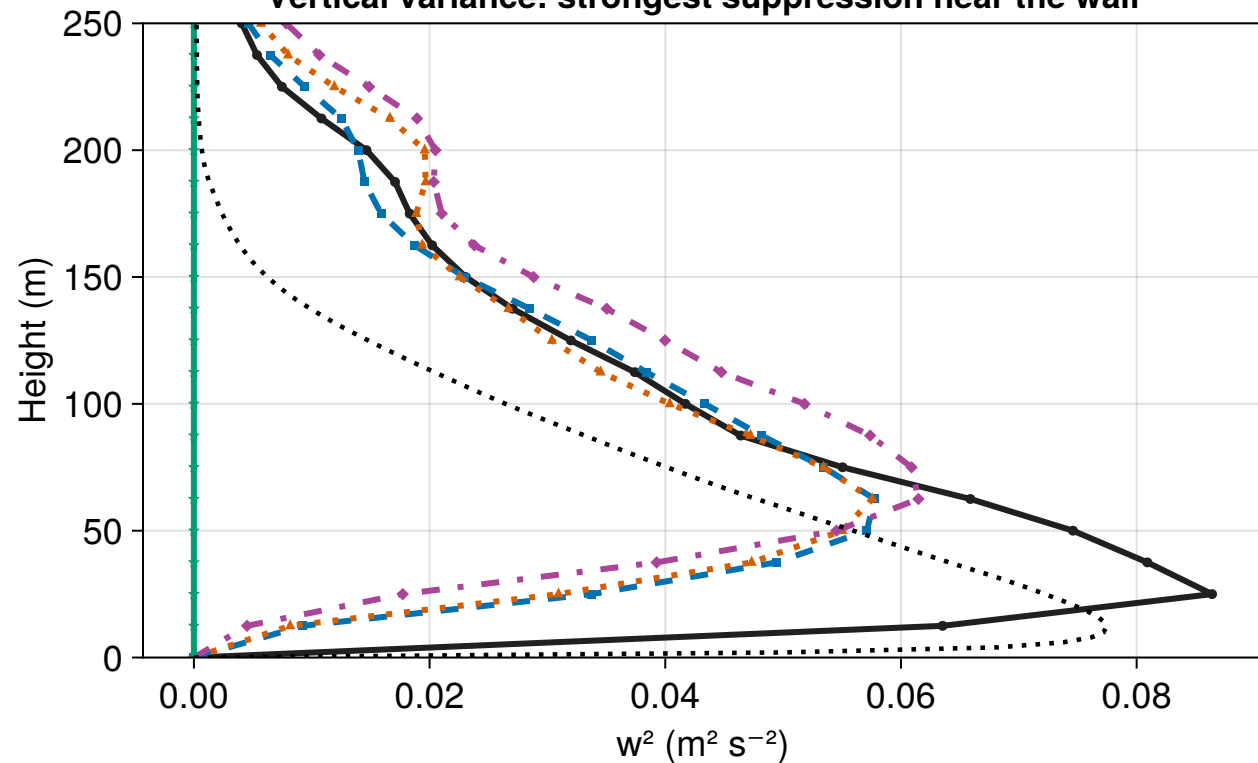
Wind: added near-wall mixing changes the profile



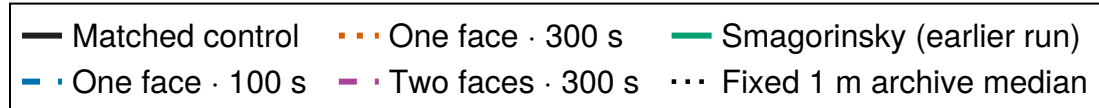
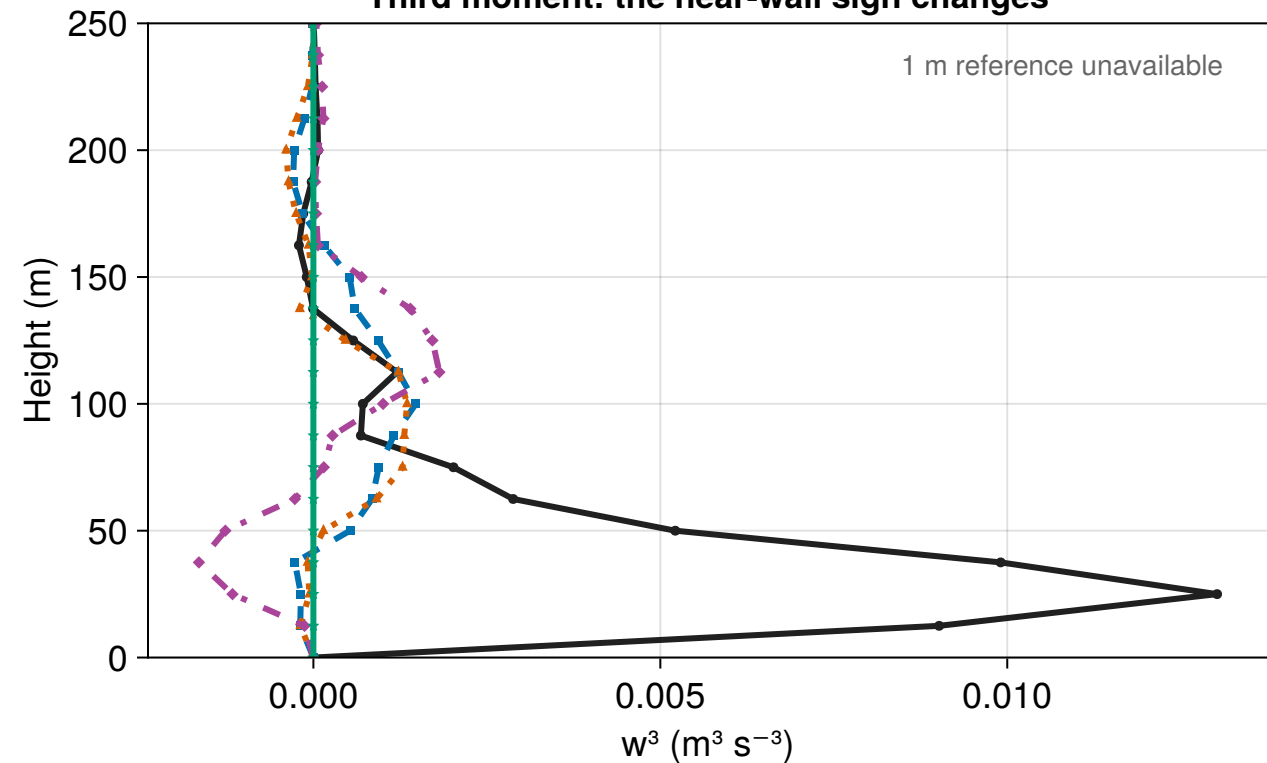
Temperature: smaller changes above the first levels



Vertical variance: strongest suppression near the wall



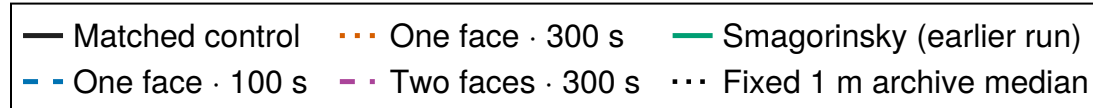
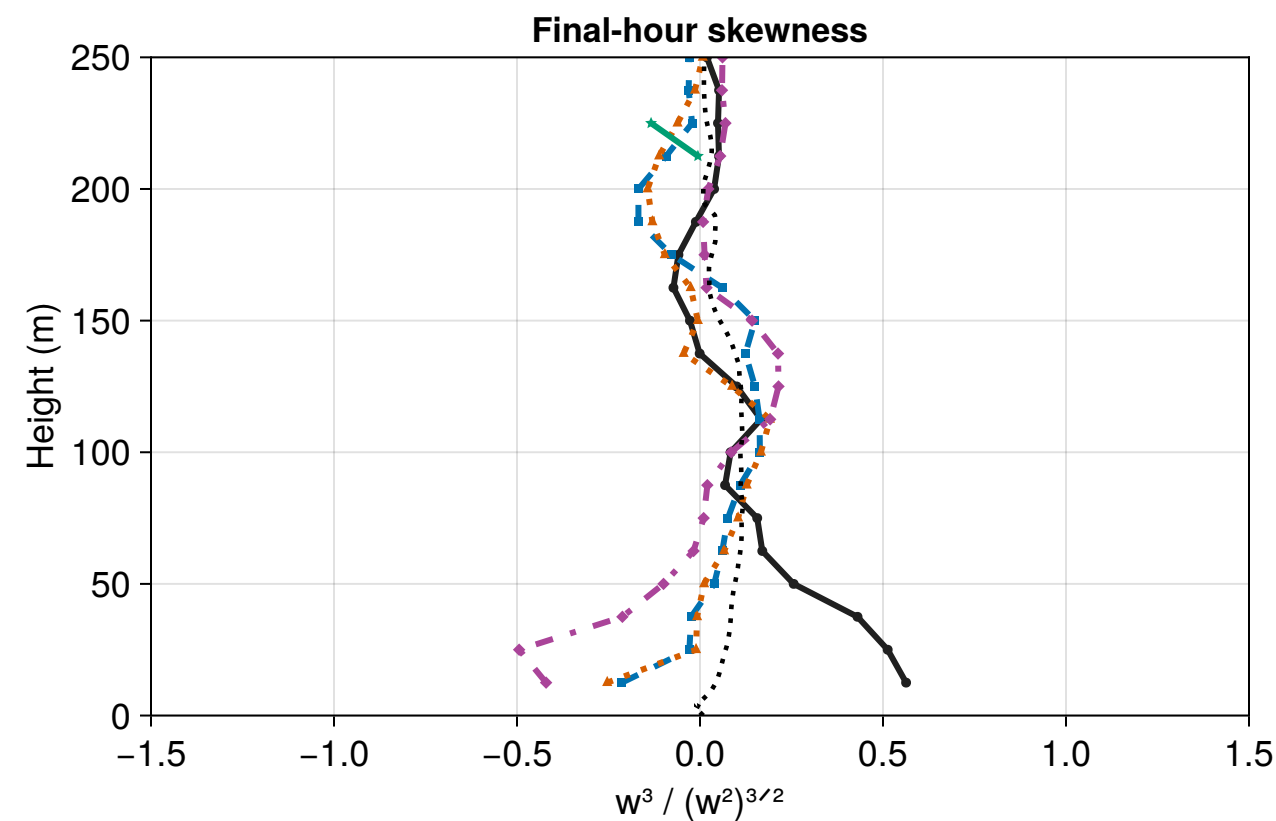
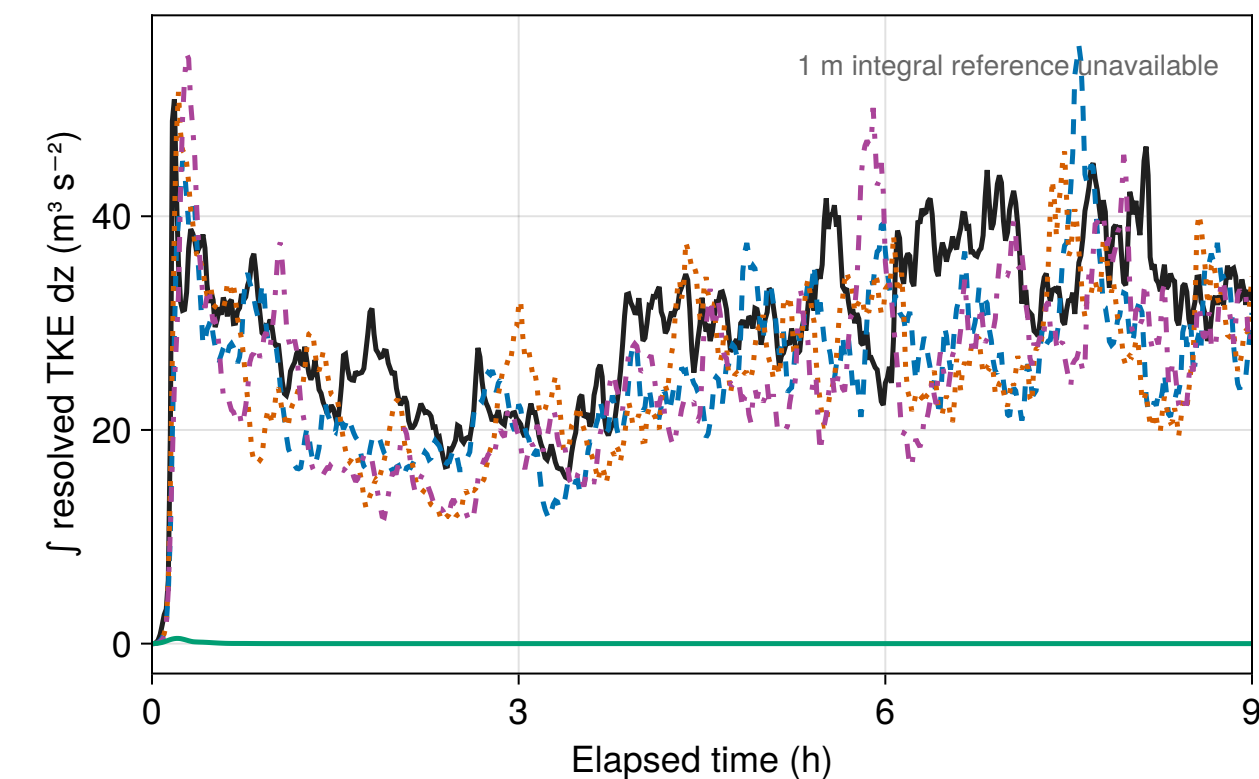
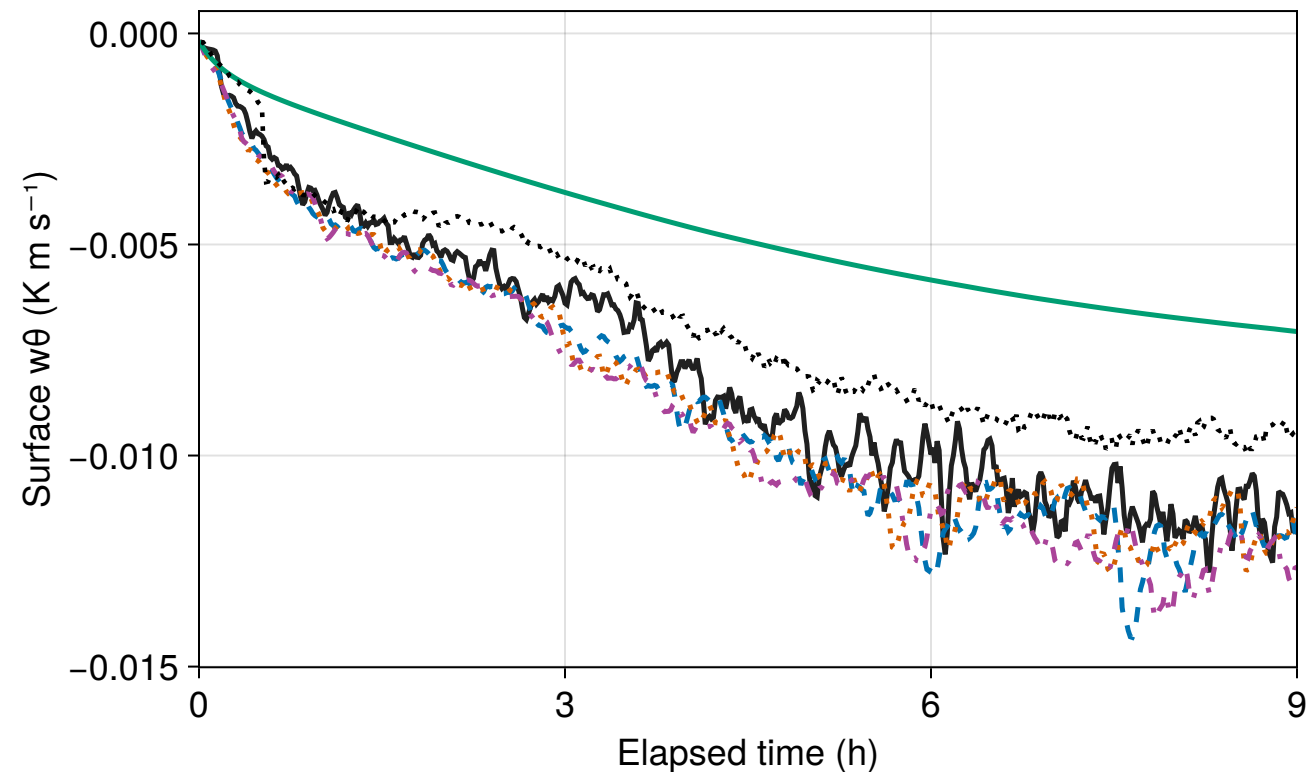
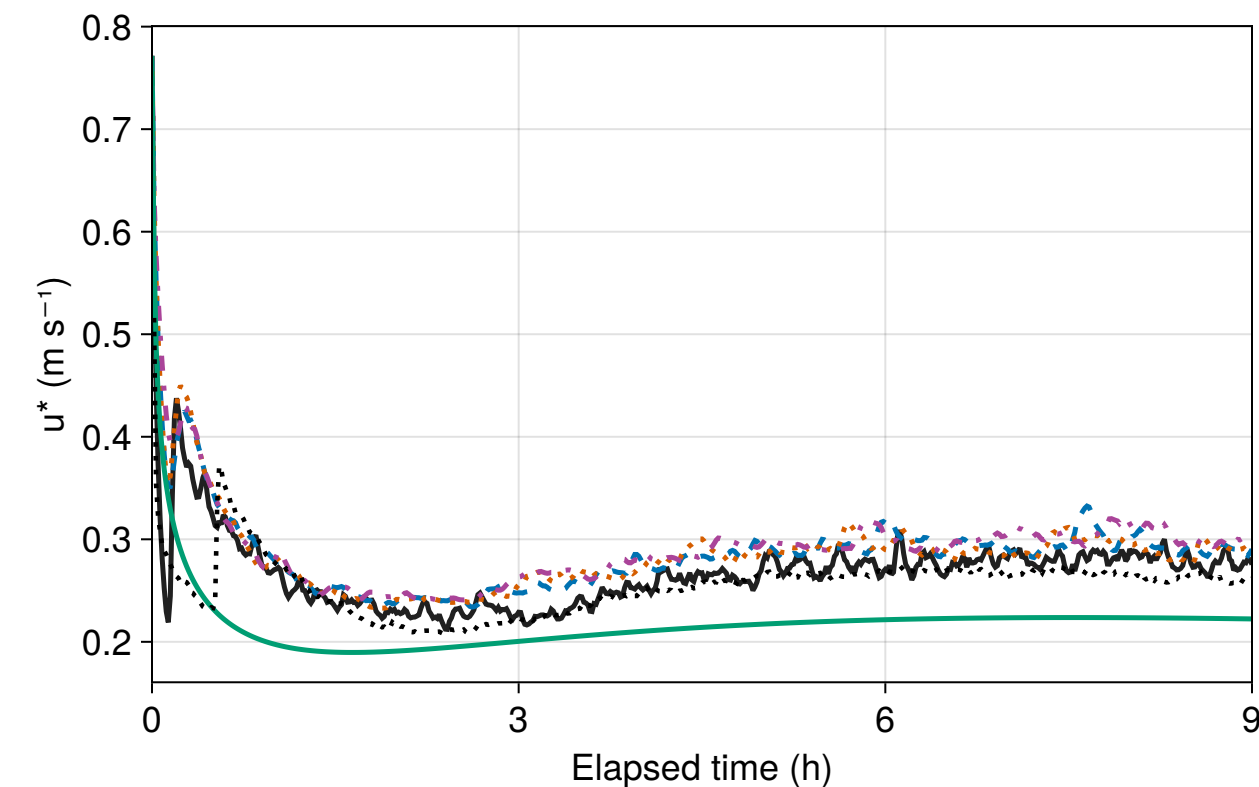
Third moment: the near-wall sign changes



Teal stars: earlier Smagorinsky source; its w^2 and w^3 nearly coincide with zero on these linear axes.
Black dotted: fixed 1 m LES median. Matched runs use the corrected, GPU-validated flux diagnostics.

Follow the response beyond the mean profile

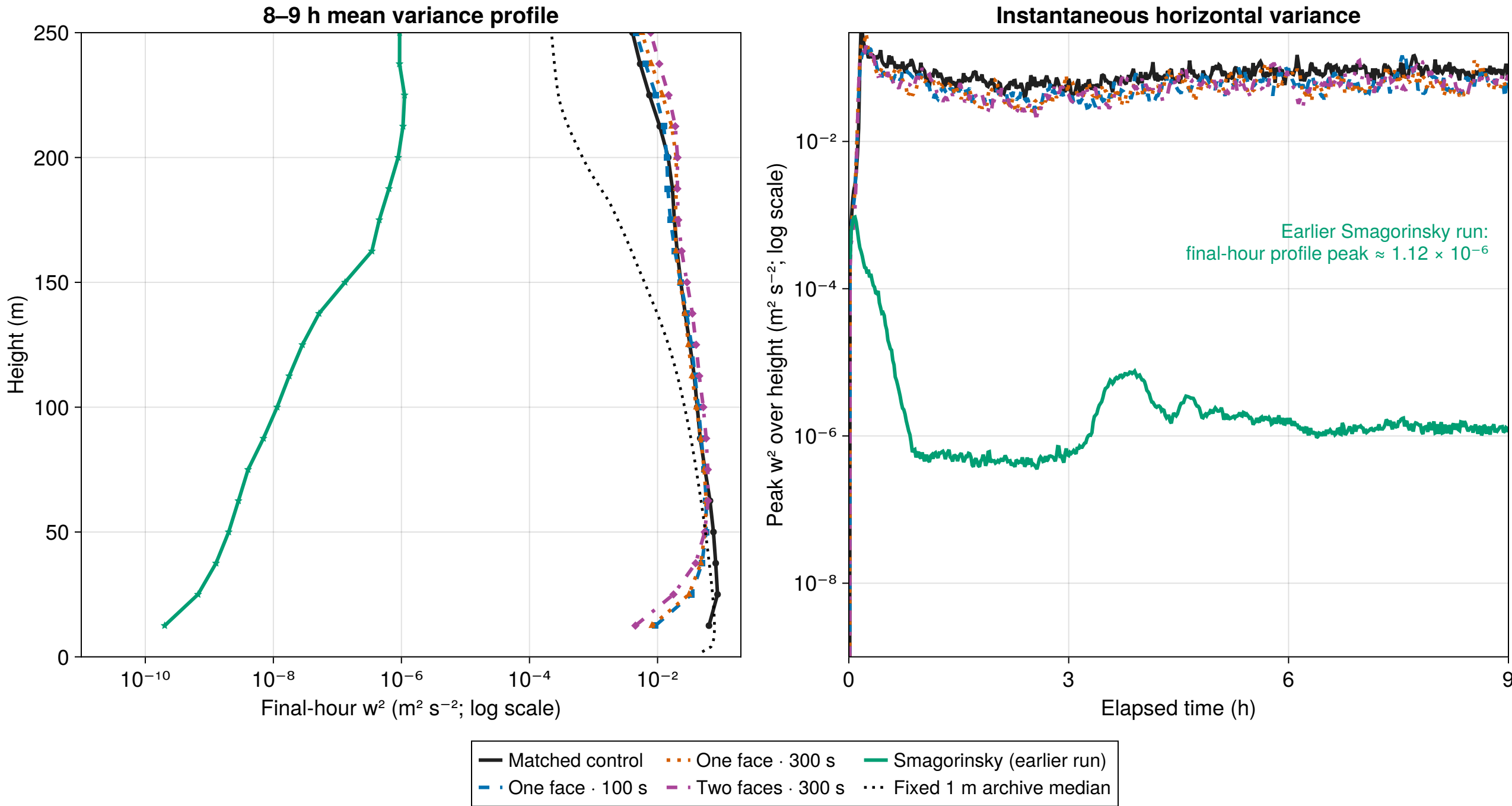
Surface exchange through 9 h; skewness from the final-hour native-face moments



Wall flux remains usable; resolved TKE excludes SGS energy. Teal: earlier Smagorinsky source.
Skewness omits $w^2 < 10^{-6} \text{ m}^2 \text{s}^{-2}$, masking most Smagorinsky levels; a ratio of tiny moments is not robust.

Smagorinsky nearly eliminates resolved vertical turbulence

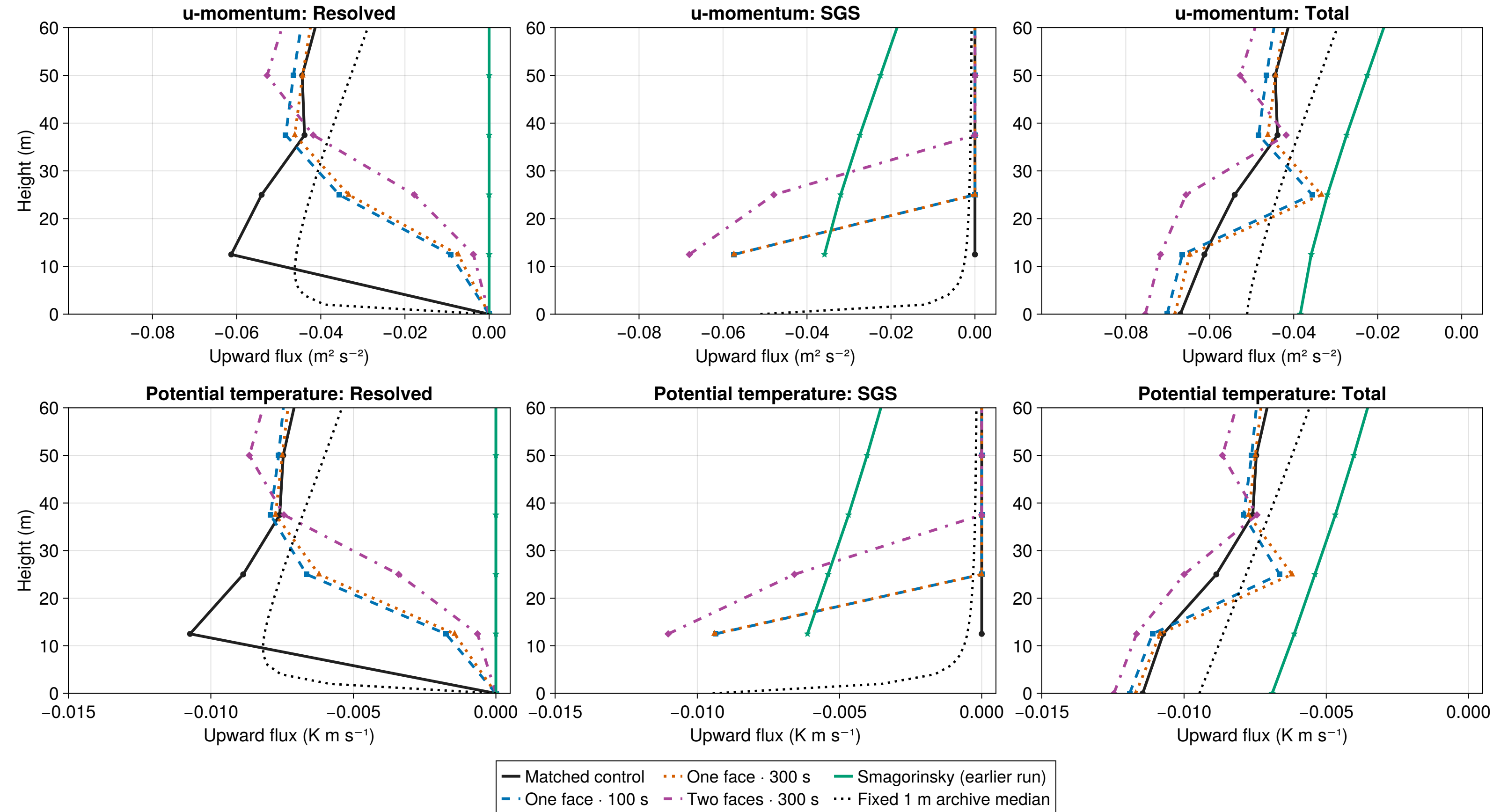
12.5 m GABLS1 · WENO9 · log axes expose the difference hidden by linear scales



Same grid and advection; Smagorinsky is historical context, not a same-revision controlled treatment.
Zeros are omitted on log axes. The left panel averages first; the right panel takes each instantaneous maximum.

SGS transport replaces most of the resolved near-wall flux

GABLS1 · 12.5 m · 8–9 h means · native faces in the lowest 60 m · corrected diagnostics



Colors, line styles and markers identify cases. SGS wall zeros are omitted; total uses the prescribed wall flux. Interior total = resolved + SGS. This split does not quantify WENO numerical mixing. Teal is historical context.

Resolved-flux factor 2: limited response

Matched sensitivity | 21 September 2026 | GABLS1, 12.5 m, WENO9, one interior face, 300 s filter, 9 h, one seed.

Result: crediting twice the resolved transport reduces the added diffusivity, but does not restore resolved turbulence. In the final hour, first-level w^2 rises 7.1%, while peak w^2 falls 6.3% and vertically integrated resolved TKE falls 6.4%. Both factors remain far below the earlier no-closure near-wall variance.

The intervention: factor a multiplies only the signed local filtered resolved flux inside the closure deficit $\max(0, 1 - a F_{\text{resolved}}/F_{\text{target}})$. The same factor is used for momentum and heat. Factor 2 assumes an additional numerical flux equal and aligned with resolved transport. It neither doubles viscosity nor rescales diagnostic fluxes. The experiment does not measure numerical transport or establish that assumption.

Mechanism: final-hour first-face viscosity falls from 1.321 to 1.118 m^2/s (15.4%); heat diffusivity falls from 1.216 to 0.774 m^2/s (36.3%). The first-face SGS share of u-momentum transport remains large: 88.7% versus 86.8%. Reduced closure coefficients therefore produce only a modest change in the resolved/SGS partition.

Moments: at 12.5 m, w^2 changes from 0.008171 to 0.008754 m^2/s^2 , versus 0.06353 without interior closure. The third moment changes from $-1.872\text{e-}4$ to $-1.186\text{e-}4$ m^3/s^3 , and skewness from -0.253 to -0.145. Skewness is the ratio of hourly averaged moments, not an average of instantaneous skewness.

Mean profiles and exchange: u^* falls from 0.2886 to 0.2831 m/s and sensible heat flux changes from -15.54 to -15.38 W/m^2 . Against the fixed 1 m median at native heights $0 < z \leq 200$ m, final-hour RMS errors change by +0.3% for u , -9.2% for θ , and -7.3% for w^2 . Smaller profile errors coexist with lower resolved energy; they do not demonstrate restored turbulence.

Hour sensitivity matters: in 7–8 h, first-level w^2 instead falls 5.1%, integrated TKE falls 10.4%, and the temperature-profile error increases 8.0%. Peak variance and integrated energy decrease in both windows. The near-wall variance increase and temperature improvement in the last hour are not consistent across the two hours. One seed and neighboring hours do not provide an uncertainty estimate.

Comparison integrity: two fresh cases passed strict admission with matching source, grid, seed, initialization and exact 19-profile/541-series schedules. The factor-1 run reproduces every exported profile and time-series value from corrected array 7293's one-face300 case bit for bit. Native physical flux partition, support and coefficients were checked. Fixed 1 m medians are LES references, not observations.

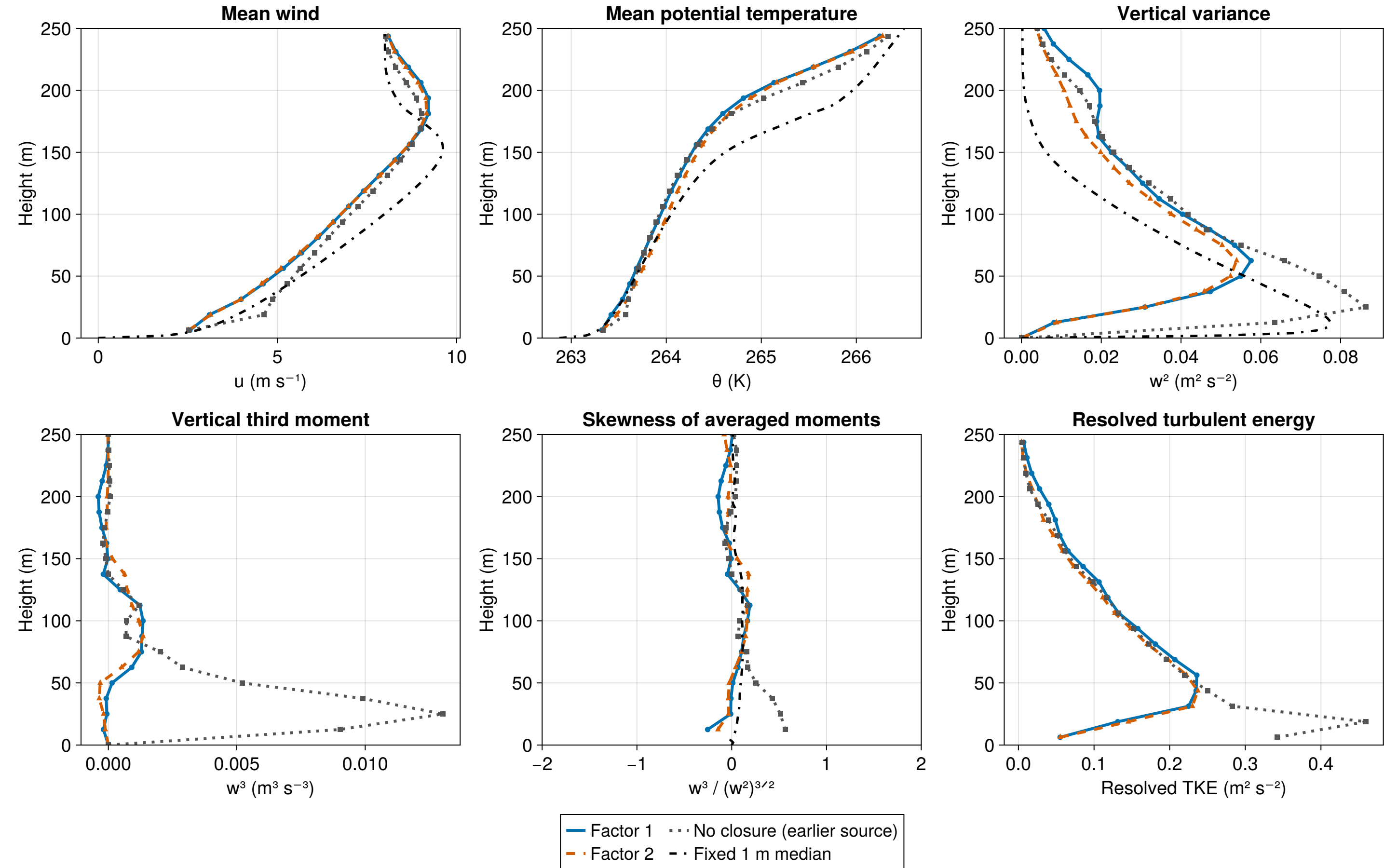
Resolved-flux factor 2: limited response

Execution provenance: both model children completed 32400 s and exited zero. Slurm nevertheless marked the factor-2 batch exit as one. A separate harmless login-shell probe reproduces that pattern through failing `clear_console` in `.bash_logout` under `set-e`; actual batch shell depth was not recorded. The discrepancy and canceled downstream job are preserved. Admission relies on independently complete outputs, durable child exits, matching sources and logs, not a relabeled scheduler success.

Conclusion: this factor-2 test changes coefficients and exchange modestly, but provides no evidence of broad turbulence recovery. It does not identify the numerical flux or establish a calibrated correction factor. All earlier DYCOMS, GABLS1 and SurfaceLayerDiffusivity results remain in the master report. GABLS3's separate humidity-boundary validation is still pending; no new GABLS3 science is included.

Crediting twice the resolved flux: does turbulence recover?

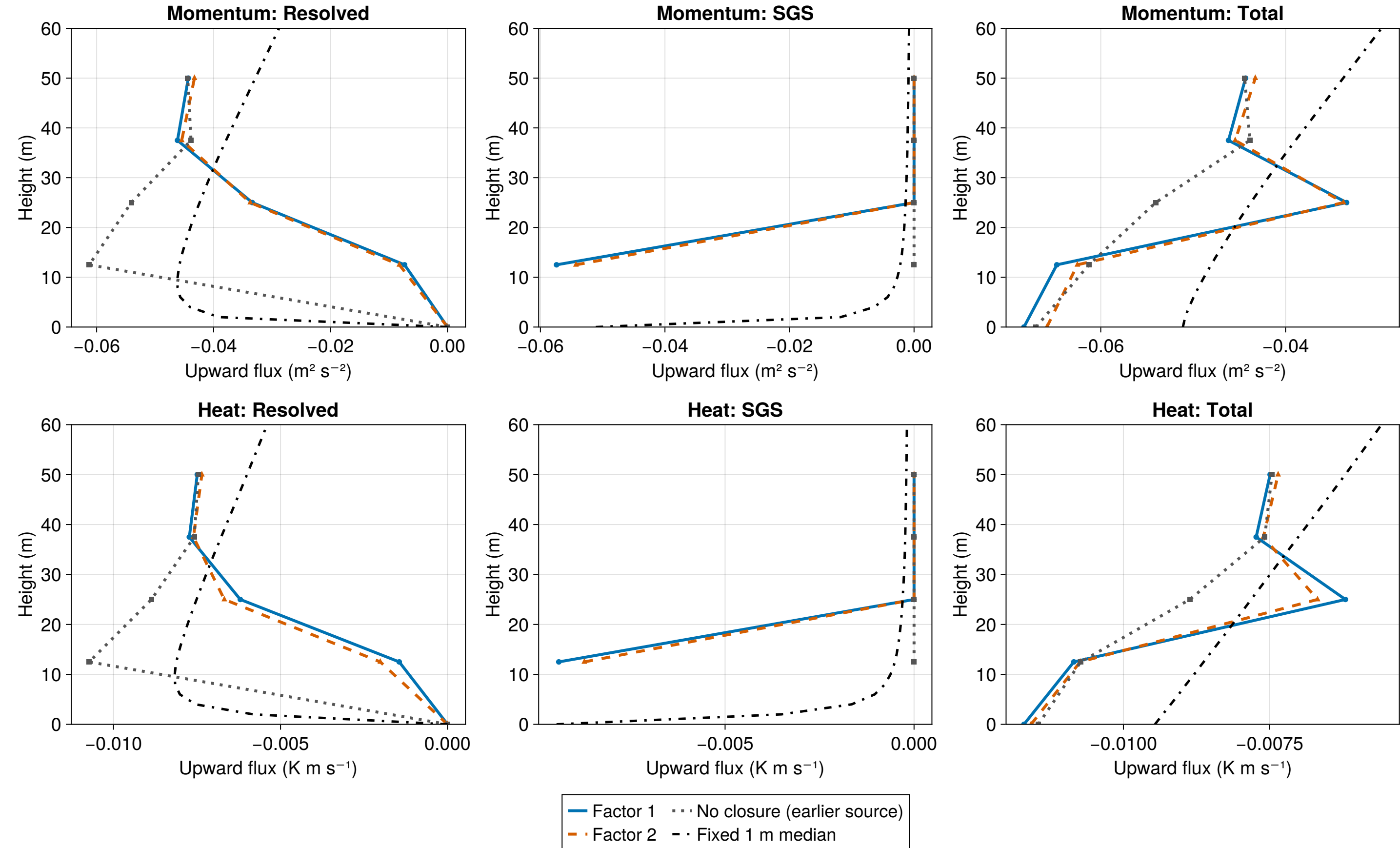
GABLS1 · 12.5 m · WENO9 · one face · 300 s filter · 8–9 h mean



Factor changes the closure deficit for momentum and heat; displayed physical fluxes are unscaled.
Earlier no-closure run is context. This tests an assumed numerical contribution, not a measured numerical flux.

Resolved and SGS transport near the surface

Same final-hour average · native flux faces · physical fluxes remain unscaled



Interior total = resolved + SGS. At the wall, total includes the prescribed flux; artificial SGS wall zeros are omitted.
Each panel includes the fixed 1 m median wherever that reference quantity is available.

Factor 10 mostly switches the closure off

GABLS1 factor-10 sensitivity | 21 September 2026 | 12.5 m, WENO9, one interior face, 300 s filter, 9 h, seed 123. All earlier factor-1/2 results are retained separately.

Result: factor 10 largely removes the added surface-layer mixing after spin-up. Resolved turbulence then resembles the earlier no-closure run. In 8-9 h, first-face w^2 is $0.064671 \text{ m}^2/\text{s}^2$, versus 0.00817056 for factor 1, 0.00875396 for factor 2, and 0.0635343 without closure. Factor 10 is 7.92 times factor 1 and only 1.79% above the no-closure value. This is recovery of resolved turbulence, not proof of accurate total transport.

Mechanism: the multiplier acts only inside the signed local deficit, $\max(0, 1 - a F_{\text{resolved}}/F_{\text{target}})$, for momentum and heat. With $a=10$, aligned resolved transport carrying 10% of the target can satisfy the deficit. Countergradient transport increases it. Crediting ten times the resolved flux assumes a missing contribution nine times as large; the experiment does not measure that contribution. Physical covariance diagnostics are never multiplied.

The saved fractions explain the response. In the final hour, momentum viscosity is exactly zero at 99.9934896% of sampled horizontal point-times; heat diffusivity is zero at 100%. The corresponding raw-zero-deficit and guard-valid-zero-deficit fractions agree. Momentum and heat guards remain valid at 100%, with zero coefficient-cap activity, so this is deficit satisfaction rather than guard failure. Each value averages 60 one-minute samples over all 32×32 horizontal points. It does not imply continuous switch-off between samples: true averaged SGS heat flux is tiny but nonzero, $-1.26136\text{e-}08 \text{ K m/s}$. These six fractions were not saved for factors 1/2 and remain unavailable there.

The closure acts during spin-up. During 0-1 h, mean viscosity is $0.5168 \text{ m}^2/\text{s}$ and heat diffusivity is $0.6870 \text{ m}^2/\text{s}$; zero-coefficient fractions are about 81.54% and 82.98%. By 1-2 h, momentum is zero at 99.9837% of sampled points and heat at 100%. Small intermittent reactivations remain later. In 8-9 h the mean viscosity is $2.72361\text{e-}05 \text{ m}^2/\text{s}$, versus 1.32118 and 1.11797 for factors 1 and 2; sampled mean heat diffusivity is zero.

Moments and energy: factor-10 first-face w^3 is $+0.00977604 \text{ m}^3/\text{s}^3$ and skewness $+0.59443$, compared with negative values for factors 1/2 and $+0.56311$ without closure. Final-hour peak w^2 is 0.0877192 , and integrated resolved TKE is $35.1207 \text{ m}^3/\text{s}^2$, 28.83% above factor 1. The response persists in 7-8 h: first-face w^2 is 0.0668137 , skewness $+0.58543$, and integrated TKE 35.2916 . Neighboring hours and one seed are not an uncertainty estimate.

Factor 10 mostly switches the closure off

Transport and exchange: final-hour SGS carries 0.012852% of first-face u-momentum covariance-plus-SGS transport, versus 88.66% and 86.82% for factors 1/2. Factor-10 u^* is 0.279088 m/s and surface sensible heat flux -15.2328 W/m², close to the no-closure values 0.279164 and -15.2131. The plots' quantity called total is covariance plus actual constitutive SGS. It is NOT the scheme-native WENO advective flux plus SGS: the numerical correction to covariance has not been reconstructed. Agreement of that plotted quantity with a reference cannot validate the missing numerical contribution.

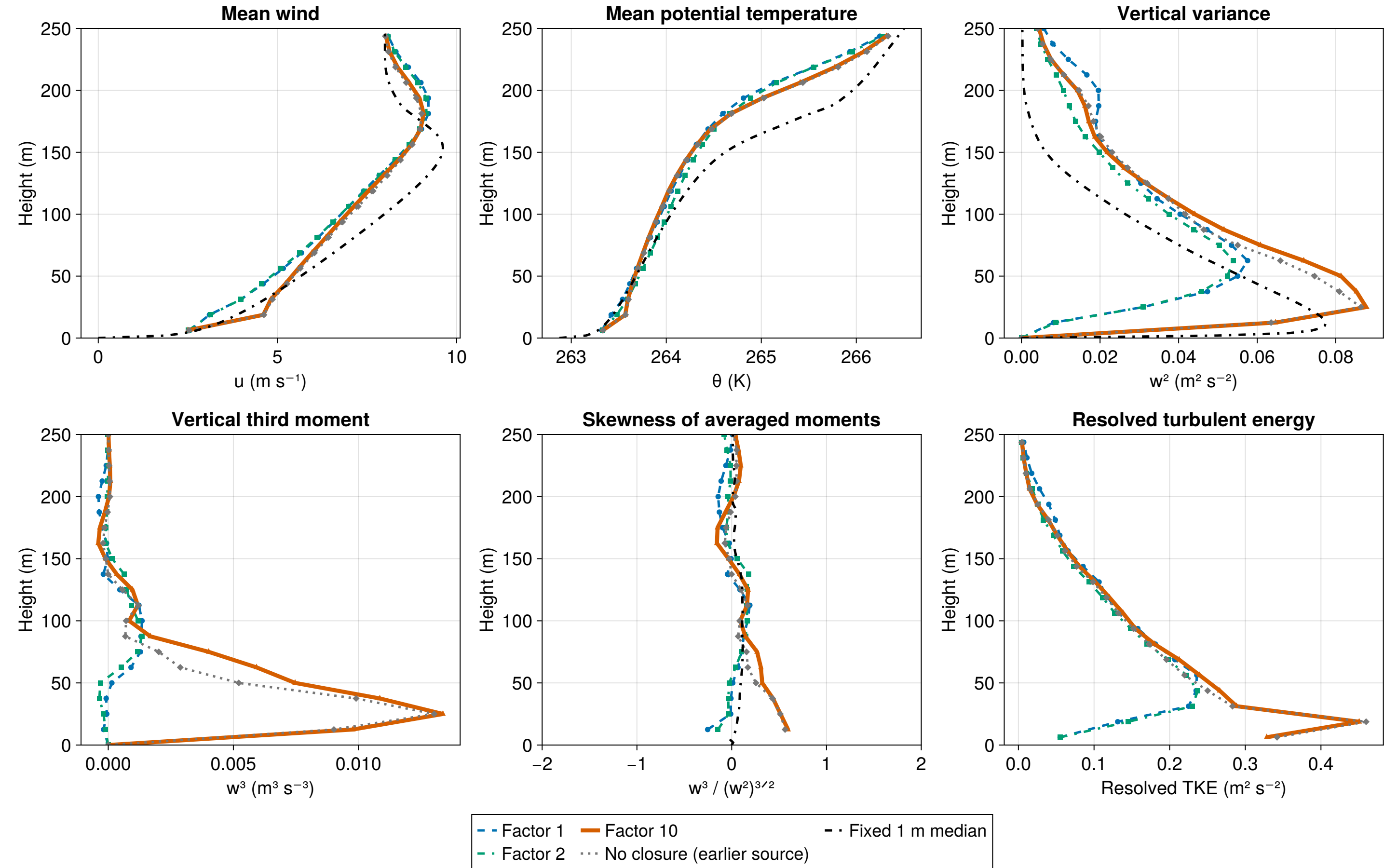
Reference comparison: relative to factor 1, final-hour native-level RMS errors against the fixed 1 m LES median fall 17.33% for u , 9.354% for θ , and 27.43% for w^2 . Nevertheless, factor-10 errors remain larger than the no-closure errors for all three quantities in 8-9 h. The reference is an LES ensemble median, not observations. This strong sensitivity chiefly approaches the no-closure regime; it does not establish factor 10 as a calibrated or faithful correction.

Comparison integrity: factor10 passed separate strict admission (1 admitted, 0 rejected), with 19 native profiles, 541 exact series times, finite values, supported-face SGS and covariance-plus-SGS consistency. Factors 1/2 retain their original admission. Breeze physics commit is identical; factor10's evaluation revision adds read-only diagnostics, with source-specific GPU validation (4,462 checks). Initial native profiles match exactly. Fresh factor1 already reproduces every corrected historical one-face300 profile and series value bit for bit. The earlier no-closure run is labeled as source context. Job7339 used a non-login wrapper and its durable child record verifies exit zero. The preserved older factor2 scheduler discrepancy is documented separately.

Definitions: plotted 8-9 h profiles average the two true half-hour means ending at 30600 and 32400 s; 7-8 h uses 27000 and 28800 s. Series metrics average samples strictly after the hour start through its end. Skewness is the ratio of averaged third moment to averaged variance^(3/2), omitted for $w^2 < 1e-6$. Native model heights are retained; only the fixed 1 m median is interpolated for unweighted RMS errors over $0 < z \leq 200$ m. No WENO flux reconstruction or additional experiment is included.

Factor 10: mean profiles and resolved turbulence

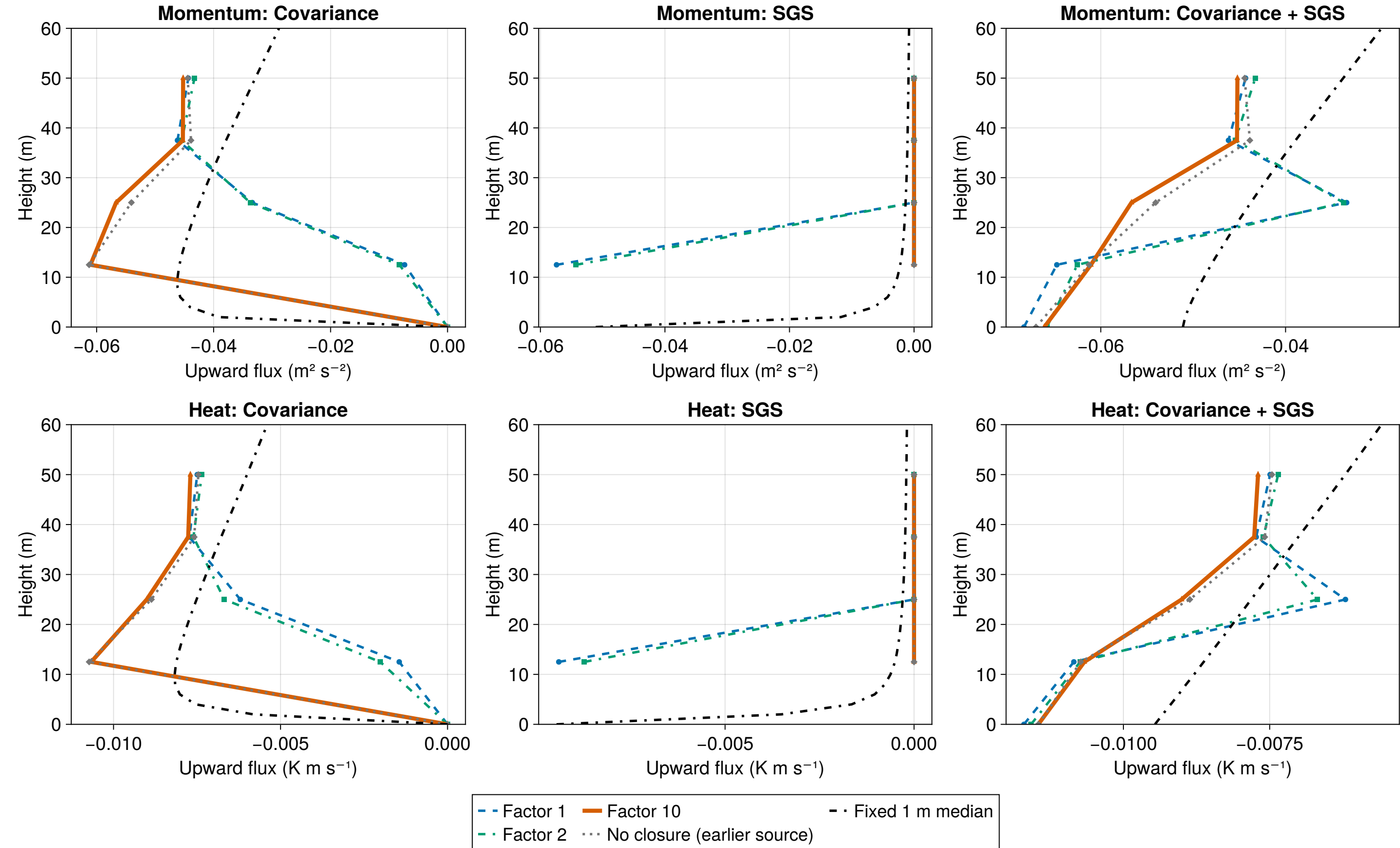
GABLS1 · 12.5 m · WENO9 · one face · 300 s filter · 8–9 h mean



Factor 10 is vermillion, solid, triangles. Reference medians shown where available.
Native heights and physical moments retained; the added numerical-flux credit is an assumption.

Covariance and SGS transport near the surface

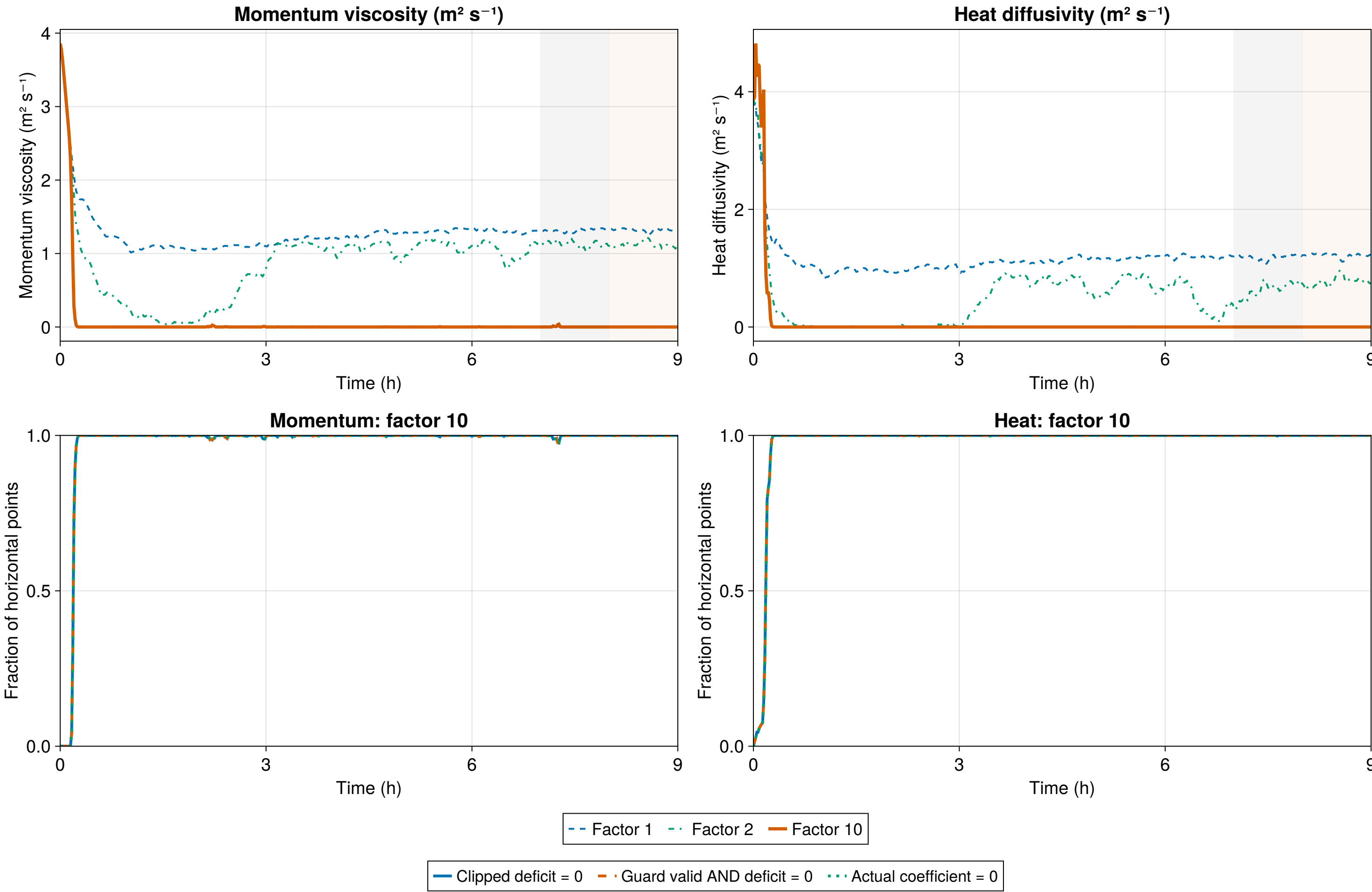
Same final-hour average · native flux faces · physical fluxes remain unscaled



Interior total = covariance + actual SGS. This is NOT scheme-native WENO transport: its numerical correction is missing.
Wall total is prescribed; artificial SGS wall zeros omitted. Fixed 1 m medians shown where available.

What the factor changes: coefficients and local switch-off

One supported face at 12.5 m · coefficients from all three factors · switch-off fractions saved only for factor 10



All fraction denominators include every horizontal point. Exact overlap is meaningful; quantitative values are in the report.
Active/guard validity is separate from positive mixing. These six fractions were not saved for factors 1 and 2.

Factor 3 mostly switches the closure off

GABLS1 factor-3 sensitivity | 22 September 2026 | 12.5 m, WENO9, one interior face, 300 s filter, 9 h, seed 123. New plots compare only factors 1 and 3, with fixed 1 m LES reference medians where available. Earlier reports and data are preserved.

Result: factor 3 largely switches off the added near-surface closure after spin-up, while resolved turbulence increases strongly. In 8-9 h, first-face w^2 rises from 0.00817056 to 0.060734 m^2/s^2 , a factor of 7.433. Integrated resolved TKE increases 16.38%. First-face w^3 and skewness change from negative to positive. This is a large response to the assumed missing-transport credit, not a calibrated accuracy result.

The intervention remains narrow: multiply signed, locally time-filtered resolved flux by three only inside $\max(0, 1 - a F_{\text{resolved}}/F_{\text{target}})$, for momentum and heat. Aligned resolved transport reaching one-third of the target can eliminate that local deficit; countergradient transport increases it. The assumed extra transport is twice the resolved contribution. Coefficients respond dynamically; neither viscosity nor plotted covariance is simply multiplied by three.

Closure response: final-hour mean first-face viscosity falls from 1.32118 to 0.00417626 m^2/s ; heat diffusivity falls from 1.21593 to 7.01368e-05 m^2/s . Factor-3 viscosity is exactly zero at 98.473307% of sampled horizontal point-times, and heat diffusivity at 99.975586%. The timeline shows substantial initial mixing followed by mostly inactive coefficients, with intermittent reactivation. This is a change in the closure's operating regime, not simply a small uniform reduction in its strength.

Fraction interpretation: each hourly value averages 60 one-minute samples over all 32×32 horizontal points. Raw zero deficit, guard-valid zero deficit and actual zero coefficient are separately retained for momentum and heat; all six fractions appear in the detailed table. All three definitions agree in the final hour. Momentum and heat guards remain valid at 100% and 100%; coefficient-cap fractions are 0 and 0. Thus deficit satisfaction explains the switch-off, rather than invalid targets or active coefficient caps. Sampled zeros do not prove continuous switch-off between outputs. Factor 1 did not save these fractions; its values remain unavailable, not zero.

Moments and hourly sensitivity: factor-3 first-face w^3 is 0.00890504 m^3/s^3 and skewness 0.59496; peak w^2 is 0.0839403 and integrated resolved TKE 31.7259 m^3/s^2 . In 7-8 h, first-face w^2 is 0.0623155, w^3 0.00902423, skewness 0.580117, and integrated TKE 34.7786. The earlier-hour zero-viscosity/zero-heat-diffusivity fractions are 98.18034% / 99.96582%. The two neighboring windows show time dependence; they do not quantify ensemble uncertainty.

Factor 3 mostly switches the closure off

Transport and exchange: the SGS share of first-face u-momentum covariance-plus-SGS transport falls from 88.6586% with factor 1 to 1.14829% with factor 3. Factor-3 u^* is 0.272265 m/s and sensible heat flux -14.5912 W/m², versus 0.288603 m/s and -15.5382 W/m² for factor 1. Every plotted interior total is covariance plus actual constitutive SGS. It is NOT scheme-native WENO advective transport plus SGS: the numerical correction to covariance remains unmeasured. The factor experiment cannot identify or validate that missing flux.

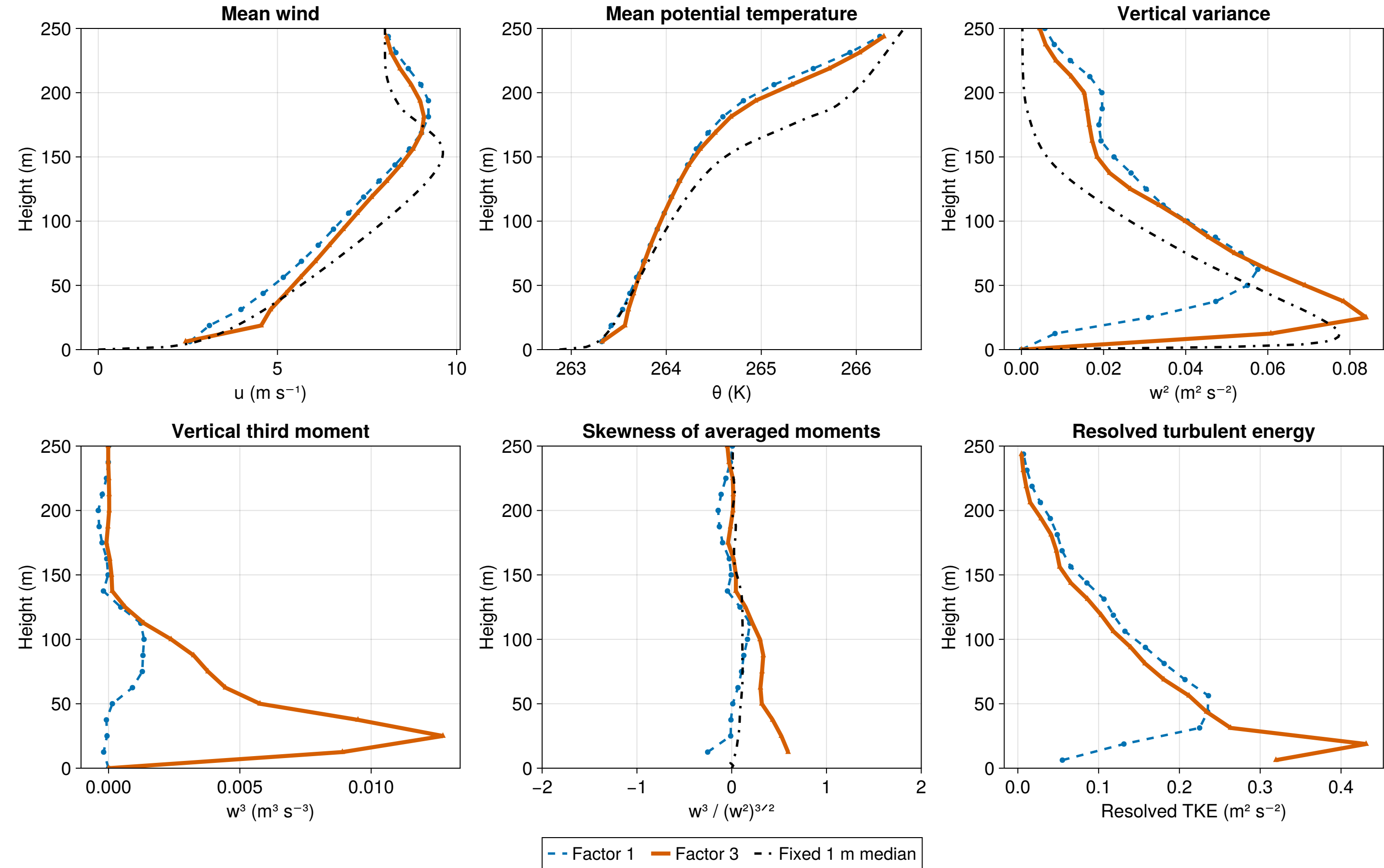
Reference comparison: at native heights $0 < z \leq 200$ m, factor-3 final-hour RMS errors against the fixed 1 m LES median are 0.720275 m/s for u , 0.376886 K for θ , and 0.0132648 m²/s² for w^2 . Changes relative to factor 1 are -23.19%, -9.761%, and -45.94%. The reference is an LES median, not observations. Greater resolved variance or a smaller mean-profile error alone does not establish faithful total transport.

Comparison integrity: factor 3 passed its source-specific strict admission (1 admitted, 0 rejected), with identical Breeze physics and initial native profiles, 19 exact profile times and 541 exact series times. Checksums, finite values, native vertical locations, one-face support, covariance-plus-SGS consistency and fraction bounds were verified. Factors 1/2/10 retain their original admissions. Factor 1 reproduces the corrected historical one-face300 output bit for bit; the no-closure case remains labeled as earlier-source context. The durable child record for non-login job 7348 verifies exit zero; no unavailable batch status is inferred.

Definitions: 8-9 h profiles average true half-hour means ending at 30600 and 32400 s; 7-8 h uses 27000 and 28800 s. Series averages exclude the hour start and include its end. Skewness is averaged w^3 divided by averaged w^2 to the three-halves power, masked for $w^2 < 1e-6$. Native model heights are retained; only the reference is interpolated for unweighted RMS errors. No scheme-native flux reconstruction, new closure design, or additional simulation is included.

Factor 3: mean profiles and resolved turbulence

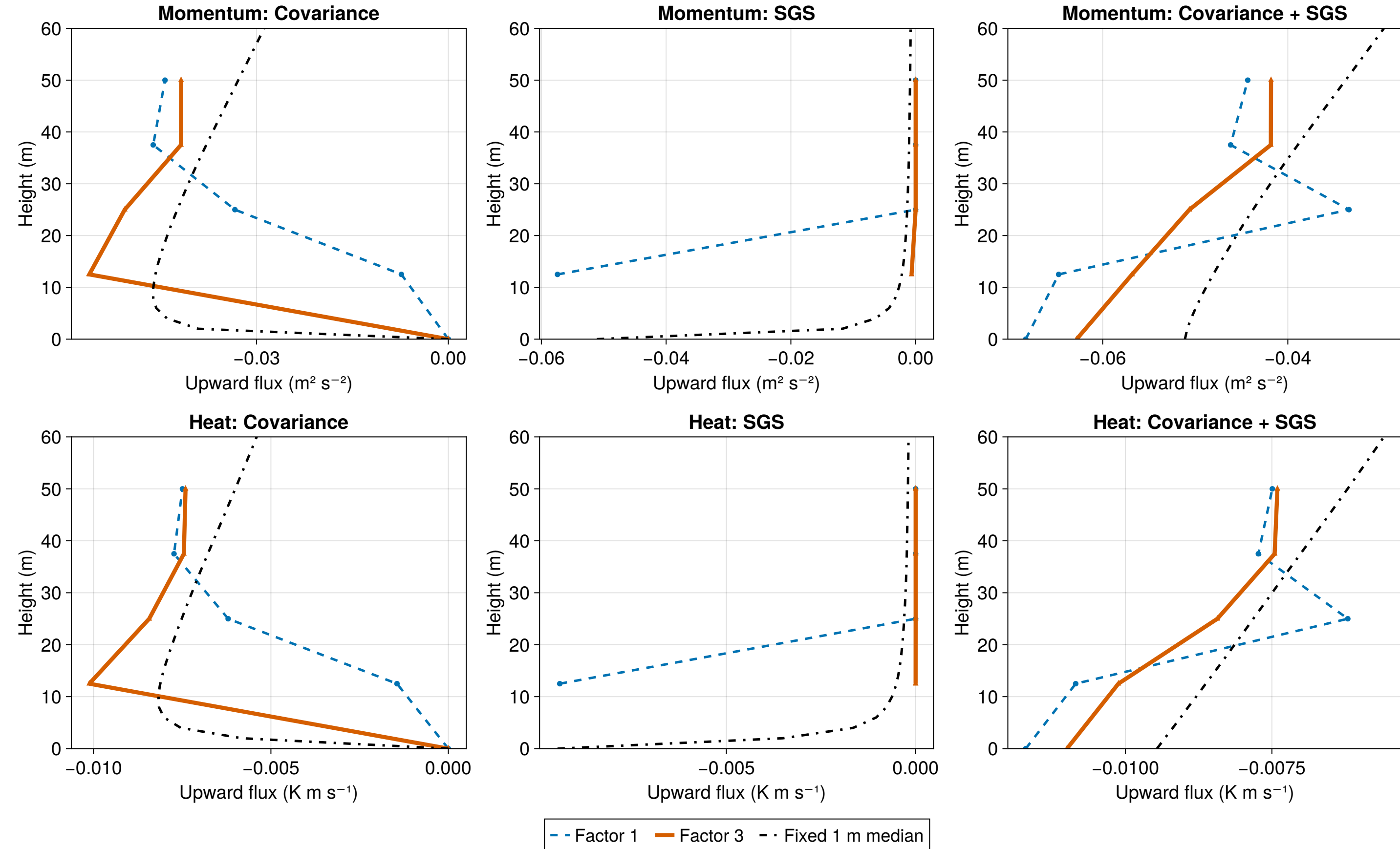
GABLS1 · 12.5 m · WENO9 · one face · 300 s filter · 8–9 h mean



Factor 1 is blue, dashed, circles; factor 3 is vermillion, solid, triangles.
Native heights and physical moments retained. Fixed 1 m reference medians shown where available.

Covariance and SGS transport near the surface

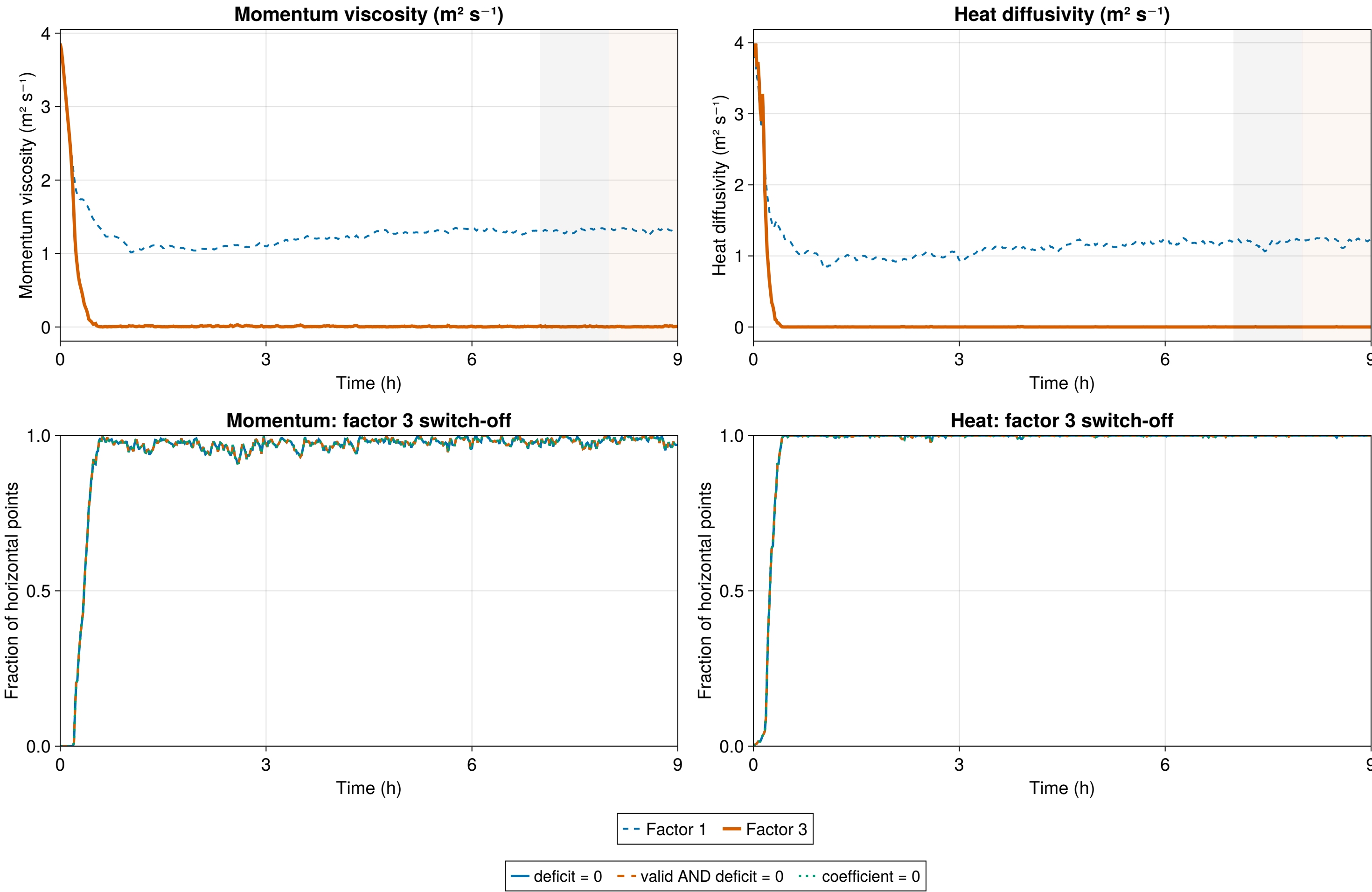
Same final-hour average · native flux faces · physical fluxes remain unscaled



Interior total = covariance + actual SGS. This is NOT scheme-native WENO transport: its numerical correction is missing.
Wall total is prescribed; artificial SGS wall zeros omitted. Fixed 1 m medians shown where available.

What the factor changes: coefficients and local switch-off

One supported face at 12.5 m · factors 1 and 3 · factor 1 switch-off fractions unavailable



All fraction denominators include every horizontal point. Exact overlap is meaningful; quantitative values are in the report.
Active/guard validity is separate from positive mixing. These six fractions were not saved for factor 1.

GABLS1: measured scheme-native WENO flux

GABLS1 scheme-native WENO transport | 22 September 2026. Matched 12.5 m, 32³, 400 m, WENO9, one face, 300 s filter, factor one, seed 123, nine-hour LES. The old admitted factor-one covariance run is the control.

At the first interior face, the measured 8–9 h WENO numerical correction to u-momentum flux is $-0.000228 \text{ m}^2/\text{s}^2$, versus covariance $-0.007749 \text{ m}^2/\text{s}^2$ (2.95%). The heat correction is $1.4\text{e-}10 \text{ K m/s}$, effectively zero versus covariance -0.001472 K m/s . The saved covariance plus correction equals reconstructed flux at both diagnostic faces for momentum and scalars. These measured corrections are far below the assumed extra flux behind factors two and three in this case.

Mean 8–9 h first-face momentum viscosity falls 1.3214 to 1.2958 m^2/s (-1.9%); heat diffusivity rises 1.2160 to 1.2642 m^2/s (+4.0%). Integrated resolved TKE rises 3.8%, peak w^2 rises 3.1%, diagnosed boundary-layer height falls 7.4 m. One evolving realization cannot distinguish a systematic response from trajectory divergence. The profiles remain near the covariance control and show substantial coarse-run departures from the fixed 1 m LES median in resolved turbulence.

Source and data audit: immutable Breeze/Evaluation manifest SHA-256 72f1bc03 and 792 source-file hashes passed; paired initial-theta SHA-256 1f5db33f matched. H100 gate 7496 passed; science 7497 had durable exit zero and CASE_DONE at 32400 s. Raw output passed exact 1/19/541 initial/profile/series schedules, finite values, native 32/33-level heights, and flux decomposition on both faces. Figures use Julia. The reference is an LES median, not observations. Historical factor and control outputs were not rerun.

GABLS1: shear and surface-layer scaling

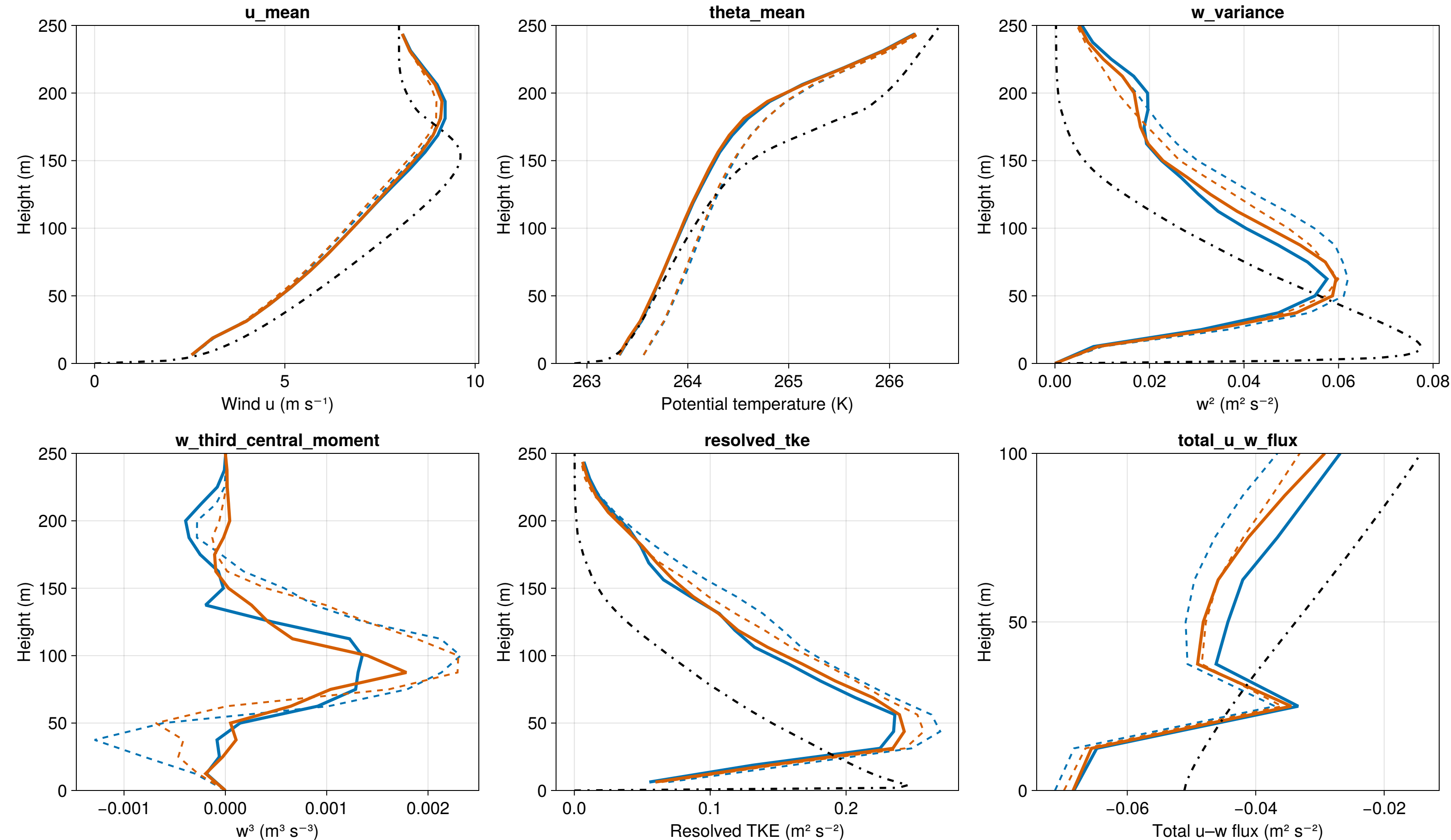
The final-hour mean horizontal-wind vector shear across 6.25-18.75 m is 0.0522 s^{-1} for scheme-native SLD and 0.0511 s^{-1} for covariance SLD, versus 0.1004 s^{-1} from the fixed 1 m median profiles; the no-closure case is high at 0.1861 s^{-1} . The pattern persists during 7-8 h, but differences change sign higher up. All model gradients use adjacent 12.5 m cell-center levels. Reference componentwise median u/v profiles are sampled at those heights and differentiated identically; this is not the median of member shears or a wall gradient.

A restricted comparison of the requested treatments gives first-layer vector shear 0.1861 s^{-1} without closure, 0.0522 s^{-1} with scheme-native SLD, and 0.1173 s^{-1} with Smagorinsky, versus 0.1004 s^{-1} from the fixed 1 m median profiles. Smagorinsky is from an earlier source revision and is contextual. The SLD deficit is concentrated near the surface; it is not a uniform shear bias over the full layer.

The 1 m median is not a neutral log-law profile over 2-30 m: horizontal speed has fitted log-height slope 1.123 m/s , whereas mean u^*/κ gives 0.654 m/s . Stable GABLS1 requires a positive Monin-Obukhov correction. That correction reduces, but does not eliminate, the departure for both the reference and scheme-native SLD. The wall coefficient couples the surface to the first velocity level; SLD acts at one interior face. Neither imposes a log or stable-similarity profile throughout the column. Time-averaged fluxes and independently median u/v components need not obey a pointwise similarity identity.

GABLS1: scheme-native transport versus covariance

12.5 m · WENO9 · one face · 300 s · factor 1 · same seed; dashed 7–8 h, solid 8–9 h

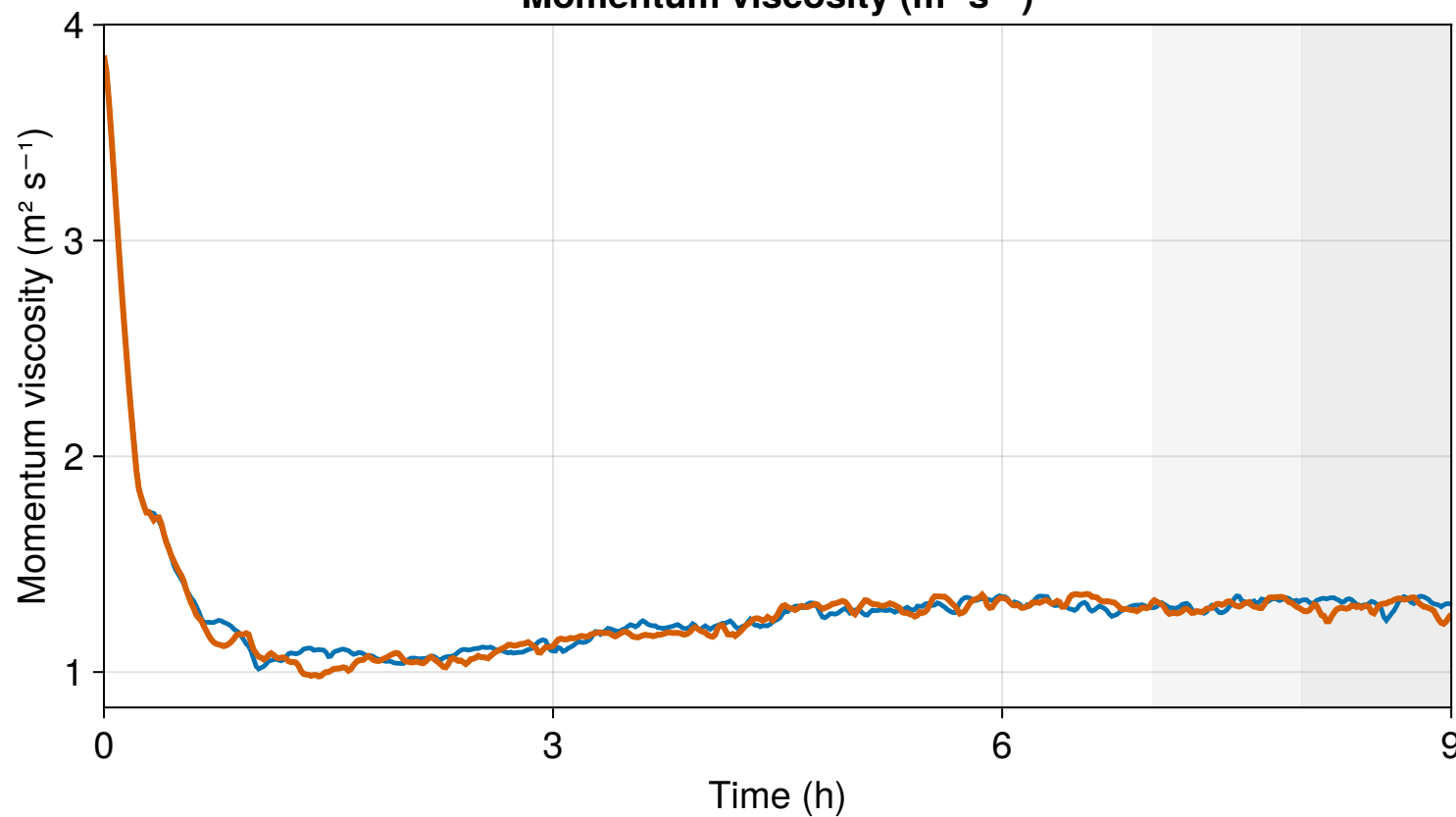


- - Covariance control 7–8 h - - Scheme-native 7–8 h - · Fixed 1 m LES median
 — Covariance control 8–9 h — Scheme-native 8–9 h

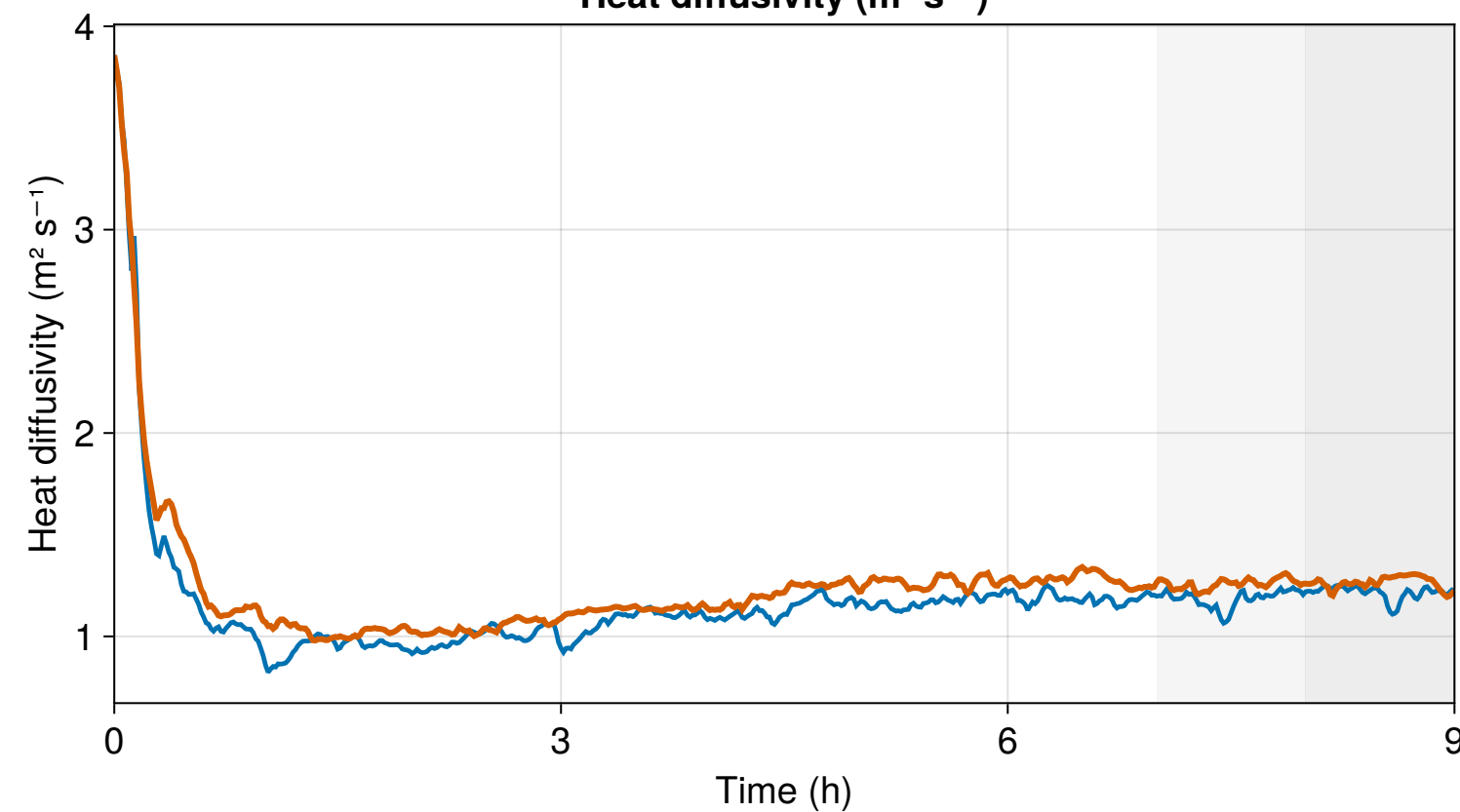
What the transport estimator changes at the first interior face

Same nine-hour forcing and discretization · 12.5 m supported face · shaded final two hours

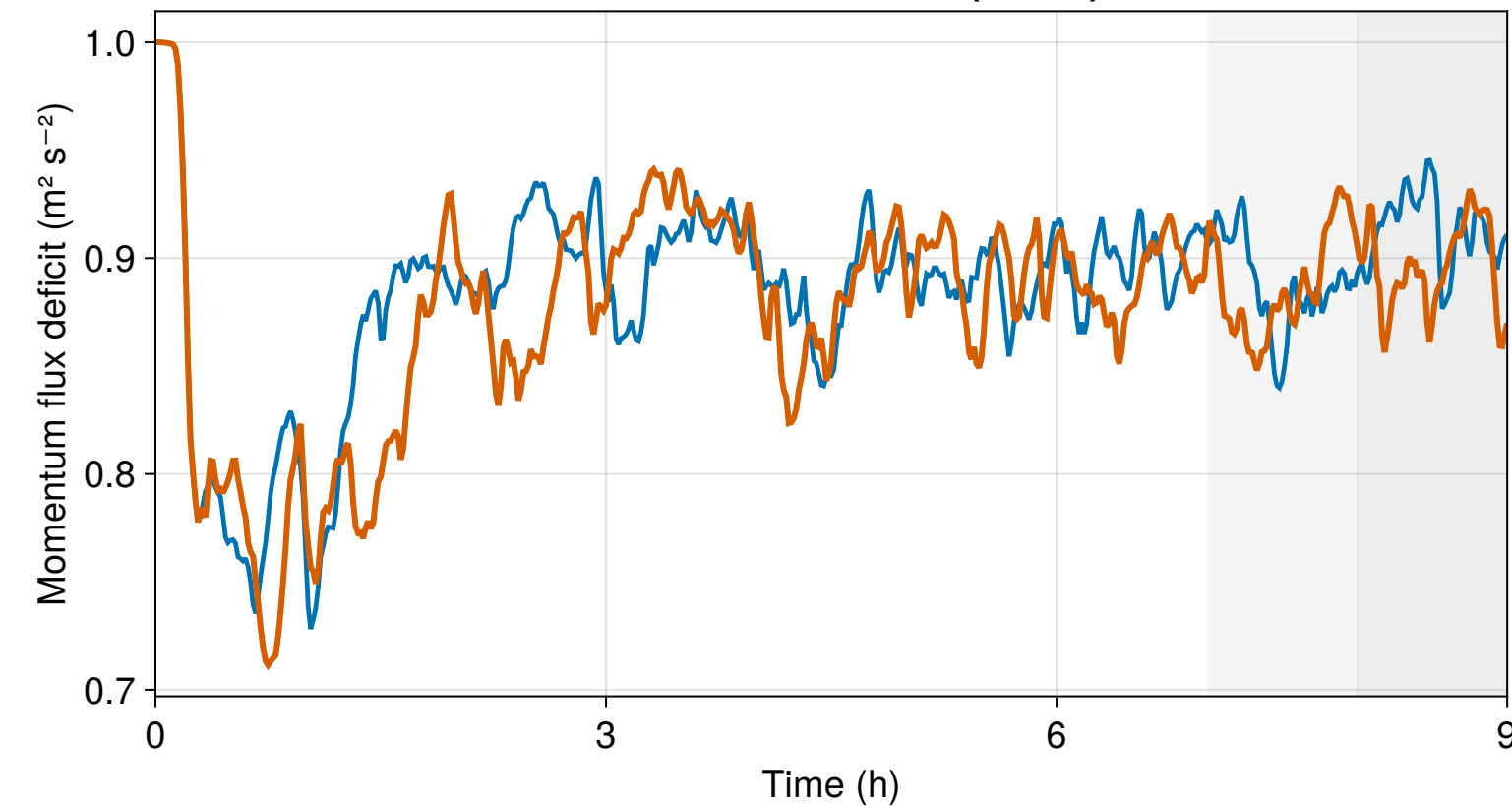
Momentum viscosity ($\text{m}^2 \text{s}^{-1}$)



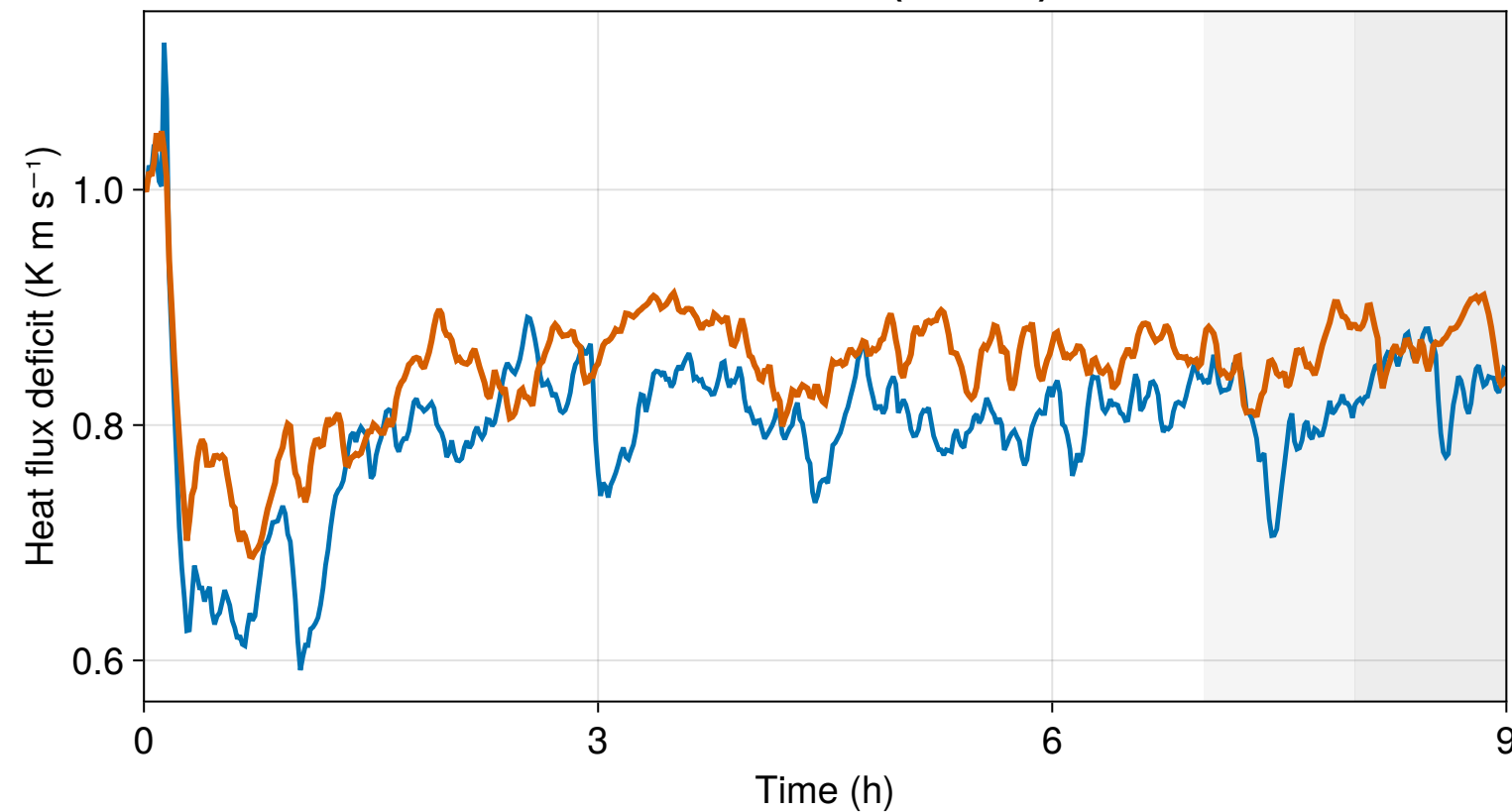
Heat diffusivity ($\text{m}^2 \text{s}^{-1}$)



Momentum flux deficit ($\text{m}^2 \text{s}^{-2}$)



Heat flux deficit (K m s^{-1})

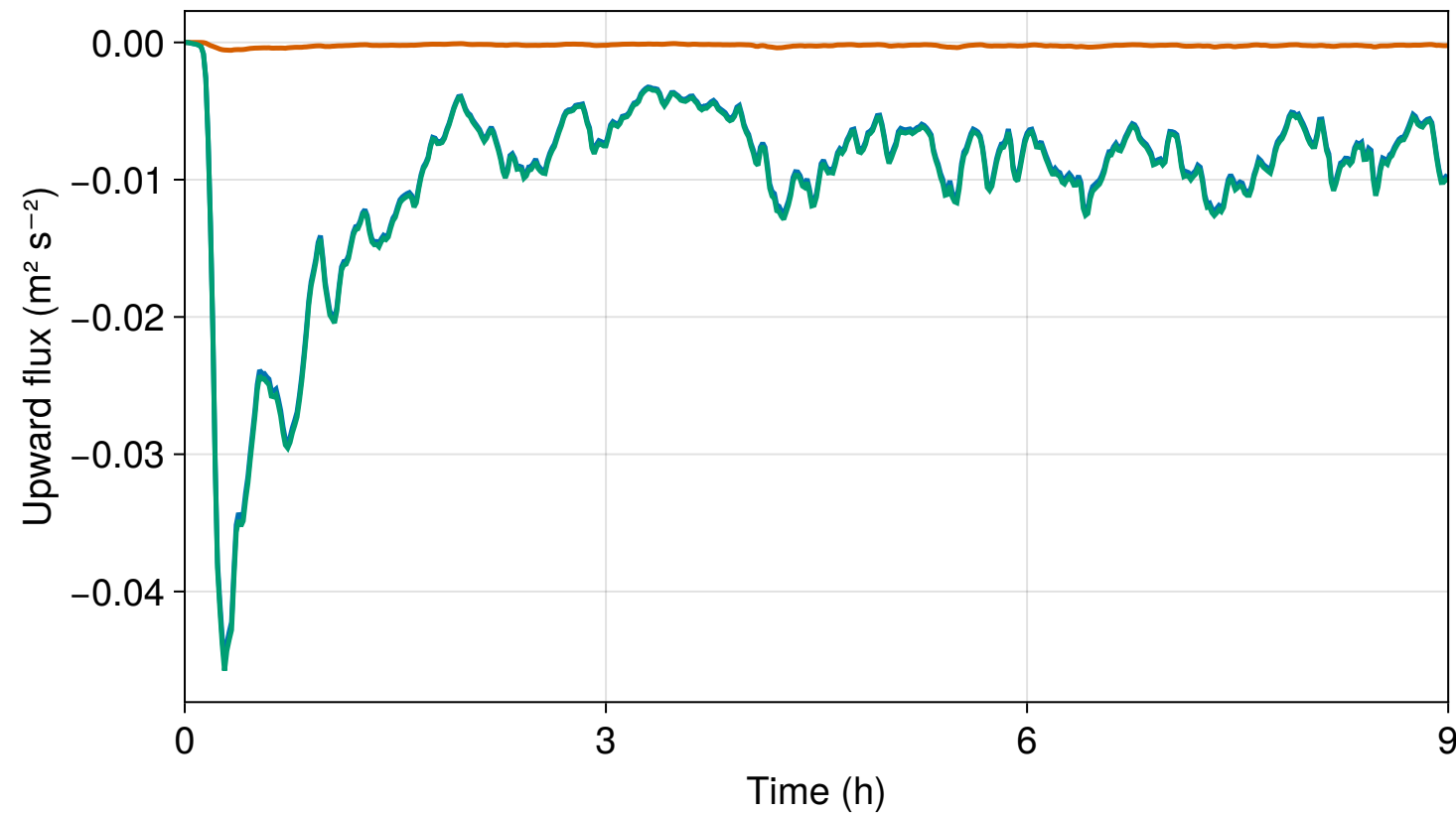


— Covariance control — Scheme-native

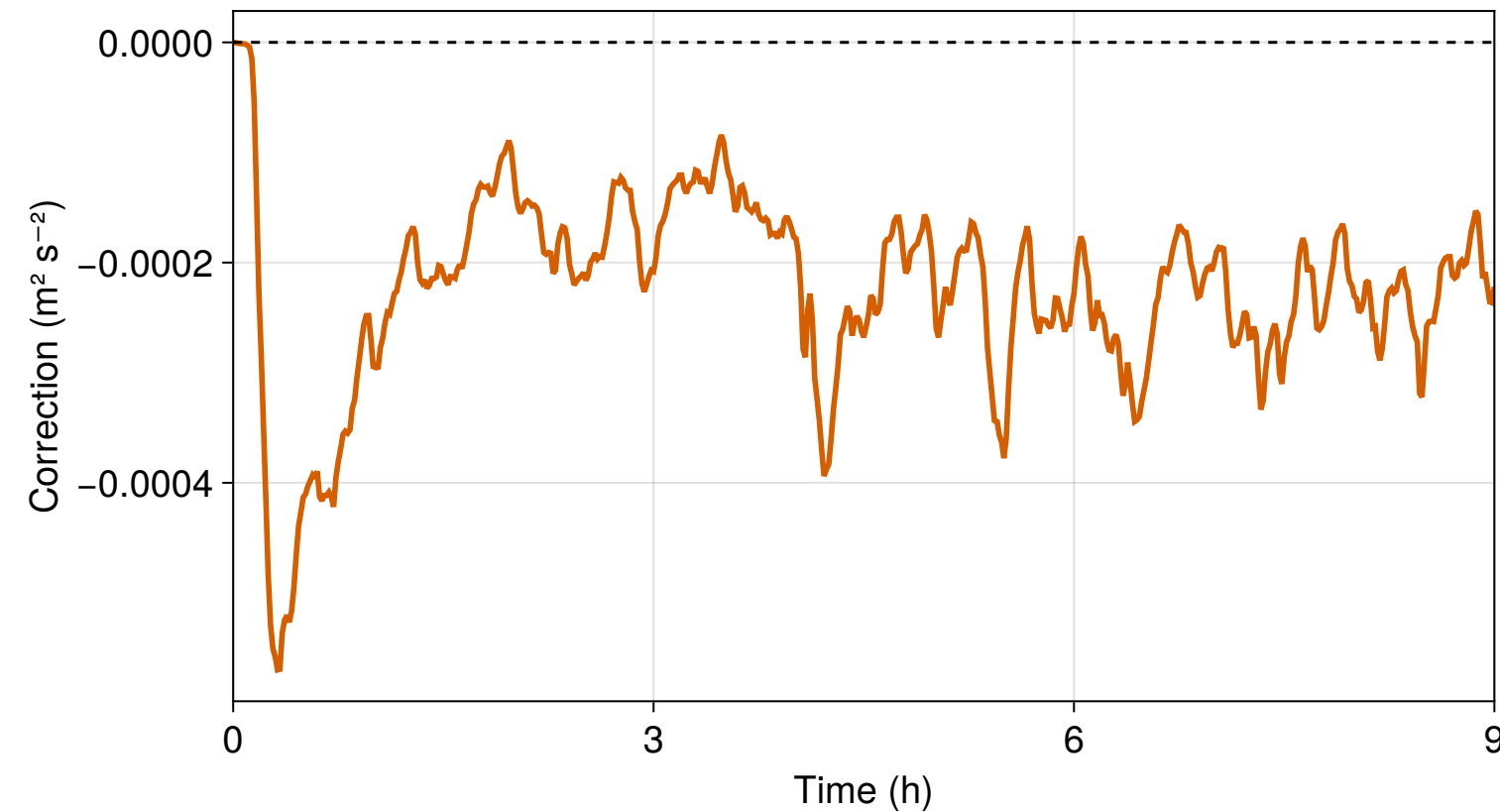
Measured WENO transport correction

Scheme-native run only · covariance + numerical correction = reconstructed flux

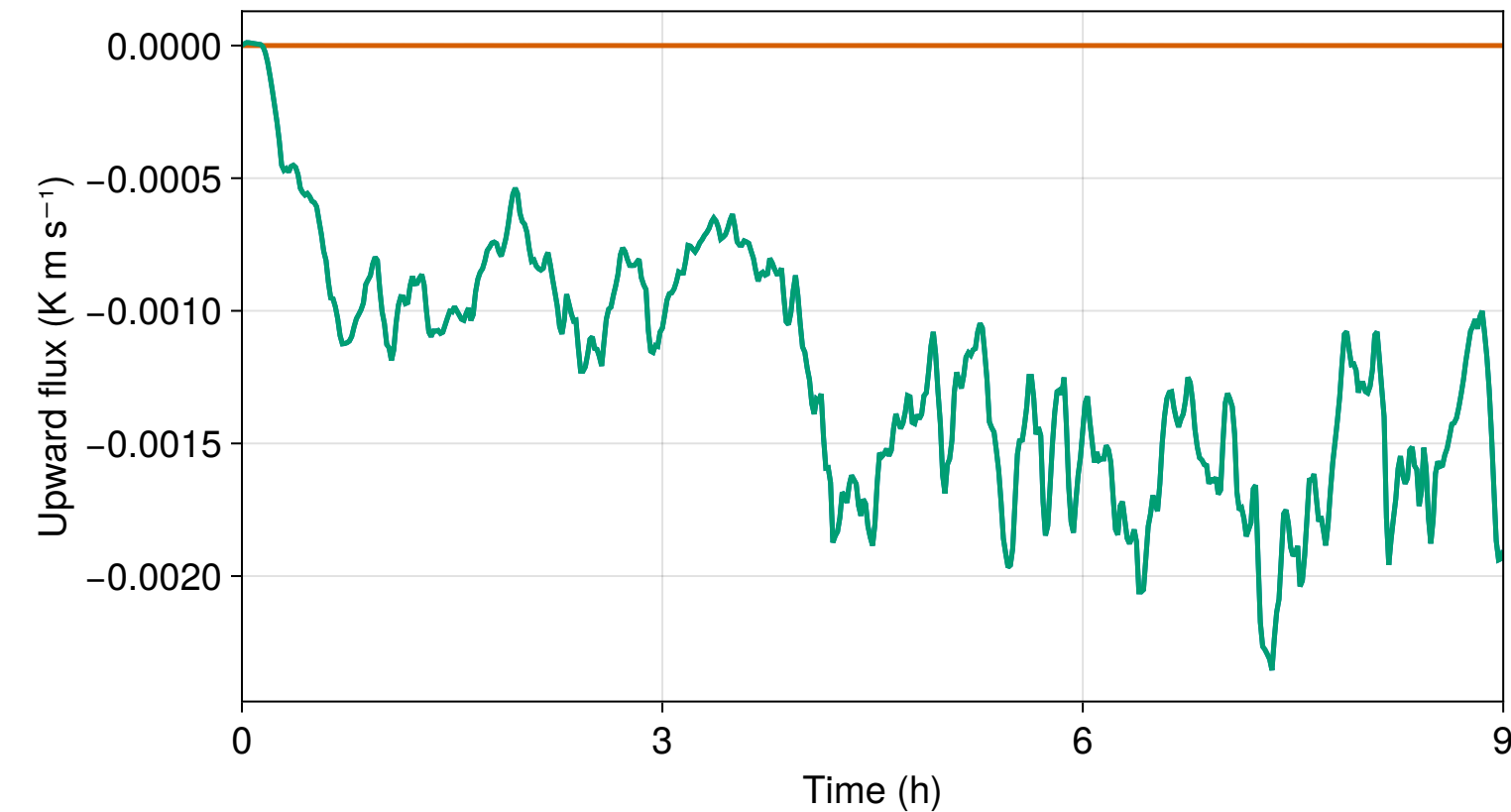
Momentum flux



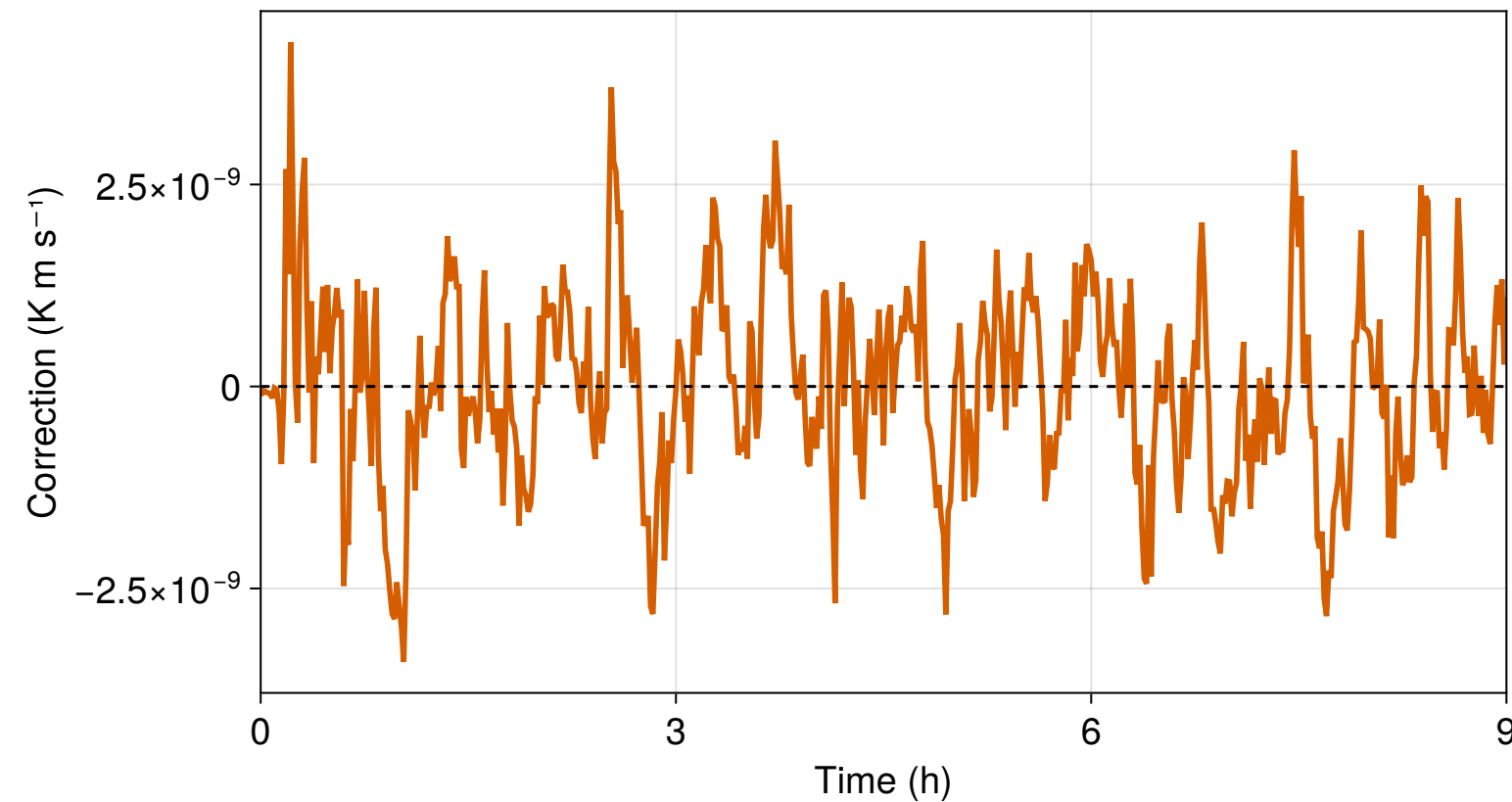
Momentum numerical correction



Heat flux



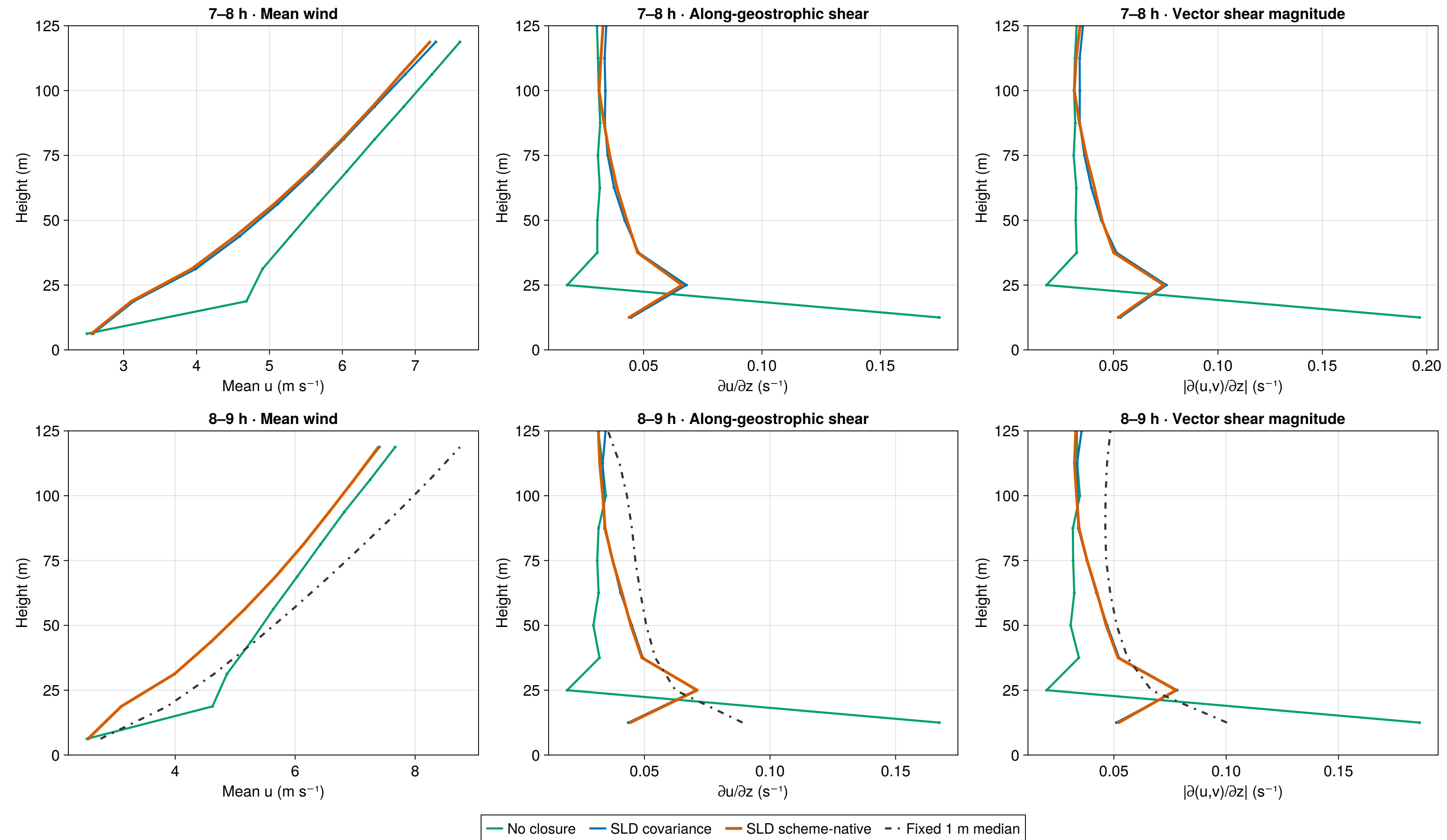
Heat numerical correction



— Covariance — Numerical correction — Reconstructed

GABLS1: vertical shear of the mean horizontal wind

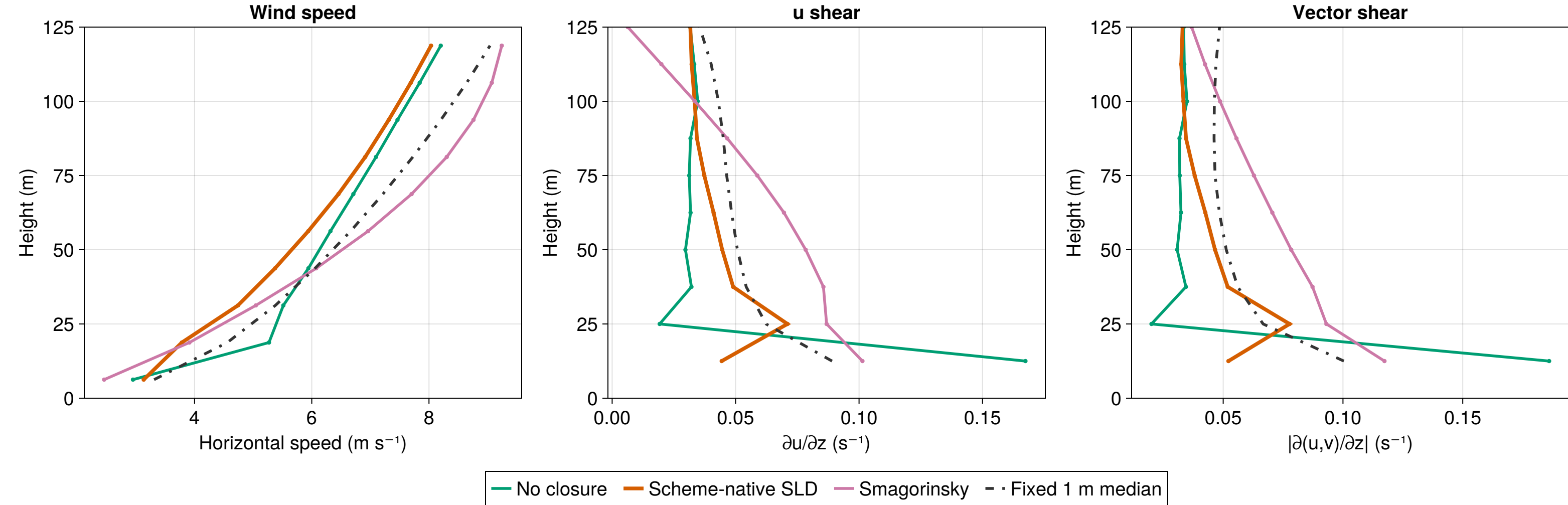
12.5 m WENO9 · same 400 m cube · derivatives between adjacent cell-center heights



Reference is for 8–9 h only: componentwise median u and v profiles sampled at 12.5 m cell centers, then differentiated identically.
The reference shear is not the median of member shears. Each curve is a layer slope at 12.5, 25, 37.5... m, not a wall gradient.

GABLS1 mean wind and shear: three requested treatments

8–9 h · 12.5 m WENO9 · 32^3 · same 400 m cube; fixed 1 m LES median for context

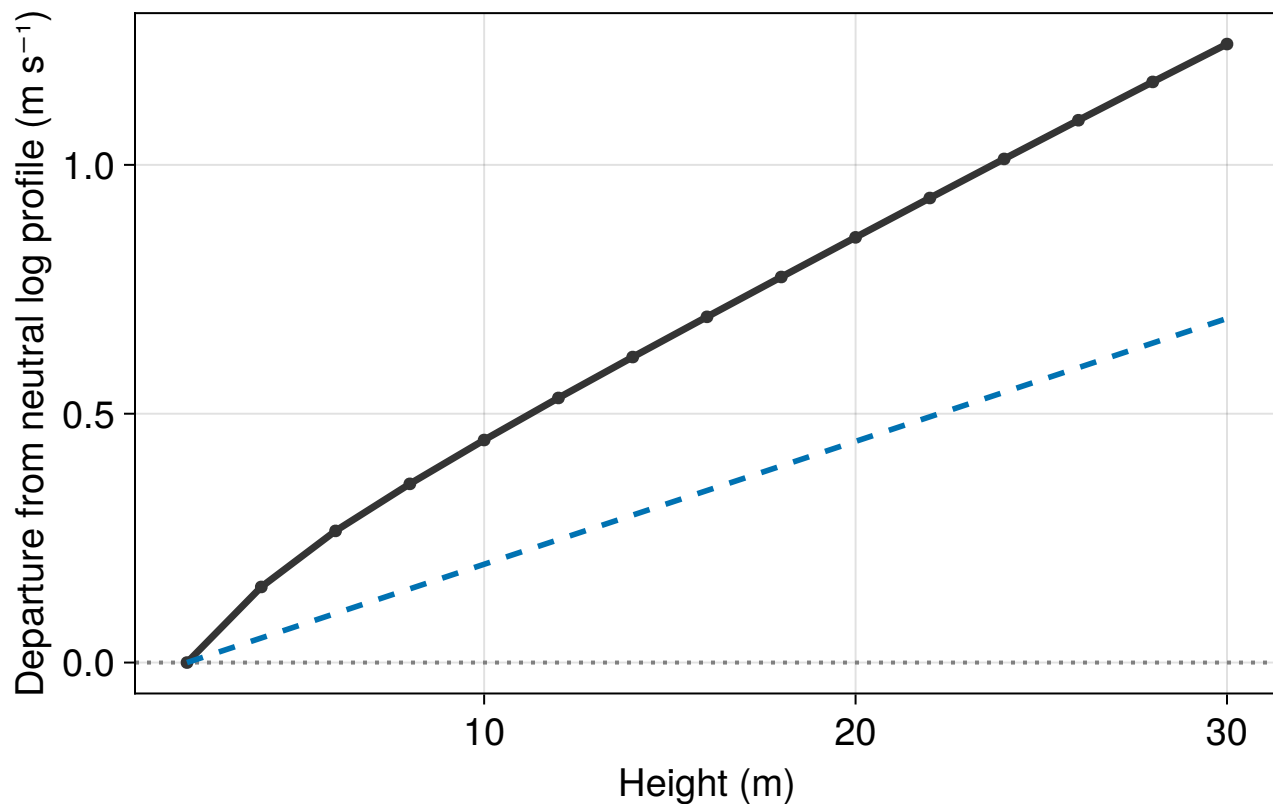


Reference componentwise median u/v sampled at 12.5 m centers before differentiation. Shear at 12.5 m spans 6.25–18.75 m; it is not a wall gradient.
Smagorinsky used an earlier source revision and is contextual rather than an exact same-revision treatment.

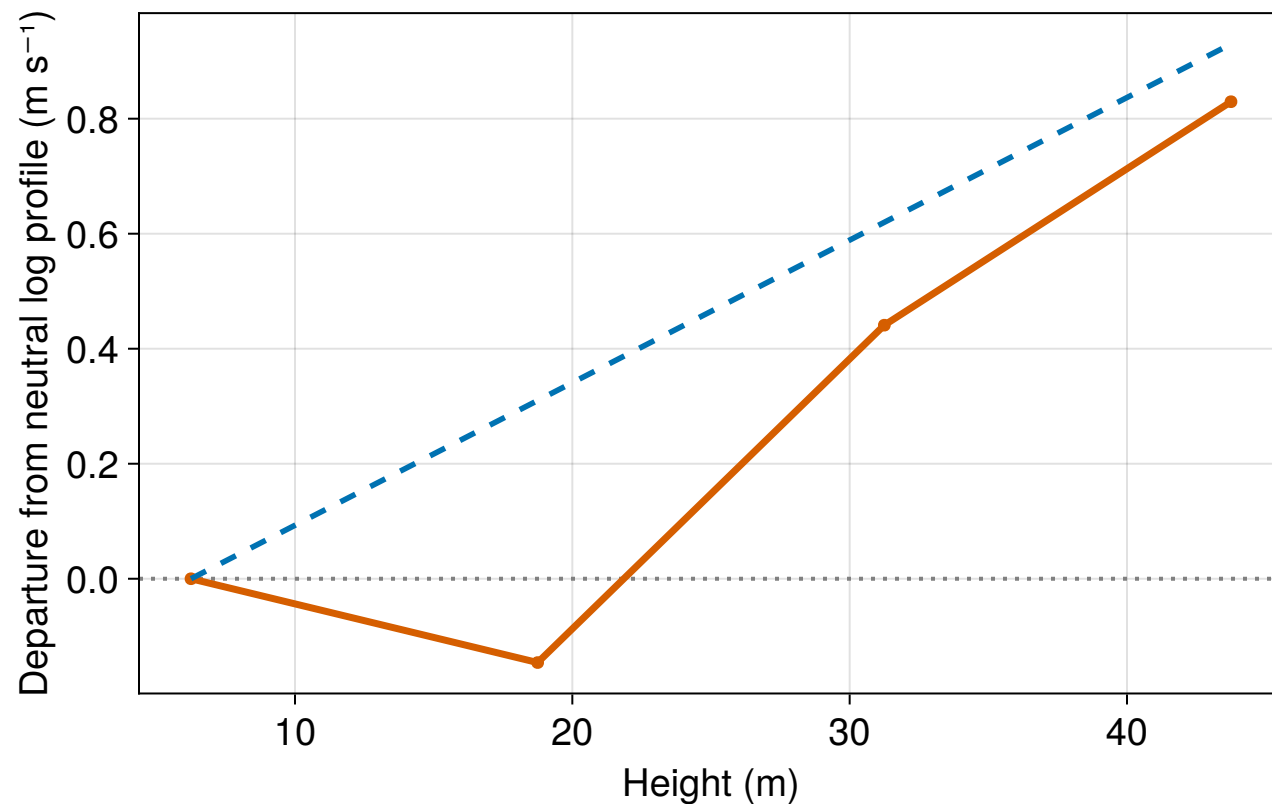
Neutral log-law versus stable surface-layer scaling

Residual from a neutral log profile anchored at the first plotted level; zero would be neutral-log behavior

Fixed 1 m LES median · 2–30 m



Scheme-native SLD · 6.25–43.75 m



···· Neutral log-law — Saved wind profile - - Stable MOST term

MOST term = $(u^*/\kappa) \beta_m(z-z_1)/L$, $\beta_m=4.8$; each curve uses its mean 8–9 h u^* and L . Reference L is derived from median u^* and heat flux.
Averaged profiles and fluxes need not satisfy pointwise MOST exactly; the law is a surface-layer expectation, not a constraint imposed through the column.

GABLS1: filtered wall drag

GABLS1 filtered rough-wall drag | 23 September 2026. Two new 32³, 12.5 m, WENO9 cases use a 300 s temporally filtered wall state, matched against earlier unfiltered no-closure and scheme-native one-face SLD controls. All use a 400 m cube, seed 123, and nine hours of the same cooling and geostrophic forcing. The fixed 1 m LES median is a model intercomparison reference, not observations.

Filtering improves no-closure near-wall shear but leaves the SLD deficit almost unchanged. In 8–9 h, mean-wind vector shear across 6.25–18.75 m falls from 0.1861 to 0.1541 s⁻¹ without an interior closure, versus 0.1004 s⁻¹ for the fixed 1 m median profiles. With scheme-native SLD it changes only from 0.0522 to 0.0530 s⁻¹. The same ordering appears during 7–8 h. These are adjacent-cell-center gradients, not wall gradients; reference u and v medians were sampled at those heights before differentiation.

In the no-closure pair, filtered wall drag raises final-hour mean u^* from 0.2794 to 0.3019 m/s, makes mean kinematic surface heat flux more negative (-0.01146 to -0.01314 K m/s), and raises peak resolved w^2 from 0.0864 to 0.1013 m²/s². For SLD, u^* rises from 0.2901 to 0.2949 m/s and peak w^2 from 0.05945 to 0.06276 m²/s². Both SLD cases continue to suppress resolved near-wall w^2 and w^3 relative to no closure. A single turbulent realization cannot establish a systematic effect beyond this matched sensitivity test.

Filtering barely changes SLD's first-face mean momentum viscosity (1.2958 to 1.2892 m²/s) or heat diffusivity (1.2642 to 1.2396 m²/s). Filtered SLD's measured WENO u-flux correction is -0.000256 m²/s² versus covariance -0.009609 m²/s², about 2.7%; its heat correction is approximately zero. The small numerical correction does not explain the persistent SLD shear deficit.

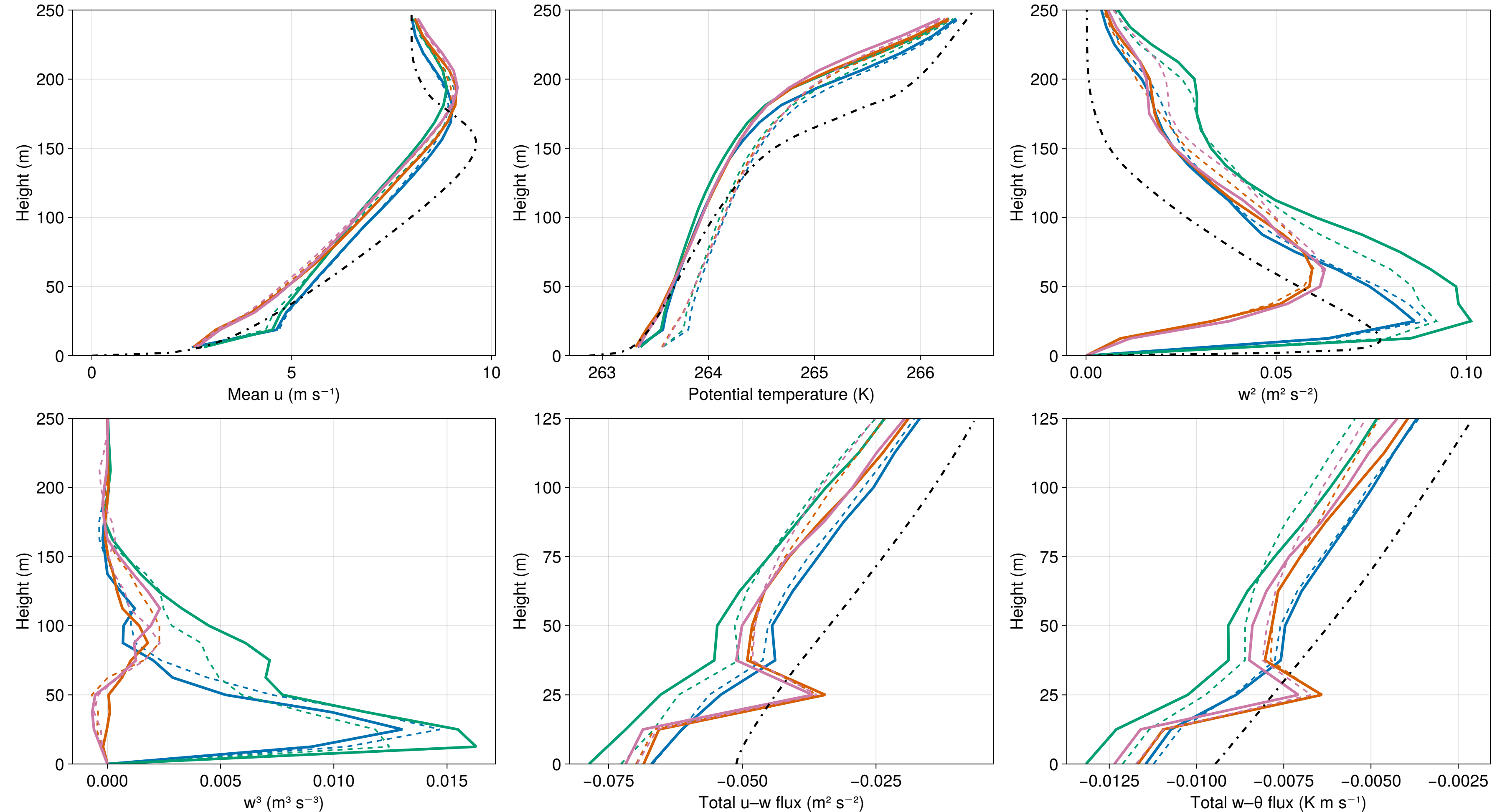
Validation: corrected H100 gate 7536 passed 19/19 no-closure and 21/21 SLD checks. Both science-array 7537 tasks ended with durable exit zero and CASE_DONE at 32400 s.

Immutable source hash manifest

61a5a464b9d100ff2f302ad822972af9ee47e9340a25bbae0ff597890911367e was rechecked; initial theta digest matched across cases. Independent Julia raw audits verified exact 1/19/541/55 initial/profile/series/filter schedules, native 32/33 vertical levels, finite output, evolving filtered state, density-consistent wall fluxes, and both time windows. Both-face reconstructed flux identities passed for SLD. The failed first gate 7534 launched no science and remains historical evidence.

GABLS1: filtering the rough-wall bulk drag

32³ · 12.5 m · WENO9 · 300 s filtered wall state · same seed · dashed 7–8 h, solid 8–9 h

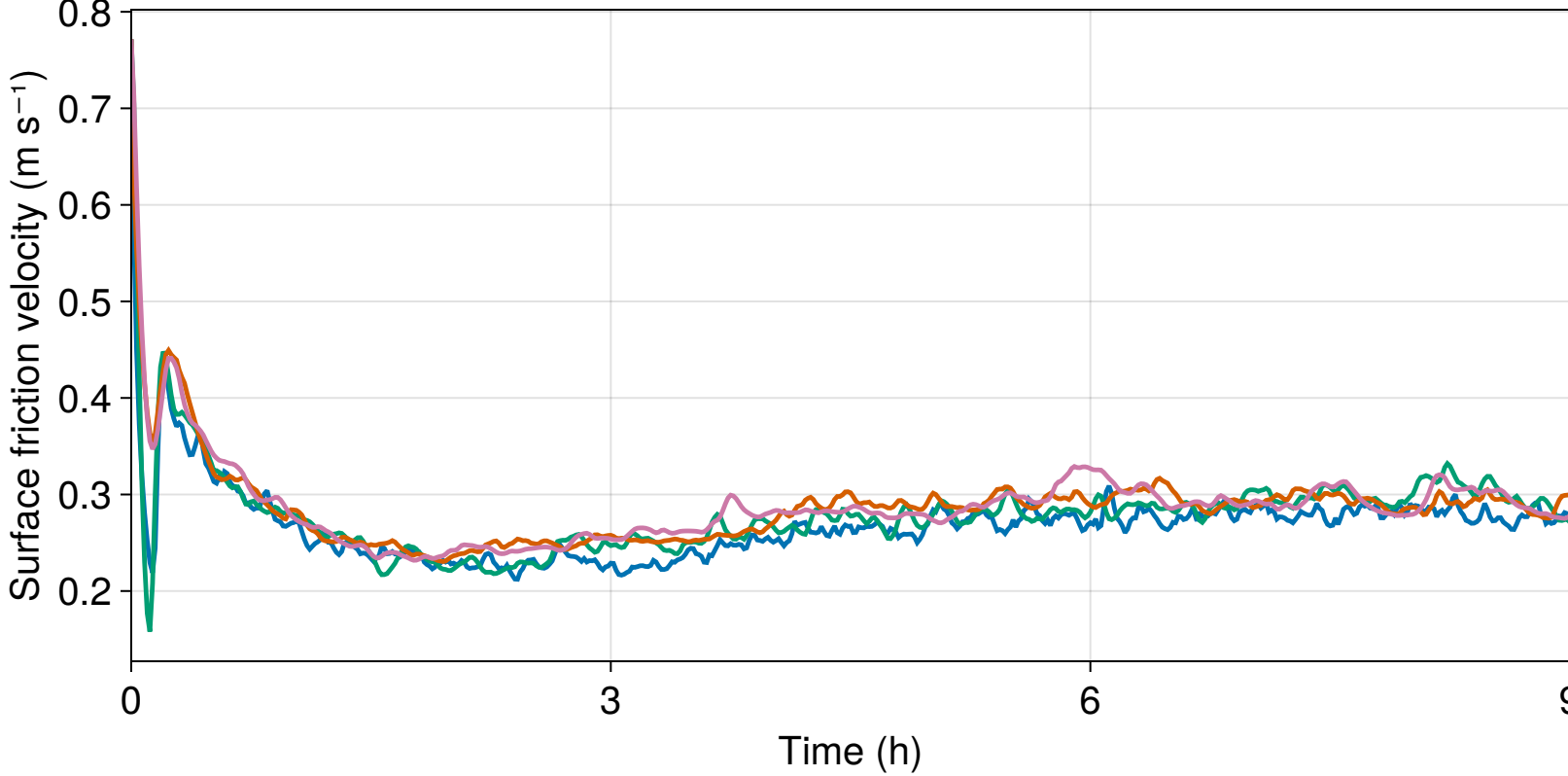
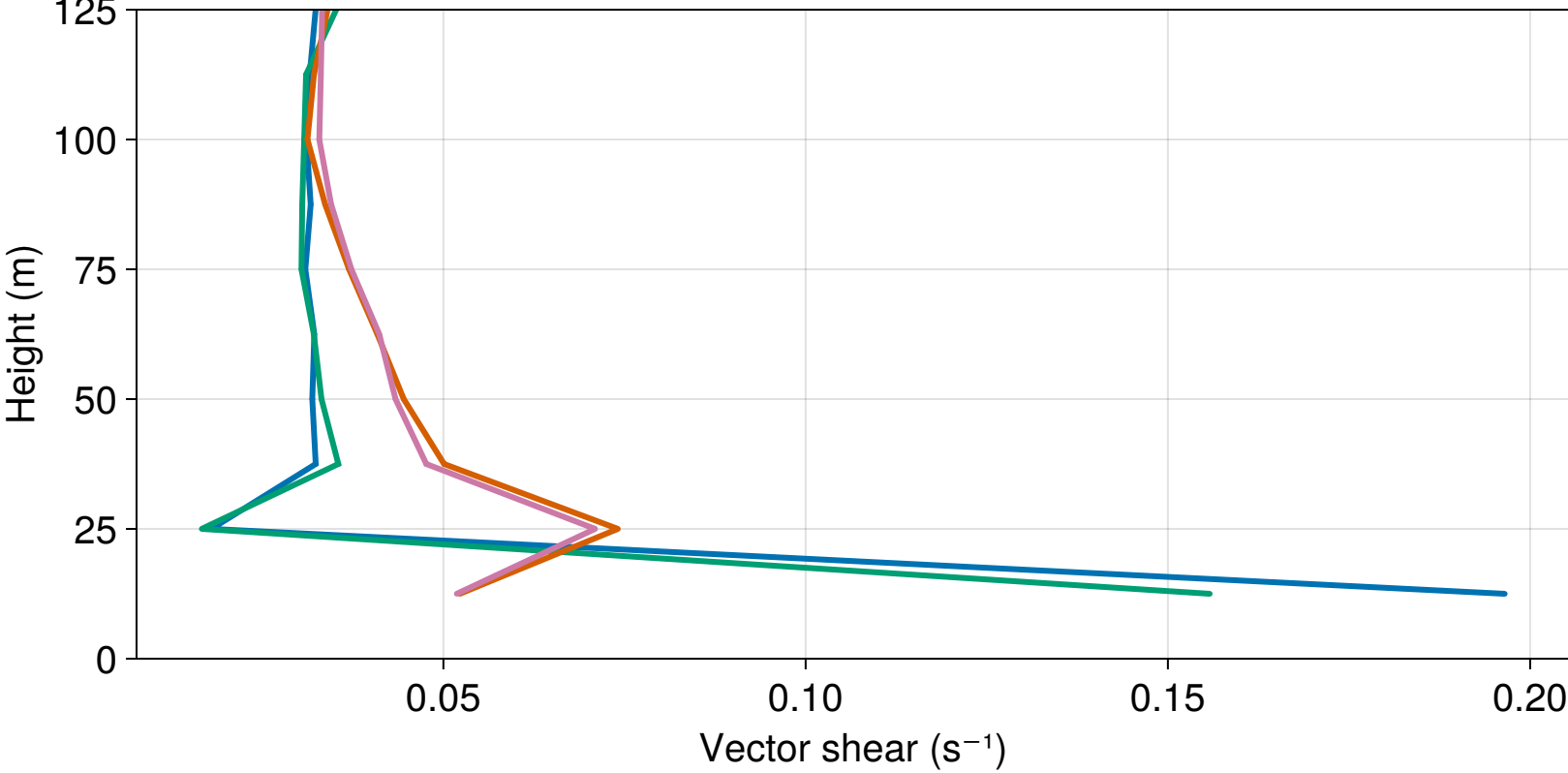


— Unfiltered · no closure — Unfiltered · SLD - - Fixed 1 m LES median
 — Filtered · no closure — Filtered · SLD

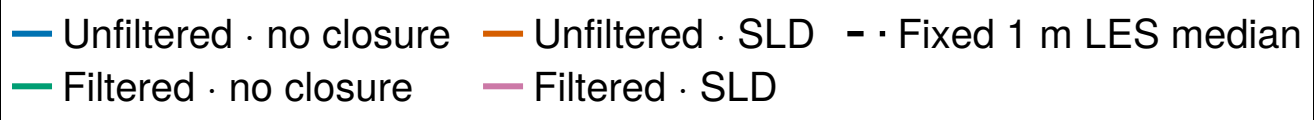
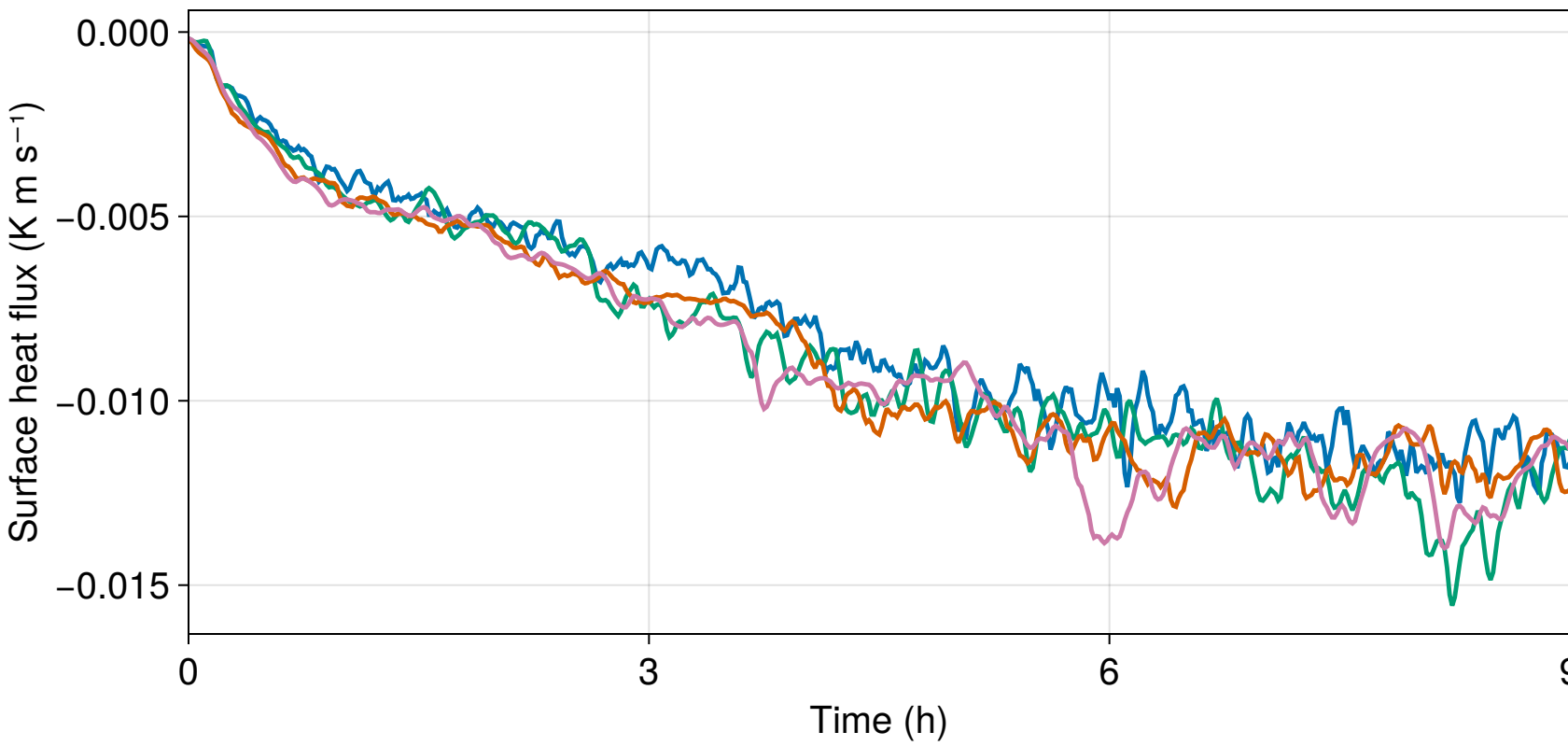
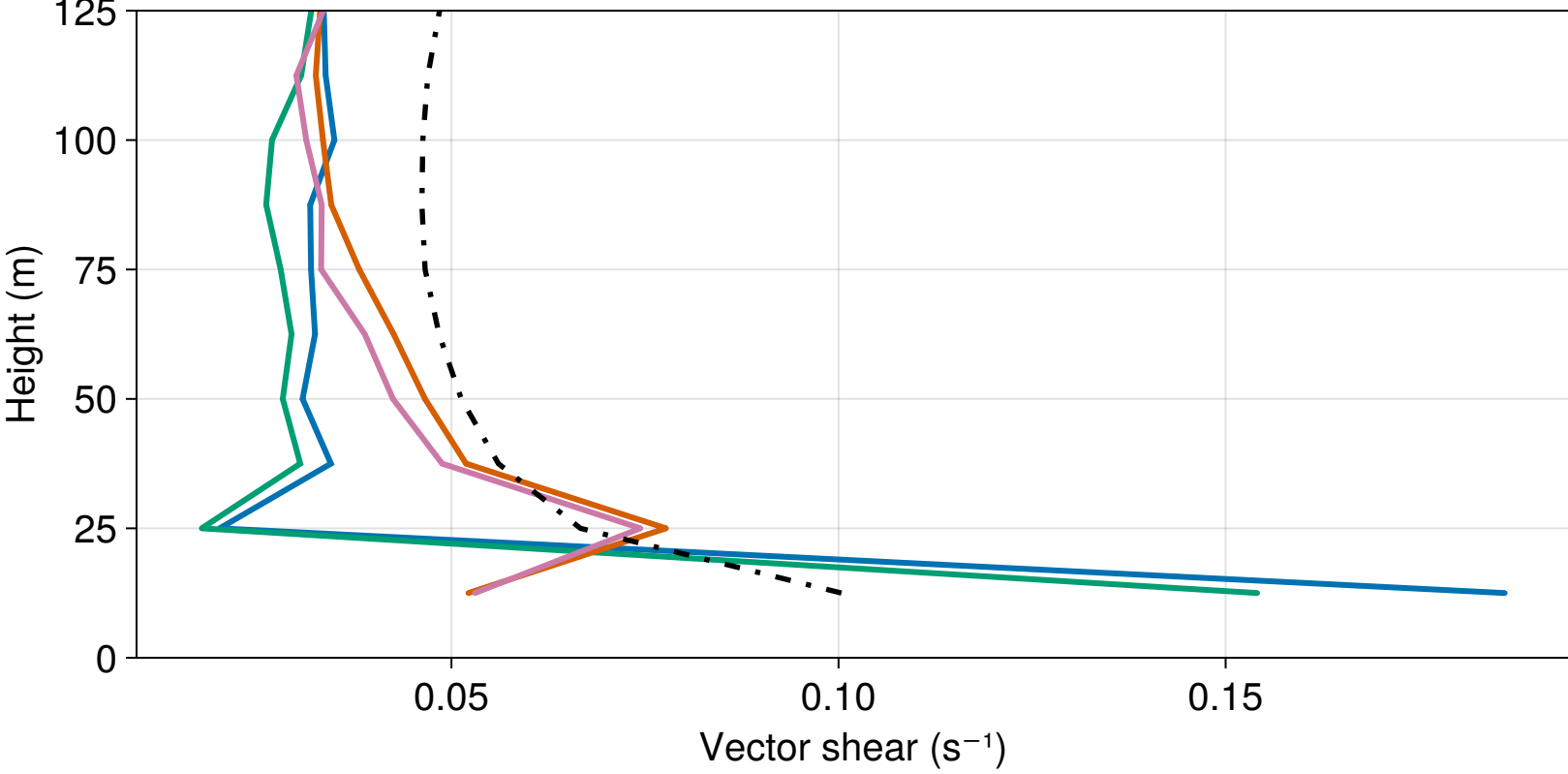
Mean-wind shear and wall flux

Shear between adjacent 12.5 m cell centers; fixed 1 m median sampled at the same heights

7–8 h



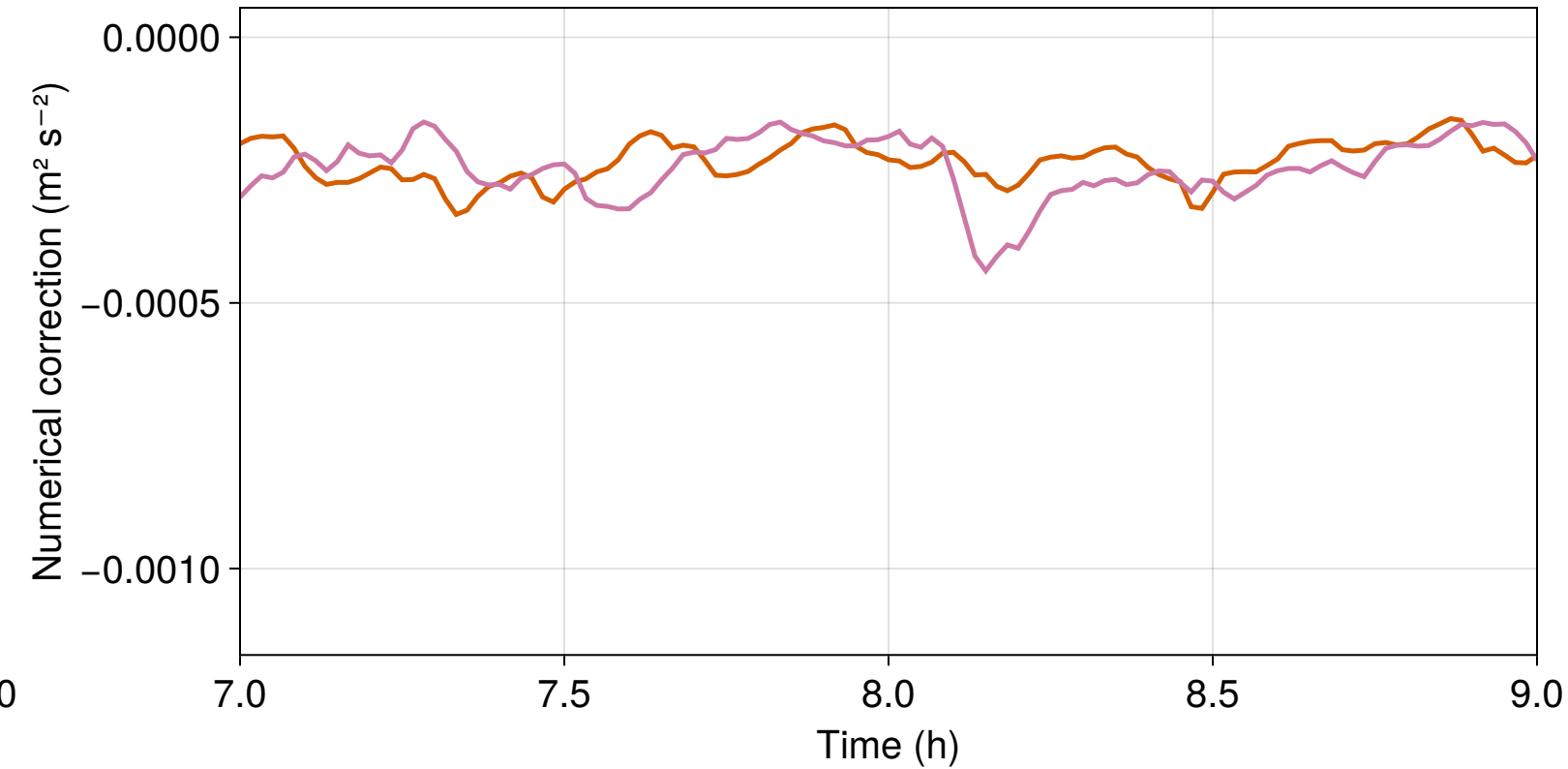
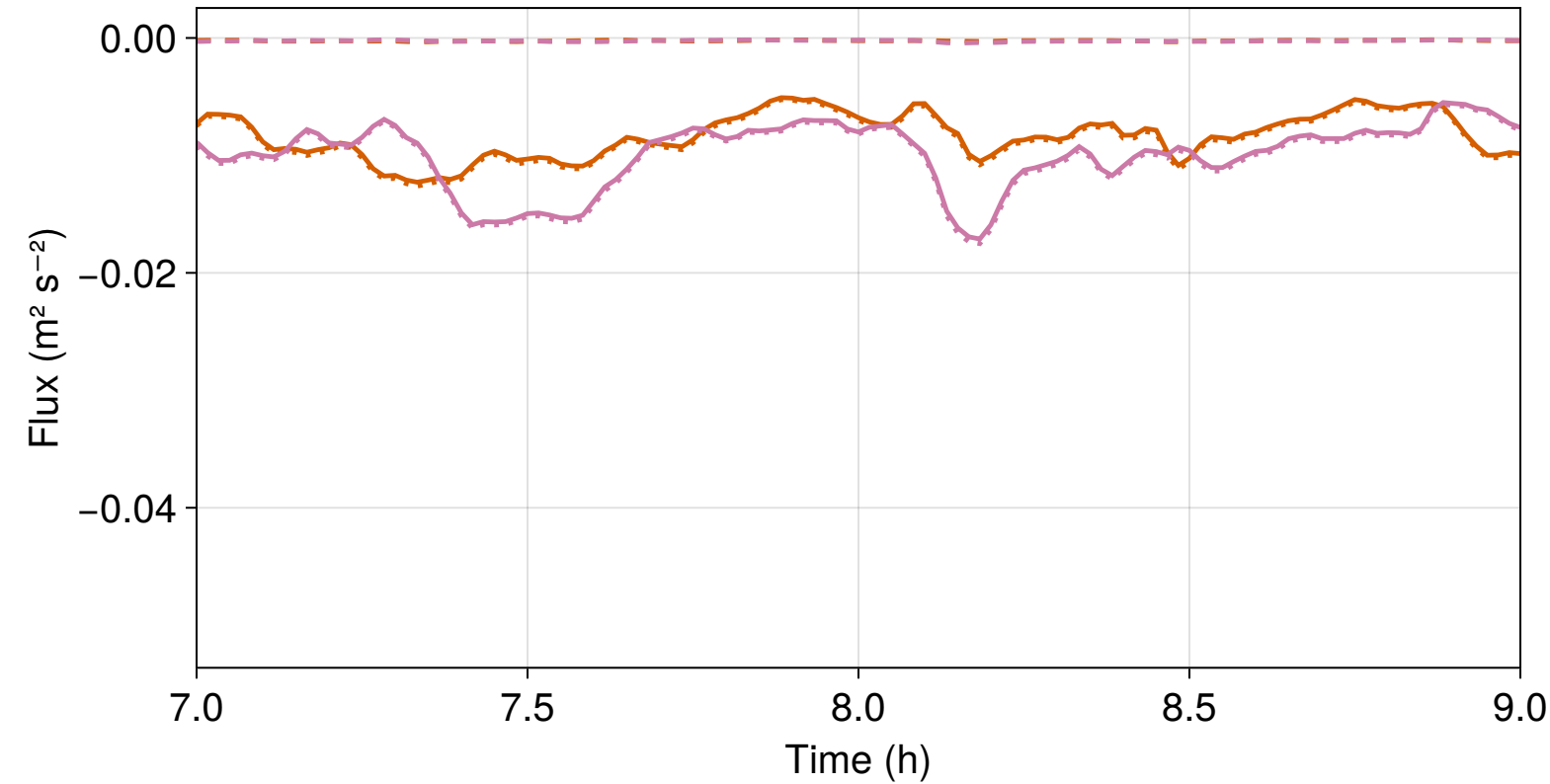
8–9 h



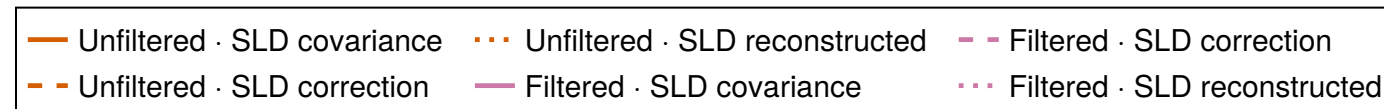
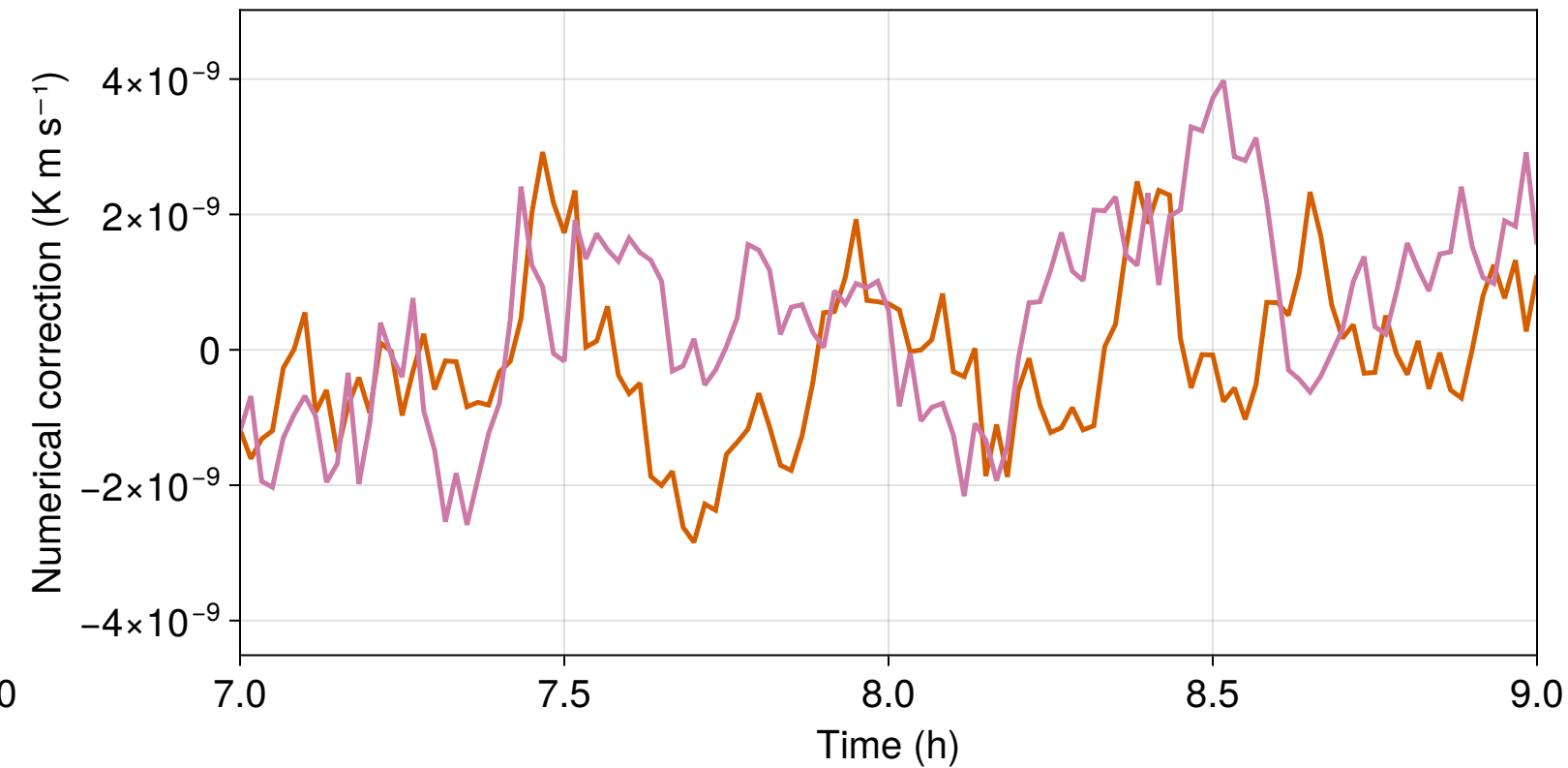
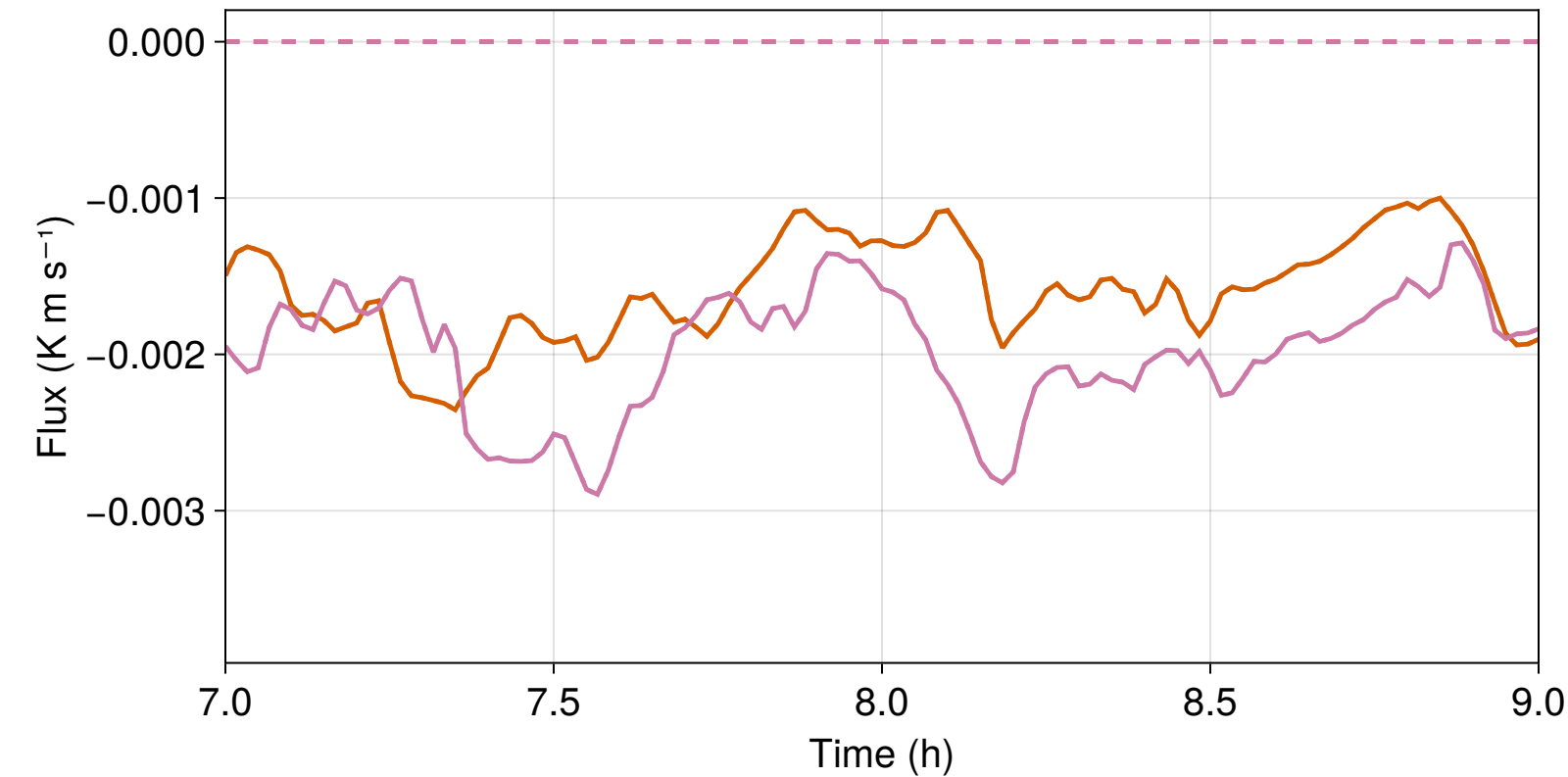
Scheme-native SLD: first-face transport partition

Filtering changes the wall state; reconstructed = covariance + numerical correction

Momentum

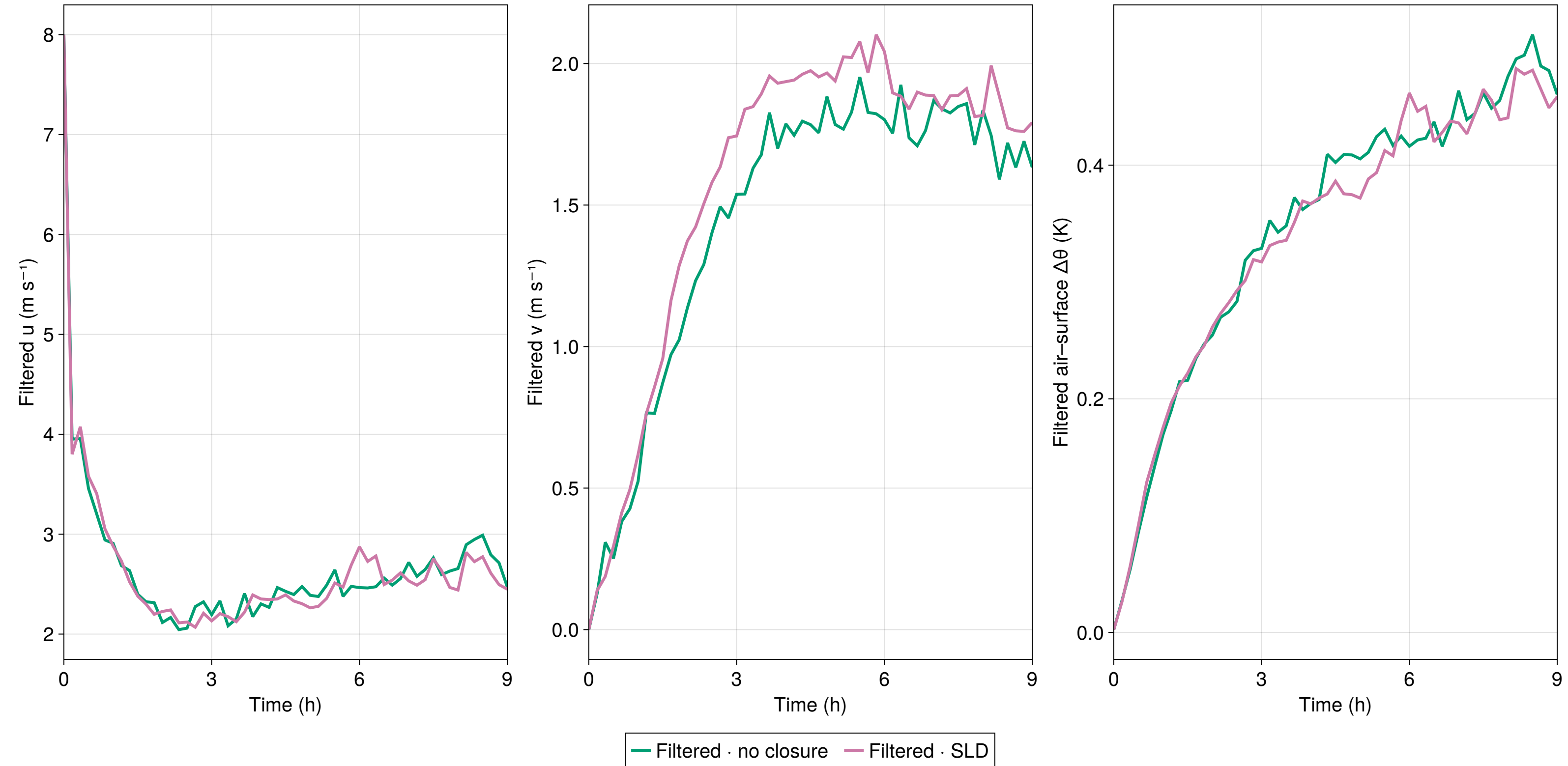


Heat



Saved filtered wall state

Horizontal mean of the actual 300 s filtered fields, sampled every 600 s



Part III: GABLS3 setup and evaluation

GABLS3: observational evaluation and case preparation

Status update, 21 September 2026: replacement SurfaceLayerDiffusivity runs are held for validation of a surface-humidity boundary-condition correction identified by independent review. Existing GABLS3 attempts remain excluded from scientific comparison; no new observational or LES curves are implied by this status update.

Status: preparation, with no GABLS3 production results yet. All DYCOMS and GABLS1 material remains in this master report. GABLS1's fixed 1 m ensemble reference is retained in its own comparison.

Published LES case

The original GABLS3 LES experiment covers 00-09 UTC on 2 July 2006 at Cabauw. It uses an 800 m cube with a common 6.25 m grid. Initial profiles combine tower, wind-profiler and sounding measurements. Geostrophic wind and advective tendencies vary in time and height. Prescribed potential temperature and humidity at 0.25 m replace interactive land-surface and radiation schemes; aerodynamic roughness is 0.15 m. Basu, Holtslag and Bosveld (2012), sections 2-3.

For 03-04 UTC, the ensemble reproduced jet height, strength, directional shear and fluxes reasonably well, but near-surface shear was deficient. A 1 m simulation mixed less and produced a stronger jet. Observed submesoscale variability and a turbulence burst were missed. Published LES variances are resolved, whereas observations include total variance. The ensemble percentiles pool models and five-minute samples; tower whiskers show hourly minima/maxima, not confidence intervals. Basu et al. (2012), sections 4-5 and figures 1-4.

The related 24-hour SCM experiment begins at noon on 1 July and couples the atmosphere to land-surface and radiation schemes. Its input tables must not silently replace the LES specification. Bosveld et al. (2014), case selection and setup.

Breeze experiment under preparation

Our proposed resolution comparison keeps the 800 m cube fixed: 12.5 m (64 cubed), 6.25 m (128 cubed), and 3.125 m (256 cubed). These nine proposed cases use exactly the user's three configurations: WENO9 without interior closure, WENO5 without interior closure, and WENO9 with Smagorinsky. The added coarse and fine grids are our experiment design, not a claim about the historical intercomparison matrix. Case preparation and validation precede production scheduling; no GABLS3 production run is claimed here.

Part III: GABLS3 setup and evaluation

We will verify the initial profiles, forcing tables, UTC origin, units, moisture convention, roughness lengths, near-surface reference height, wall stability functions through sunrise, damping and perturbations before freezing the setup. Surface exchange remains active in runs without an interior closure. Unresolved parameters stay explicit; GABLS1 defaults cannot substitute for missing GABLS3 inputs.

The comparison will retain observed points and their sampling ranges, published ensemble ranges, and any individual high-resolution LES as distinct references. A single 1 m GABLS3 run is not the two-model GABLS1 median and is not observational truth. Resolution agreement and agreement with observations will be assessed separately.

The author's revised instructions specify instantaneous plane-averaged profiles every five minutes, and surface/point time series every ten seconds. We will preserve that sampling, rather than applying GABLS1 half-hour output averaging. The 0.25 m scalar reference height is distinct from momentum roughness; the wall treatment must handle both stable nighttime and unstable morning conditions. Revised participant specification, archived in the author's repository.

Planned output includes mean u/v /potential temperature/humidity, velocity variances, vertical-velocity third moment and skewness, resolved/SGS/total momentum, heat and moisture fluxes, available TKE terms, jet height and speed, and surface fluxes, friction velocity and Obukhov length. We will compare the paper's 03-04 UTC interval and the complete night-to-morning evolution. Missing observed third moments or SGS quantities will remain unavailable. Numerical budget residuals will not be labeled dissipation without a closed budget.

Preparation checks cover forcing interpolation and jumps, units, profile shapes, surface fluxes and time stepping. GPU smoke tests retain the two-job limit and GABLS1 priority. Scientific admission requires source/input hashes, completion evidence and audited exports.

Implementation and validation status: 20 September 2026

The GABLS3 runner and diagnostics are committed. One-second CPU checks passed for the 64 cubed WENO9/no-closure, WENO5/no-closure, and WENO9/Smagorinsky configurations. The WENO9 checks exercised diagnostic writers; the Smagorinsky check produced nonzero explicit heat and moisture fluxes. Native-face vertical-velocity moments retain 65 levels, and centered moisture retains 64. Moist Obukhov length has an explicit validity flag; its zero fallback must be masked when invalid.

Part III: GABLS3 setup and evaluation

Separate CPU fixtures passed six writer checks at exactly 300 s and six morning-event checks, including a callback at exactly 21600 s and the contemporaneous surface humidity. These fixtures initialize the clock just before the event to test scheduling; their states are deliberately nonphysical and are excluded from scientific comparisons. They establish output and callback behavior, not nine-hour stability. The case-local environment still uses a mutable Breeze path during development; an immutable dependency snapshot and bounded GPU validation remain required before scientific runs.

A separate SurfaceLayerDiffusivity study is also authorized: at 12.5 m, a matched WENO9/no-closure control and three treatments (one interior face with 100 s or 300 s filtering, and two faces with 300 s filtering). The same four-case comparison is planned for GABLS1 at 32 cubed and GABLS3 at 64 cubed. This adds a coarse-grid study while retaining the original campaigns. No results from this new closure are available yet. Implementation and evaluation plan.

Persistent evaluation repository

Public evaluation repository: `glwagner/BreezeEvaluation.jl`. Migration, GABLS3 preparation and scientific completion are tracked separately. The combined report and all plots use Julia.

The repository now retains the 15 completed DYCOMS and 12 completed GABLS1 cases as audited compact comparison data, alongside case/source snapshots, reference data, Julia plotting/report code, figures and this report. Compact profile subsets are explicitly distinguished from full scientific exports. A hash-verifying restoration workflow recovered all 27 complete profile histories in a separate pcluster analysis copy; the original GABLS1 admission checks accepted all 12 restored cases. Frozen production, failed attempts and raw fields remain on pcluster with checksummed manifests. Original simulation identities and hashes are preserved. Third-party papers are linked.

All analysis and plots use Julia. Existing completed experiments and running GABLS1 jobs continue unchanged during migration. Future evaluations will live in `BreezeEvaluation.jl`, with their Breeze dependency revision pinned explicitly.

Validated replacement campaign — 22 September 2026

Part III: GABLS3 setup and evaluation

The corrected surface-humidity boundary passed source-specific GPU validation (7335, 1,750 checks). A fresh four-case, 12.5 m GABLS3 comparison is now queued/running as array7343: WENO9 with no interior closure, one-face100 s, one-face300 s, and two-face300 s SurfaceLayerDiffusivity. Cases run sequentially on one GPU; the original 1 m GABLS1 Smagorinsky run continues separately. All use the corrected frozen evaluation131ad9b/Breeze02a1647, seed20260702, and nine-hour duration. These are status statements, not scientific results; old7166 outputs remain excluded.

Queue update — 22 September 2026

The corrected GABLS3 control (7343_1) completed nine hours with a durable zero child exit. The three closure cases were briefly held to prioritize the requested factor3 sensitivity; those holds are now released. They await cluster resources. Full campaign admission and scientific comparison remain pending, and no completed-control curve is presented as a four-case result. The original GABLS1 matrix is now complete: all15 cases passed local admission.

Completed corrected comparison — 22 September 2026, 05:29 UTC

The queue notes above are historical. All four corrected array7343 runs completed nine hours and passed strict collection admission (4 accepted, 0 rejected), followed by an independent raw/export hash, native-coordinate, schedule and physical-flux audit. Each has 109 profile times and 3,241 series/point times. The following results section compares nocturnal and morning responses, with no observational-reference overlay. All15 original GABLS1 and all15 DYCOMS cases are also preserved in the public repository. Analysis-only revision b144835 documents final-record precision for three prescribed forcing scalars without changing their values or any simulation source.

SurfaceLayerDiffusivity: GABLS3 results

Four matched 9-hour GABLS3 runs | 12.5 m isotropic grid, 64³ cells, 800 m cube | WENO9 | seed 20260702.

The corrected control and three SurfaceLayerDiffusivity variants have completed and passed independent export and physical-flux audits: 109 instantaneous profile times, 3241 series and 3241 point records per case. The comparison varies one-face support with 100 or 300 s filtering, and two-face support with 300 s filtering. All use the same frozen simulation source.

During 03–04 UTC, SLD suppresses first-face vertical variance by 88–96% relative to the matched no-interior-closure control. At $z=12.5$ m, w^2 decreases from 0.02320 to 0.00260, 0.00270 and 0.00104 m²/s². Skewness changes from +0.344 to -0.112, -0.097 and -0.186. Domain-integrated resolved TKE decreases by 32–36%. These near-wall responses resemble the GABLS1 comparison.

The morning response differs. During 08–09 UTC, SLD first-face w^2 remains 28–35% below the control, but domain-integrated resolved TKE is 30–36% higher. Peak w^2 is 7–11% higher. Mean sensible heat flux is 82.6–84.1 W/m² upward, compared with 80.2 W/m² in the control. Local suppression therefore does not imply suppression of the whole convective layer.

One-face 100 and 300 s results are relatively close in these diagnostics: first-face w^2 differs by about 4% during 03–04 UTC and 9% during 08–09 UTC. Two-face support produces substantially stronger nocturnal near-wall suppression. These are single-seed results without sampling-uncertainty estimates; differences cannot be attributed solely to direct damping of w because surface exchange and mean shear also respond.

Profiles average twelve instantaneous outputs in each hour: 11100:300:14400 s or 29100:300:32400 s. This differs from GABLS1's saved half-hour means. Skewness is the ratio of averaged central moments. The transition figure masks skewness where w^2 is below 10⁻⁵ m²/s²; full diagnostic figures retain native data.

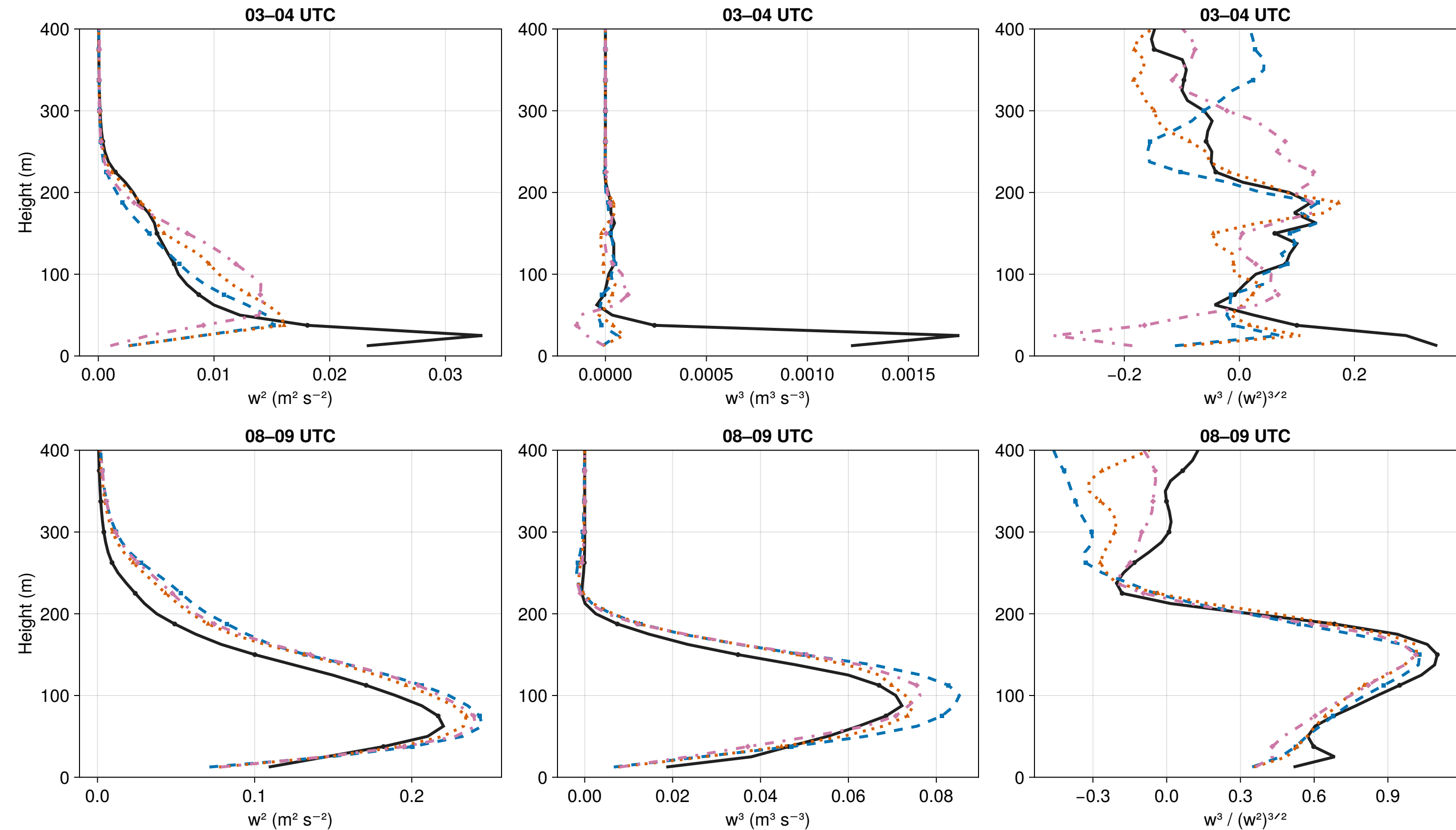
These plots compare Breeze variants, not fidelity against observations. No GABLS1 median is used as a GABLS3 reference. GABLS3 observational and LES sources remain described in Part III of the master. Quantitative observational agreement, boundary-layer growth relative to the 600 m sponge onset, and sampling uncertainty remain to be assessed before claiming improvement.

SurfaceLayerDiffusivity: GABLS3 results

Flux definitions: interior total means resolved covariance plus explicit SGS transport; it excludes unmeasured WENO numerical transport. Bottom values are wall-prescribed fluxes and are audited separately. The corrected humidity boundary condition and exact forcing schedule passed the source-matched GPU gate before production. Historical GABLS3 outputs with the earlier humidity treatment remain excluded.

Provenance: array 7343, GPU gate 7335; simulation core Evaluation 131ad9b and Breeze 02a1647. Analysis b144835 preserves the exact raw values while documenting Float32 storage in the final record of three prescribed surface-forcing scalars; preceding records are Float64. Raw files and simulation source were not changed. All four exports and the collection passed hash, native-height, schedule and physical flux checks.

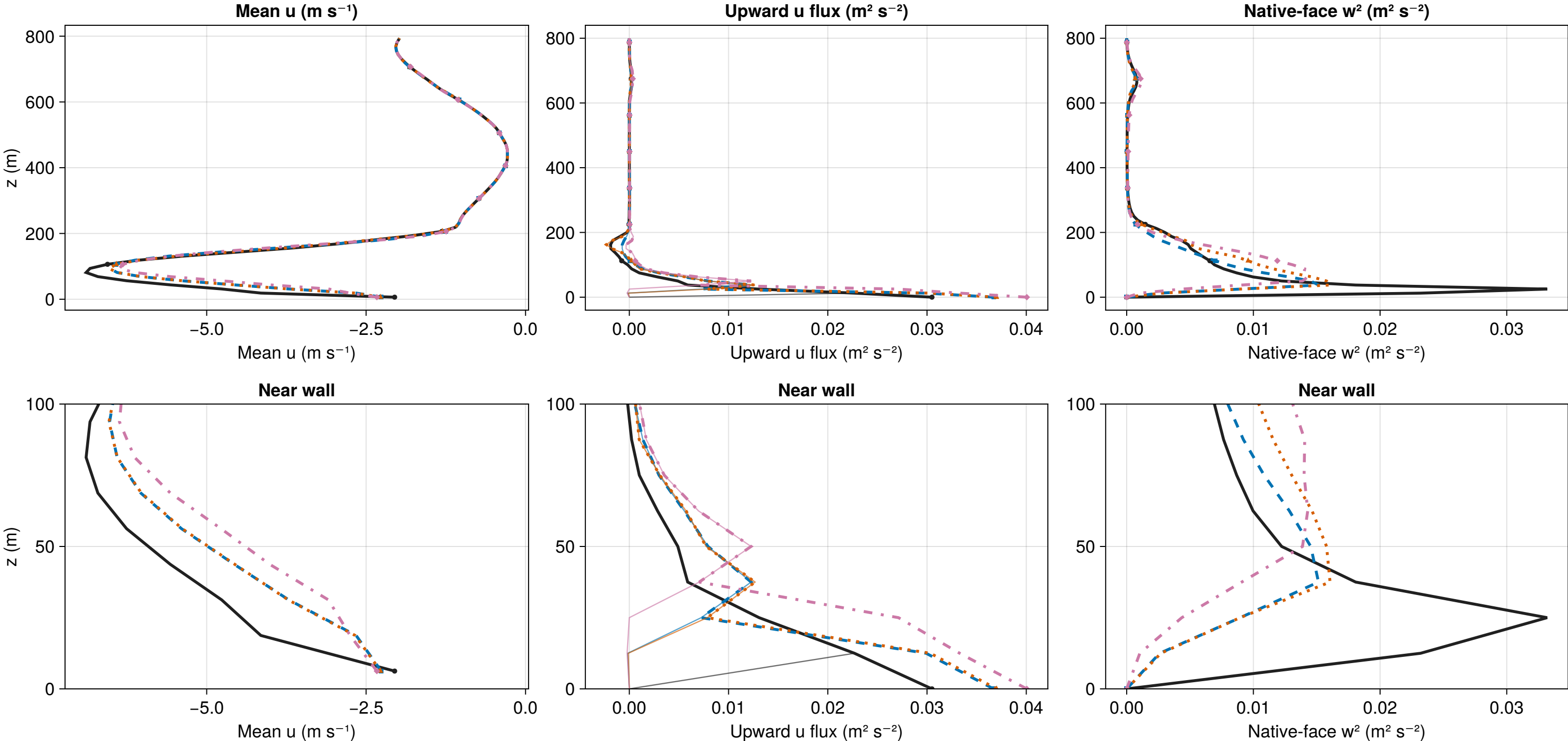
GABLS3: from nocturnal cooling to morning convection



Native-face moments; 12 instantaneous profiles per hour. Skewness omitted where $w^2 < 10^{-5} \text{ m}^2 \text{s}^{-2}$.
Matched 12.5 m WENO9 cases, one seed. This comparison has no observational reference overlay.



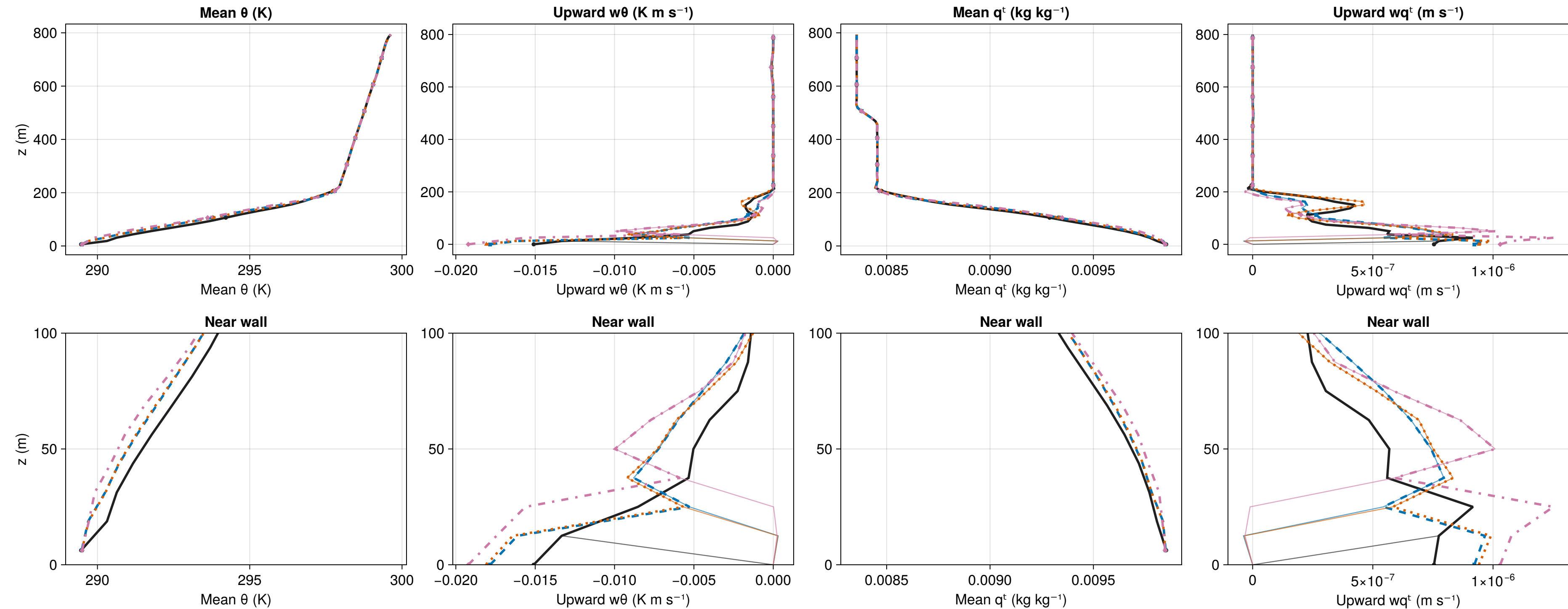
GABLS3 03:00–04:00 UTC matched comparison — source times 11100.0–14400.0 s



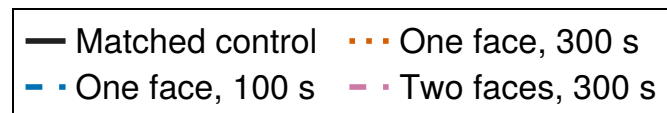
Interior thick flux = resolved + explicit SGS; thin flux = resolved. Boundary flux is wall-prescribed. WENO numerical transport is unmeasured. Native Center/Face heights retained.



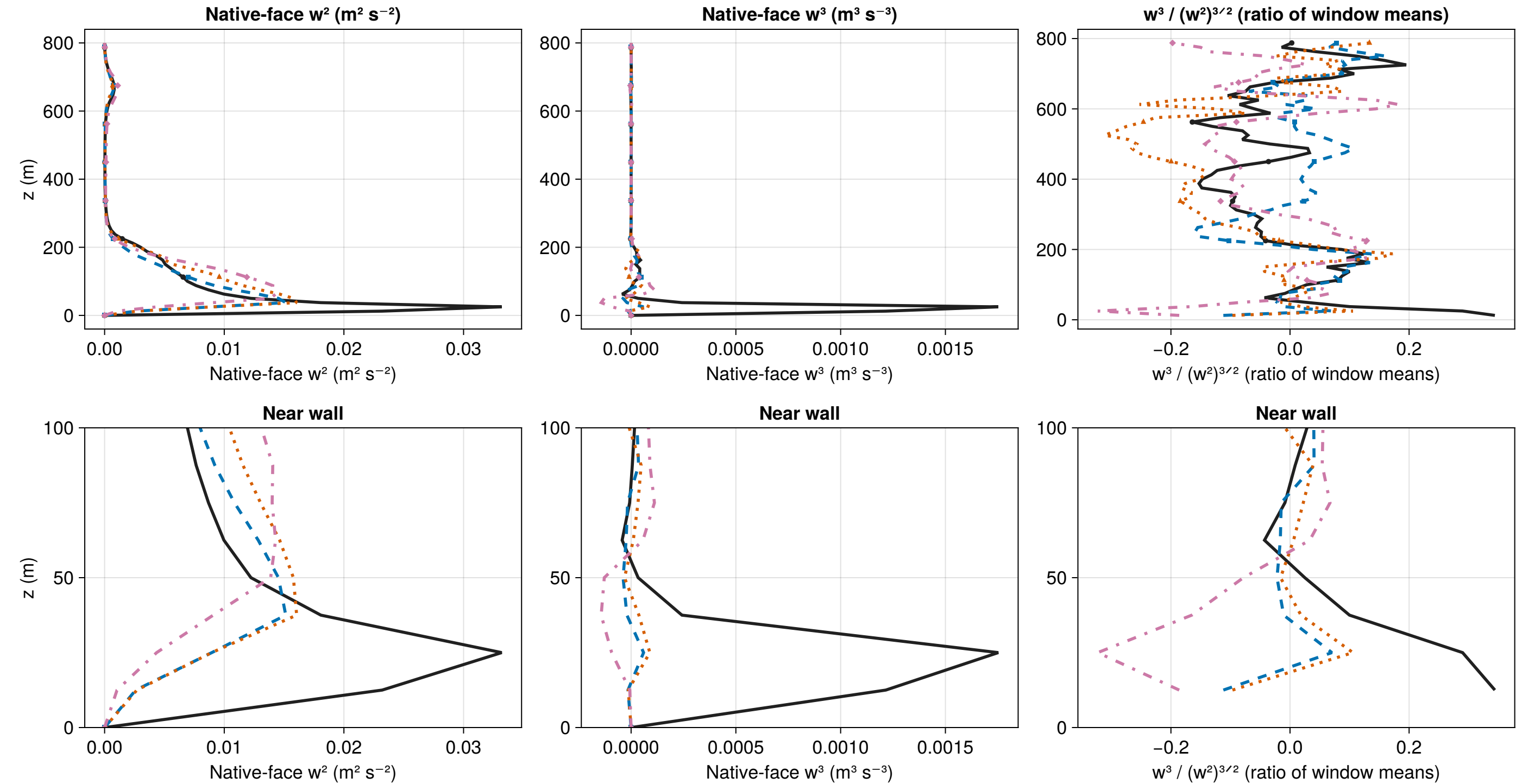
GABLS3 03:00–04:00 UTC thermodynamics and scalar flux — source times 11100.0–14400.0 s



No interchangeable GABLS1 reference is used for GABLS3. Native Center/Face heights and physical units are retained.



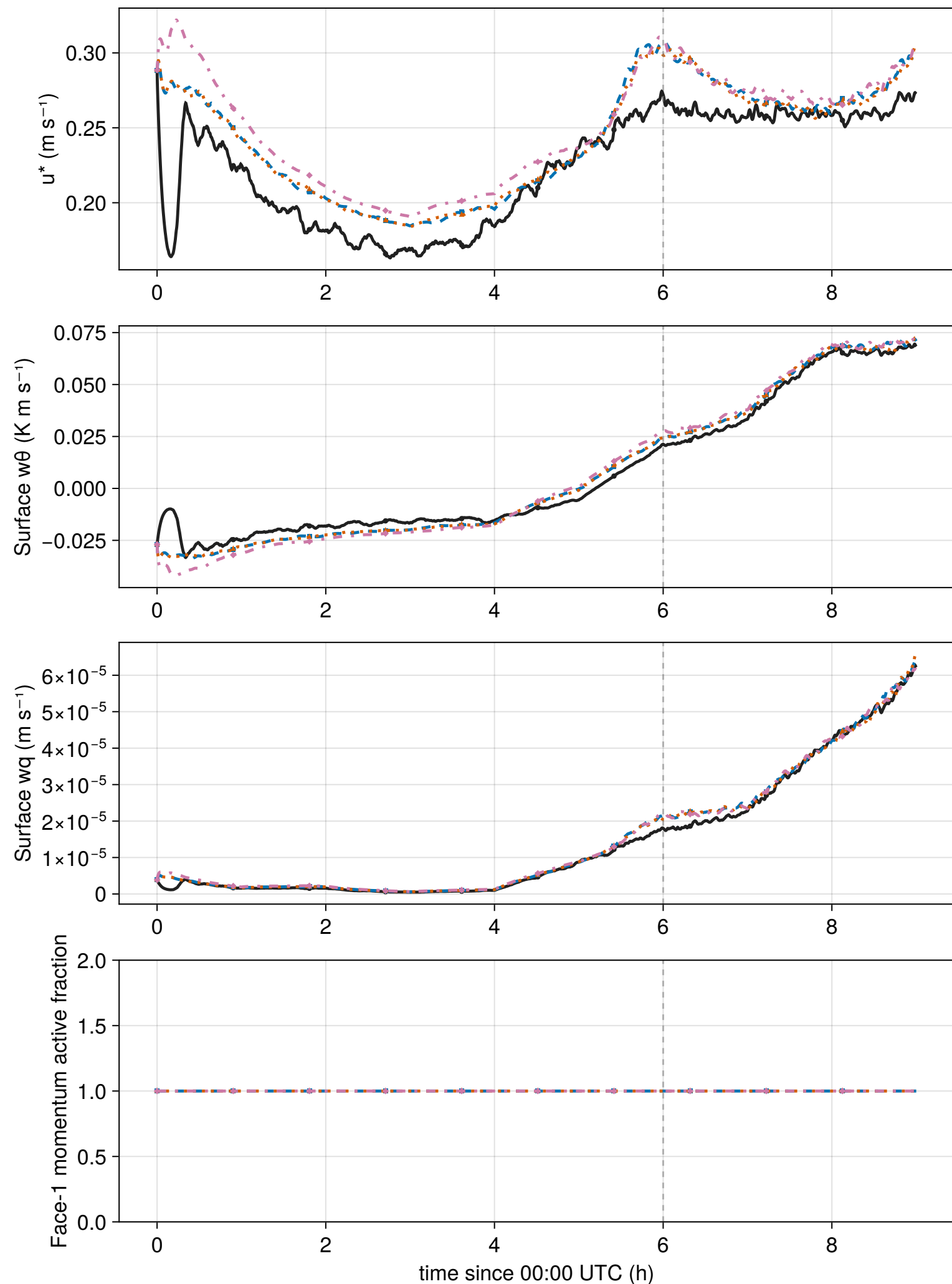
GABLS3 03:00–04:00 UTC native-face vertical-velocity moments — source times 11100.0–14400.0 s



No interchangeable GABLS1 reference is used for GABLS3. Native Center/Face heights and physical units are retained.

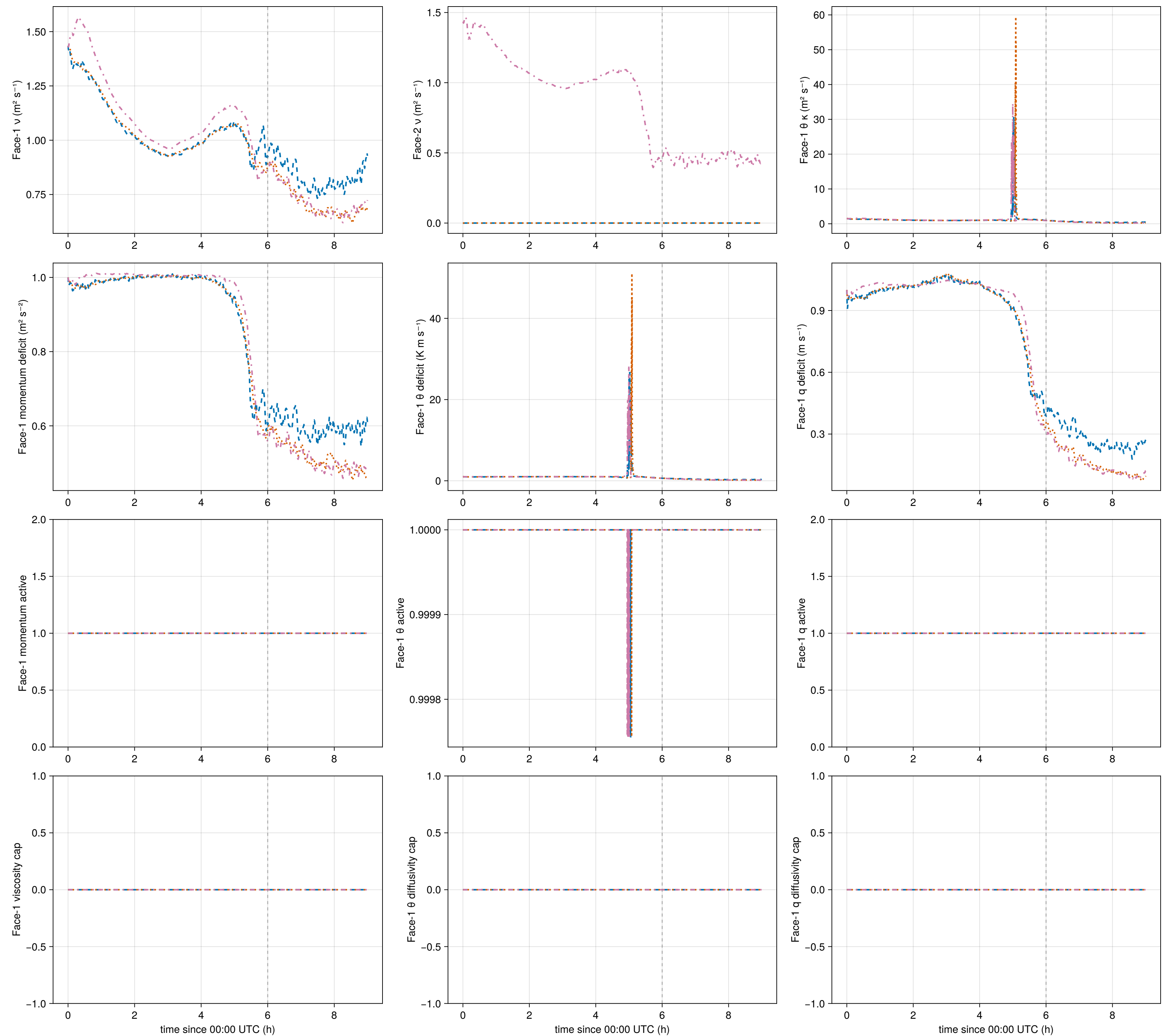


GABLS3 admitted comparison: surface and boundary-layer evolution



— Matched control - - One face, 100 s - . One face, 300 s - . Two faces, 300 s

GABLS3 SLD coefficients, deficits, activity and caps



Face-2 values require support weight > 0; one-face variants have no active face-2 closure. Zero active fraction is not a valid scalar-flux measurement. Exact support heights/weights and guards are in each admitted manifest.

— One face, 100 s . One face, 300 s - Two faces, 300 s

SurfaceLayerDiffusivity: neutral ABL

Two matched five-hour neutral ABL runs: WENO9 with no interior closure versus one-face SurfaceLayerDiffusivity with a 300 s filter. Both use a 96^3 grid in a $3000 \times 3000 \times 1000$ m domain (31.25 m horizontal, 10.42 m vertical), seed 1994, identical forcing and initial state. The surface prescribes local $u^* = 0.5$ m/s and zero heat flux. This is a fixed-stress test, not a test of a predicted drag coefficient or MOST surface exchange.

During 4–5 h, SLD reduces first-interior-face vertical variance from 0.09209 to 0.03061 m^2/s^2 , a 66.8% decrease at $z = 10.42$ m. The third central moment changes from +0.01718 to -0.000720 m^3/s^3 , and skewness from +0.615 to -0.134. This repeats the near-wall suppression and asymmetry reversal seen in the stable cases, even though the neutral scalar guard is inactive.

The response is not uniform through the layer. Peak w^2 rises from 0.1915 to 0.2448 m^2/s^2 (27.8%), while vertically integrated resolved TKE rises from 260.0 to 319.8 m^3/s^2 (23.0%). Near-wall suppression therefore coexists with more resolved energy elsewhere. These differences cannot be attributed solely to direct w damping: mean shear, momentum transport and turbulence organization also respond.

The neighboring 3–4 h window has similar first-face w^2 (0.09272 control, 0.03074 SLD) and skewness (+0.588, -0.147), but integrated TKE is lower (202.5, 231.7 m^3/s^2). The layer is still evolving. Five hours and one seed do not establish equilibrium, sampling uncertainty, or improved accuracy. No external neutral LES or observational reference is included.

Each plotted final-hour profile is the equal mean of six true 600 s averages ending at 15000:600:18000 s. Moments retain native vertical-face coordinates. Skewness is the ratio of averaged central moments, masked where $w^2 \leq 10^{-5} \text{ m}^2/\text{s}^2$. Interior total flux means covariance plus SGS and excludes unmeasured WENO numerical transport. The wall law fixes local stress magnitude; averaging differently directed vectors can lower the magnitude of the mean stress and its diagnosed u^* .

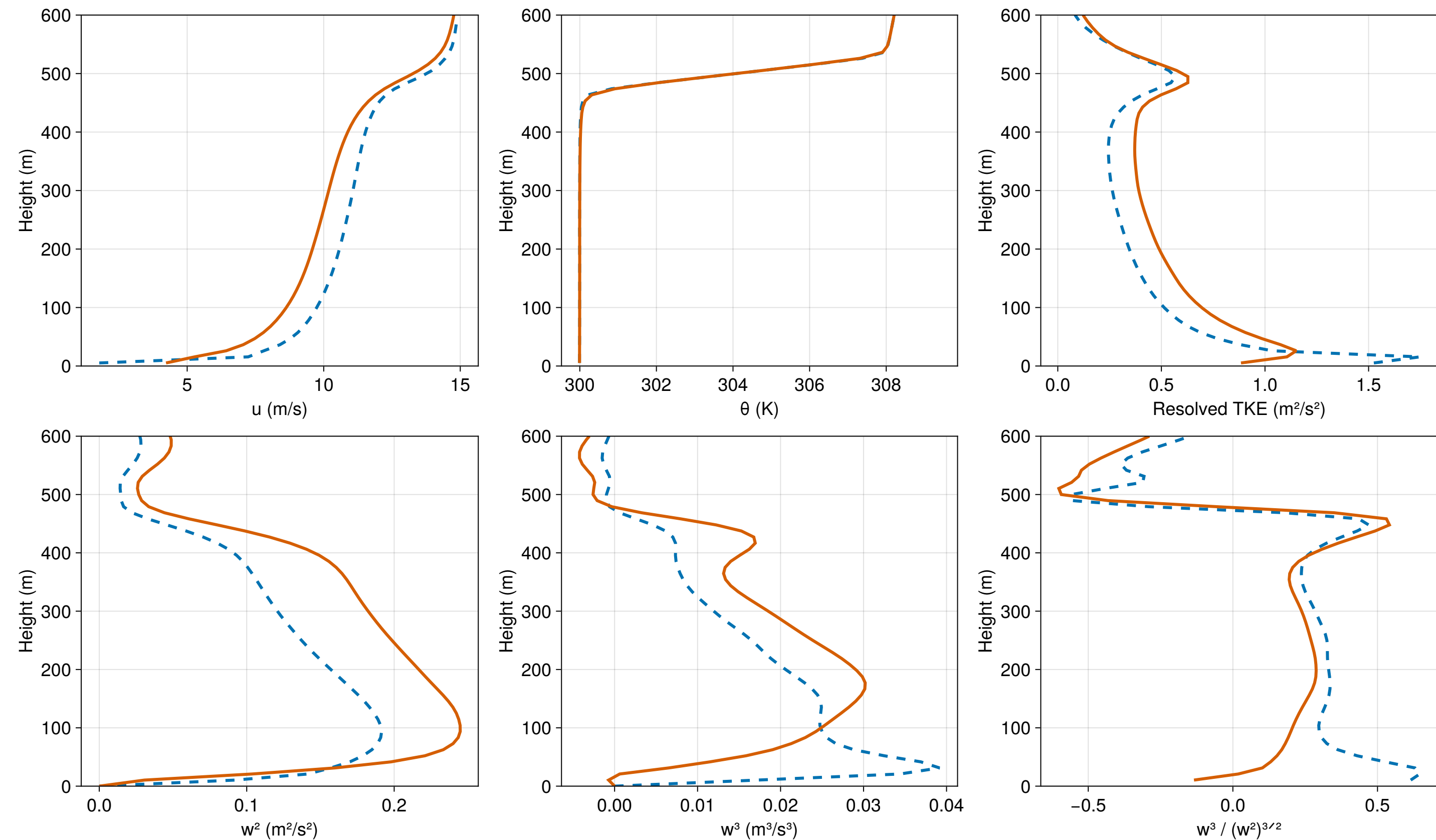
Both cases passed full saved-output scientific admission: 30 averaged profiles plus separate $t=0$ data, 301 series and bounds records, six finite hourly checkpoints, source and output hashes, native coordinates, zero heat/scalar guard, and momentum flux checks. The original batch wrappers retain code-3 failure records because a post-run `rg` command was missing. Independent audit of the pinned wrapper and logs establishes that each solver returned zero first; a separate Codex-root acceptance records this distinction. No GPU rerun or raw-data edits were made.

SurfaceLayerDiffusivity: neutral ABL

Measured run! wall times were 6.506 min (control) and 6.725 min (SLD), including run initialization; complete attempt durations were about 14.1 and 15.5 min including process startup and other setup. These single timings are not a robust performance benchmark. Simulation source: Evaluation a14c358 / Breeze 02a1647; analysis c9220b1; original array 7367; collection 2 admitted, 0 rejected. All earlier DYCOMS and GABLS material is preserved.

Neutral ABL: matched five-hour comparison

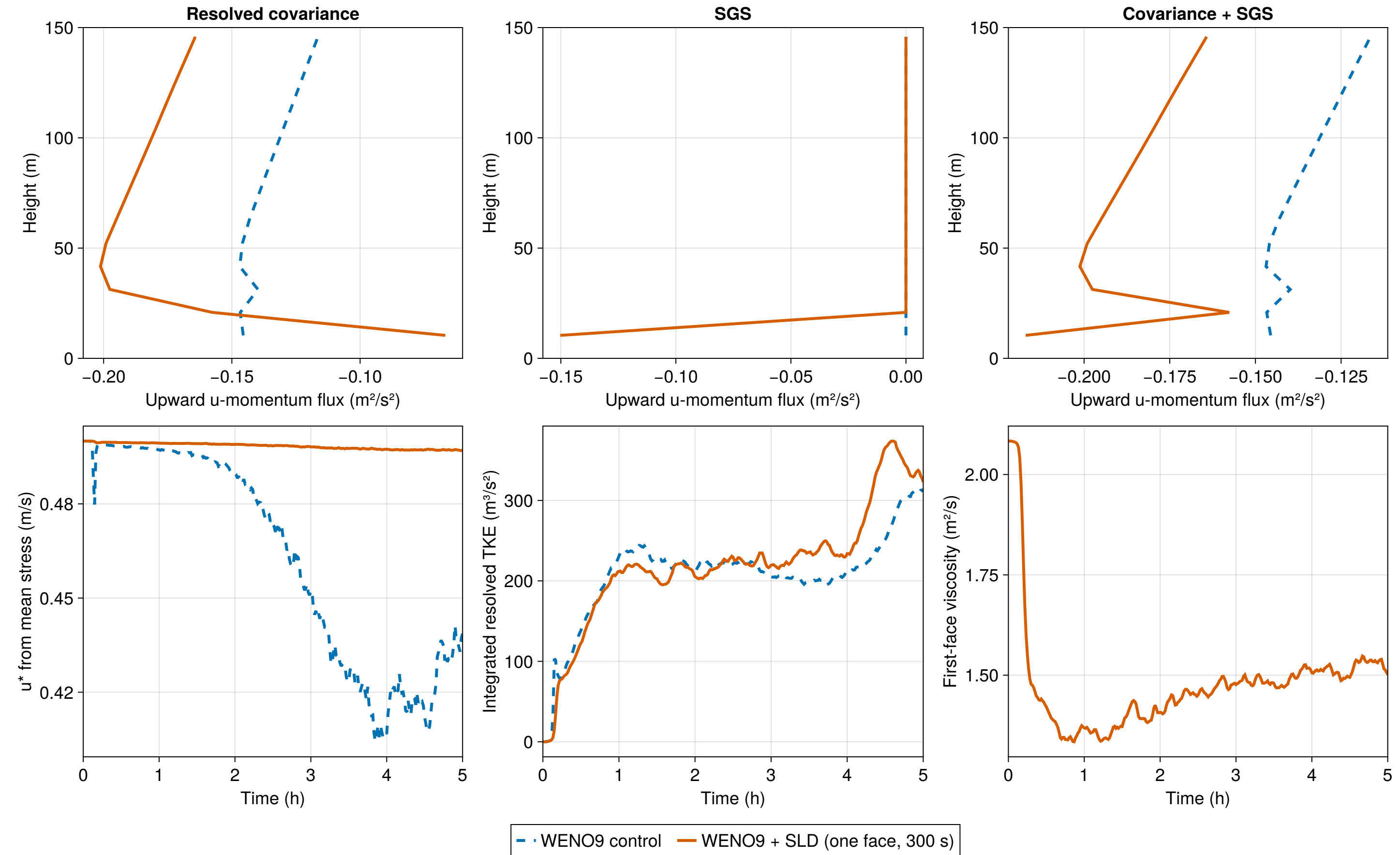
$96^3 \cdot \Delta x = 31.25 \text{ m} \cdot \Delta z = 10.42 \text{ m} \cdot \text{fixed local } u^* = 0.5 \text{ m/s} \cdot 4\text{--}5 \text{ h means}$



— · — WENO9 control — WENO9 + SLD (one face, 300 s)

Six true 600 s mean profiles per hour. Native-face moments; skewness masked for $w^2 \leq 10^{-5} \text{ m}^2/\text{s}^2$.
One seed, no observational or LES reference: this tests response, not accuracy.

Neutral ABL: momentum transport and evolution



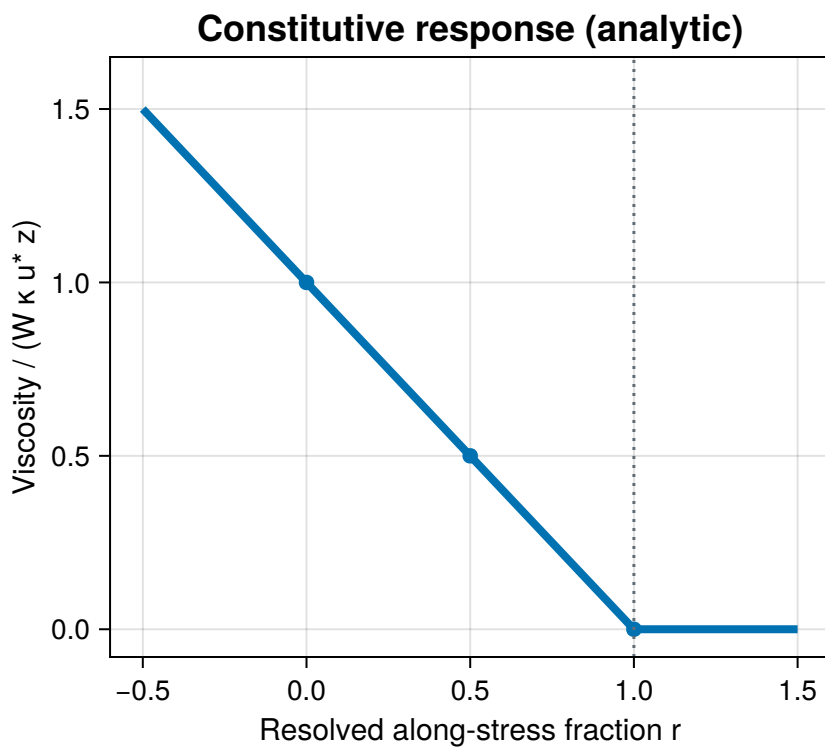
Interior fluxes exclude unmeasured WENO numerical transport. Wall values omitted from these flux panels.
 Local prescribed stress magnitude is 0.25 m²/s²; averaging stress directions can reduce the mean-vector magnitude.

Can a thin surface closure improve coarse LES?

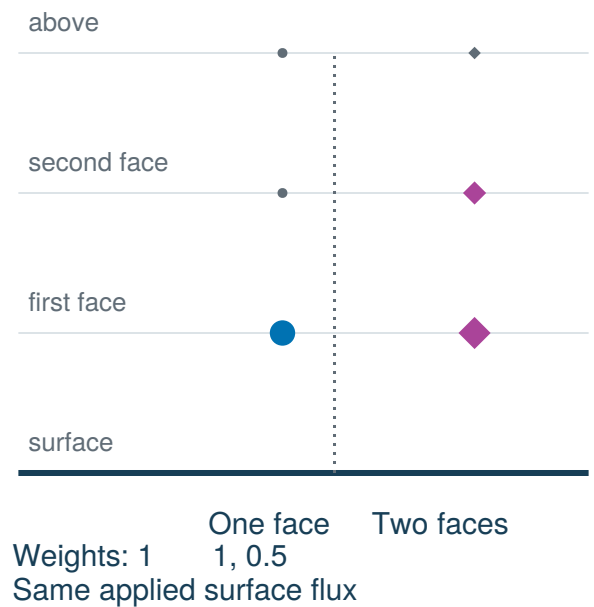
Supply unresolved near-wall transport while preserving resolved turbulence.
Four coarse GABLS1 comparisons admitted; transfer tests ongoing.

A local estimate of missing transport

Filter velocity, scalar, and wall-flux samples in physical time. Use their centered covariance to estimate the resolved share of the surface flux.



Shallow vertical support



What this formula does

No resolved stress: full neutral similarity viscosity. Half resolved: half the viscosity. Fully resolved or more: the correction turns off.
Opposite-sign transport can give a factor greater than one.

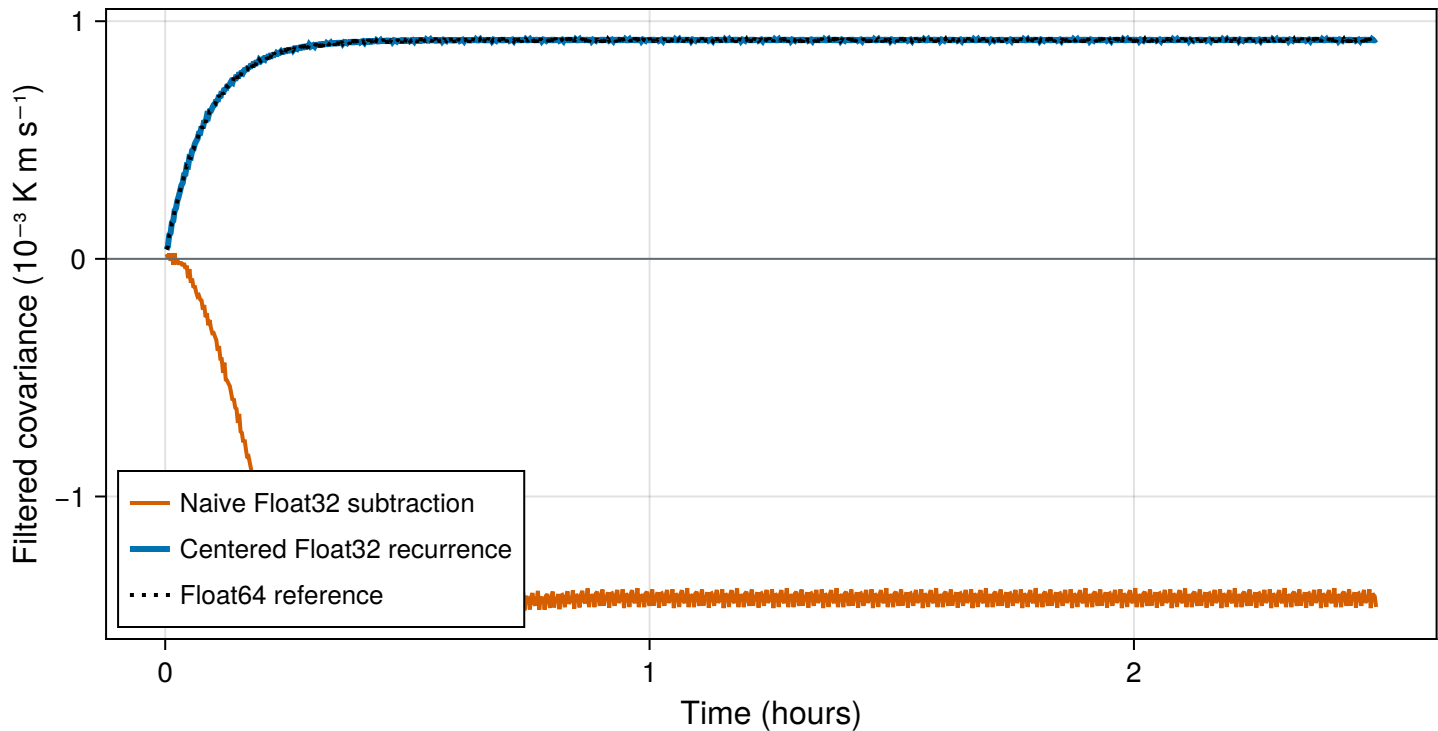
What it does not establish

The added flux still depends on the instantaneous vertical gradient.
The formula does not force exact flux balance, include stability functions, or measure the numerical transport produced by WENO.

One-face height: 12.5 m in both planned GABLS comparisons.
Scalar correction uses signed flux ratios; zero/tiny wall flux disables it.

A small covariance can get the wrong sign in Float32

Synthetic arithmetic test: 90,000 samples, $\Delta t = 0.1$ s, filter time = 300 s.
Identical Float32 input samples are used in the Float64 reference.



Final covariance: naive -1.465 | centered 0.925 | reference 0.924

Units: 10^{-3} K m s⁻¹. Centered recurrence relative error: 0.020%.

Why this matters for the closure

Subtracting two much larger filtered products loses the small turbulent flux. Here it reverses its sign. A centered covariance update retains the signal that controls the added diffusivity.

Implementation checks support the arithmetic result

The original 119-check CPU suite covers filtering, signs, guards and conservation. Later halo/restart checks and the 1,678-check GPU suite support the admitted run source. Their evidence files retain each revision.

Boundary of the evidence: this arithmetic example is synthetic.
Physical fidelity is assessed separately using the completed GABLS1 runs.
Reproducer, exact plotted CSV, and source hashes accompany this section.

Judge transport, turbulence, and the mean state together

One controlled intervention, three complementary tests of physical behavior. See the accompanying GABLS1 results for the first completed comparison.

01 Did the near-wall transport change as intended?

Lead with resolved, added, and combined momentum/heat flux profiles. Zoom the lowest four levels; mark every active face. Pair flux mismatch with viscosity, diffusivity, guard activity, and mean vertical transport.

02 Did the boundary layer improve without losing turbulence?

Place mean wind and temperature beside w^2 , w^3 and skewness. Show surface flux through time; withhold invalid depth diagnostics. Retain the fixed 1 m GABLS1 median; missing reference quantities remain explicit.

03 Is the result robust to filter time and support?

Use the same four styles in every GABLS panel. Compare with a freshly run, same-revision control, then examine 7-8 h versus 8-9 h. Report changes in physical units; avoid a composite score that hides tradeoffs.

-  Matched WENO9 control
-  One face / 100 s
-  One face / 300 s
-  Two faces / 300 s

Independent challenges strengthen the conclusion

GABLS3: moist nocturnal evolution and the sunrise flux sign change.
Neutral ABL: fixed-stress transport, without a surface-drag prediction.
Both add evidence; neither substitutes for the GABLS1 reference comparison.

GABLS1: stable SLD correction

GABLS1 stability-corrected surface-layer diffusivity | 23 September 2026. A matched nine-hour 32-cubed, 12.5 m, WENO9 run changes only the interior SLD stable-similarity strength from $\lambda=0$ to $\lambda=1$. Both use one-face scheme-native flux, 300 s wall and SLD filters, a 400 m cube, and seed 123. The fixed 1 m LES median is an intercomparison reference, not observations or a tuning target.

The correction raises 8-9 h first-layer vector shear from 0.05302 to 0.07387 per second. The fixed 1 m median sampled at the same two cell-center heights is 0.10038 per second. About 44 percent of the neutral SLD shortfall is closed, but the corrected profile still has too little near-wall shear. The 7-8 h shear moves similarly, from 0.05184 to 0.07162 per second.

At the first interior face, mean local stable gradient functions over 8-9 h are $\phi_m=1.437$ and $\phi_h=1.710$; every saved column is on the stable branch. The evolved SLD viscosity falls from 1.2892 to 0.8239 square meters per second, and heat diffusivity from 1.2396 to 0.6045. Mean friction velocity changes modestly, from 0.29485 to 0.28907 meters per second, as does kinematic heat flux, from -0.012333 to -0.011838 kelvin meters per second.

Peak resolved w -squared increases only from 0.06276 to 0.06676 square meters per second squared, below the filtered no-closure value of 0.10125. At 18.75 m, w -cubed changes from -0.000295 to +0.000102 cubic meters per second cubed, versus +0.0163 without closure. The stable correction improves mean shear but does not recover strong near-surface turbulent asymmetry. One seed at one resolution establishes a matched sensitivity, not an optimal universal parameter.

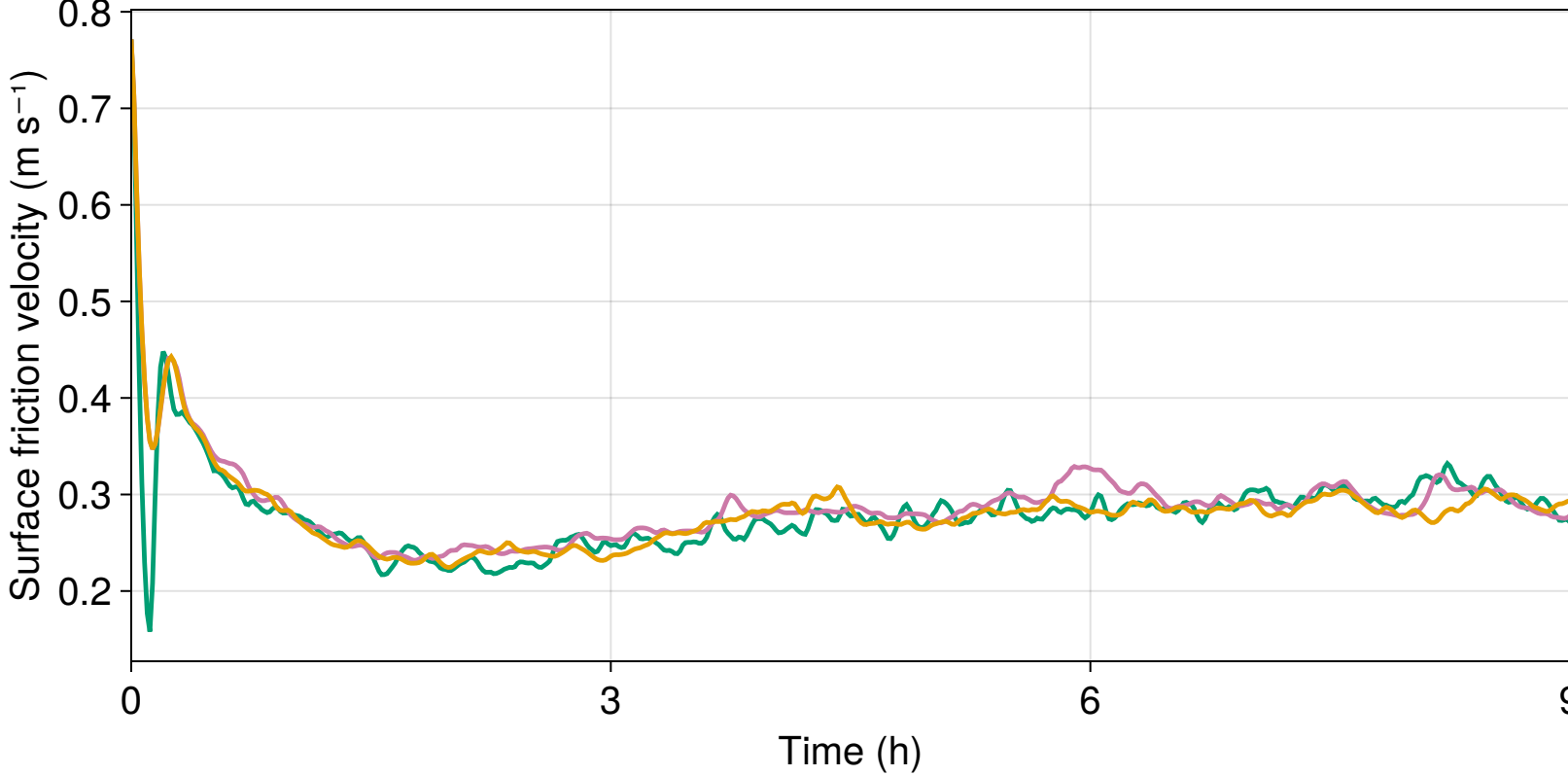
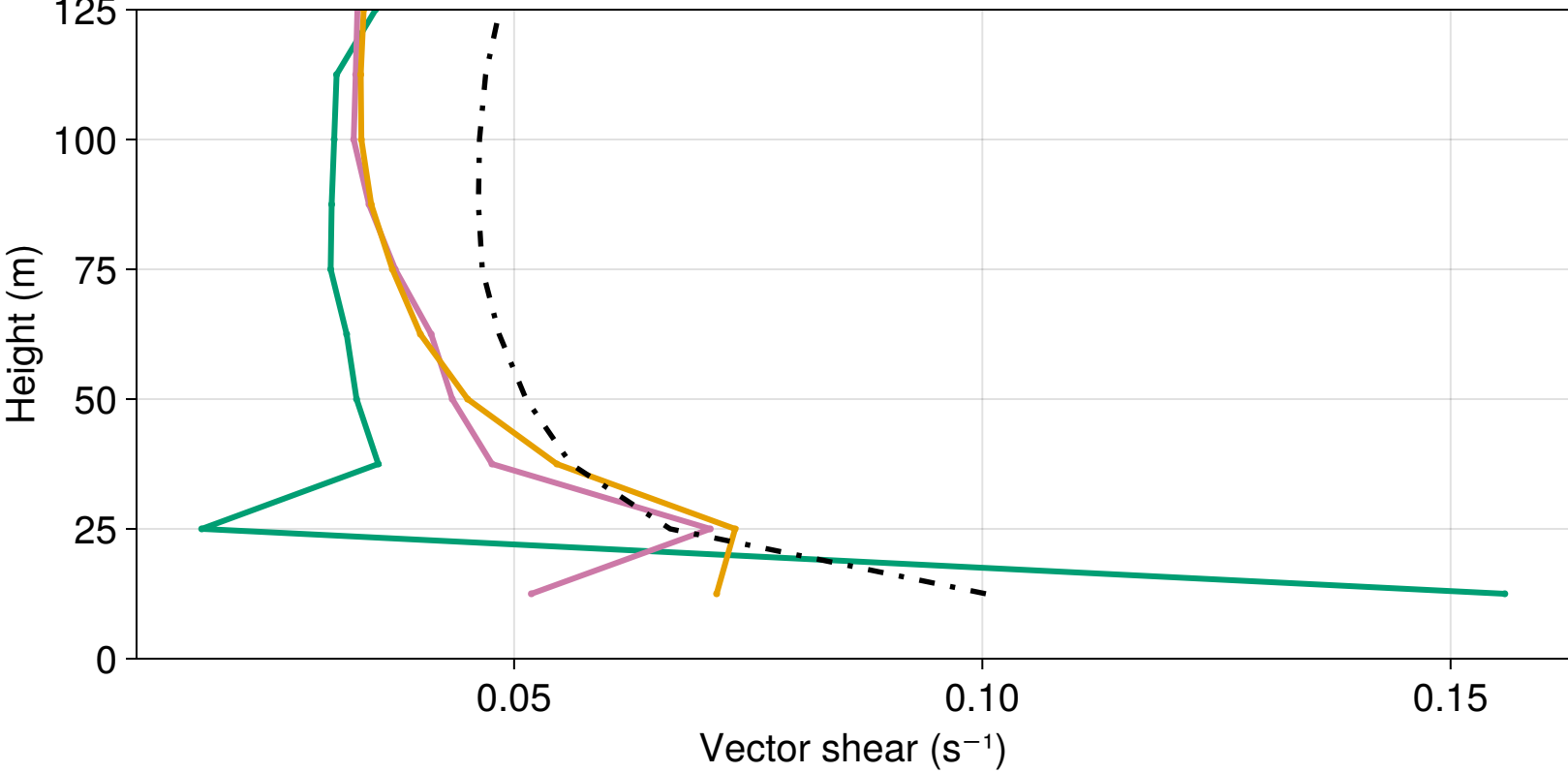
Transport is separated into resolved covariance, the measured WENO numerical correction, and the SLD constitutive flux. At 8-9 h, first-face resolved u - w becomes more negative (-0.00991 to -0.01461), while SLD u - w becomes less negative (-0.05866 to -0.05222) in square meters per second squared. The numerical correction is small compared with the change in the SLD transport.

Validation: H100 gate 7580 passed CUDA parity 34/34 and the 1,800 s GABLS1 check 48/48. Science 7581 ended with durable exit zero and CASE_DONE at 32,400 s. Independent Julia audits verified the immutable source manifest, paired initial theta digest, exact schedules, 46 native-height profiles and 135 series, finite data, wall flux density consistency, local inverse Obukhov length and ϕ identities, and scheme-native transport accounting. No upward-flux fallback occurred after one hour. The source freeze omits .git; its verified SHA-256 manifest and recorded commits provide provenance.

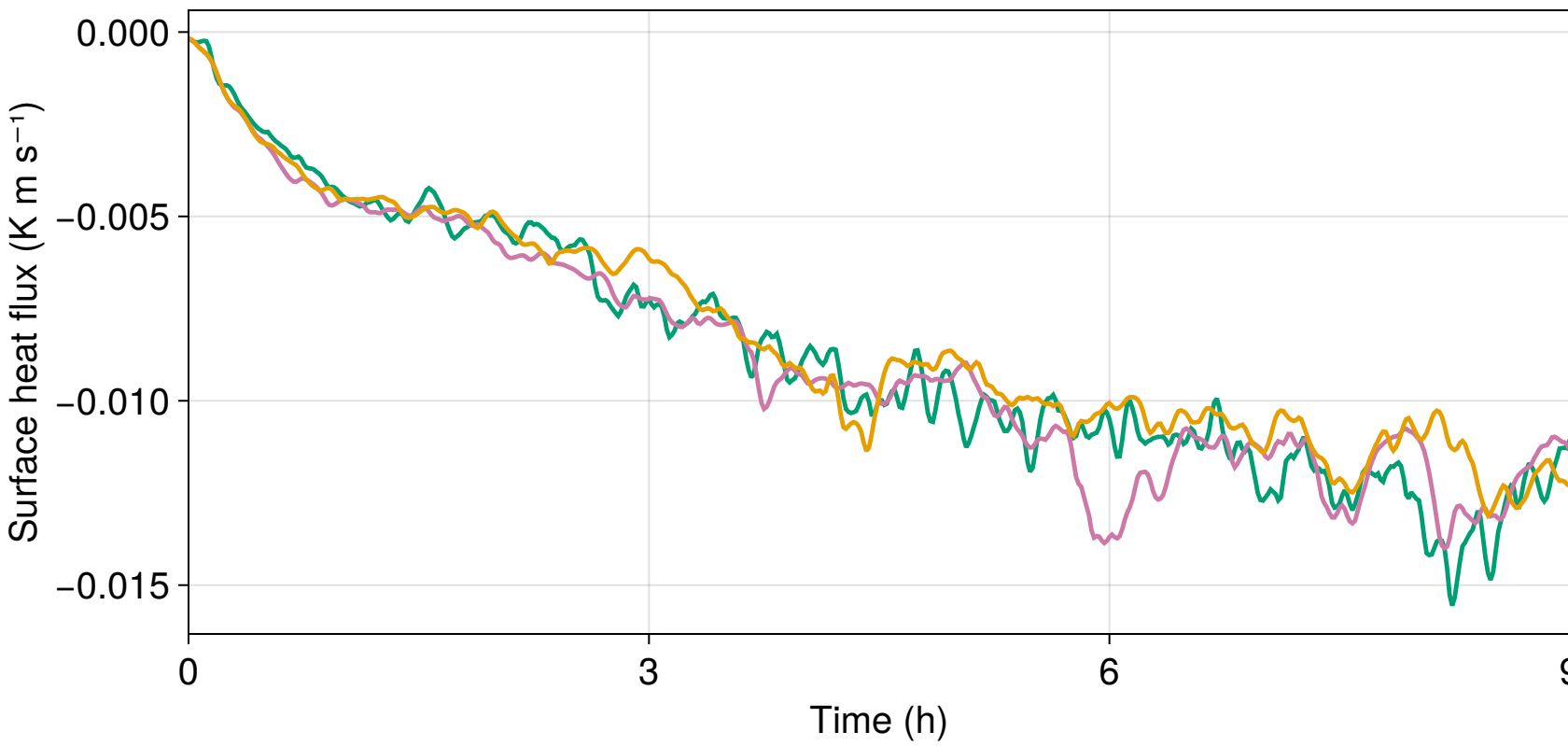
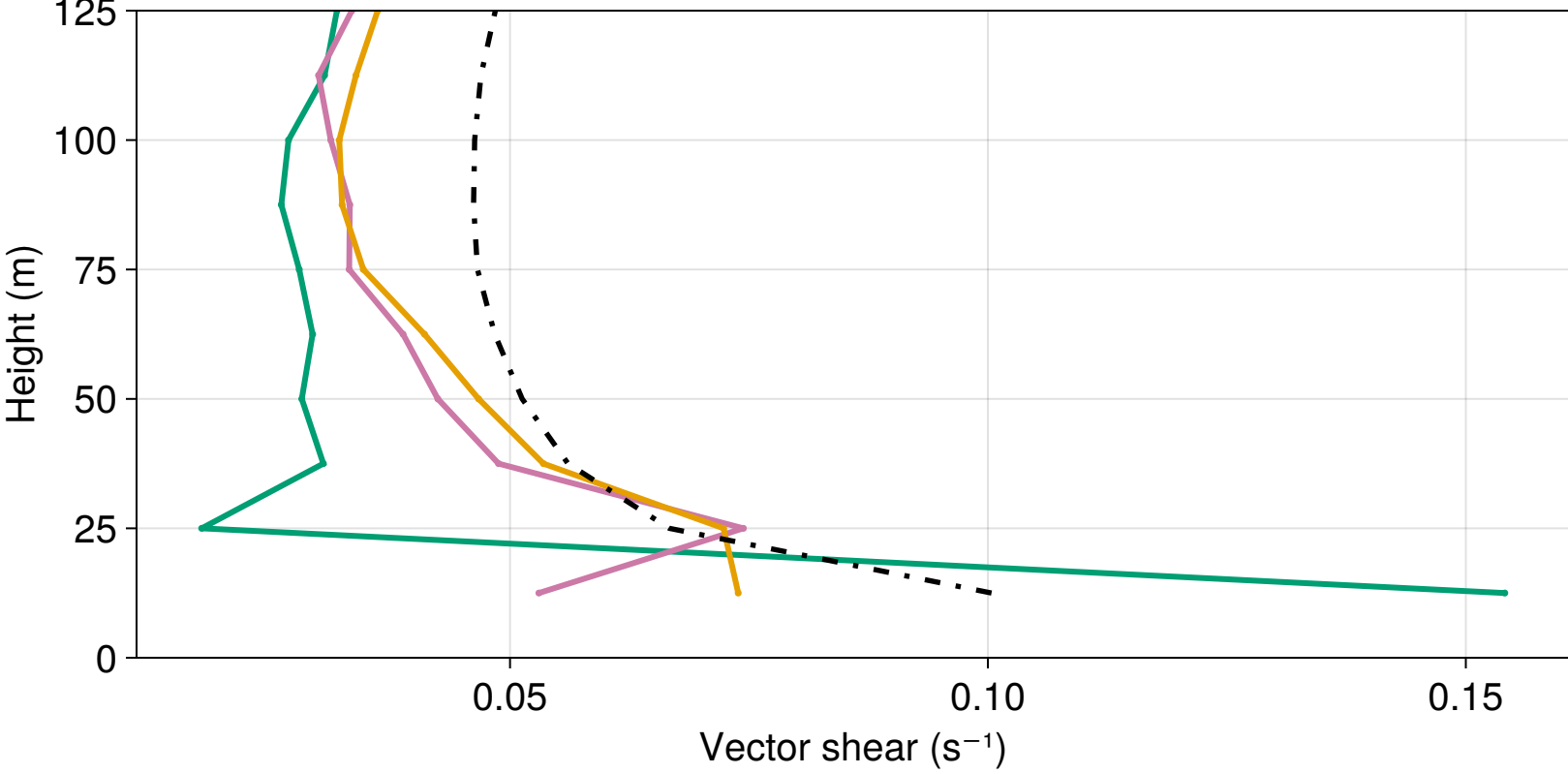
Mean-wind shear and wall flux

Shear between adjacent 12.5 m cell centers; fixed 1 m median sampled at the same heights

7–8 h

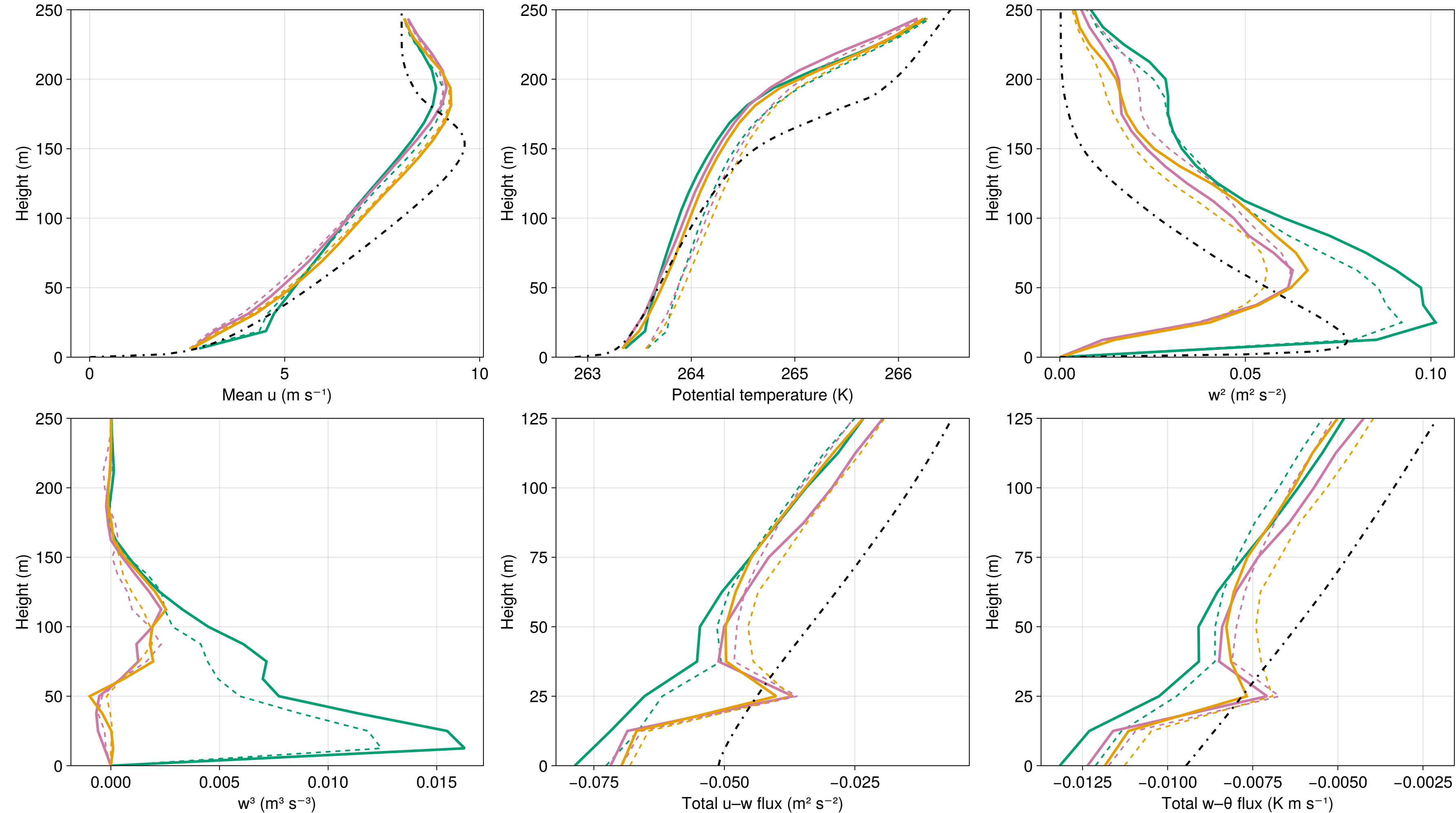


8–9 h



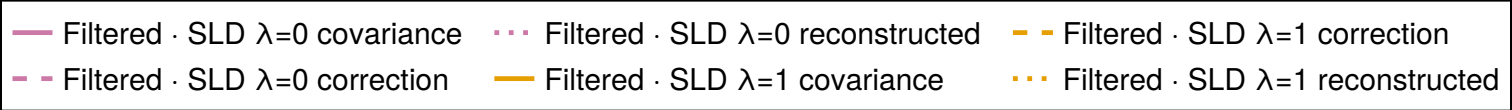
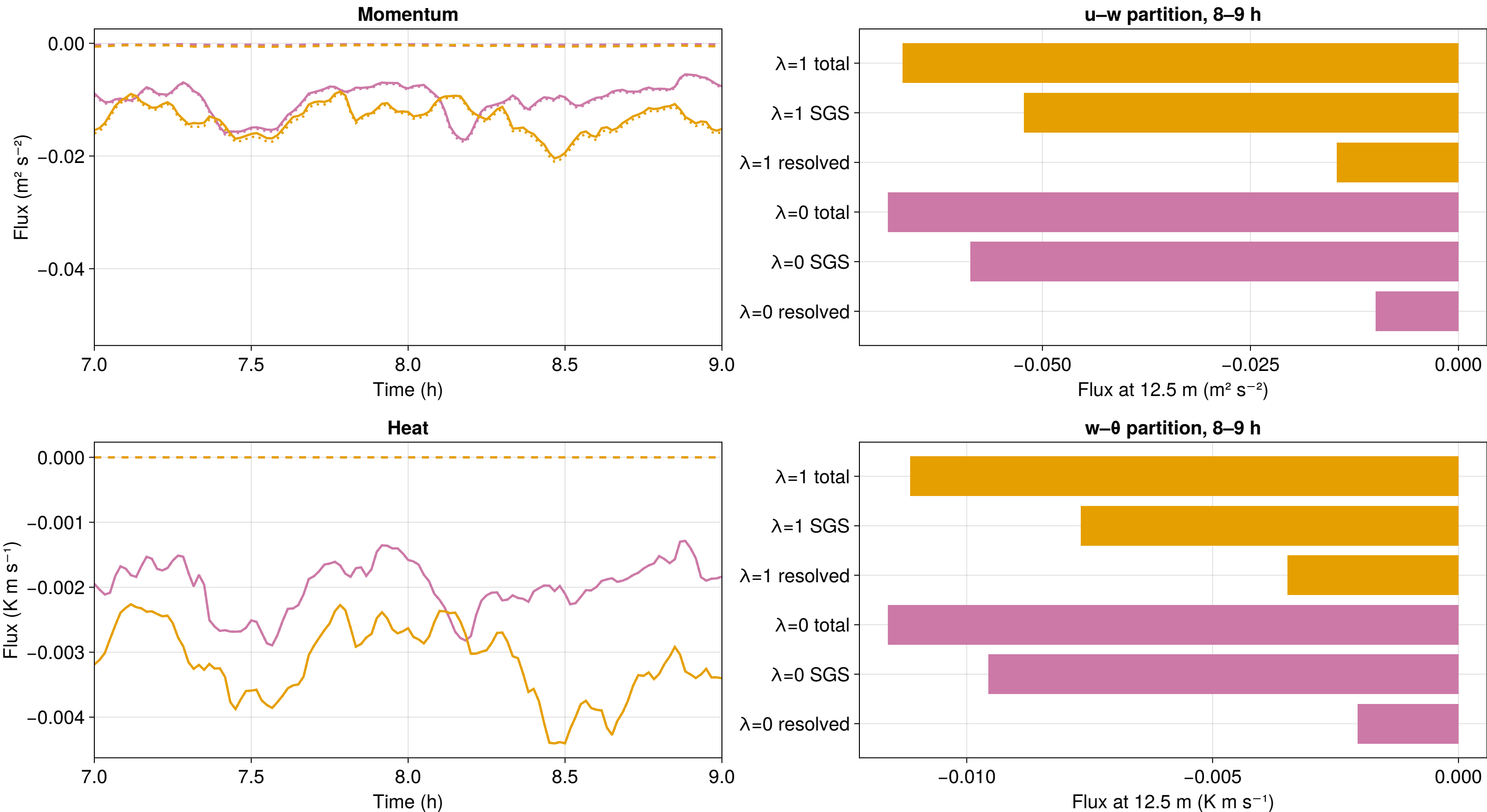
GABLS1: stable Monin–Obukhov correction of SLD

32³ · 12.5 m · WENO9 · filtered wall · scheme-native one-face SLD · $\lambda=0$ vs $\lambda=1$ · dashed 7–8 h, solid 8–9 h



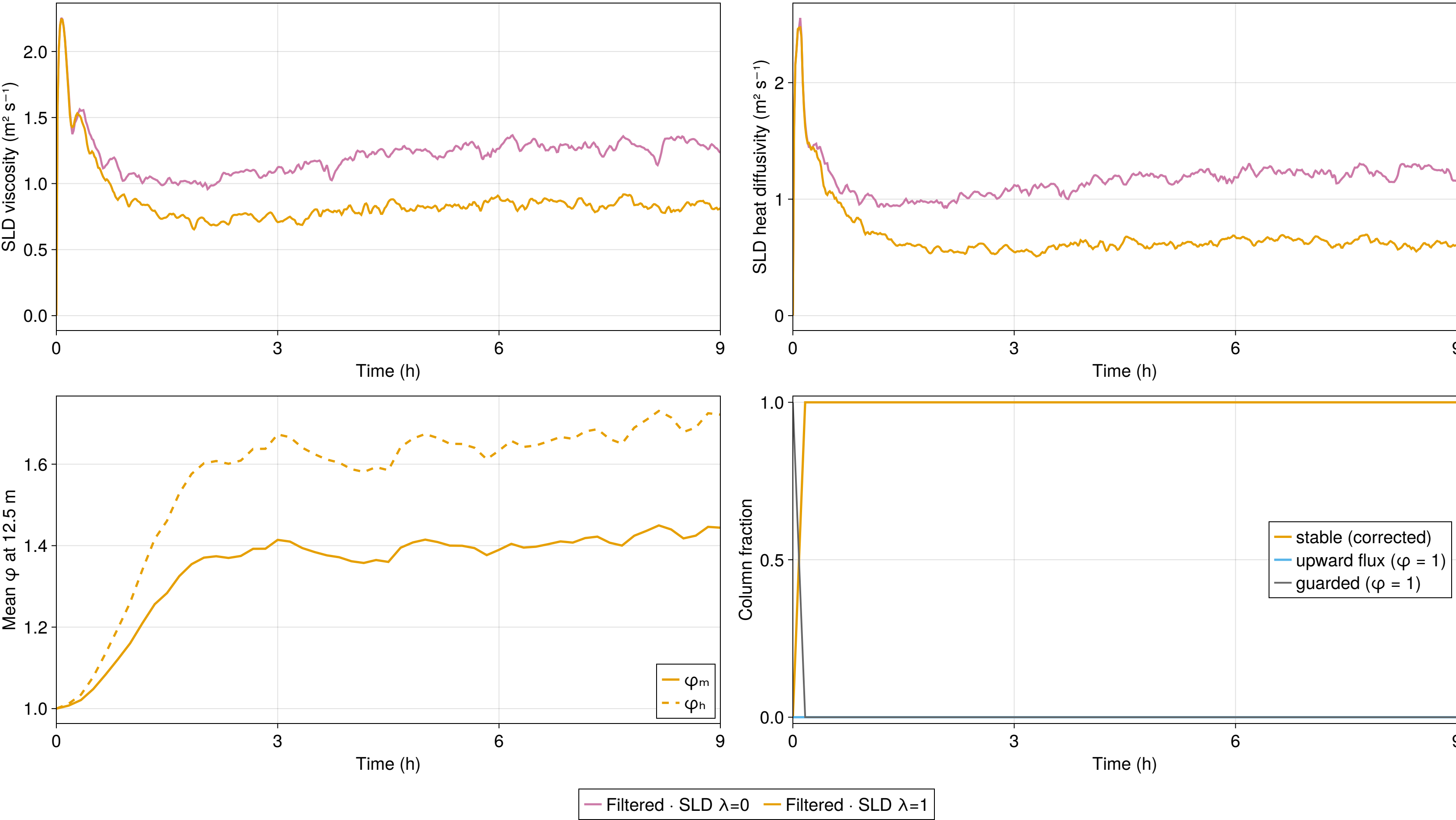
Scheme-native SLD: first-face transport partition

Reconstructed resolved = covariance + numerical correction; SGS is the SLD constitutive flux



First-face SLD coefficients and local stability

Horizontal means of local quantities at $z = 12.5$ m; φ is never evaluated from a mean L



GABLS1: SLD $\gamma=4$

GABLS1 stable surface-layer diffusivity, $\gamma=4$ | 23 September 2026. A matched nine-hour, 32-cubed, 12.5 m WENO9 case changes only the interior SLD stable-similarity strength from $\gamma=1$ to $\gamma=4$. Neutral $\gamma=0$ and filtered no-closure are admitted controls. The fixed 1 m LES median is an intercomparison reference, not observations.

During 8-9 h, first-layer vector shear is 0.12911 per second at $\gamma=4$, versus 0.05302 at $\gamma=0$, 0.07387 at $\gamma=1$ and 0.10038 for the reference sampled at the same heights. $\gamma=4$ overshoots the reference by 0.02873 per second; $\gamma=1$ undershoots by 0.02651. The two strengths bracket the median, with nearly equal absolute shear errors. The 7-8 h $\gamma=4$ shear is 0.11921 per second.

Peak resolved w -squared during 8-9 h is 0.08575 square meters per second squared, up from 0.06276 for $\gamma=0$ and 0.06676 for $\gamma=1$. At 18.75 m, w -cubed is +0.005932 cubic meters per second cubed, versus -0.000295 and +0.000102. These moments are closer to the reference profile, while the overshoot mean shear limits the overall improvement. $\gamma=4$ w -squared and w -cubed rose between 7-8 and 8-9 h, so statistical equilibration remains uncertain.

At the first interior face, mean $\gamma=4$ local similarity functions over 8-9 h are $\phi_m=2.7169$ and $\phi_h=3.7900$. Every saved column is on the stable branch, with no upward-flux fallback. The evolved viscosity is 0.24517 square meters per second, versus 1.2892 at $\gamma=0$ and 0.82387 at $\gamma=1$. Heat diffusivity is 0.088672 versus 1.2396 and 0.60453. The wall law and filters are unchanged.

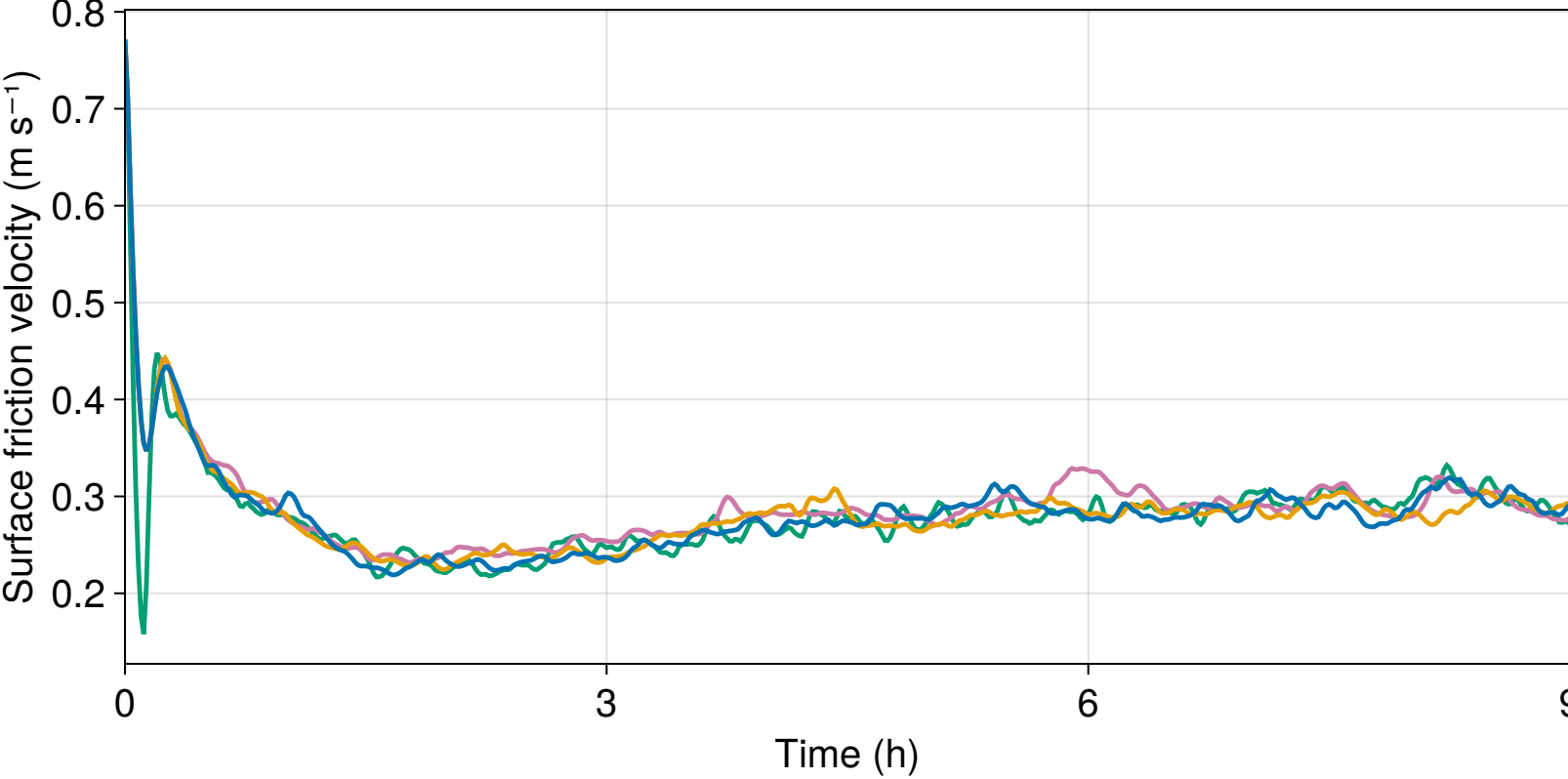
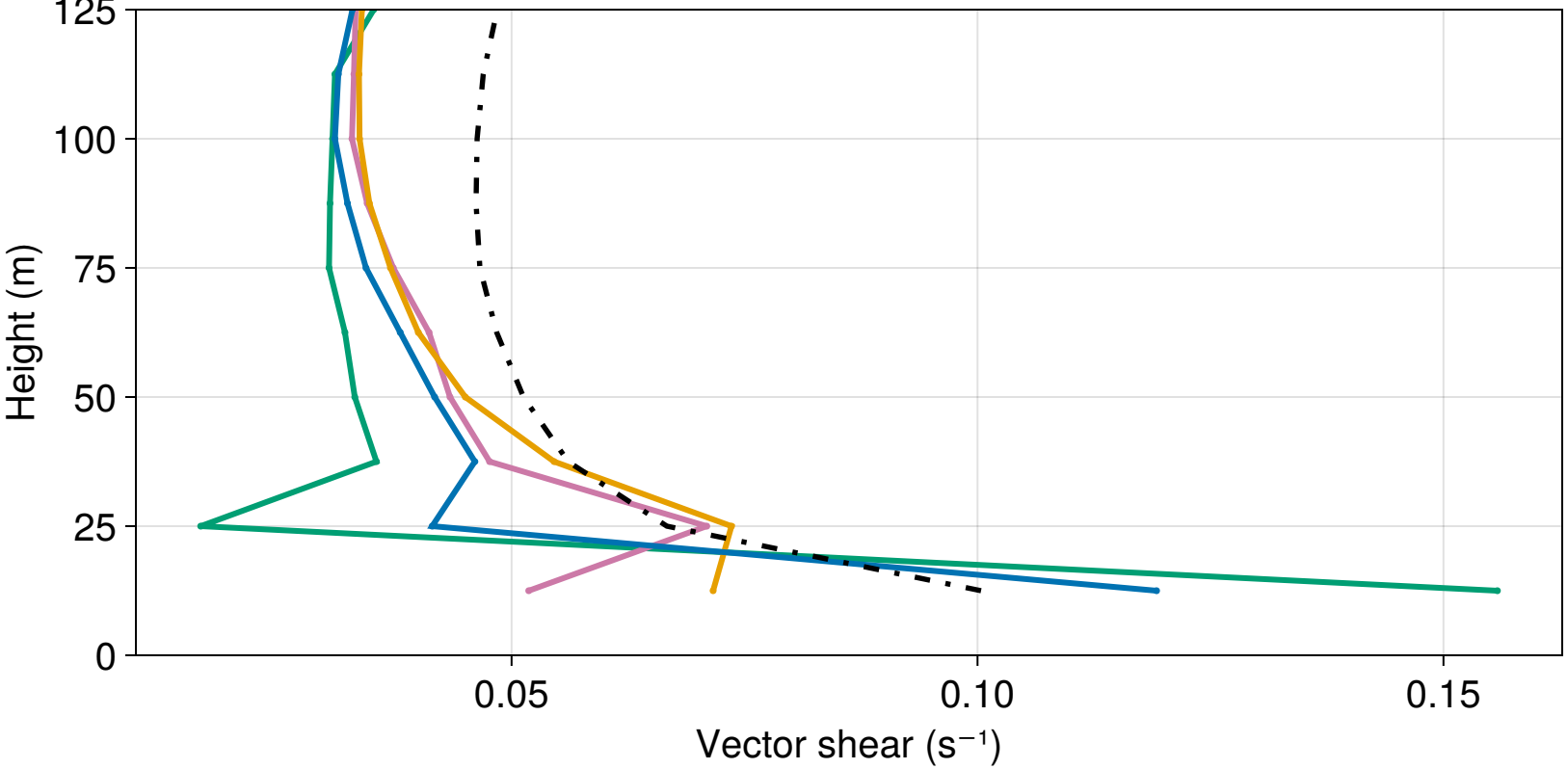
Reduced closure changes the transport partition: first-face resolved $u-w$ is -0.045186 and SGS $u-w$ is -0.027868 square meters per second squared at $\gamma=4$, compared with -0.014608 and -0.052218 at $\gamma=1$. Resolved w -theta becomes -0.010566 and SGS w -theta -0.001608 kelvin meters per second. The measured WENO u -flux correction is -0.001743, tracked separately from covariance.

Validation: CPU smoke 7583 passed 48/48; H100 gate 7584 passed CUDA parity 34/34 and the 1800 s case 48/48. Science 7585 ended with durable exit zero and CASE_DONE at 32400 s. Independent Julia audits verified the immutable source manifest, paired initial-theta digest, 46 native-height profiles, 135 series, finite values, exact schedules, filter evolution, density-consistent wall flux, local inverse Obukhov length and similarity identities at 55 times, and reconstructed transport accounting. The result is one seed at one coarse resolution, not a universal parameter calibration.

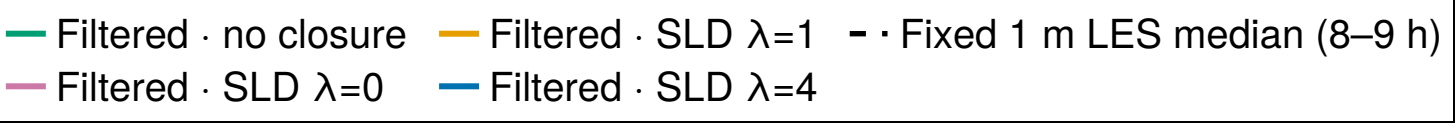
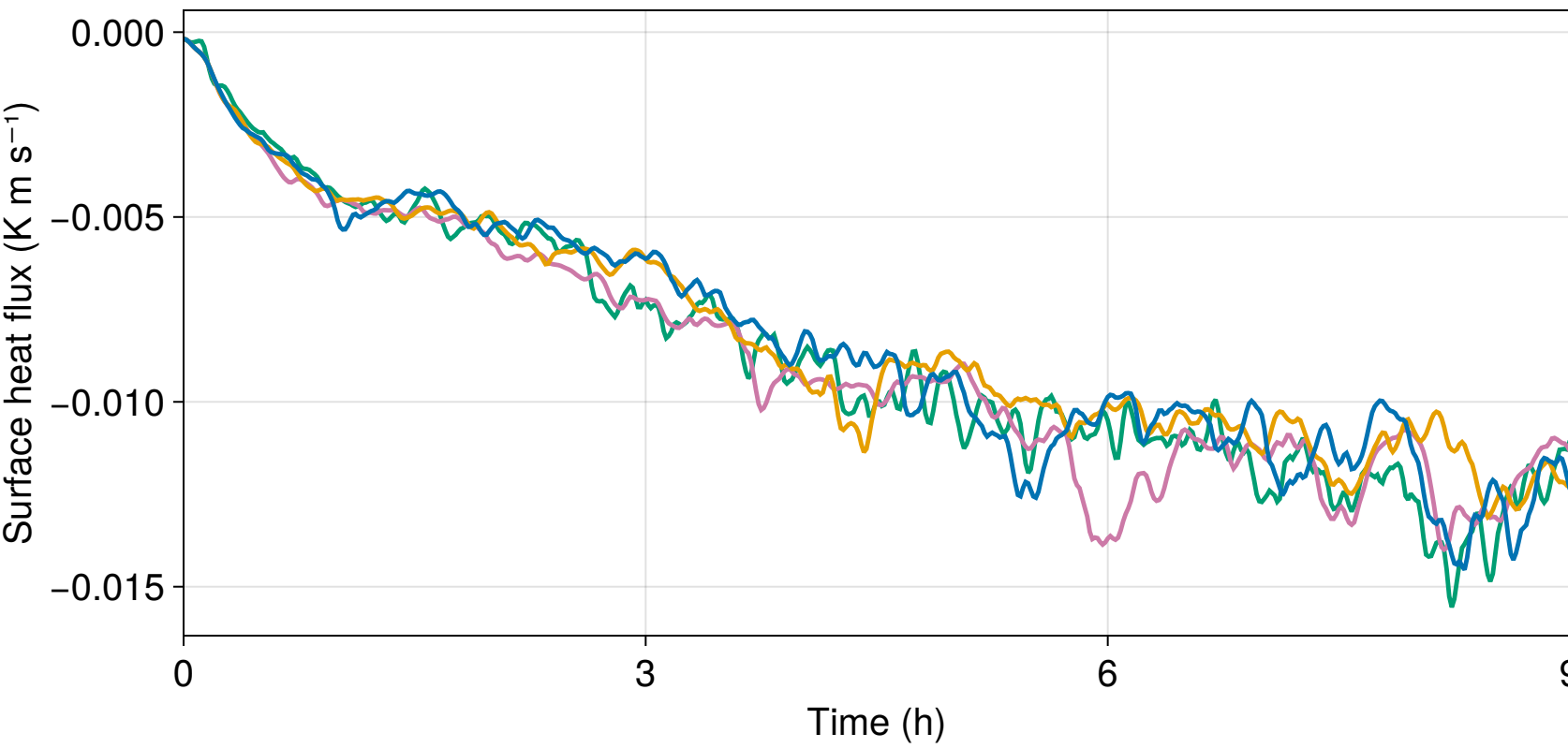
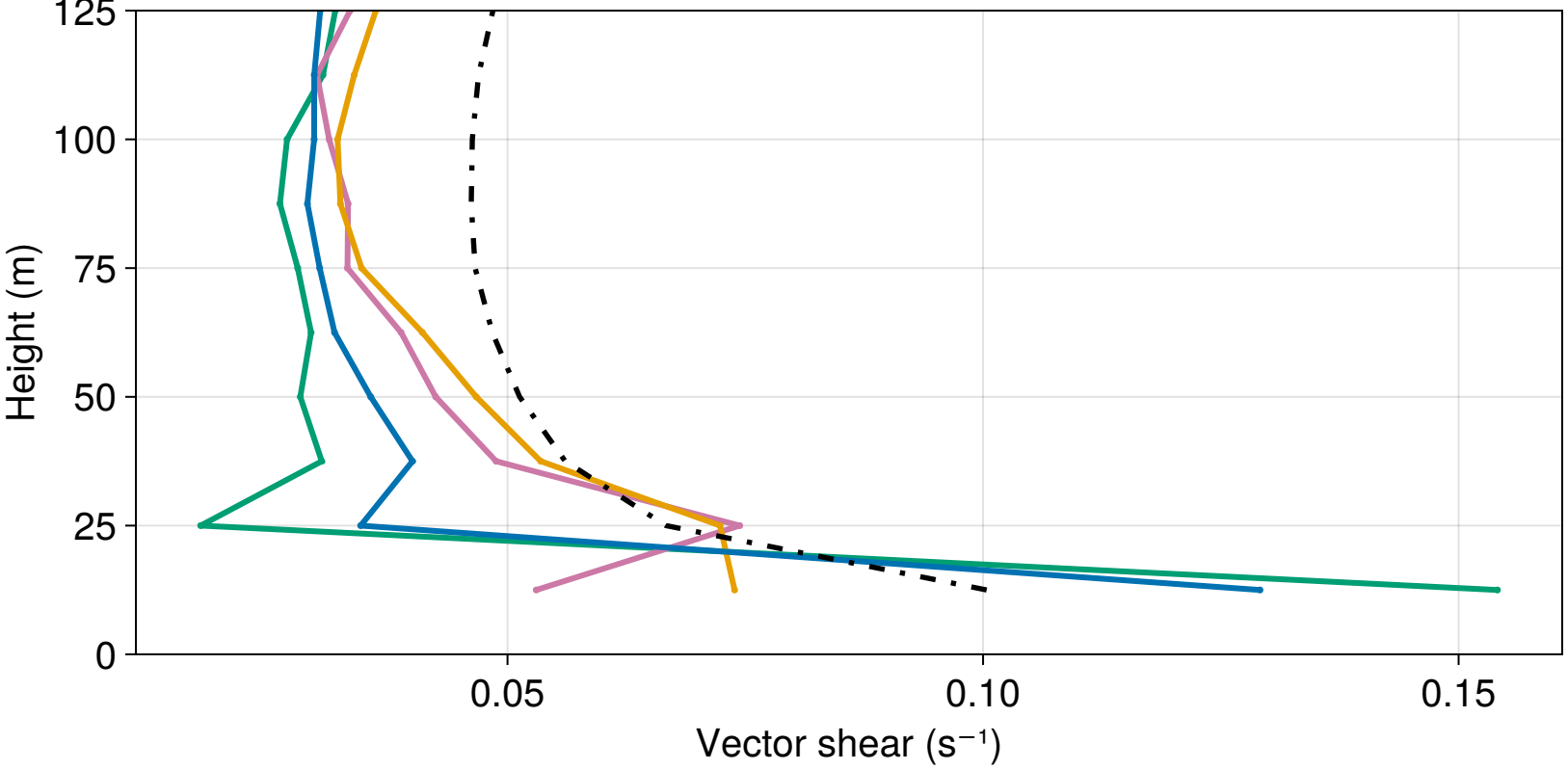
Mean-wind shear and wall flux

Shear between adjacent 12.5 m cell centers; fixed 1 m median sampled at the same heights

7–8 h

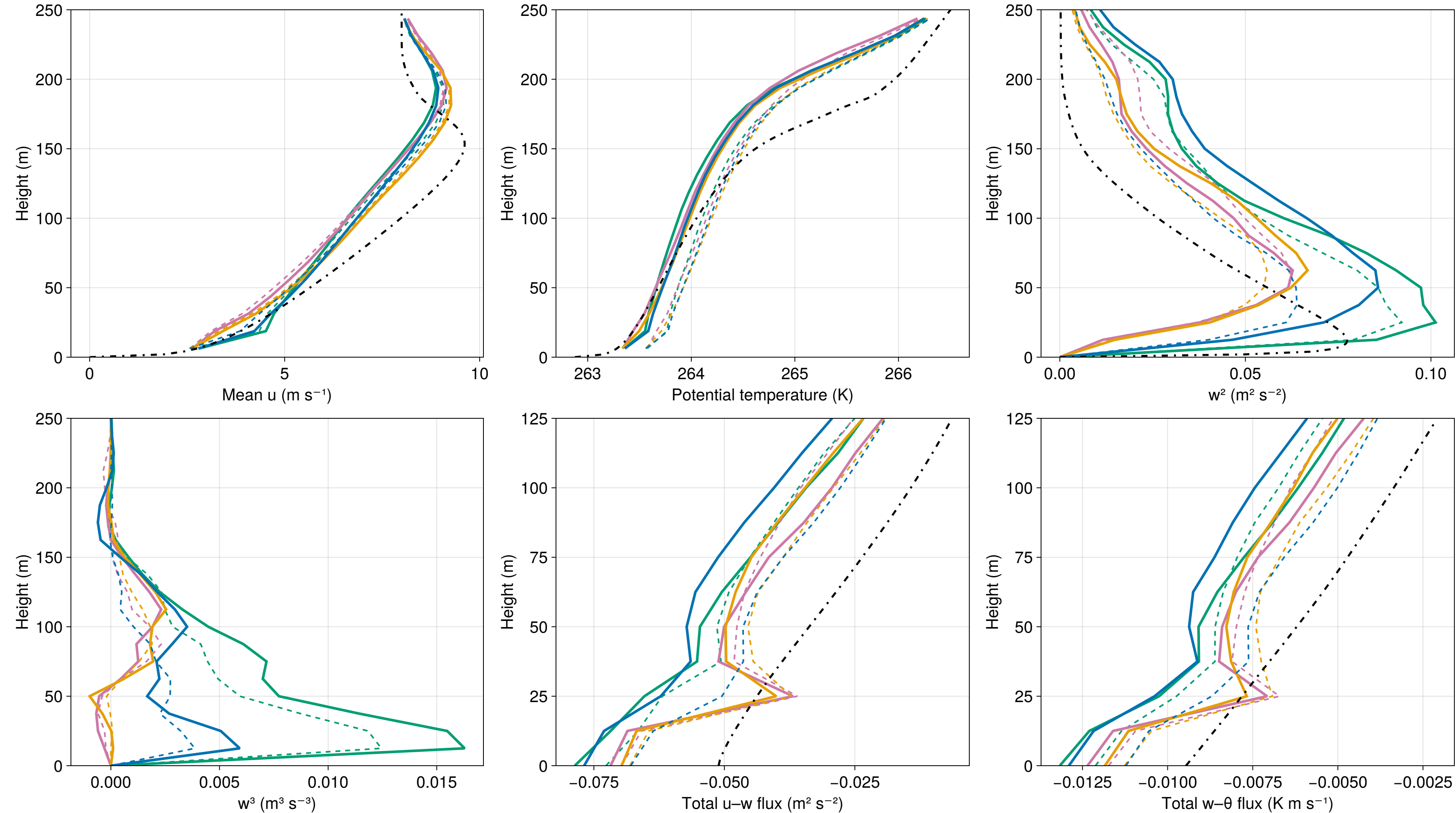


8–9 h



GABLS1: stable Monin–Obukhov correction of SLD

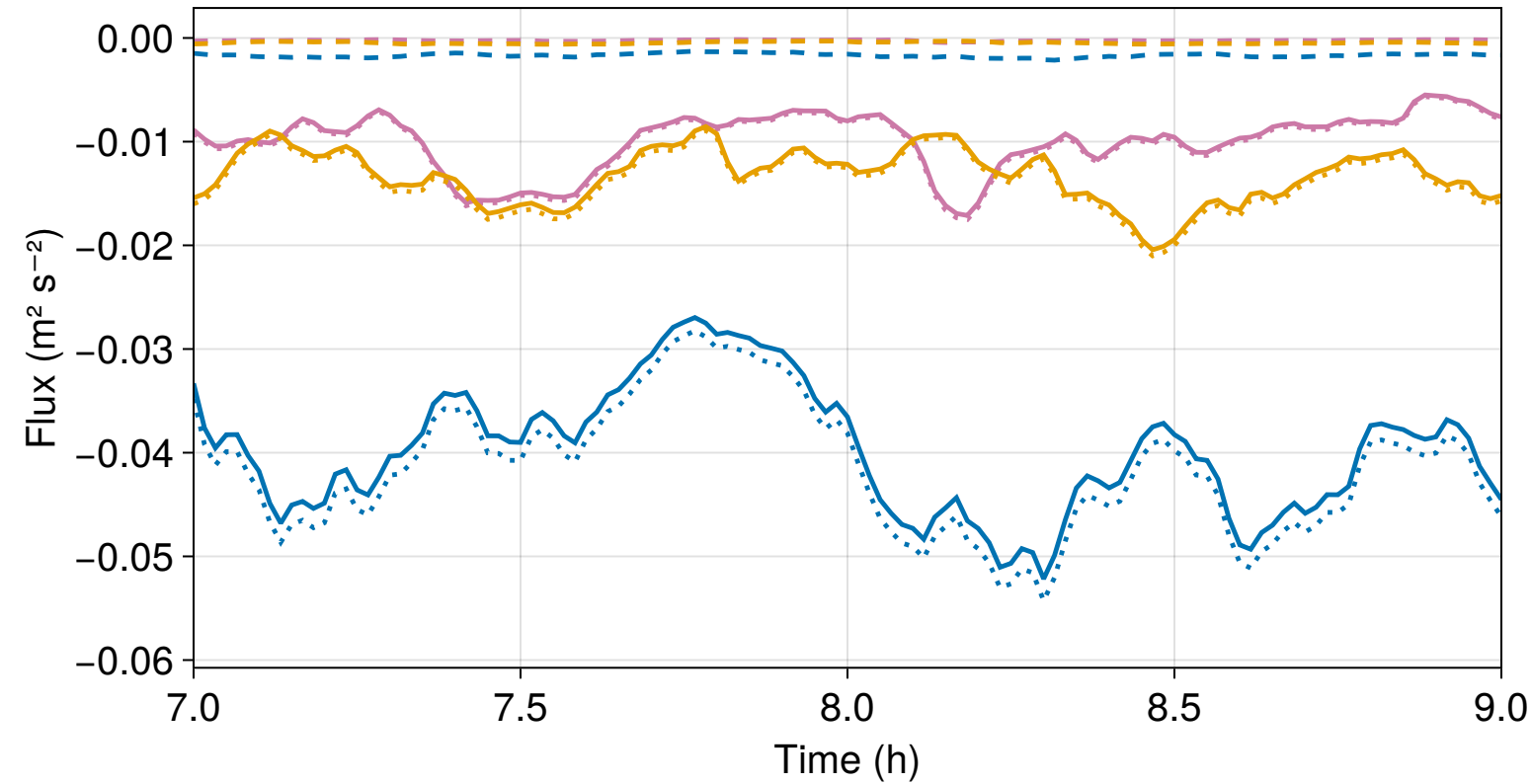
32³ · 12.5 m · WENO9 · filtered wall · scheme-native one-face SLD · $\lambda=0, 1, 4$ · dashed 7–8 h, solid 8–9 h



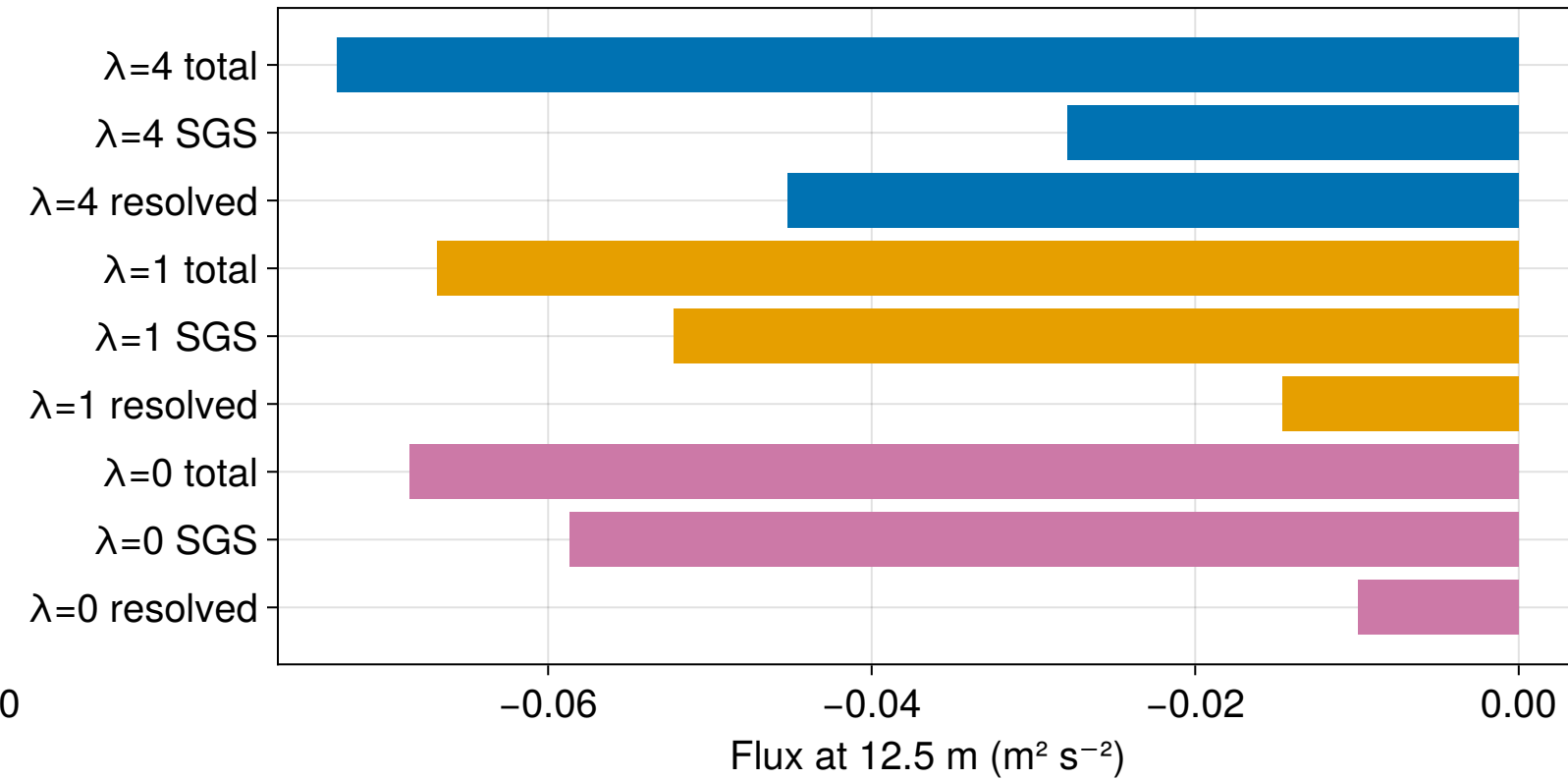
Scheme-native SLD: first-face transport partition

Reconstructed resolved = covariance + numerical correction; SGS is the SLD constitutive flux

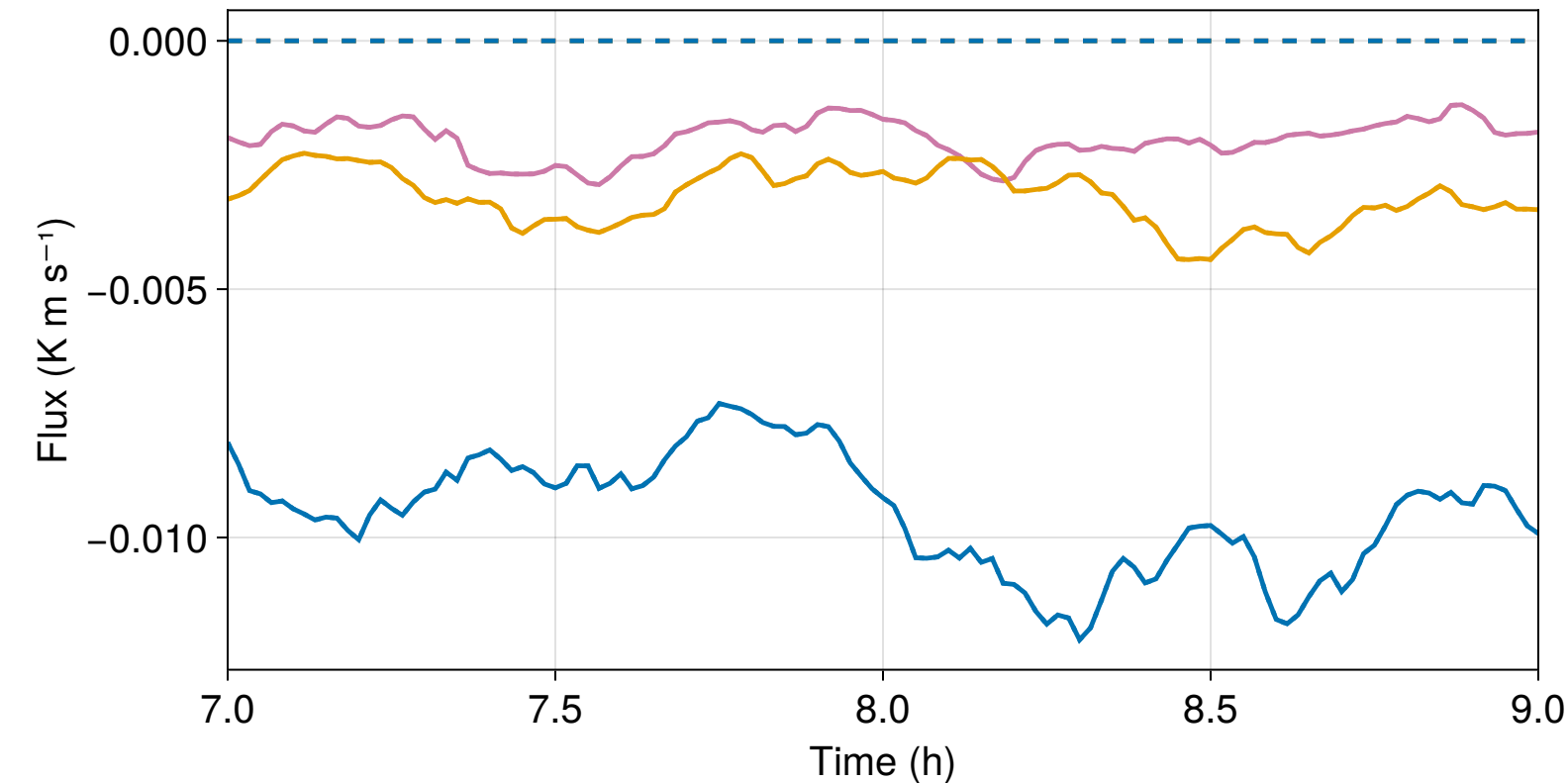
Momentum



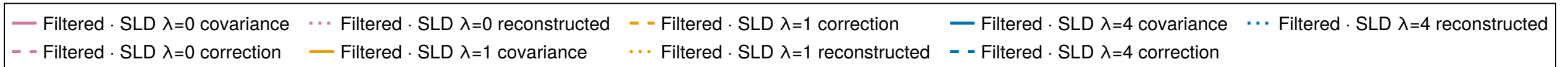
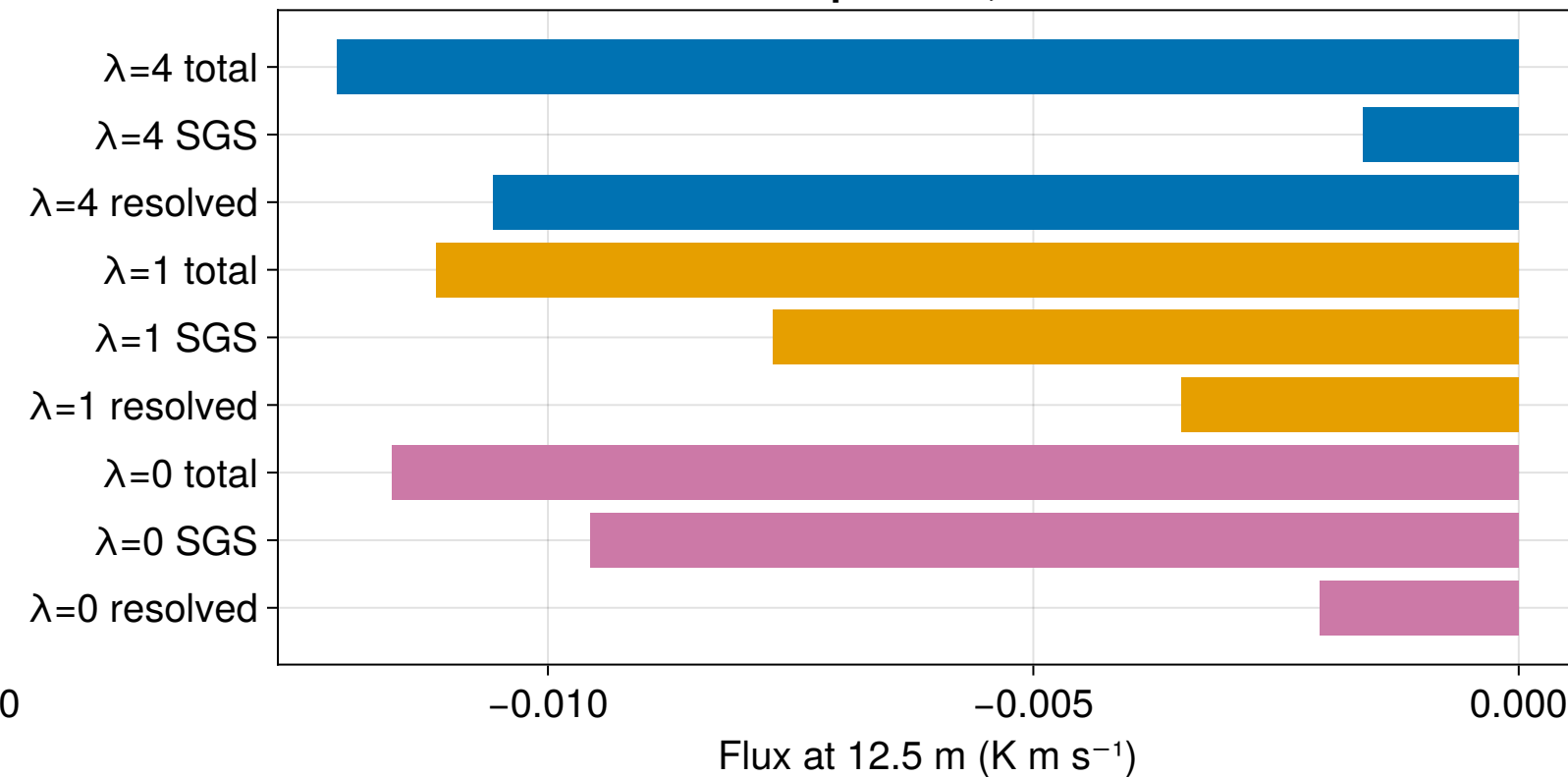
u-w partition, 8–9 h



Heat

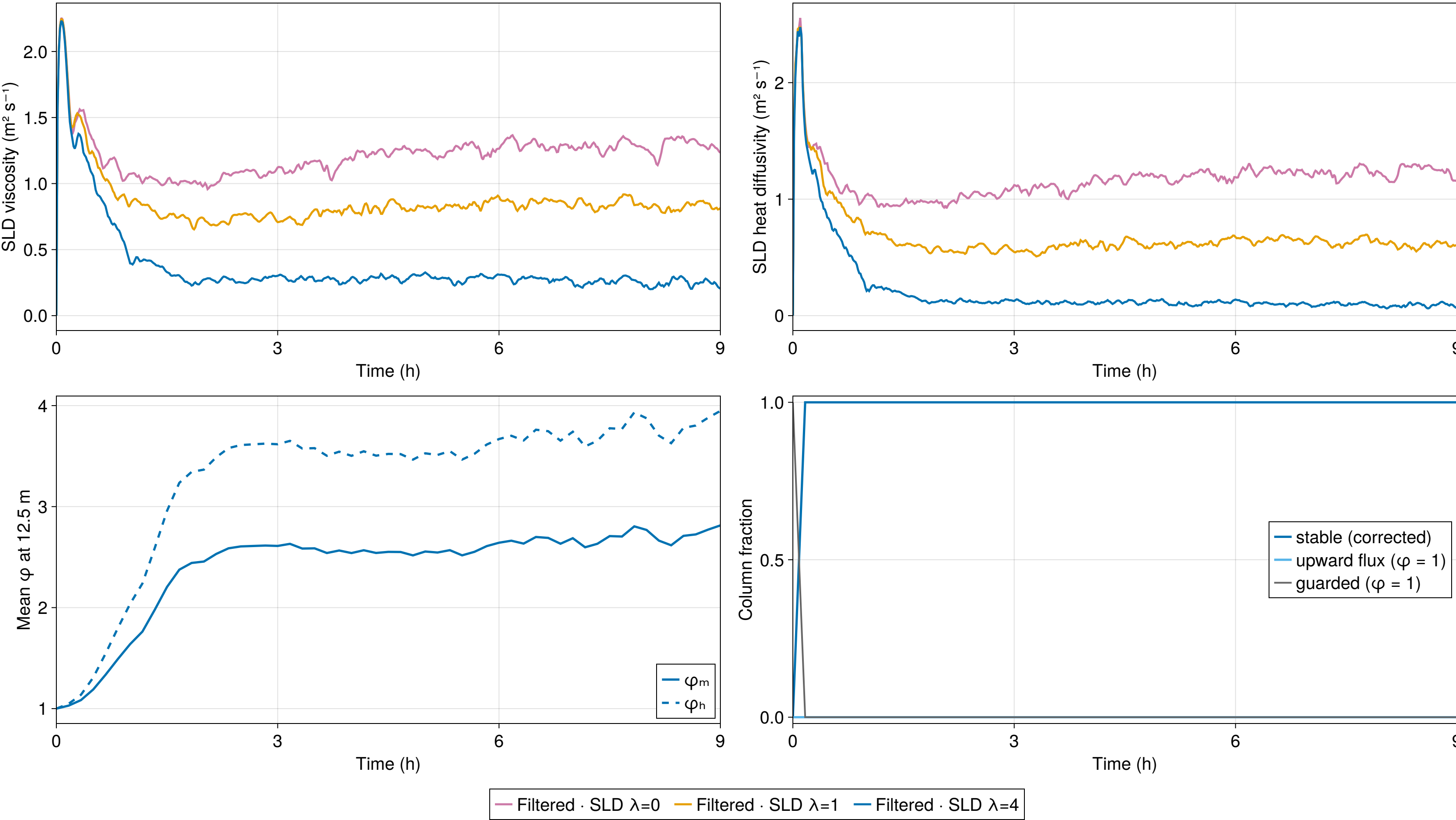


w-θ partition, 8–9 h



First-face SLD coefficients and local stability

Horizontal means of local quantities at $z = 12.5$ m; φ is never evaluated from a mean L



GABLS1: SLD $\gamma=2$

GABLS1 stable surface-layer diffusivity, $\gamma=2$ | 23 September 2026. This matched nine-hour 32-cubed, 12.5 m WENO9 case changes only interior SLD stability strength. The same 400 m domain, seed 123, filtered wall, one-face support, 300 s filters, and scheme-native factor-one flux apply to $\gamma=0, 1, 2$, and 4. The fixed 1 m LES median is an intercomparison reference, not observations.

The 8-9 h vector shear at 12.5, 25, and 37.5 m is 0.09152, 0.06140, and 0.05235 per second at $\gamma=2$, versus 0.10038, 0.06666, and 0.05607 in the median profile. These values decrease with height and remove the pronounced 25 m dip of $\gamma=4$, whose first three shears are 0.12911, 0.03456, and 0.04001. $\gamma=1$ gives 0.07387, 0.07235, and 0.05350. $\gamma=2$ first-layer shear changed only from 0.08910 in 7-8 h to 0.09152 in 8-9 h.

Shear improvement does not imply improvement in all statistics. Peak resolved w -squared is 0.05588 square meters per second squared at $\gamma=2$ in 8-9 h, below $\gamma=1$ at 0.06676, $\gamma=4$ at 0.08575, and filtered no closure at 0.10125. At 18.75 m, w -cubed is +0.000937 cubic meters per second cubed, versus +0.005932 for $\gamma=4$ and +0.01629 with no closure. The $\gamma=2$ surface friction velocity is 0.27977 meters per second and heat flux is -0.010968 kelvin meters per second.

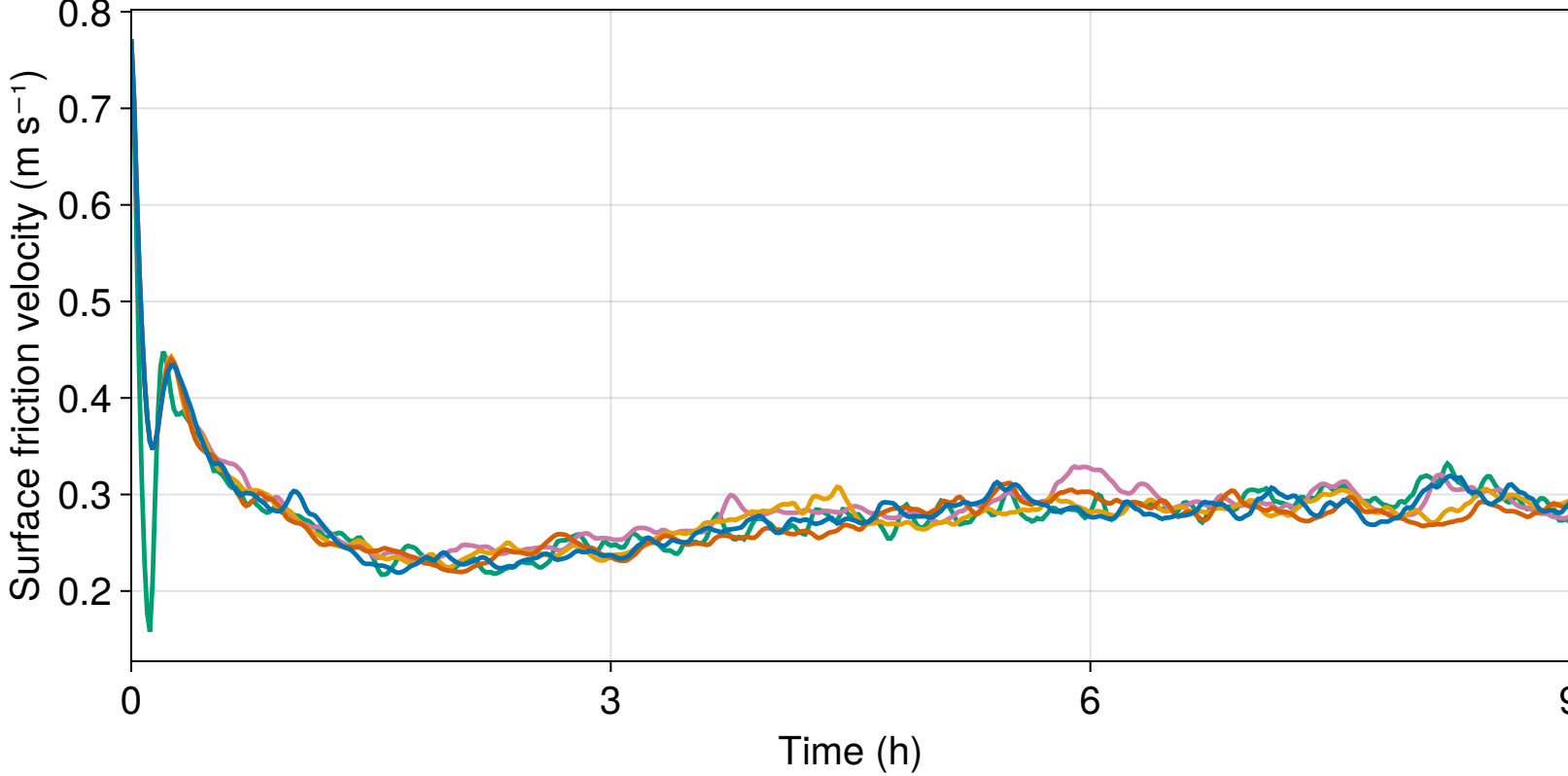
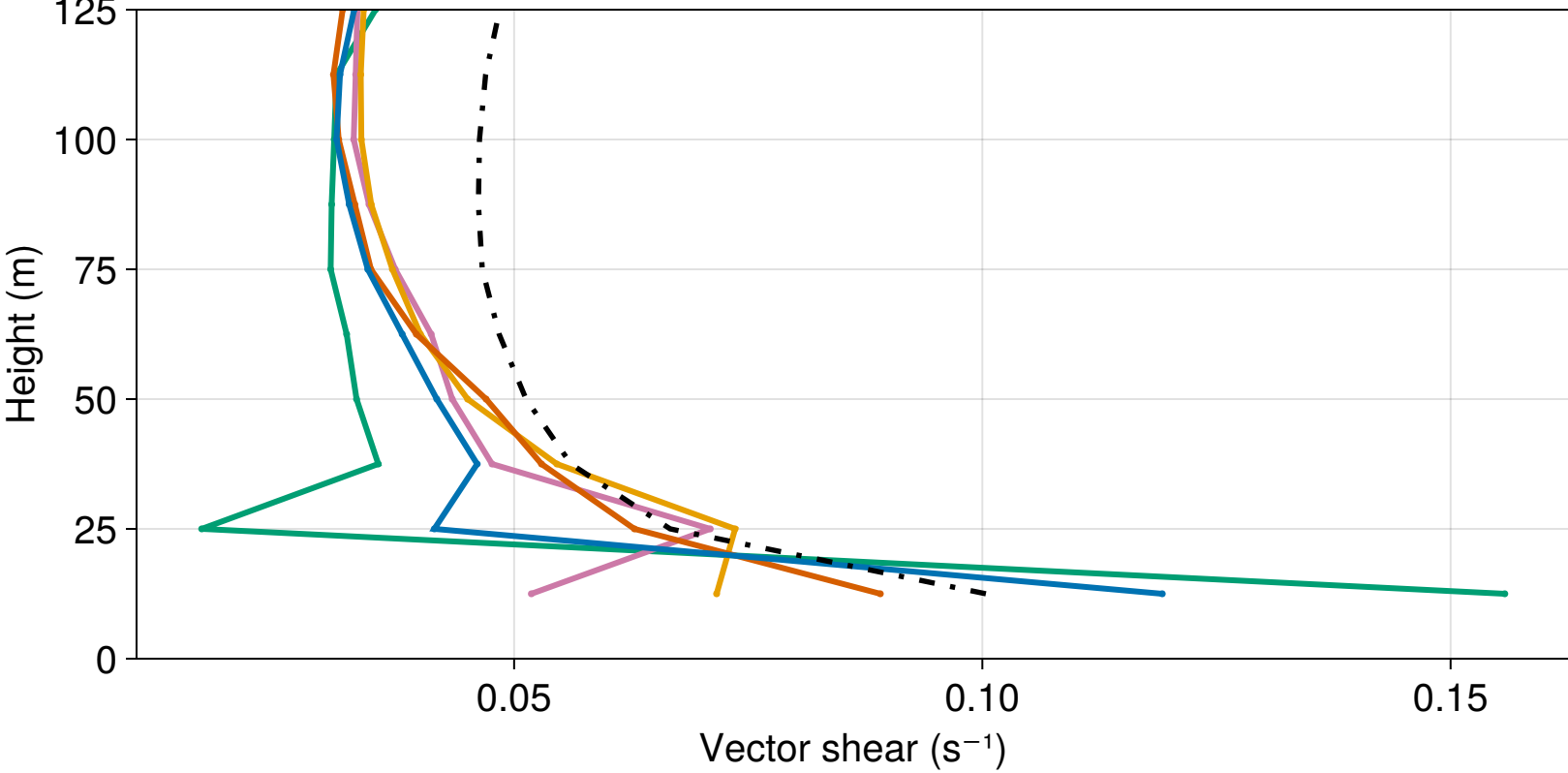
At the first interior face, $\gamma=2$ mean local $\phi_m=1.8930$ and $\phi_h=2.4512$ during 8-9 h; all saved columns stay stable and no upward-flux fallback occurs. Evolved momentum viscosity is 0.52303 square meters per second, between $\gamma=1$ at 0.82387 and $\gamma=4$ at 0.24517. Resolved $u-w$ is -0.021102 and SLD $u-w$ is -0.040393 square meters per second squared. The measured WENO numerical correction is -0.00069141 and is kept separate from covariance.

Validation: CPU smoke 7595 passed 48/48; H100 gate 7596 passed CUDA parity 34/34 and its 1800 s case 48/48. Science 7615_1 ended with durable exit zero and CASE_DONE at 32400 s. Independent Julia audits verified frozen source, paired initial-theta digest, exact schedules, 46 native-height profiles, 135 series, finite fields, filter evolution, density-consistent surface flux and scheme-native transport. Local inverse Obukhov length and similarity identities passed at 55 saved times. One seed and one coarse grid do not establish universal calibration; a paired 64-cubed study follows.

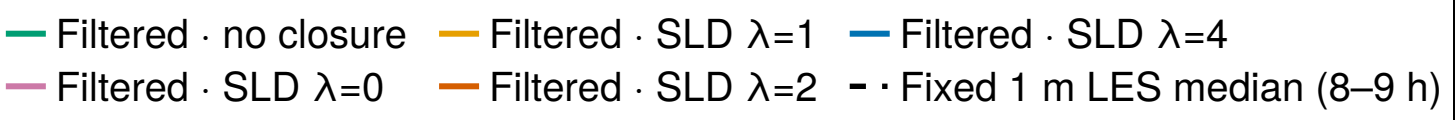
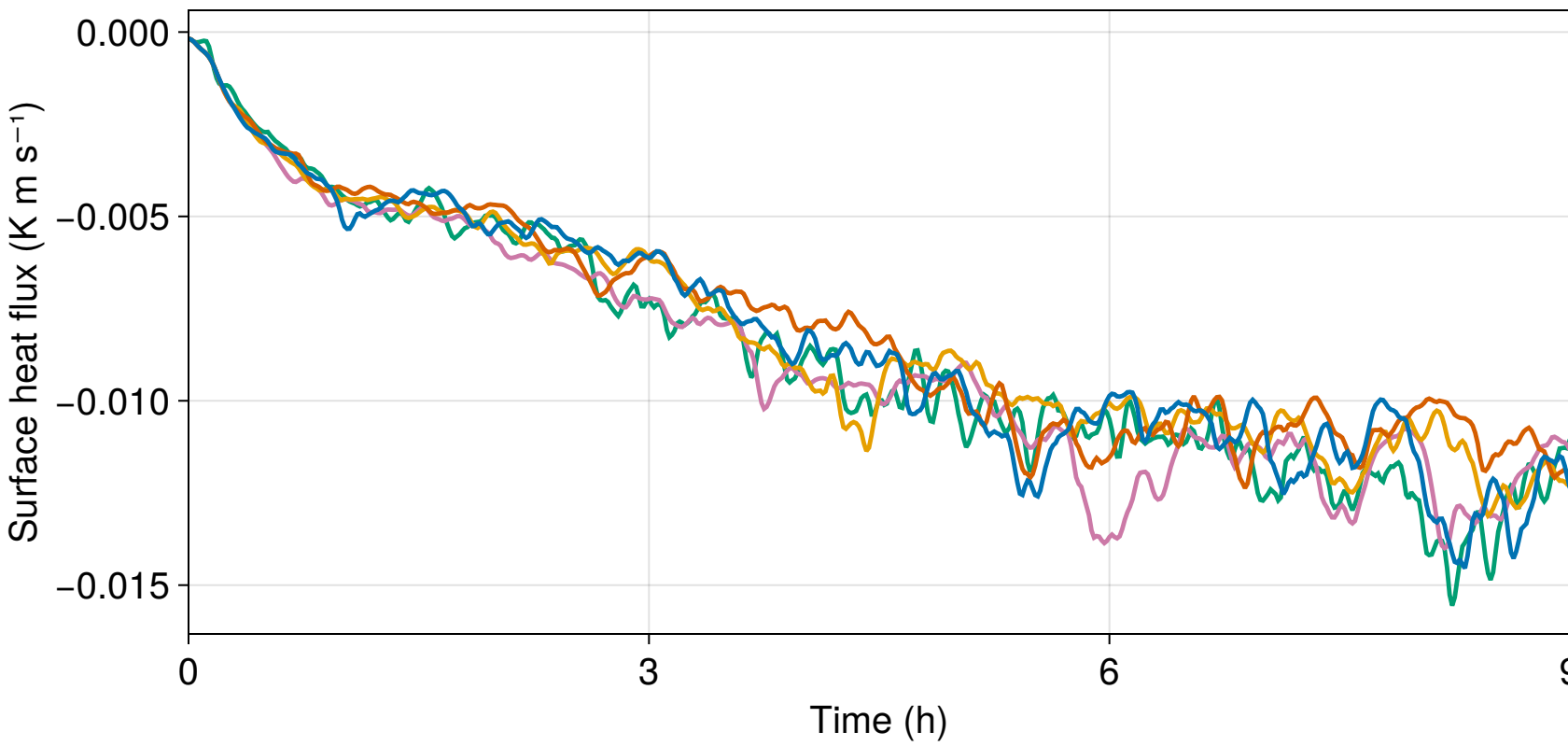
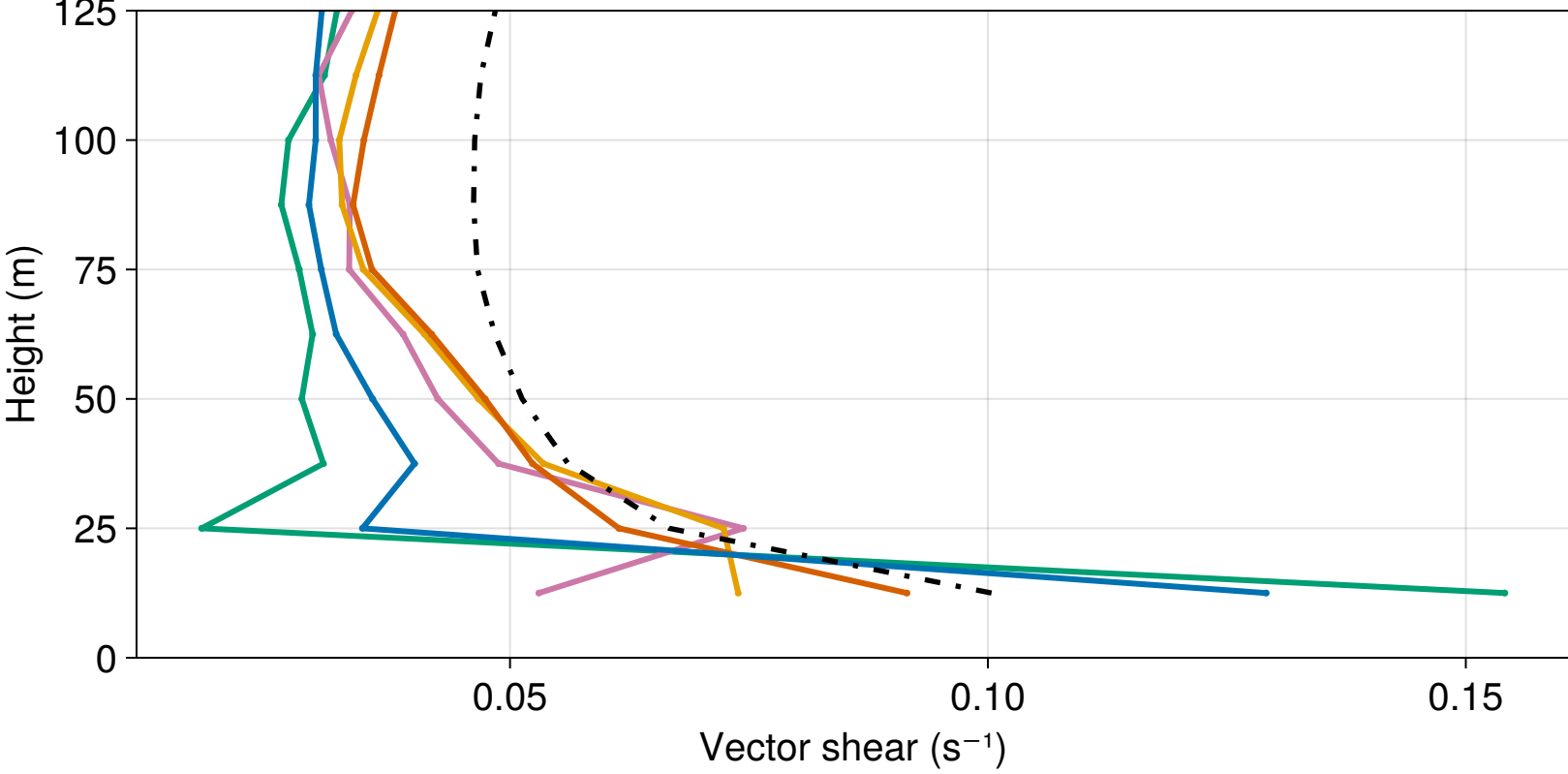
Mean-wind shear and wall flux

Shear between adjacent 12.5 m cell centers; fixed 1 m median sampled at the same heights

7–8 h

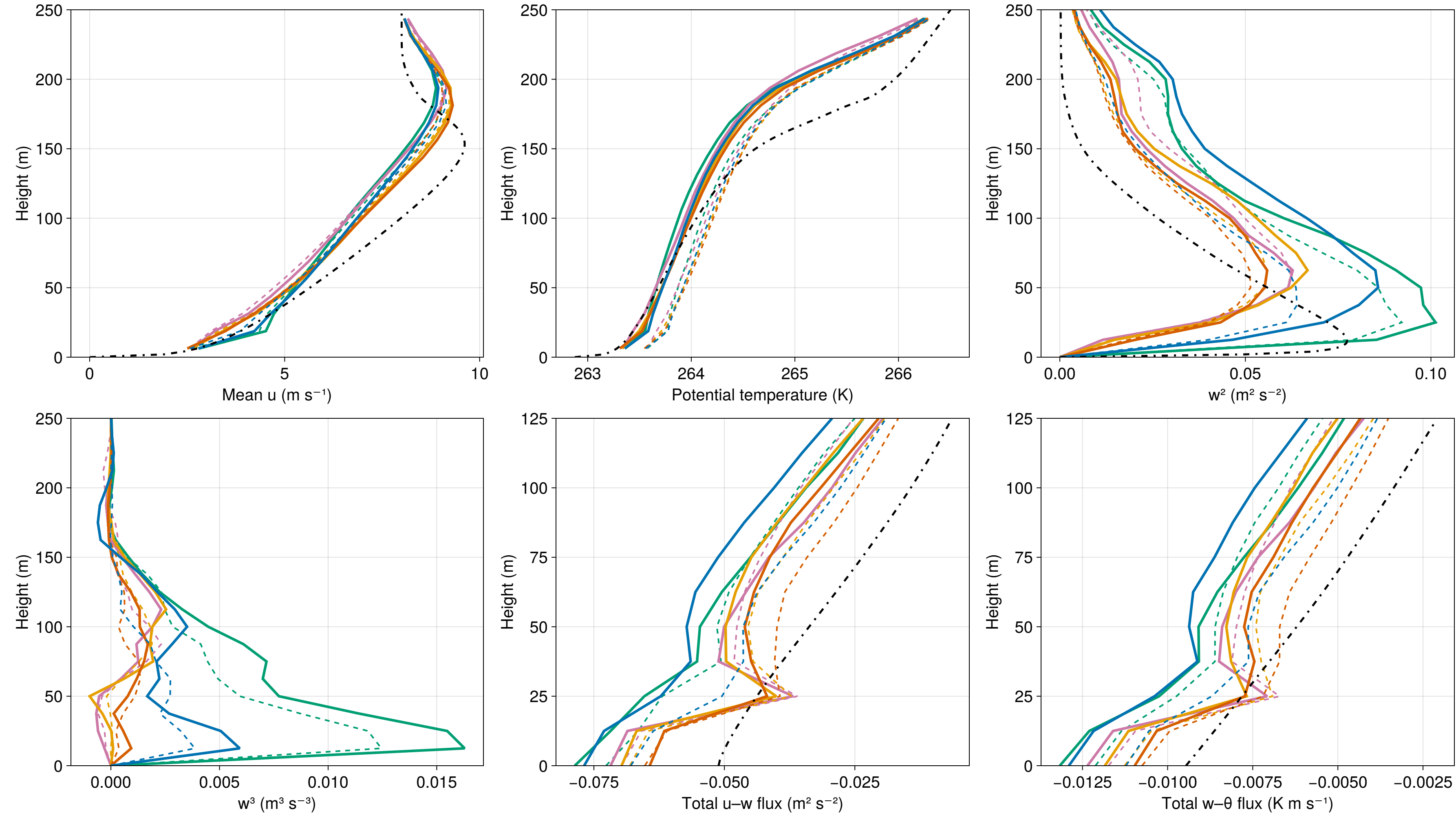


8–9 h



GABLS1: stable Monin–Obukhov correction of SLD

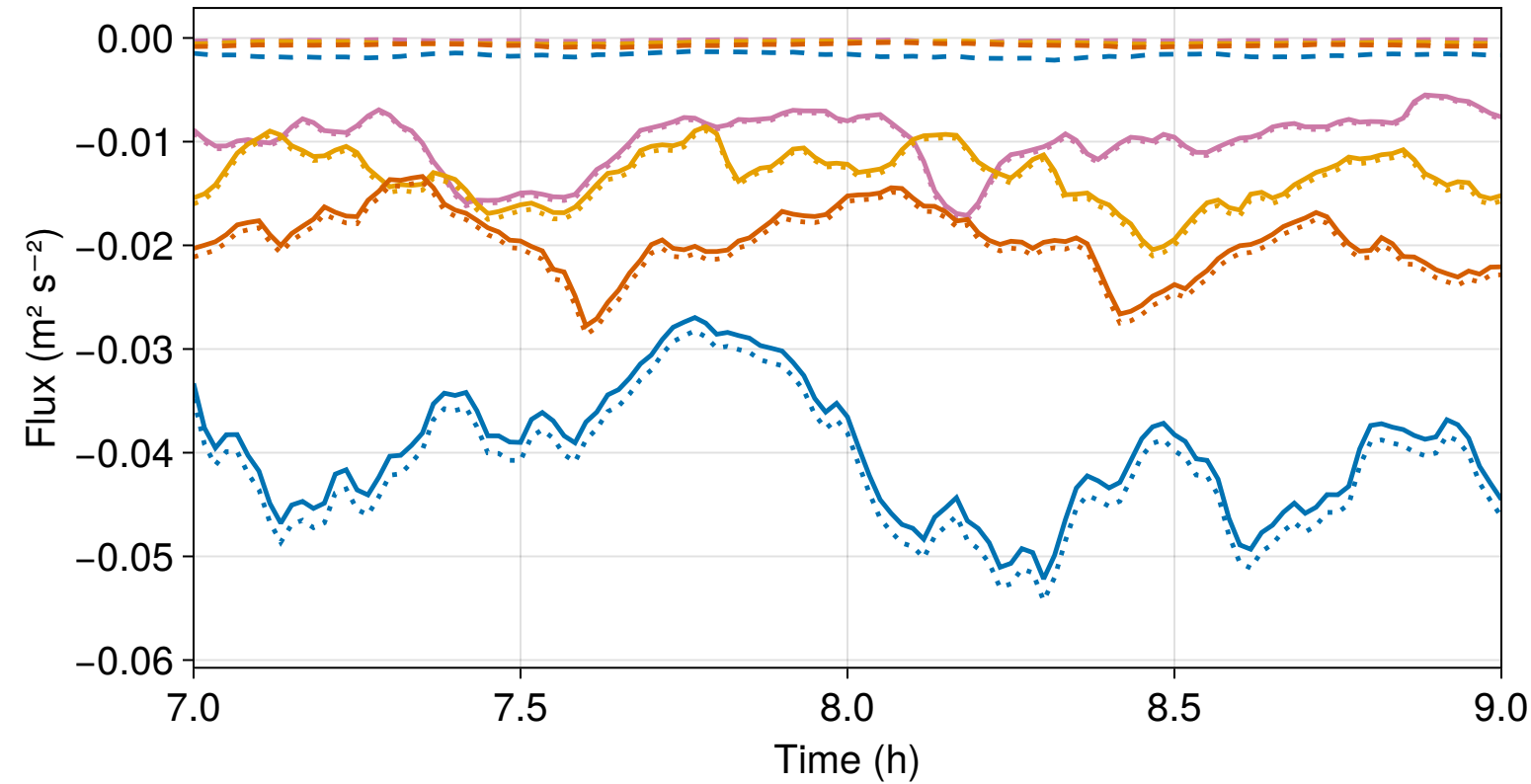
32³ · 12.5 m · WENO9 · filtered wall · scheme-native one-face SLD · $\lambda=0, 1, 2, 4$ · dashed 7–8 h, solid 8–9 h



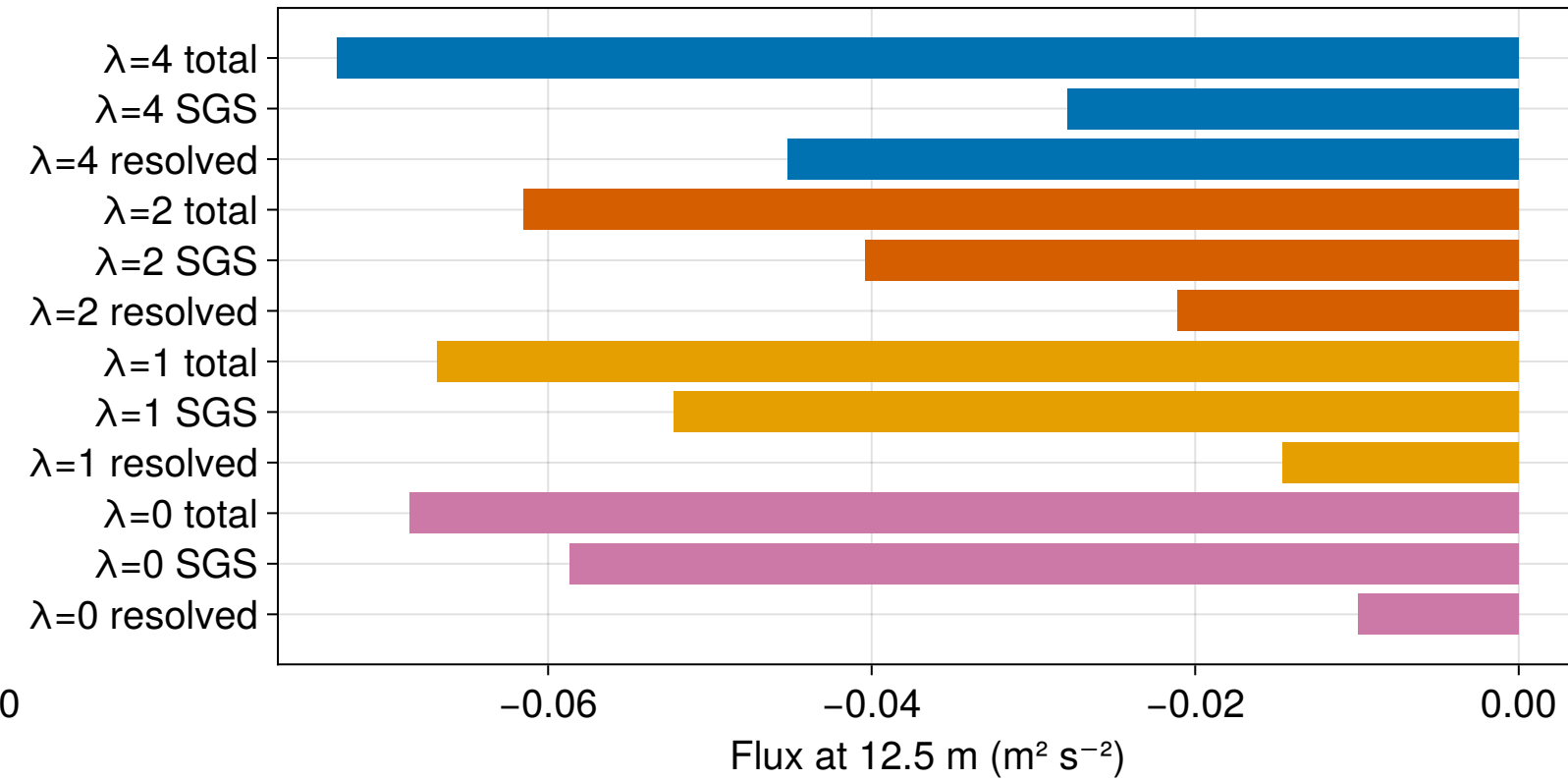
Scheme-native SLD: first-face transport partition

Reconstructed resolved = covariance + numerical correction; SGS is the SLD constitutive flux

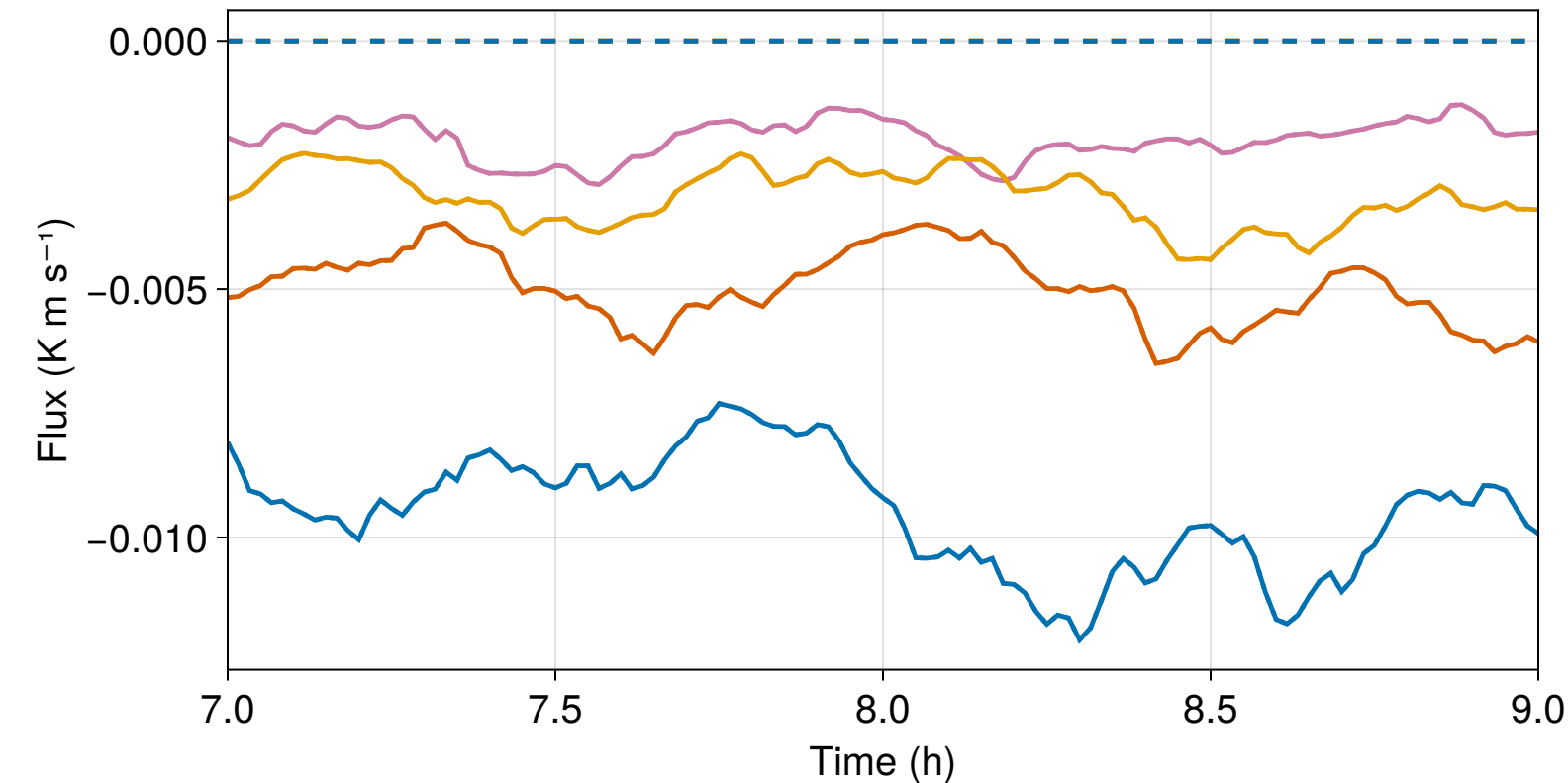
Momentum



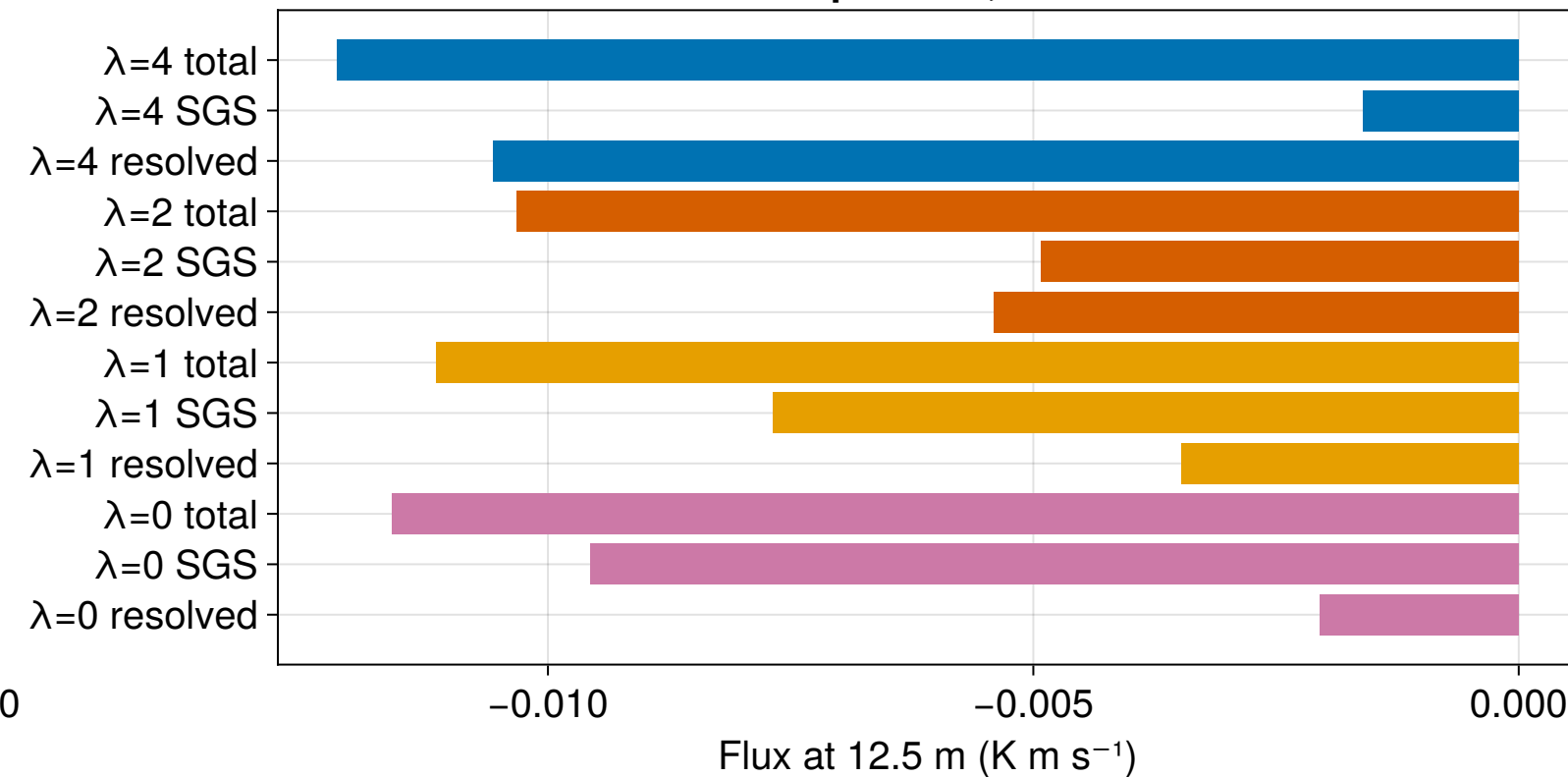
u-w partition, 8–9 h



Heat



w-θ partition, 8–9 h

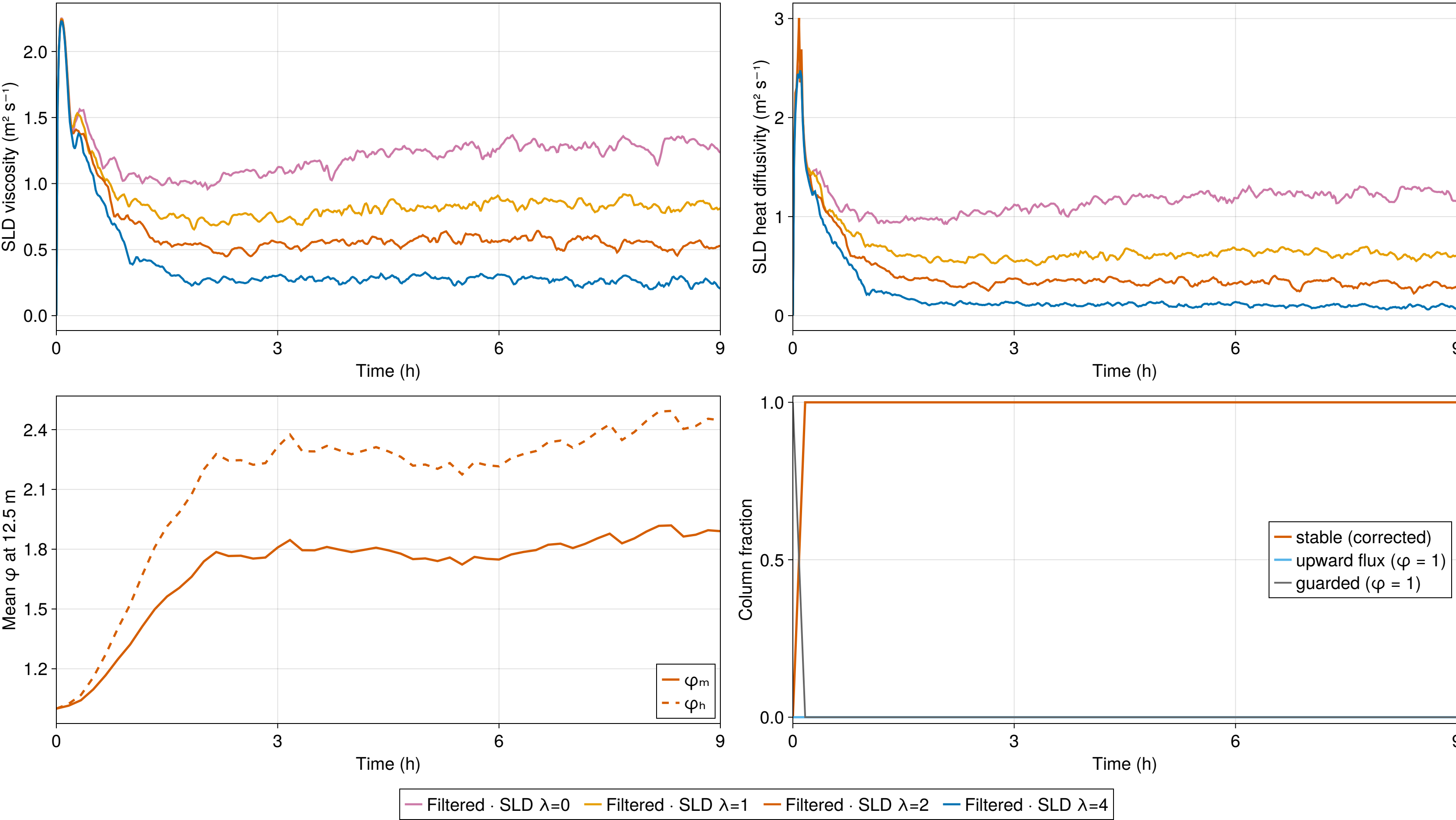


— Filtered · SLD $\lambda=0$ covariance
 - - - Filtered · SLD $\lambda=0$ reconstructed
 — Filtered · SLD $\lambda=1$ covariance
 - - - Filtered · SLD $\lambda=1$ reconstructed
 — Filtered · SLD $\lambda=2$ covariance
 - - - Filtered · SLD $\lambda=2$ reconstructed
 — Filtered · SLD $\lambda=4$ covariance
 - - - Filtered · SLD $\lambda=4$ reconstructed

— Filtered · SLD $\lambda=0$ correction
 — Filtered · SLD $\lambda=1$ correction
 — Filtered · SLD $\lambda=2$ correction
 — Filtered · SLD $\lambda=4$ correction

First-face SLD coefficients and local stability

Horizontal means of local quantities at $z = 12.5$ m; φ is never evaluated from a mean L



GABLS1: 64³ resolution study

GABLS1 32-cubed to 64-cubed resolution sensitivity | 23 September 2026. Matched nine-hour WENO9 cases use a 400 m cube, seed 123, filtered rough-wall drag, and paired initial perturbations. The gamma=2 scheme-native SLD uses a 300 s filter. Fixed 1 m LES medians are intercomparison references, not observations. Shear below is computed on common 12.5 m physical-height intervals after interpolating both u and v to coarse-grid centers; native shear is shown separately.

During 8-9 h, the 64-cubed one-face SLD shear at 12.5, 25, and 37.5 m is 0.10144, 0.05600, and 0.03951 per second. The two-face values are 0.10348, 0.05945, and 0.03972. The fixed 1 m median is 0.10038, 0.06666, and 0.05607. The second face improves 25 m agreement modestly but does not repair the 37.5 m deficit. Filtered no closure at 64-cubed nearly matches the first reference shear, 0.09953, yet is low at the next two heights. Two grids reveal sensitivity, not established convergence.

Peak resolved w-squared in 8-9 h rises from 0.05588 square meters per second squared for 32-cubed one-face SLD to 0.09688 for 64-cubed one-face and 0.08933 for 64-cubed two-face, near the 64-cubed no-closure value 0.10113. At 18.75 m, interpolated w-cubed changes from +0.000752 to -0.000777 and -0.001543 cubic meters per second cubed respectively. The fixed 1 m archive lacks a w-cubed median; this is not a fidelity ranking. Both time windows appear in the figures.

At the physical face 12.5 m, 32-cubed one-face and 64-cubed two-face have different transport partitions even though support height matches. Mean 64-cubed two-face viscosity is 0.09525 square meters per second, resolved u-w is -0.04946, SGS u-w is -0.00776, and separate WENO correction is -0.01178 square meters per second squared. The corresponding 32-cubed viscosity is 0.52303. One-face at 64-cubed acts at only 6.25 m, so its support height is different. All comparisons retain the same wall law and physical forcing.

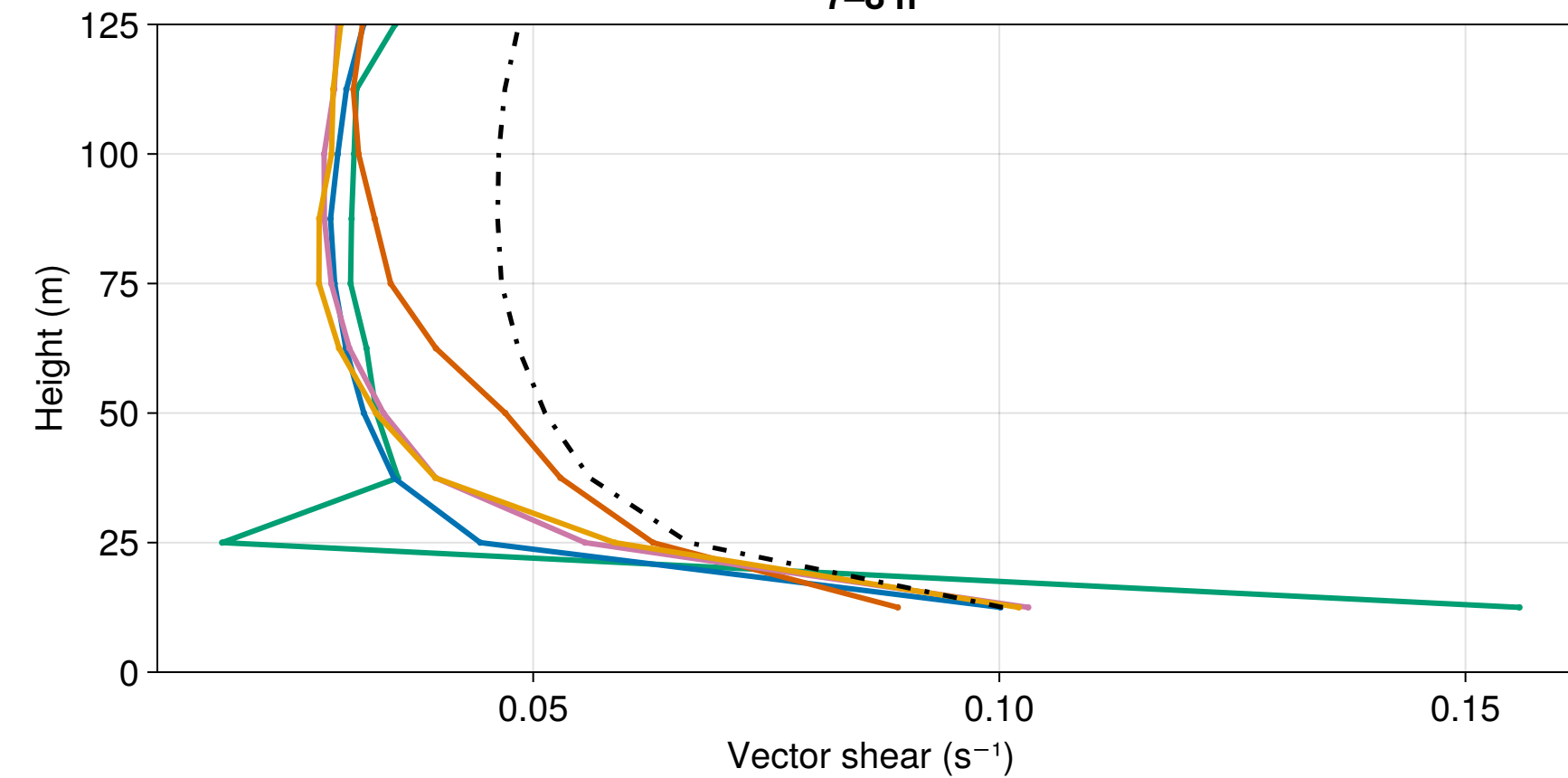
Validation: immutable 64-cubed source-manifest SHA-256

19f41d02f5bd249fe32a5edf2d8956e9886b1f58347a98907ec83b1c23d0fbd7. H100 gate 7627 passed CUDA parity and 15/15, 32/32, 38/38 case checks. Science array 7640 tasks 1-3 ended with durable exit zero and CASE_DONE at 32400 s. Independent frozen Julia audits verified source, paired initial state, native 64/65 levels, exact schedules, finite fields, evolving wall filter, density-consistent surface flux, and SLD flux identities. Both SLD cases passed 55 saved local 1/L and similarity records through nine hours. Compact exports and scripts accompany this chapter. A paired 128-cubed extension follows; current SLD supports at most two faces, so it cannot retain 12.5 m support at 128-cubed without additional code.

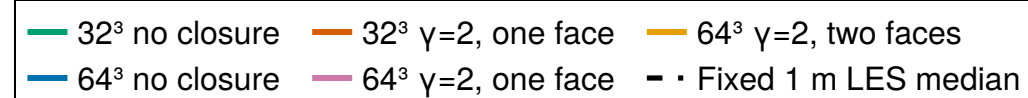
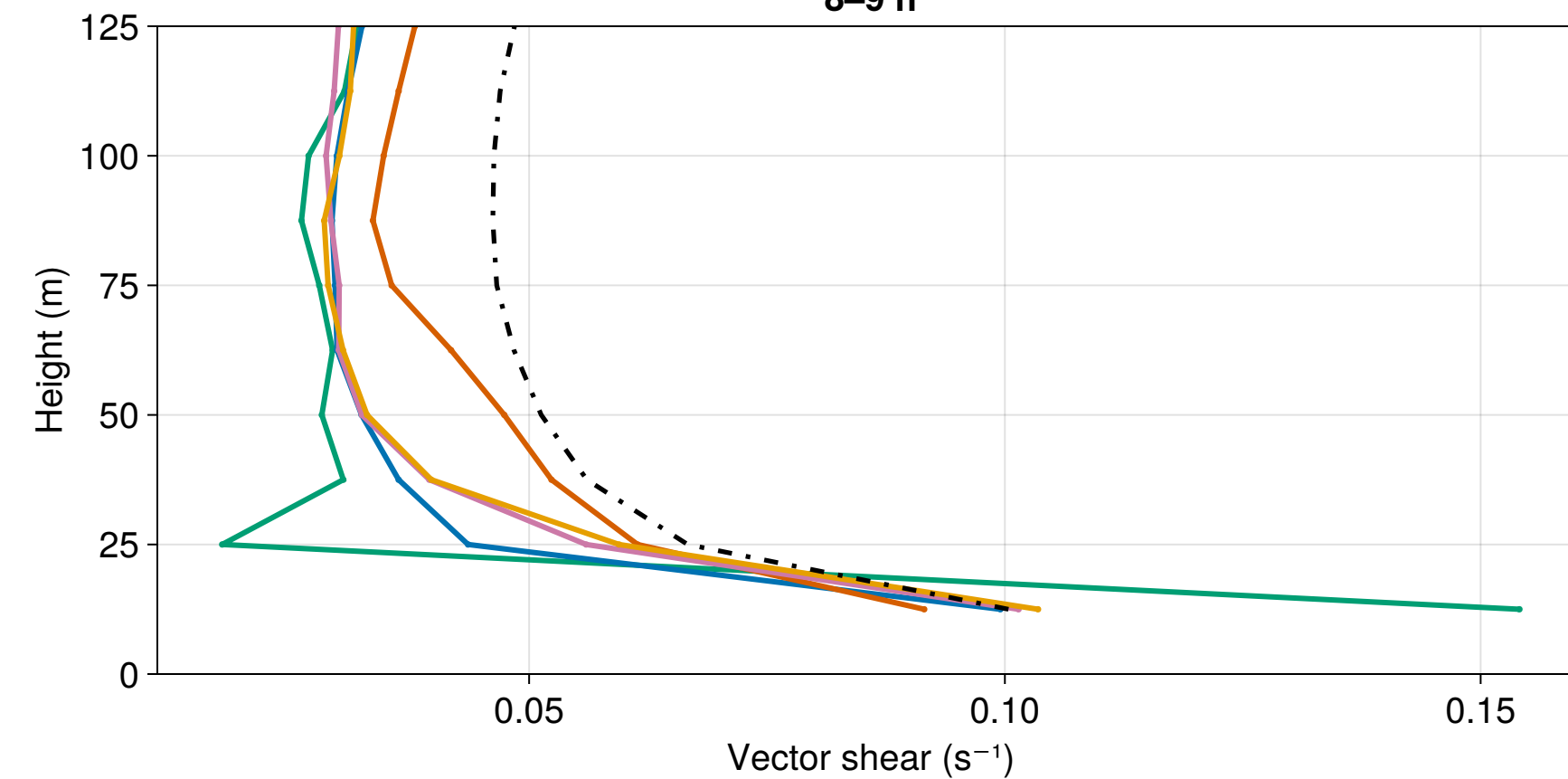
Vector shear at matched physical heights

u and v interpolated to 32^3 cell centers (6.25, 18.75, ... m), then differenced

7–8 h

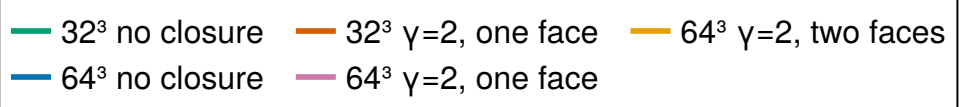
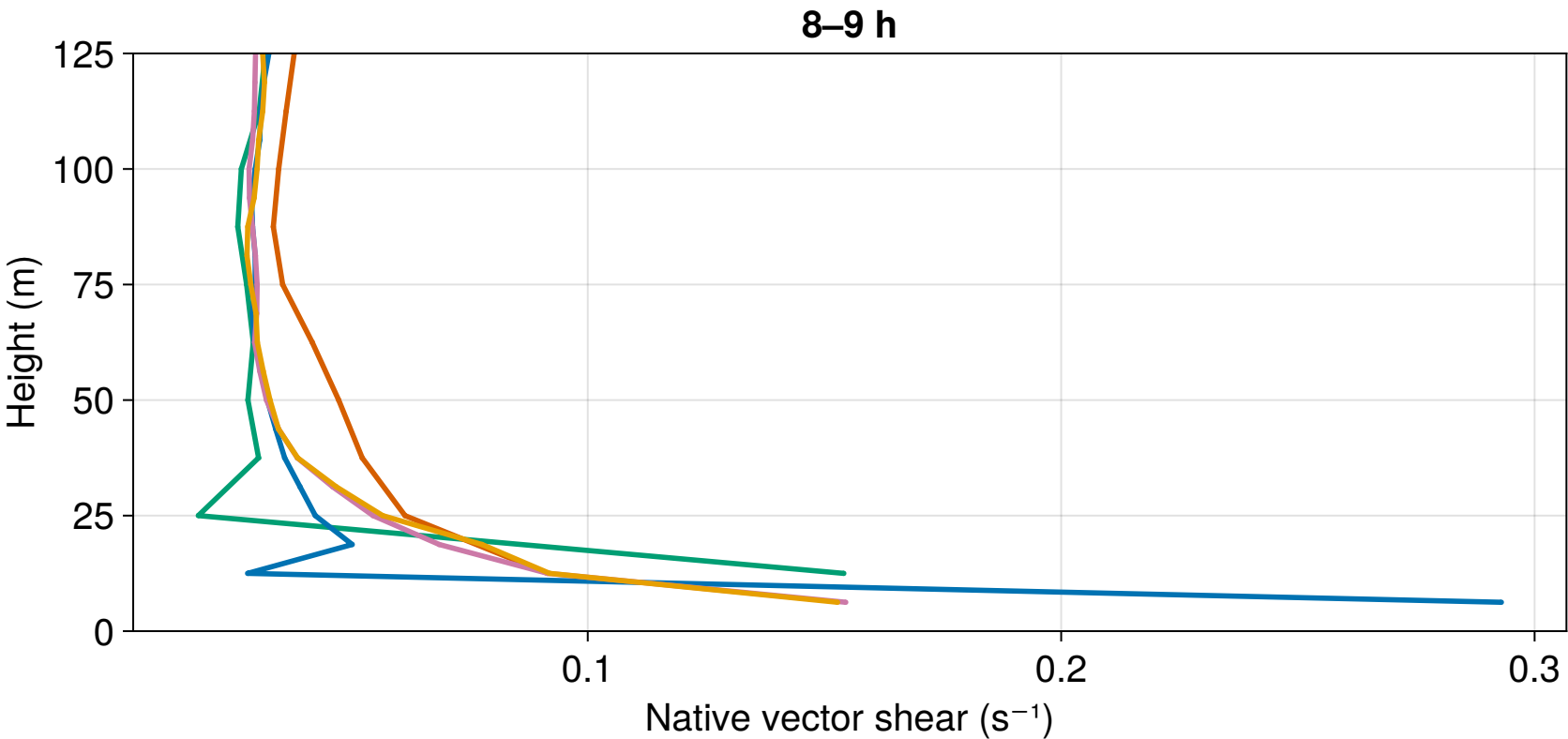
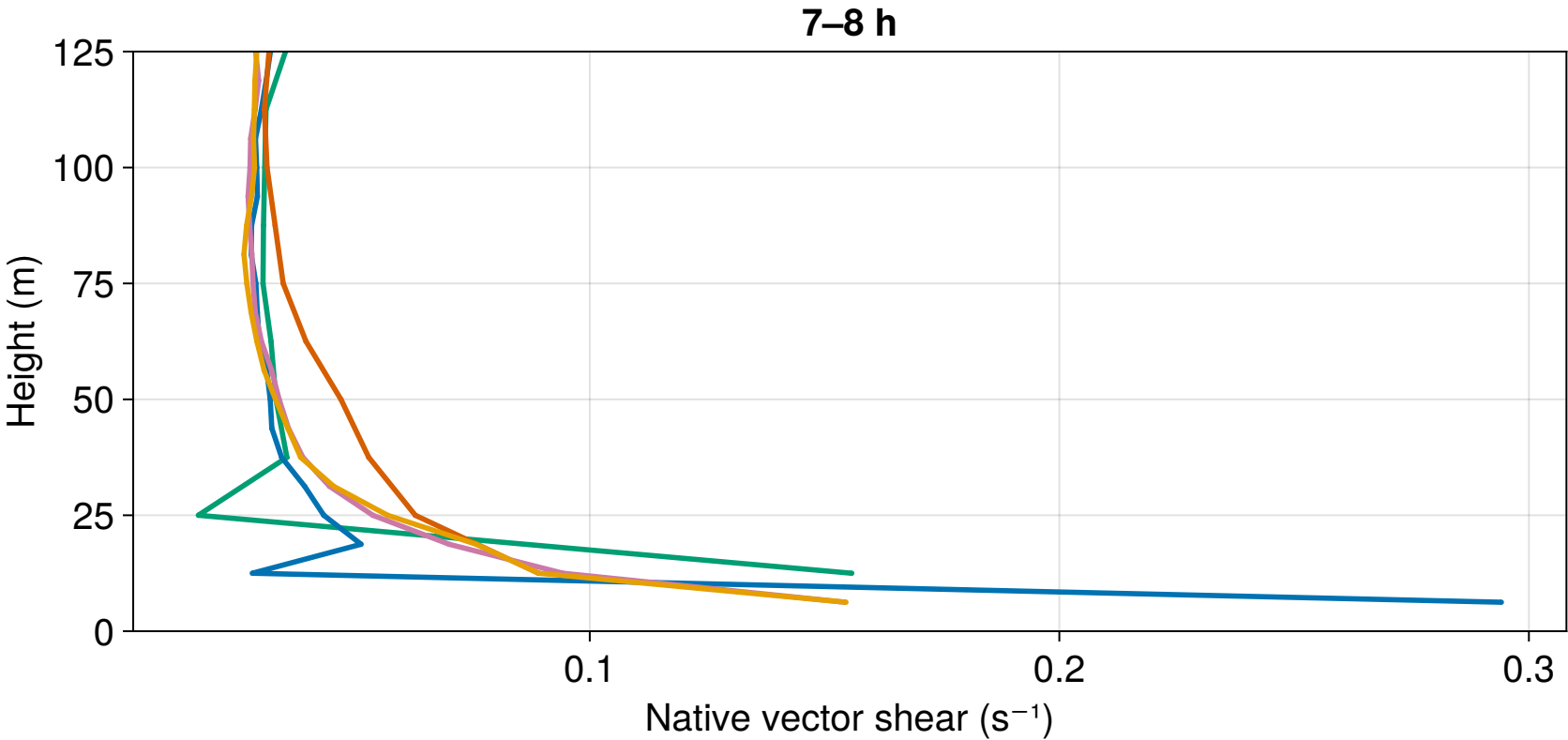


8–9 h



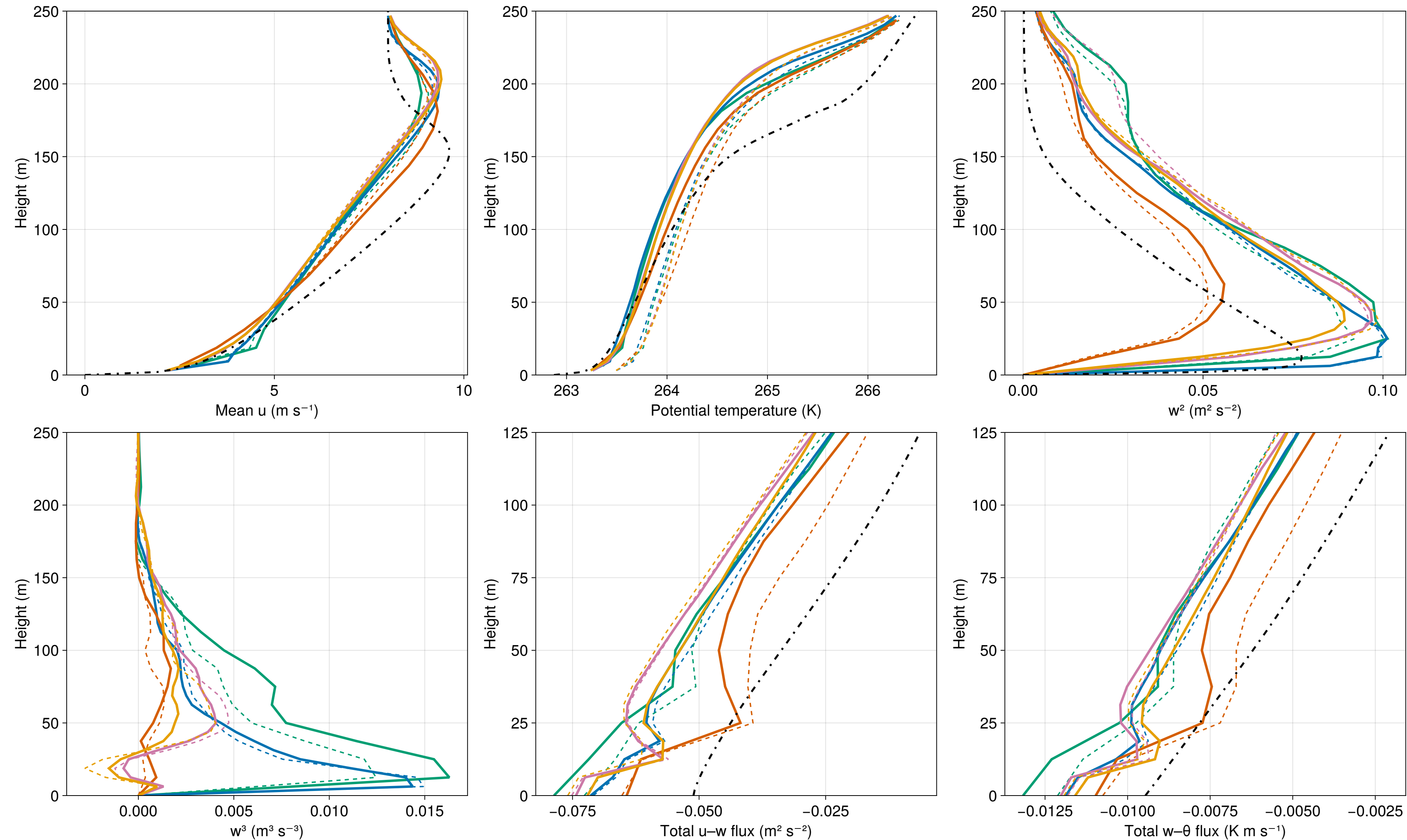
Native-grid vector shear

64³ shear spans 6.25 m; 32³ shear spans 12.5 m. Compare with matched-height figure.



GABLS1 $32^3 \rightarrow 64^3$: matched stability-corrected SLD

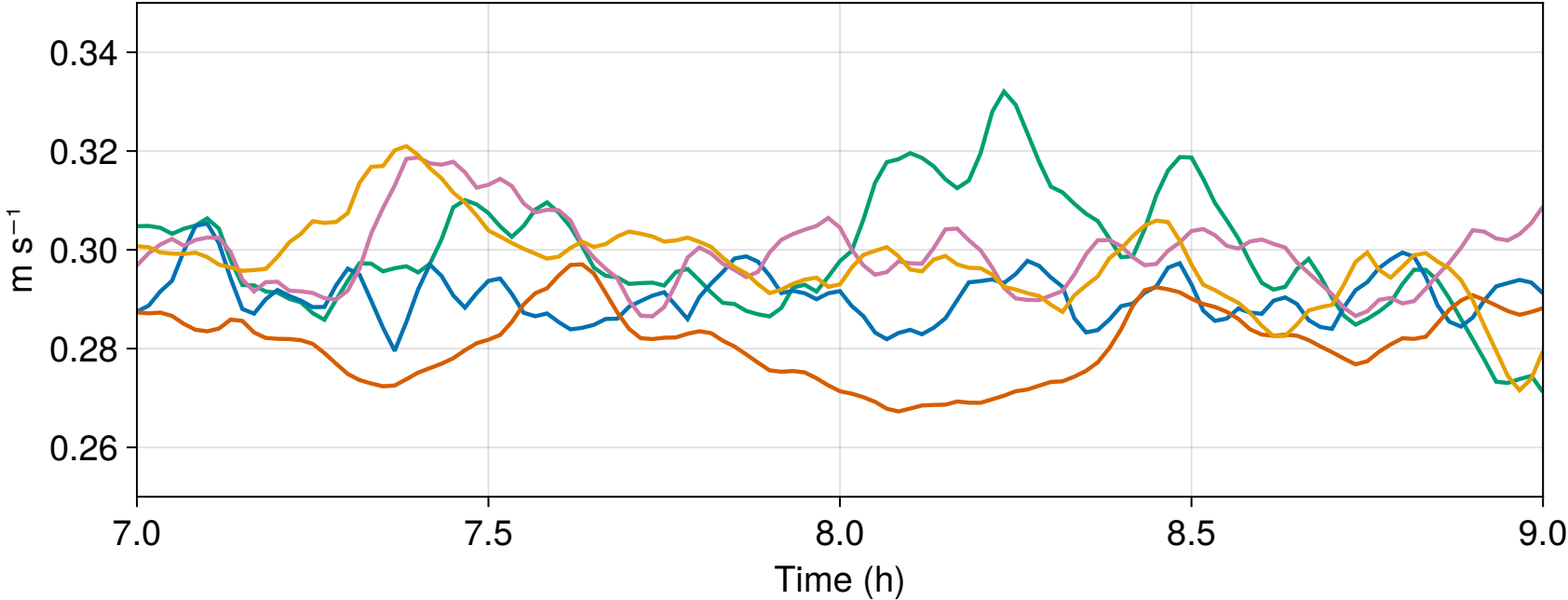
400 m cube · WENO9 · $\gamma=2$ · 300 s wall/SLD filters · dashed 7–8 h, solid 8–9 h



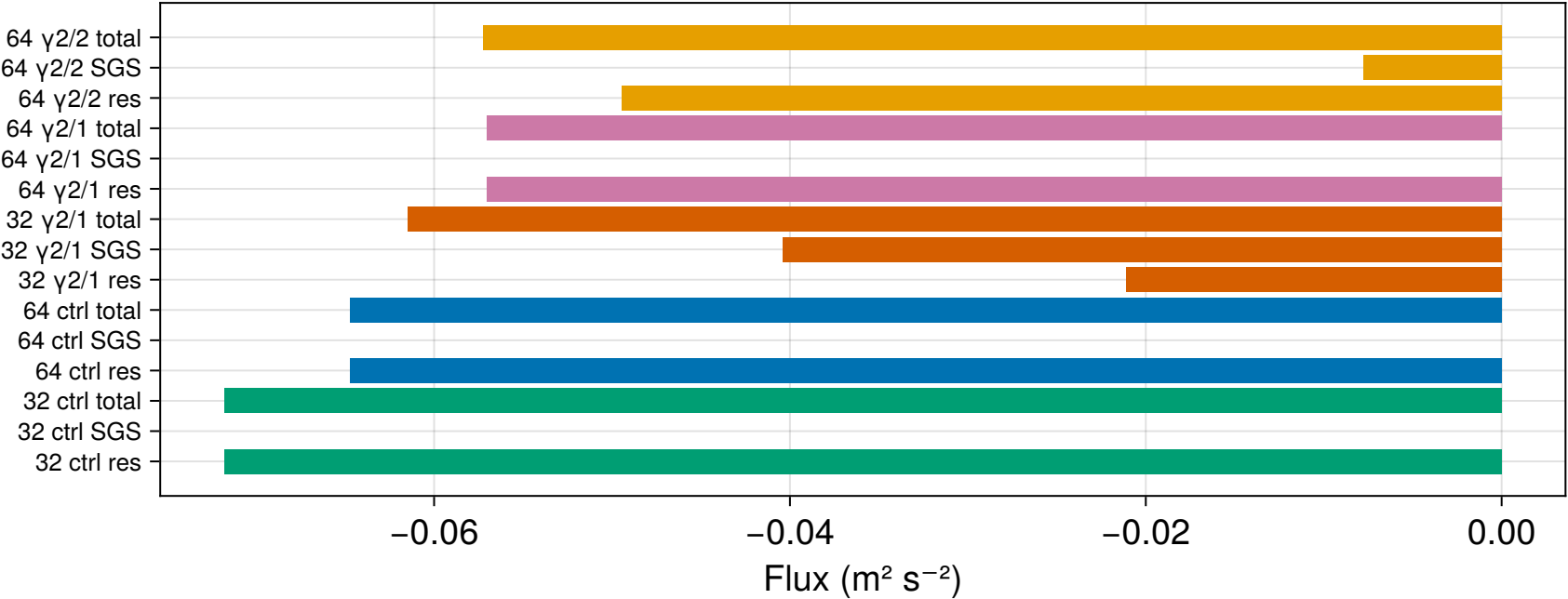
— 32^3 no closure
 — 32^3 $\gamma=2$, one face
 — 64^3 $\gamma=2$, two faces
— 64^3 no closure
 — 64^3 $\gamma=2$, one face
 - - Fixed 1 m LES median

Wall response and vertical-transport partition

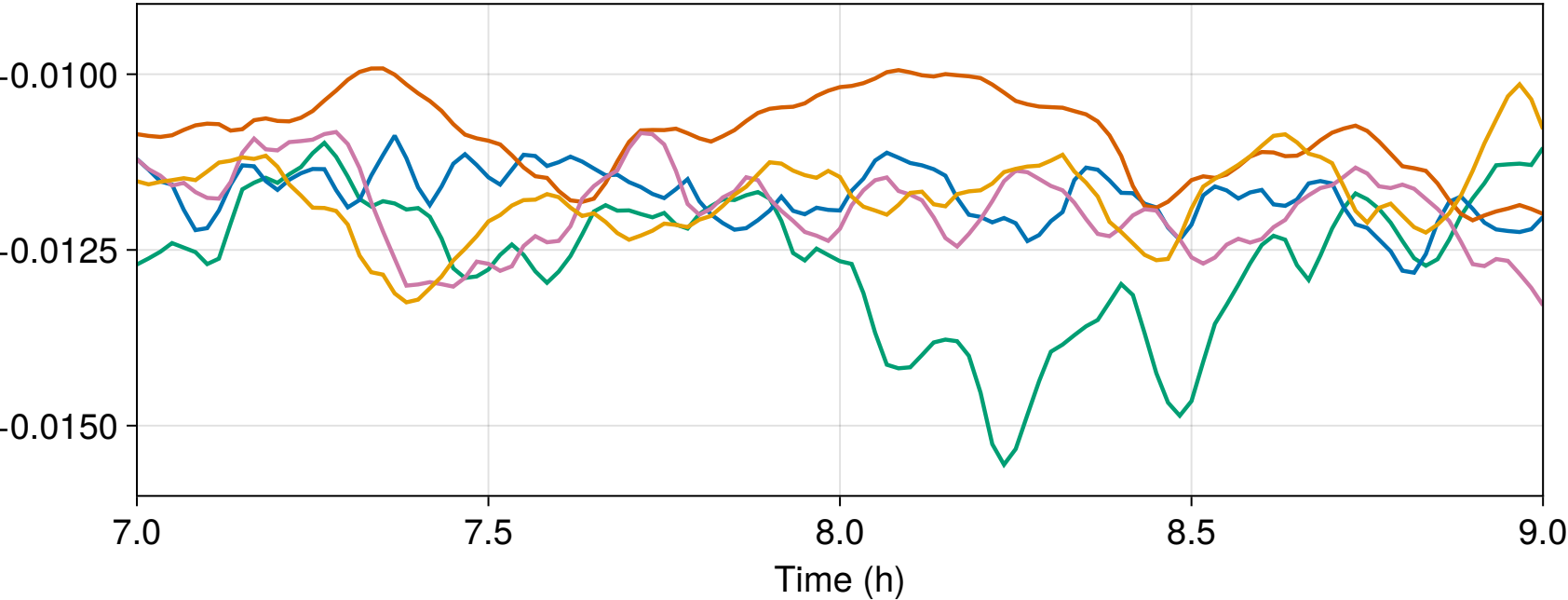
Friction velocity



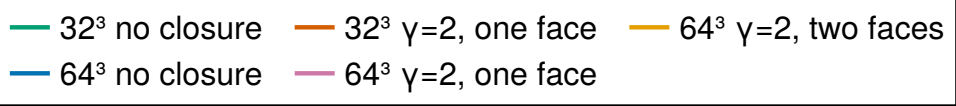
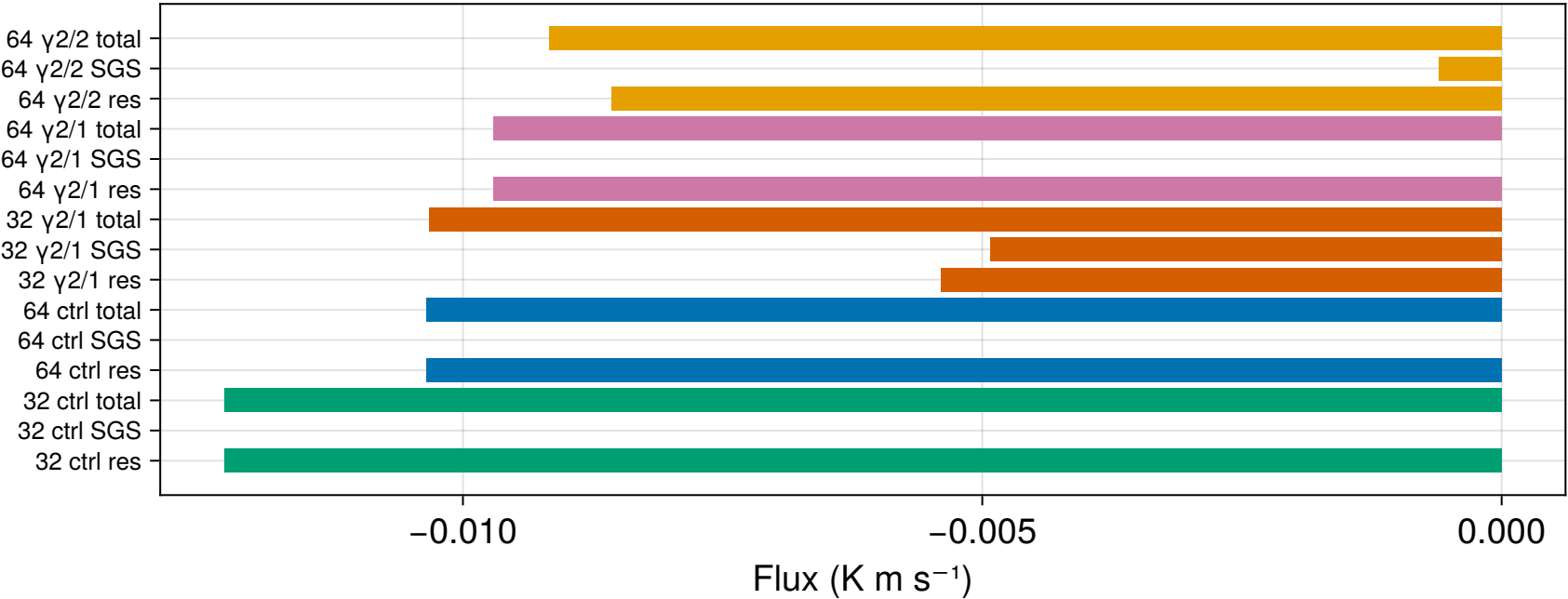
u-w at z=12.5 m



Surface heat flux



w-θ at z=12.5 m



GABLS1: 128³ resolution study

GABLS1 32-cubed to 128-cubed resolution sensitivity | 24 September 2026. Matched nine-hour WENO9 cases use a 400 m cube, seed 123, paired initial perturbations, filtered rough-wall drag, and gamma=2 scheme-native SLD with a 300 s filter. The fixed 1 m LES medians are intercomparison references, not observations. The 128-cubed grid spacing is 3.125 m. Shear below is computed on common 12.5 m intervals after interpolating both u and v to 32-cubed cell centers. Native shear is shown separately.

During 8-9 h, one-face SLD shear at 12.5, 25, and 37.5 m is 0.09152/0.06140/0.05235 per second at 32 cubed, 0.10144/0.05600/0.03951 at 64 cubed, and 0.09080/0.04759/0.03967 at 128 cubed. The fixed 1 m median is 0.10038/0.06666/0.05607. The 128-cubed two-face result is 0.09242/0.04739/0.04083. The third grid does not establish convergence or eliminate the upper-shear deficit. Both 7-8 h and 8-9 h windows appear in the figures.

Peak resolved w-squared in 8-9 h is 0.05588/0.09688/0.11556 square meters per second squared for 32/64/128-cubed one-face SLD; 128-cubed no closure gives 0.11063, and two-face SLD gives 0.11171. Resolved turbulence remains grid-sensitive. At 18.75 m, interpolated w-cubed is +0.002489 and +0.002144 cubic meters per second cubed for 128 one/two-face, compared with negative values at 64 cubed. No w-cubed median is available, so the sign change is sensitivity, not a fidelity ranking.

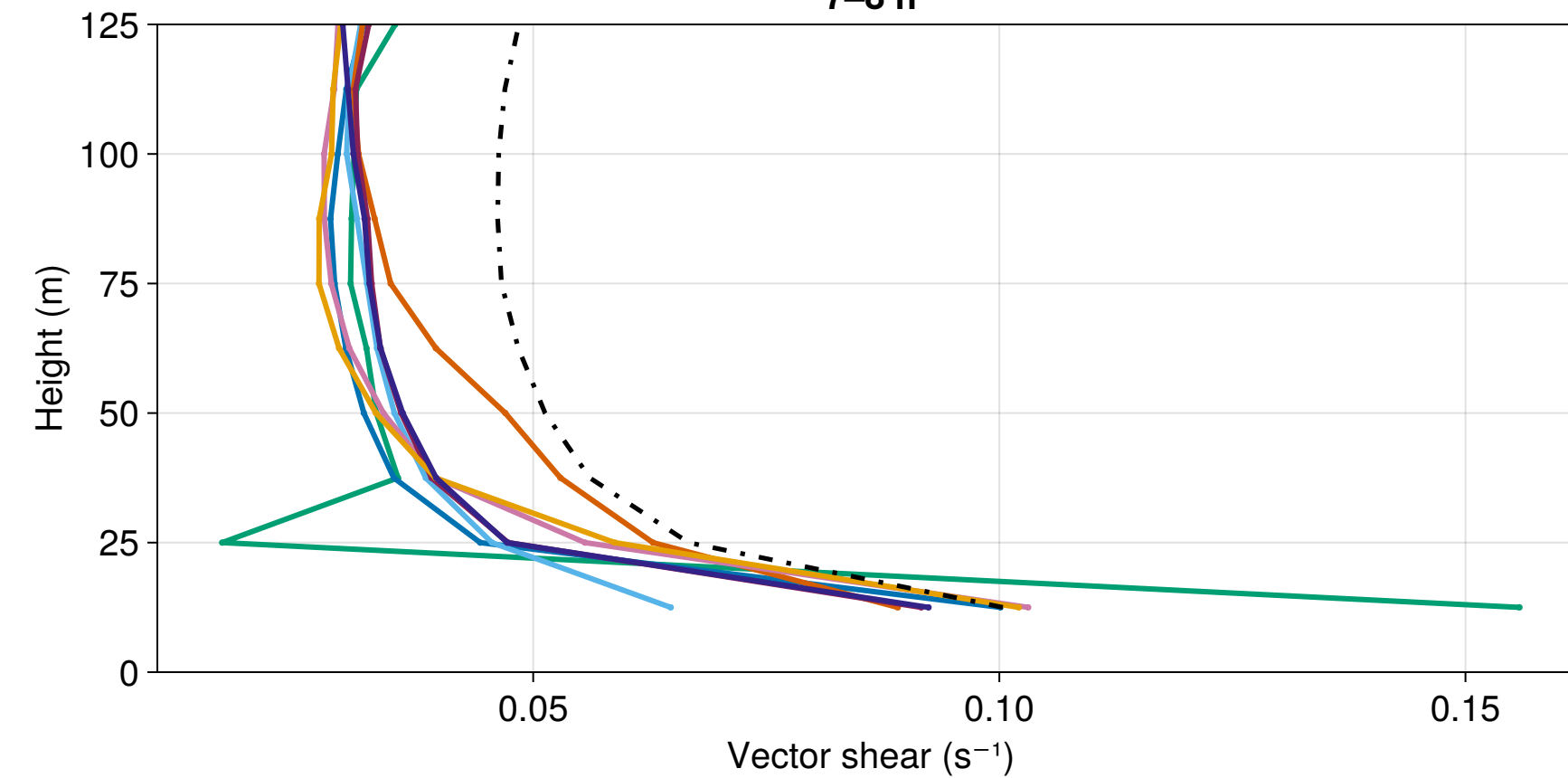
The one-face closure reaches $z=12.5/6.25/3.125$ m at 32/64/128 cubed. Two-face support reaches 12.5 m at 64 cubed but only 6.25 m at 128 cubed. The current implementation supports at most two faces; none of the 128-cubed cases retains the 12.5 m physical footprint. Refinement therefore changes closure reach as well as grid spacing. At 128 face two, mean viscosity is 0.04693 square meters per second, resolved u-w is -0.05477, SGS u-w is -0.00619, and the separate WENO correction is -0.01079 square meters per second squared.

Validation: immutable 128-cubed source-manifest SHA-256
dfdcd8846a756b73b7006b67a0bdcd565a2df5400bf769e64f9d7d5345e3e233. H100 gate 7674 passed CUDA parity34/34 and 15/15, 32/32, 38/38 short-case checks. Science array7688 tasks1-3 each ended with durable exit zero and CASE_DONE at 32400 s. Independent Julia audits verified source, paired initial state, native128/129 levels, exact schedules, finite fields, evolving wall filter, density-consistent surface flux, and scheme-native flux identities. Both SLD cases passed 55 local1/L and stability records through nine hours. An audit-only face-two height was corrected to 6.25 m; the frozen simulation was unchanged. Compact exports, scripts and evidence accompany this chapter. Three grids show a trend, not a formal convergence proof.

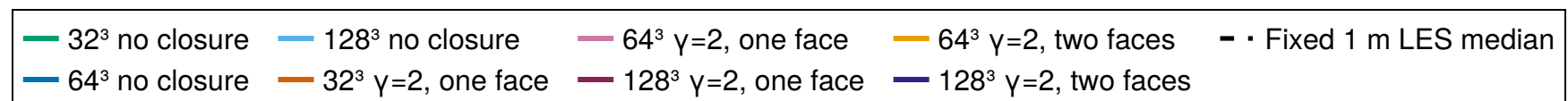
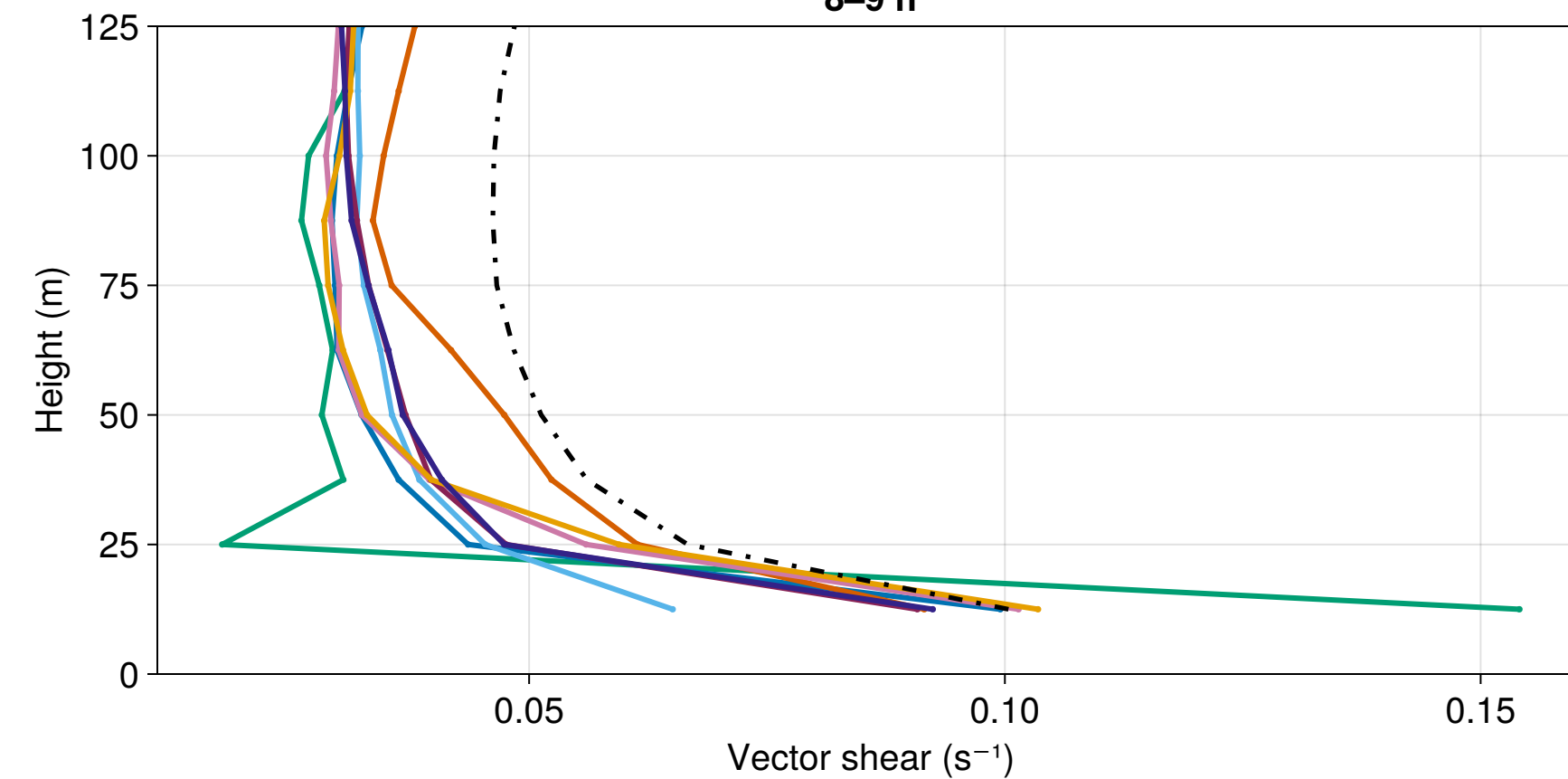
Vector shear at matched physical heights

u and v interpolated to 32³ cell centers (6.25, 18.75, ... m), then differenced

7–8 h



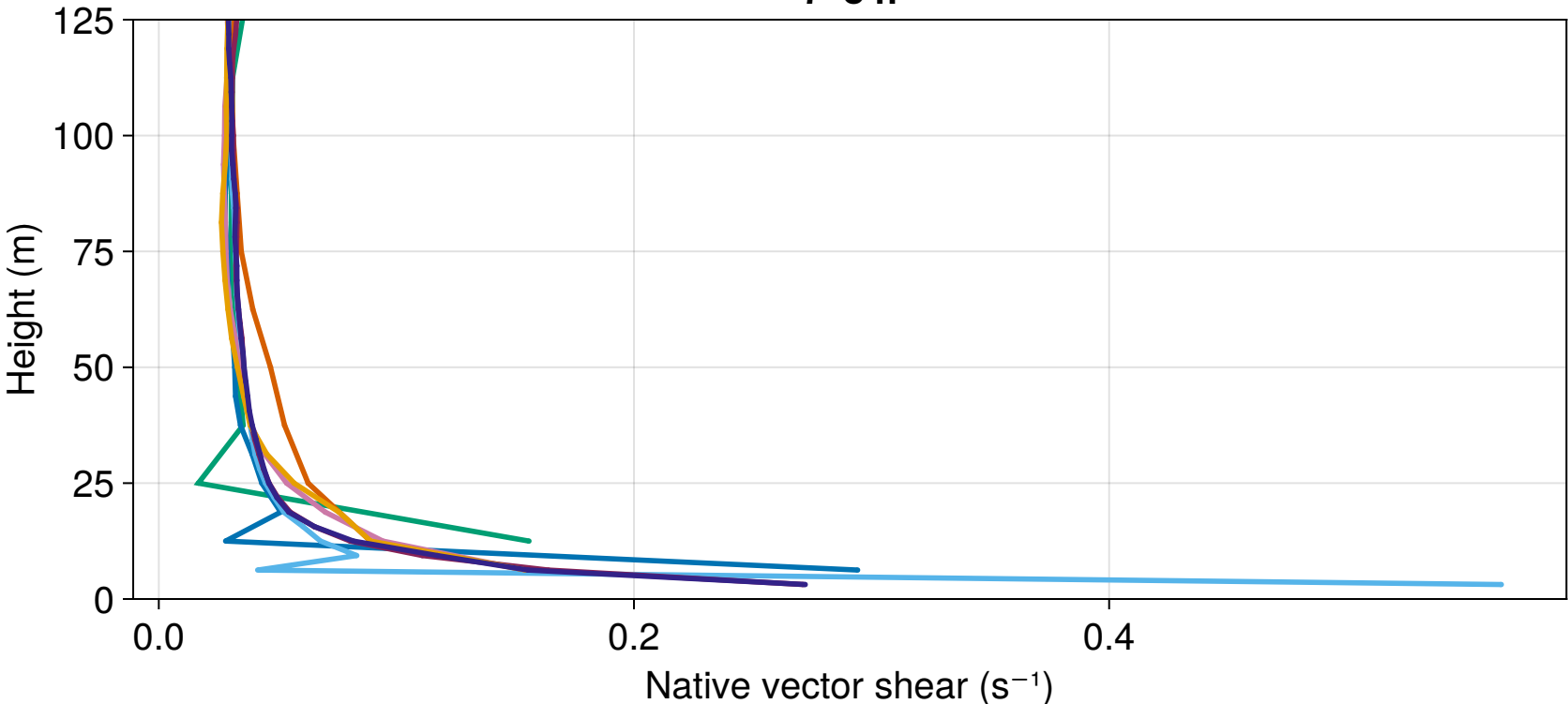
8–9 h



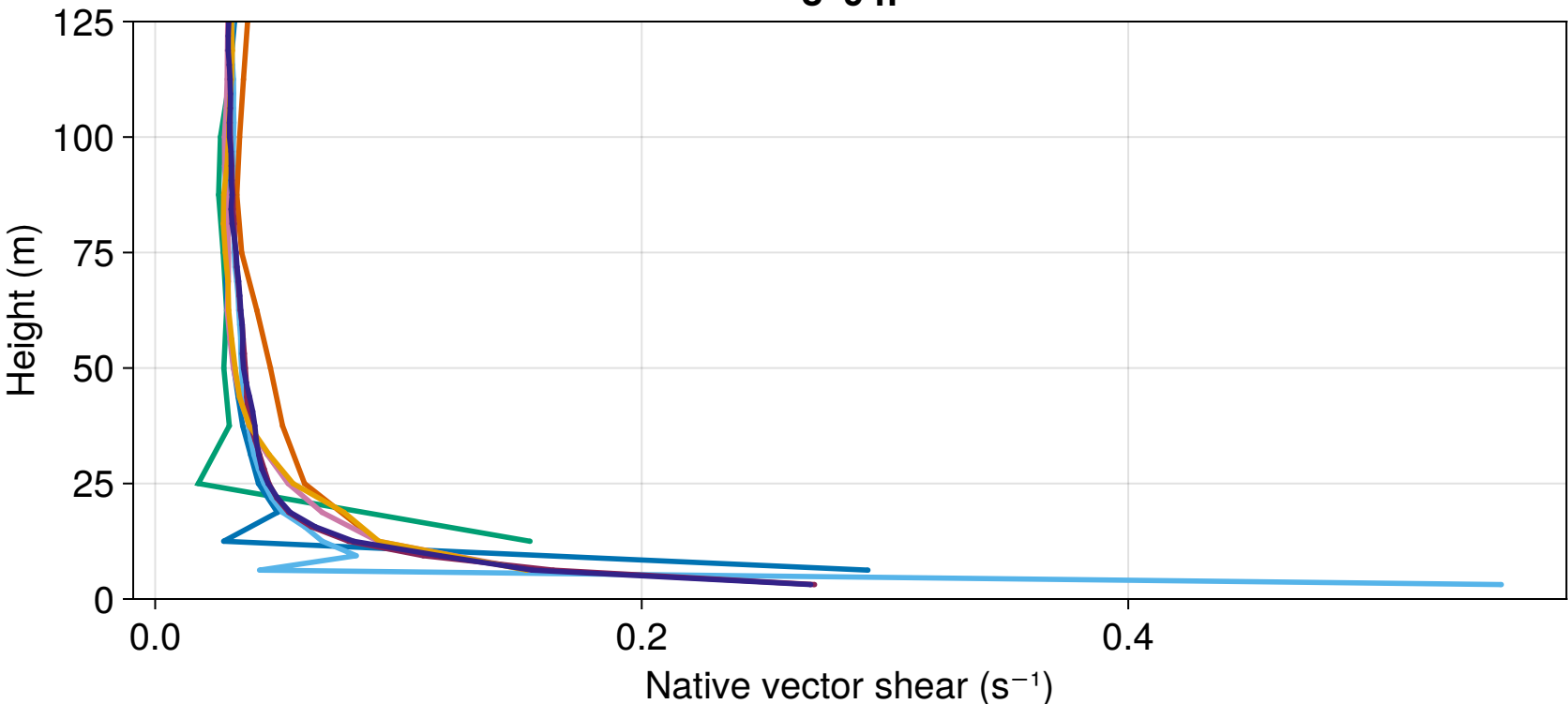
Native-grid vector shear

128³/64³/32³ native differences span 3.125/6.25/12.5 m; compare common-height shear.

7–8 h

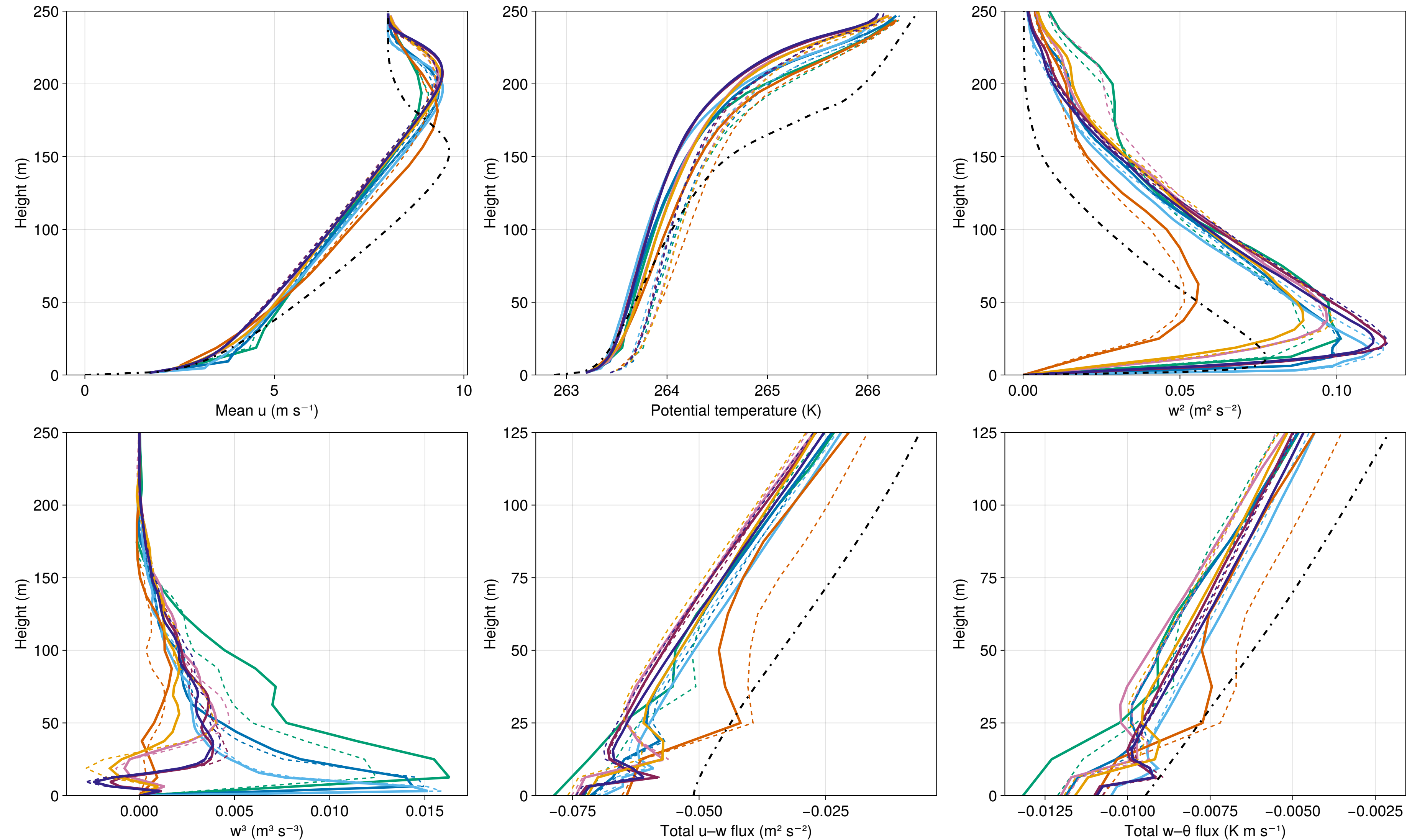


8–9 h



GABLS1 $32^3 \rightarrow 64^3 \rightarrow 128^3$: matched stability-corrected SLD

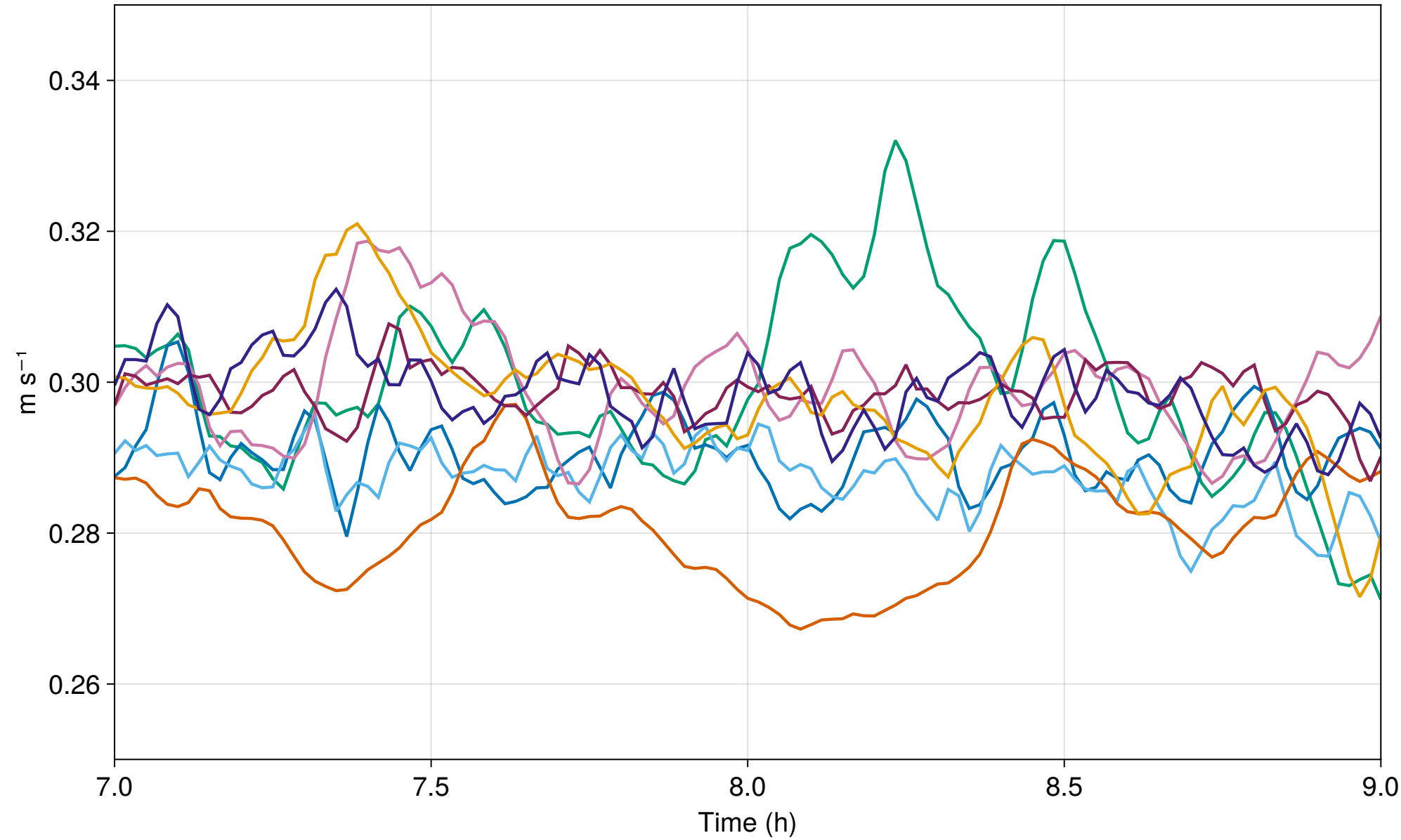
400 m cube · WENO9 · $\gamma=2$ · 300 s wall/SLD filters · dashed 7–8 h, solid 8–9 h



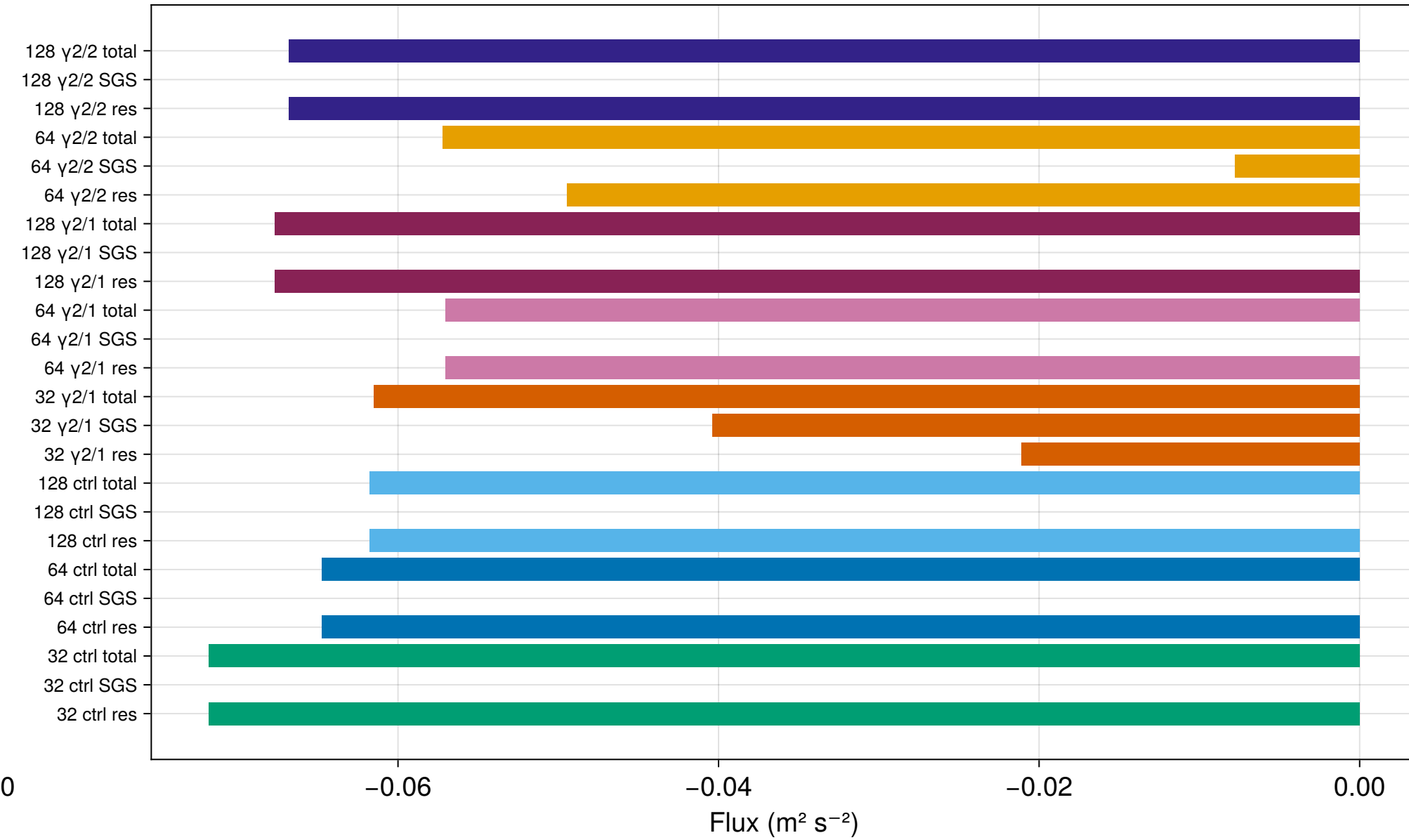
Wall response and vertical-transport partition

Common face $z=12.5$ m · 128³ SLD support ends at 3.125 or 6.25 m

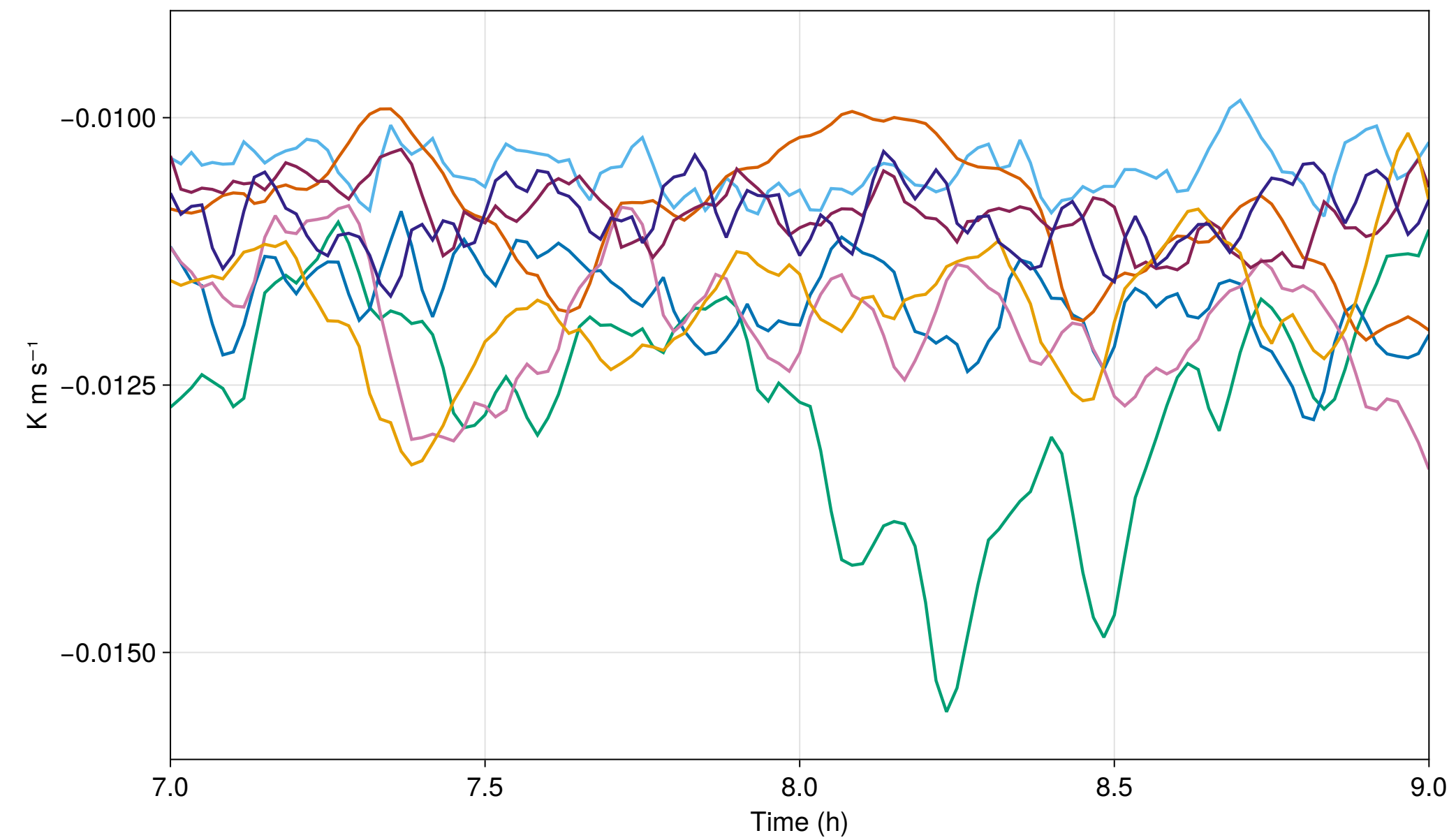
Friction velocity



$u-w$ at $z=12.5$ m



Surface heat flux



$w-\theta$ at $z=12.5$ m

